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SECOND EDITION

GEOFF CHAPLIN

Credit Derivatives

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Credit Derivatives

Trading, Investing and Risk Management

Second Edition

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WILEY

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To my parents, my partner and my children

Contents

Preface to the First Edition

This book arose out of several courses and training sessions given to finance professionals. Those courses, and this book, are aimed at traders in the credit and other derivatives, quants wishing to develop some product knowledge in this area, risk managers including the corporate treasurer, investors and others requiring both product knowledge and a grasp of valuation and risk. The aim is to develop an understanding of the various credit derivatives products. In order to achieve this it is necessary for the reader to have a certain amount of financial background – and a certain level of maturity when it comes to understanding structured products and the management of a portfolio of risks. I have tried to cover key credit background in Part I of the book. Readers with experience in this area can skip through most of the topics covered here – though generally the role of repo, and the difference between asset swap spreads and z-spreads, do not appear to be well understood, and the reader is advised to study these sections at least. Other areas that may be unfamiliar are the calibration of transition matrices to market data (spreads and volatilities), and the generation of correlated spread moves using the Normal 'Copula'.

The most common credit derivative product is the credit default swap (CDS), and this is covered along with other 'single name' credit derivatives in Part II. A relatively simple 'deterministic' pricing model is described, together with a stochastic model in section 9.4. A model is of little use without data, and various data-related issues are discussed, together with detailed analysis of model shortcomings, potential improvements, and the setting of reserves. Chapters 10 to 12 analyse a variety of trades: credit curve trades, cross-currency hedges, spread and event risk, valuation and unwind of default swap positions, and a detailed breakdown of the basis between CDS premiums and bond spreads. Non-vanilla default swaps, options, and total return swaps are also covered in Part II. Chapter 17 draws together many of the aspects of valuation, risk calculation and hedging with a discussion of book management applied to the single name credit risk book. Finally we end this section with a look at pricing single name default swaps via default time simulation. This introduces some of the techniques used in analysing portfolio products in the relatively simple context of default swaps.

Part III looks at nth-to-default products and CDOs. The latter form a very extensive topic in their own right: I have given a descriptive analysis of many of the different types of CDO structure, and product applications though, when it comes to using the pricing model, we concentrate on synthetic static CDOs (including standard CDO products such as iTraxx). Nevertheless the model described is appropriate to both the standardised products and the more complex waterfall structures seen in some cashflow CDOs. Examples are given to develop an intuitive understanding of the correlated 'default time' modelling approach, and of default time correlation. This part quotes valuation results extensively to illustrate important points. In addition, software provided allows the user to reproduce quoted results and to experiment with different portfolios, structures and situations to learn more about both the product and the consequences of the modelling approach. Product risks and model deficiencies are discussed in detail as is the question of reserves. A section is devoted to the correlation matrix. The preferred interpretation of the model is as an 'interpolation rule', driven by market prices – giving rise to 'implied correlation' – and then applied to non-traded products. Implied ('compound') correlation and 'base' correlations are covered here. Alternative Copulas are discussed and applied. This part concludes with a discussion of the management of a portfolio of 'correlation' products and high- and low-risk hedging techniques.

We revisit the topic of 'single' name credit default swaps in Part IV where we take account of counterparty risk. In addition, the topic of protection on counterparties is briefly covered.

In writing the book I have been conscious that CDO products, and methods of analysis, have been developing rapidly. Although the book is illustrated using some of these products, their precise form has changed in the past and will continue to change in the future. I have not been able to cover all aspects of credit derivatives – neither equity default swaps nor constant maturity default swaps have been discussed, and some model approaches have not been covered (for example 'claim of market value' models, and 'blended correlation'). I have chosen to concentrate on core products, general principles and features that are likely to continue to apply in the future. Nevertheless the book reflects the situation at the time of writing (late 2004): products, methodologies and interpretations will evolve.

Throughout the book I have tried to illustrate ideas and principles using real world examples. In addition there are a variety of thought experiments and exercises to encourage the reader to test theories against real life, and to develop a deeper understanding of the various concepts through their application. A prerequisite to effective application and management of credit derivative positions is an understanding of the products, and an ability to discuss those products in a meaningful way. This book is intended to help in this process and is therefore not aimed solely at the trader, or the risk manager, or the investor, but is intended to be read by all. Consequently I have tried to separate the mathematics from product discussion and application. Of course a good understanding of one cannot be achieved without some understanding of the other. To this end I have used as simple a version of the theory as possible, yet one which is capable of capturing the key aspects of the problem. Generally simplifications amount to ignoring the details of premium payment (frequency and day-count conventions) and often using a constant hazard rate curve. I have also given many calculated results to illustrate points arising out of the theory, and provided software to allow the reader to examine the implications of the model.

Preface to the Second Edition

In writing the second edition I have tried to keep to the objectives and style of the first edition while updating the content and, additionally, broadening the standpoint from which the book is written. Specifically, the examples have been updated and enhanced, and Parts I and II have been amended where relevant to bring them up to date. Additional text has been included to cover syndicated loans, LCDS, up-front premiums, and other changes to the CDS market, and further background has been included in Part I on terminology, investment perspectives and the ‘credit crunch’. Part III has been restructured and largely rewritten, including material on LCDOs, cashflow CDOs and structurings as well as significantly enhanced descriptions of the standard and bespoke CDO products and valuation and risk methodology. New sections (Part V and VI) have been added covering aspects of model testing and implementation, and the credit crisis respectively.

Acknowledgements

In preparing this second edition I have also asked for input and different perspectives from several individuals, and their contributions have been included. In particular, my thanks go to my colleagues Robert Reoch and Robert Baker (Reoch Credit Partners LLP), Darren Smith (WestLB) and Onur Cetin (Calypso) for considerable contributions included as separate chapters or sections in the book and attributed to them. I would also like to thank Markit Group Limited for their substantial assistance, and, again, Bloomberg, Moody's and ISDA. Contributed material and helpful comments in the first edition continue to be reflected in this edition and again I thank JP Morgan, Mizuho International, Morgan Stanley, Richard Flavell, Phil Hunt, Lee McGinty (JP Morgan) and Chris Finger (RiskMetrics Group). I would also like to thank Dr Jonathan Staples (fund manager, Charles Taylor Investment Management Company Limited) and Rafik Mrabet (Structured Credit Trading, Mizuho International) for their significant help in collecting data and research material and in reviewing parts of the book.

Remaining errors are my responsibility and the views expressed are entirely my own.

Disclaimer

1. Use of company names in examples and illustrations in no way indicates the actual involvement of that company in the example deal or any similar deal. Use of such names is purely for a sense of realism.
2. Software is provided to illustrate certain points in the text only. It is in no way intended for commercial use and, indeed, commercial use of the software, examples, or underlying code in any form is strictly forbidden.

Most examples are in Excel (2002) spreadsheets, some requiring the dll.

A few applications are in MathCad[1] sheets (and run under MathCad 11). Sometimes the code and the application are in separate sheets. MathCad users will need to copy these sheets to a single directory and re-establish the reference to the code sheet in the application sheet. For non-MathCad users the sheets are also copied in rtf format so that the code can be read and converted to their preferred 4GL.

INSTRUCTIONS FOR THE 'NDB PRICER' AND THE 'CDO PRICER'

- These spreadsheets require the 'Xlcall32.lib' and the 'bookdemo.xll' to be installed in the directory in which the spreadsheets are run. The user will also need to create the directory '"C://CDO intermediate output/'.
- All other Excel applications should be closed down and calculation should be set to manual (with no 'recalculate on save'). Do not subsequently open up any other spreadsheets unless they are also set to manual calculation – doing so can cause Excel to recalculate the CDO sheet and this will lead to a crash.
- Begin by opening the xll and enabling macros.
- Open the spreadsheet (either CDS or NDB) enabling macros, and go to the 'CDS calibration' sheet. Enter data in the yellow areas only – Name, Currency, Seniority, Notional Size, rating, CDS premium, and recovery rate. Hit shift-F9 to recalculate the sheet. Implied default rates and survival probabilities are shown in AN-BF.

[1] MathCad and MathSoft are registered trademarks of Mathsoft Engineering and Education, Inc., http://www.mathsoft.com.

NDB Pricer

- Enter the NDB sheet
- Set up the effective date and maturity date in C7 and D7. Enter cell E7 and recalculate that cell only.
- Enter the interest rate in cell F7 (either 0 or 4).
- Enter the number of names in the basket (2 or 3 only) in cell C11. Click the 'setup number of names' button.
- From the CDS calibration sheet copy the first 2 (or 3) names, currency, seniority and spreads and 'paste special / values' into cells B23–E24 (B23–E25).
- Cells C13 to C19 have been disabled.
- Choose the Copula required (cell C20).
  - Normal or Student's: enter a single correlation number in cell F15 and click 'set up parameters'. This creates a 2 × 2 (3 × 3) correlation matrix with a single non-trivial correlation. (Alternatively – for 3 names – a correlation matrix can be pasted into cells I23–K24).– Student's: enter the degrees of freedom in cell F17.
- Select the number of simulations required in cell B7. (A total of one million simulated defaults can give quite accurate pricing – 500,000 simulations for 2 names.)
- Click the 'Price'. DO NOT HIT SHIFT-F9.
- Cell E9 will produce identical sequences of random numbers if set to –1, different sequences if set to +1.
- Results appear in cells J8–O10 for the first, second (and third) to default.
  - Col. J: contingent benefit = expected loss = non-refundable up-front premium (% notional in C18)
  - Col. K: the value of a premium stream of 10,000 bp p.a. (% notional)
  - Col. L: current fair value premium
  - Col. M: if col. I contains the actual deal premium then col. M gives the net value of the position (% notional, excluding accrued)
  - Col. N: standard deviation of expected loss
  - Col. O: standard deviation of premium

CDO Pricer

- Enter the CDO sheet
- Set up the effective date and maturity date in C7 and D7. Enter cell E7 and recalculate that cell only.
- Enter the interest rate in cell F7 (either 0 or 4).
- (Enter the number of names in the basket (100 only) in cell C11. Click the 'setup number of names' button.
- From the CDS calibration sheet copy the 100 names, currency, seniority, size and spreads and 'paste special / values' into cells B23 × F122.
- Cells C13 to C19 have been disabled.
- Choose the Copula required (cell C20).
  - Normal or Student's: enter a single correlation number in cell F15 and click 'set up parameters'. This creates a 100 × 100 correlation matrix with a single non-trivial correlation. (Alternatively a correlation matrix can be pasted into cells I23–DD122).
  - Student's: enter the degrees of freedom in cell F17.

- Select the number of simulations required in cell B7. (A total of one million simulated defaults can give quite accurate pricing [depending on tranche] – 10,000 simulations for 100 names.)
- Define the tranche names and sizes in cols J and K. Choose waterfall 1 for column L. If the tranche premiums are known enter them into col. M.
- Click the 'Price'. DO NOT HIT SHIFT-F9.
- Cell E9 will produce identical sequences of random numbers if set to –1, different sequences if set to +1.
- Results appear in cells N8–V17 for the tranches.
  - Col. N: contingent benefit = expected loss = non-refundable up-front premium (% notional in C18)
  - Col. O: the value of a premium stream of 10,000 bp p.a. (% notional)
  - Col. P: current fair value premium. For the equity piece the premium may be a mix of an up-front amount (e.g. 500 bp) – the net value then gives the up-front premium (as a percentage of notional).
  - Col. Q: if ccol. M contains the actual deal premium then col. Q gives the net value of the position (% notional, excluding accrued)
  - Col. R: standard deviation of new premium
  - If 'write output?' is set to 1 (cell I3) then simulated default times are written to 'C:CDO intermediate output' (you will need to create this directory manually). This slows down run time.

APPLICATION RESTRICTIONS

The software is intended only to illustrate points made in the text. To restrict its possible commercial application the following limitations are imposed.

1. Only 0 or 4% flat interest rate curves
2. Only flat hazard rate curves
3. Only continuous CDS and portfolio product premiums
4. No standard maturity or coupon dates have been catered for
5. The basket pricer will only handle two or three name baskets, and the CDO pricer will only handle 100 reference names
6. The following are disabled
   (a) Hedge calculations (except for the default event hedge of 22.2)
   (b) VaR and tranche option calculations
   (c) Semi-closed form pricing – crude Monte Carlo simulation only is used.

Table of Spreadsheet Examples and Software

Part	Section	Name	Topic	Type
I	2.3	chapter 2.xls	Basic TM example adjusting for 'NR'	Excel
I	2.4	chapter 2 and chapter 15.xls	Use of TM to get spreads; modified TM to fit spreads and vol	Excel
I	6.2	chapter 6.xls	Random number to forward rating	Excel
I	6.3	chapter 6 corr spreads.xls	Correlated forward spreads	Excel
II	8.5	chapter 8.xls	Potential future exposure calculation	Excel
II	9.2, 9.3	chapter 9.xls	Recovery modelling; simple CDS and bond calibration and sensitivities	Excel
II	10.1	chapter 10 and chapter 20 hedge theory.mcd	CDS, F2D sensitivity code and tests	MathCad & rtf
II	10.1.1, 10.1.2	chapter 10.xls	Approx cross currency CDS pm calculation; spread hedge calculation	Excel
II	11.5	chapter 11.xls	Spread, LIBOR and bond yield volatility	Excel
II	12.1, 12.2, 12.3	chapter 12.xls	Forward CDS; unwind/back-to-back/valn calcn	Excel
II	13.5	chapter 22 N2D CDO CLN price and hedge.mcd and pricing app	CLN pricing	MathCad & rtf
II	13.6	chapter 13.xls	Capital guaranteed note	Excel
II	15.2	chapter 2 and chapter 15.xls		Excel
II	15.2	chapter 15 stochastic hazard rate.mcd	Hazard rate tree	MathCad & rtf
II	18.2	chapter 18.xls	Single name default time simulation; simulated correlated default times (2 names)	Excel

Part	Section	Name	Topic	Type
III	20.3	chapter 10 and chapter 20 hedge theory.mcd	First and second to default, zero and 100% correlation	MathCad & rtf
III		CDO pricer.xls	CDO pricing tool	Excel & DLL
III		N2D pricer.xls	N2D pricing tool	Excel & DLL
III	20.1	chapter 20.xls	F2D priced using modified TM	Excel
III	22.5	chapter 22 N2D CDO CLN price and hedge.mcd and pricing app	Basket, CDO and CLN pricing code and implementation	MathCad & rtf
III	20	chapter 20 Normal Copula closed form N2D pricing.mcd	Closed form pricing for correlated baskets	MathCad
III	23.3	chapter 23.xls	Correlated default times using the Clayton Copula	Excel
IV	25.1	chapter 25.mcd	Uncorrelated and correlated counterparty risk	MathCad & rtf
IV	25.3	chapter 22 N2D CDO CLN price and hedge.mcd	Correlated counterparty risk pricing code (used by 'Part IV section 1.mcd')	MathCad & rtf
		Part III test data name spreads.xls	Data for MathCad CDO pricing app	Excel

About the Author

Geoff Chaplin studied mathematics at Cambridge (MA 1972) and Oxford (MSc 1973, DPhil 1975) and trained as an actuary (FFA 1978) while working in a life insurance company. He moved to the City in 1980 and has worked for major banks (including HSBC, Nomura International, and ABN AMRO) as well as consulting to hedge funds, corporate treasurers, and institutional investment funds. He has been involved in the credit derivatives market since 1996 and has both traded portfolio products and developed risk management systems for these products. In addition to consulting and training work for the major financial institutions, Geoff has maintained strong academic interests and was a visiting (emeritus) professor at the University of Waterloo (Canada) from 1987 until 1999. He has also published many articles (in *Risk*, the *Journal of the Institute and Faculty of Actuaries*, and others), speaks regularly at conferences on credit derivatives, and recently co-authored *Life Settlements and Longevity Structures* in the Wiley Finance series. Geoff continues to be actively involved in the credit derivatives market as a consultant to a wide range of institutions, as an expert witness in dispute resolution, and is a Partner in Reoch Credit.

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Part I

Credit Background and Credit Derivatives

1

Credit Debt and Other Traditional Credit Instruments

The reader is assumed to be familiar with government bonds, with the LIBOR market and to have had some familiarity with traditional credit instruments (bonds and loans). The following sections briefly review these areas and develop some techniques for the analysis of credit risk – particularly in relation to credit portfolios.

1.1 BONDS AND LOANS; LIBOR RATES AND SWAPS; 'REPO' AND GENERAL COLLATERAL RATES

1.1.1 Bonds and Loans

A **bond** (Bloomberg definition) is a certificate of debt issued by a government or corporation with the promise to pay back the principal amount as well as interest by a specified future date. A **loan** is a broader concept than a bond – it is a sum of money lent at interest. In practice bonds are usually traded instruments (at least in principle) whereas loans are often private agreements between two or more parties (usually a corporate entity and a bank). There has been a growing market recently in **syndicated loans** as opposed to **bilateral loans**. Traditionally loans have been bilateral agreements between the borrower (typically a corporate entity) and a lender (typically a bank but often a private individual) the terms of which can be very varied including various options and restrictions on the borrower's use of the money or financial performance. A syndicated loan is offered by a group of lenders (a 'syndicate') who work together to provide funds for a single borrower and share the risk – unlike bilateral loans. Typically there is a lead bank or underwriter of the loan, known as the 'arranger', 'agent', or 'lead lender'. The term '**leveraged loan**' simply means a high-yield loan (high-risk borrower) and apart from the higher spread on such loans there are no other new features. There has also been growing market in **secondary trading** of bilateral loans – where the cashflows under the loan agreement are **assigned** (sold) to a third party. For this to be possible the loan agreement has to allow this transfer to take place; such loans are **assignable loans**. Many loans – although this is becoming less common than historically – are non-assignable (the ownership of the cashflows cannot be transferred). Some loans are non-assignable except in default. (We see later that these variations have implications for the credit derivatives market – they potentially restrict the deliverability of some debt into default swap contracts.) The syndicated loan market has sought standardisation of loan terms and restrictions – this is covered in more detail later, but syndicated loans are usually immediately callable by the borrower, contain a wide range of restrictions on the borrowers financial performance (covenants) and are often but not always issued through CLO structures to investors.

When we speak of a **credit bond** (or loan) we are explicitly recognising the risk that the payments promised by the borrower may not be received by the lender – an event we refer to as **default** (we discuss this further below).

1.1.2 BBA LIBOR and Swaps

According to the British Bankers' Association, '**LIBOR** stands for the London Interbank Offered Rate and is the rate of interest at which banks borrow funds from other banks, in marketable size, in the London interbank market'. BBA LIBOR rates are quoted for a number of currencies and terms up to one year, and are derived from rates contributed by at least eight banks active in the London market. (See http://www.bba.org.uk for more information.) LIBOR rates clearly refer to risky transactions – the lending of capital by one bank to another – albeit of low risk because of the short-term nature of the deal and also the high quality of the banks contributing to the survey.

An **interest rate swap** contract (Bloomberg) is 'a contract in which two parties agree to exchange periodic interest payments, especially when one payment is at a fixed rate and the other varies according to the performance of a reference rate, such as the prime rate'. Typically interest rate swaps are for periods of more than a year, and usually the reference rate is LIBOR. The swap itself is a risky deal although on day one the value of the fixed flows equals the value of the floating payments, so the risk is initially zero. Risk emerges as interest rate levels change, affecting the value of the floating and fixed payments differently. (Expected risk at a forward date will not be zero if the interest rate curve is not flat though will typically be small in relation to the value of one of the legs of the swap.) Swaps are low risk, although **swap rates themselves are risky rates largely because the reference floating rate itself (LIBOR) is a risky rate**.

1.1.3 Collateralised Lending and Repo

Banks often use listed securities as **collateral** (assets pledged as security) against cash they borrow to meet other needs. Such lending (of securities) and borrowing (of cash) is referred to as collateralised borrowing and the rate of interest applicable to generic collateral is the **general collateral rate** (GC). Typically such collateralised lending agreements are for short terms (they may be on a rolling overnight basis) and the GC rate itself is usually a few (2–7) basis points below LIBOR rates.

The reader should note several things at this point.

1. Any structured product created by a bank can in principle be securitised and used as collateral to obtain the cash required to finance the transaction. General collateral rates are therefore key in determining the cost of any structured deal (including credit derivatives).
2. GC rates are not readily available – they are known by the repo trading desk but not made publicly available. Some US repo rates are published on Reuters. The European Banking Federation sponsors the publication of EUREPO, a set of GC repo rates relating to European government bonds. However, LIBOR rates are very easy to obtain and are also close to GC.
3. Investment banks typically mark their positions to market by discounting off the LIBOR and swap curve (because LIBOR is the financing cost, and for arbitrage reasons – see, for example, Chapter 9).

LIBOR (and swap) rates are therefore key to the development of the pricing of credit derivatives.

We shall look at the deal underlying collateralised lending (a 'repo') in detail since an understanding of this issue is required later. A '**repo**' or **repurchase agreement** is a contract giving the seller of an asset the right, and the obligation, to buy it back at a predetermined

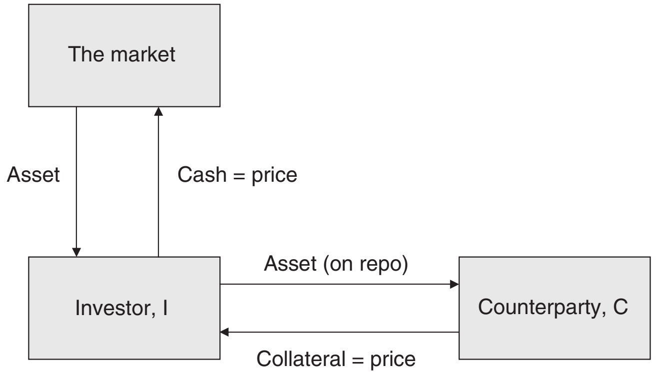

Figure 1.1 Repo deal – initial, capital and asset flows.

price on a predetermined date. The borrower of cash ('lender' of the asset) sells the asset to a counterparty under the repo contract and receives cash (equal to the market value of the asset[1]). Prior to termination of the deal any cashflows generated by the asset are passed on to the original owner of the asset, and the borrower of cash pays interest at a rate specific to that asset – the **repo rate**. On the termination of the deal the borrower of cash repays the loan and receives the asset back. In the event of default of the asset, the end date of the repo would be accelerated – the asset passes back to the original owner and the debt is repaid. (See Figures 1.1–1.3.)

We can see that, although legal ownership of the asset passes from the original owner, '**economic ownership**' remains with the original owner (i.e. the original owner of the asset receives all the cashflows from the asset in any eventuality as if he owned that asset).

Typically repo deals are short term – from overnight to a few months.

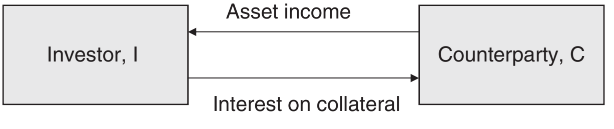

Figure 1.2 Repo deal – ongoing cashflows.

GC rates are tiered according to the class of asset. There are different GC rates for Government bonds depending on the country of issue; GC rates for corporate bonds are determined by the rating. However, a particular asset – for example, a certain bond – may go '**special**' on repo. An institution may need to borrow a particular asset (for example, it may have sold the asset short) and may be prepared to 'pay' in order to receive that asset. Under the repo deal the institution pledges (lends) cash against the asset it borrows and, if the asset were not special, would receive interest at the GC rate. But if the asset goes special, the repo rate for that asset falls *below* the GC rate – and may fall to zero or even become negative. ('Special' repo rate have an impact on the basis between bonds and default swaps (see Part II).) The borrower of the bond lends cash and therefore receives a sub-LIBOR return on that cash.

[1] Poor quality collateral may be subject to a 'haircut' where the cash lent is less than the current market value – usually by 5% or so.

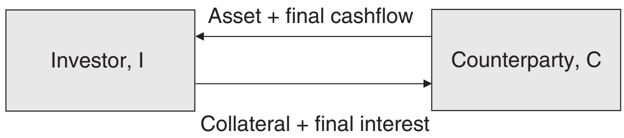

Figure 1.3 Repo deal – final capital and asset flows.

1.1.4 Repo as a Credit Derivative

A repo is not traditionally regarded as a credit derivative even if the collateral is a credit risky asset, although it is almost identical to a 'Total Return Swap' (see Part II) which *is* usually classed as a credit derivative. Furthermore, there is a significant and complex embedded credit risk if there is a correlation between the borrower of cash and the reference entity of the collateral (see Parts III and IV). A default of the borrower may cause a sudden drop in the value of the collateral in this case, so the lender of cash is taking on a non-trivial credit derivative risk. A similar risk exists in default swap contracts where the counterparty and the reference entity are correlated: indeed we can view a repo as an outright purchase of the collateral with a forward sale back to the original owner, plus a purchase of default protection on the reference entity from the borrower of cash. The question of the embedded counterparty credit risk is analysed in detail in Parts III and IV.

1.2 CREDIT DEBT VERSUS 'RISK-FREE' DEBT

We often talk of a **risk-free bond** – one where there is no risk of default – and usually identify this with government debt of certain countries issued in their own currency.[2] The yield curve associated with government debt is then often called the **risk-free curve**. This identification is, in the first instance, only approximate – any government can default on its own debt. The risk may be remote but it still exists in principle. The second problem with the identification of risk-free rates with the government curve is that trading in government debt is only a part of debt trading, and special factors – such as a heavy issuance program, a buy-back program, or regulatory requirements on banks (for example) to hold government debt – can distort the price of government debt and separate the government yield curve from 'underlying' risk-free rates.

We shall see later (Part II) that, whatever 'risk-free' rates may be, they are irrelevant to the pricing of credit derivatives.

1.3 ISSUE DOCUMENTS, SENIORITY AND THE RECOVERY PROCESS

1.3.1 Issue Documents and Default

A bond is subject to a legally binding document (the 'issue document') which is usually substantial (100 pages or so) and describes the parties involved in the issue of the bond, the

[2] In contrast debt issued by a country in a currency other than its own is typically identified as credit debt e.g. Brazilian debt in US dollars.

borrower, legal jurisdiction, etc., together with payment information such as the cashflow dates and amounts. Loans are subject to corresponding loan documentation.

One item defined in the documentation is the '**grace period**' or the number of '**days of grace**' for the bond cashflows. If a scheduled payment is not made on the due date this does not constitute default – the borrower is allowed a period of time in which to make the payment. This is intended to cover administrative errors and omissions, and other events which might make payment impossible in the very short term.

The documentation also specifically covers what constitutes default, and what recourse the lender has in the event of a default.

Typically, **default** is defined as the failure to pay a significant promised cashflow. Generally it is not just failure to pay a cashflow on that bond which causes default; failure of the borrowing entity to pay any significant loan cashflow usually triggers default on all bonds and loans issued by that entity (i.e. a **cross default clause** usually applies).

1.3.2 Claim Amount

In the event of default the lender usually has the right to claim a sum of money from the borrower, and this sum of money is usually par plus accrued coupon up to the date of default. The amount the lender can claim from the borrower is referred to as the **claim amount** or **claim value**, and in the above example the claim amount is 'par plus accrued' (usually we just say 'par' – the accrued is implicit).

The claim amount is a key element in the valuation of credit derivatives (including bonds) so we shall introduce some notation and capture the above in a formula. Define $C(t)$ to be the claim amount at time t, then the above paragraph tells us that

$$C(t) = 1 + A(t) \tag{1.1}$$

where $A(t)$ is the accrued on the bond at time t. (We shall work in par amounts of 1 rather than 100 or 1,000.)

Variations in the claim amount occur. For example, deep discount debt may have a claim amount which rises from the issue price to par at maturity, according to some formula or printed schedule. This is not necessarily the case – for example, convertible bonds usually have a low coupon but have a claim amount of par.

Some issue documents (usually only a few loan documents) say that the borrower can claim the promised cashflows in the event of default. Thus the future cashflows are not replaced by a single immediately payable sum. However, at the default date we can value these cashflows to get a financially equivalent amount. We refer to such a claim amount as 'treasury', meaning that it is financially equivalent to the value of a series of bond cashflows.

An alternative claim model is sometimes useful for risk calculations (particularly for sovereign debt – see below) so we show the formula for this case and make a few further comments. It is notationally easier to work in continuous time[3] but in this case we shall show the formula both in continuous and discrete time. Of course, the formula for continuous time

[3] Another reason we prefer a continuous time approach is that default can occur at any time – over a weekend, overnight – so a continuous time model of the default event process seems more natural.

can be made to reproduce the formula for discrete time by making the continuous cashflows have a certain function form (a sum of 'Dirac delta functions'[4]).

First, the continuous time version, let $c(t)$ be the rate of cashflow promised under the bond at time t for $t < T$ the bond maturity, and let $d(t)$ be the discount factor for a payment at time t.[5] Then the claim amount is given by

$$C(t) = \int_t^T c(x) \cdot d(x) dx \tag{1.2a}$$

Second, in the discrete time case, let the cashflow at time t_i be c_i, where i is a counter over the payment dates. For a bond, these cashflows are just the coupon payments at an ordinary coupon date, and the coupon payment plus maturity amount at the maturity date. Then we can express the claim amount as

$$C(t) = \sum_{i=1}^{n} c_i \cdot d(t_i) \tag{1.2b}$$

where n is the number of cashflows. In the typical case – claim amount of par (plus accrued) – on the default event, all debt becomes an immediately due cash amount. Thus a bond with a one year outstanding life and a bond with a 30-year outstanding life will both have exactly the same market value after default (assuming zero accrued for simplicity) – both have the same claim value (par) which is due immediately.

1.3.3 The Recovery Process and Recovery Amount

The issue document also covers the question of how a claim on one bond relates to claims on other debt. This is a complicated issue involving the law generally – usually requiring that back taxes are settled before anything else, and that employees get back pay before banks get repayment of debt etc. Settlement of claims on debt will be described in the issue document which will usually (in the case of corporate and bank borrowers) refer to the **seniority** of debt, and the order in which different seniorities are to be recompensed. Some debt may **be secured on a specific asset** – for example a property. In the event of default then settlement on this specific asset is related to the amount which the specific asset realises on sale. Typically debt is secured on residual assets of the firm generally. Terminology varies but debt commonly found in the market and in bank portfolios generally falls into one of three further levels of seniority: **loans**, **senior** secured, and **subordinated** (or **junior)** debt. It is generally the case that 'loans' differ from bonds in that they are usually more senior in addition to other points discussed in section 1.1.1.

Once a corporate entity defaults the administrators of the company seek to realise maximum value from the assets of the company. When the value of these assets is realised then cash is used in the prescribed order until it is used up. If cash is available after paying (in full) the

[4] A Dirac delta function is such that $\delta(x) = 0$ except at $x = X$, and $\int_{-\varepsilon}^{\varepsilon} \delta(x)dx = 1$. See, for example, Hoskins (1999).

[5] We understand that we are making a valuation 'now' – time $t = 0$. We could make the formulae more general (but unnecessarily so for our purposes at the moment) by making 'now' a variable too, which would allow us to write down a formula for forward claim amounts. This is necessary in an implementation of the above model for bond and CDS pricing, and is left as an exercise for the reader.

most senior creditors (such as the tax man, accountant's fees, back pay etc., and debt secured on specific assets) then cash is applied to claims on loans being the most senior debt. If this can be met in full, then the remaining cash is applied to senior unsecured bonds and, if these can be repaid in full, it moves on to junior debt. If, at any point, cash is insufficient to cover the claims of that seniority in full, then all claims receive the same proportion (the **recovery rate**) of the claim amount. The cash is then used up. Thus in a corporate default where the cash is insufficient to cover all the claims, one seniority level will receive a partial recovery; more senior levels of debt receive 100% of the claim amount, and more junior levels receive zero. Any deviation from the legal framework and the legally binding issue documents can be challenged in the courts by the creditors.

Since we shall use these concepts repeatedly we shall summarise the above in a formula. Let s denote the seniority level we are looking at, where $s = 0$ denotes junior (subordinated) debt, $s = 1$ is senior bonds, $s = 2$ is loans. (We can of course introduce more seniority levels as appropriate, but these three are generally all that is required in practice.) Then the amount recovered is

$$R_s \cdot C(t) \tag{1.3}$$

where

$$1 \geq R_2 \geq R_1 \geq R_0 \geq 0, \tag{1.4a}$$

and if $1 > R_s > 0$ for some s we have

$$R_p = 1 \quad \text{for} \quad p > s \quad \text{and} \quad R_p = 0 \quad \text{for } p < s \tag{1.4b}$$

(i.e. more senior debt gets full recovery and more junior debt gets zero recovery).

We state here that the condition (1.4b) only applies when we are looking at the actual amounts finally recovered ('ultimate recovery' – see Chapter 2). Recovery as applicable to credit derivative products is a different concept from the above (being the one-month post-default bond price) and conditions (1.4b) no longer apply (although (1.4a) remains: see Chapter 2 for further details).

The amount $1 - R_s$ is often referred to as the '**loss given default**' (**LGD**).

We shall address the question of how we estimate R prior to the default event in the following section. At present formula (1.3) and following conditions apply to a specific defaulted entity – the recovery numbers for that entity are not the same as the recovery numbers for another entity.

1.3.4 Sovereign versus Corporate Debt

Issue documents for **sovereign debt** (e.g. Argentinean debt in USD) are very similar to those of corporates. Typically the claim amount is also par. Usually there is only one level of seniority for sovereigns. The major difference between corporate and risky government debt is in the recovery process itself. Firstly there is no 'wind up' process via the courts as in the case of corporates. A 'defaulting' sovereign typically restructures its debt and investors lose value compared with their promised cashflows. The government may offer terms which are very different from the recovery levels one might expect from the issue document – typically long dated debt recovers a smaller proportional of notional than short-dated debt. However the lenders have no court they can go to in order to seek a strict implementation of the process described in the issue document. In practice this means that recovery for sovereign debt is not at the same rate for all bonds[6] – instead it is typically high for short-dated debt and low for long-dated debt.

1.4 VALUATION, YIELD AND SPREAD

Bonds are bought and sold based on a price. Often the price quoted is a clean price, and the consideration paid also takes into account the accrued interest. Given the market price of the debt, and the cashflow schedule, we can calculate the yield (internal rate of return), and we can also calculate the '**spread**'. There are various ways of measuring spread (see Chapter 5 for further discussion) – for the moment we shall define spread as the difference between the bond's yield and the interpolated yield off the LIBOR/swap curve (interpolated to the maturity date of the bond).

High-grade debt may trade close to swap rates – even sub-LIBOR for very high grade borrowers which are perceived to be less risky than the banks which define the LIBOR rate. Typically **investment grade debt** (debt rated BBB or better by the rating agencies) trades up to 300 bp over the swap curve depending on the name and varying with time and the economic cycle, sentiment, etc. Sub-investment grade debt typically trades wider – to 10 000 bp or 100 000 bp above the swap curve.

For investment grade names the market will usually talk in terms of spread rather than price. The reason for this is that the price of a 5-year Unilever bond (for example) will change moment by moment as interest rate futures tick up or down. However, the spread on the bond typically changes much more slowly and may even be static for days or even weeks.

For sub-investment grade debt the market usually talks in price terms. Where spreads are high (and bond prices may be 50% below those of low-risk debt) the main determinant of price is the perceived default risk, not interest rate levels.

1.5 BUYING RISK

A buyer of a credit bond is taking on the default risk of the underlying entity – the investor is not only buying an asset but also **buying risk**. Imagine an insurance policy which insures the par value of the bond in the event of default of the underlying name. The buyer of the insurance policy is the **buyer of protection**, and the writer of the policy is the seller of protection. We can also talk in terms of risk – the seller of protection is taking on risk, similar to the buyer of the bond itself, while the buyer of protection is also the seller of risk.

In the credit derivative market both sets of terminology are used – buyer or seller of protection or of risk. The word 'buyer' on its own conveys nothing – the buyer of protection is the seller of risk and vice versa. It is essential to be clear whether one is talking about risk or protection. Often 'selling' means selling protection when talking about single-name default swaps, but selling a tranche of a CDO usually means buying protection. Investors (asset managers, hedge funds, pension funds, etc) often talk in terms of buying and selling risk to reflect what is in effect happening if they invest in a corporate bond. Banks may use both terminologies depending on the area within the bank – traders often talk in terms of protection whereas structurers will talk in terms of risk.

[6] There are often other complications such as 'Brady Bonds' where some of the cashflows (e.g. maturity amount) will be received in full (because they are guaranteed by the US government) while other cashflows (coupons) will not be received.

In this book we shall generally talk in term of buying or selling protection when we talk about credit derivatives.

1.6 MARKING TO MARKET, MARKING TO MODEL AND RESERVES

When it is required to value a deal – whether for the purpose of calculation profit to date, for accounting, for regulatory or other reasons – the best approach in principle is to obtain a bid for the asset held. This is easy for liquid bonds for example, and is called '**marking to market**'.

For many other assets – such as credit derivatives, structured products and many option contracts – this is generally not practical. For example, consider a portfolio of equity call options of various maturities and strikes. Typically some maturities and strikes on each name trade sufficiently frequently that market prices for these maturities and strikes are easily available. We can obtain these prices and interpolate for other maturities and strikes on the same underlying asset. Usually this interpolation uses a pricing model (such as the Black–Scholes model) and an intermediate variable (volatility) is obtained. Interpolation on this variable is performed (perhaps involving a further model such as a volatility smile model) and the interpolated variable put back into the model in order to get an estimated market price for the asset. This is referred to as '**marking to model**'.

Reserves

Reserves are set against the value for products marked to model to give rise to a 'conservative' valuation of the portfolio in the institutions accounts. Even when products are marked to market they may be marked to mid: in this case a 'bid–offer' reserve will usually be held against the value to reflect the realisation value achievable on an asset.

Example 1 Suppose we mark to mid (and mid prices are easily available). Suppose we are long a unit of Asset_1, mark it to mid-price $P1$, and $S1$ is the estimate of (half) the offer–bid difference. Then the value in the books would appear as $P_1 - S_1$.

Exercise 1 If we sell the Asset_1 to a market maker, how many trades does the market maker do? The answer is (at least) two: the trade with us and a hedging trade or trades with another party. A key feature of the market maker's role is to minimise risk and lock in profits arising from bid–offer spreads rather than take views on the market direction. A market maker generally does not run unhedged positions – this role is left to other traders ('proprietary' or 'prop traders') who generally do not deal directly with investors.

Exercise 2 Suppose we are long Asset_1 above and short Asset_2, which is a very good hedge for all the risks in Asset_1. Should we mark to $P_1 - S_1 - P_2 + S_2$?

Generally if we have a hedged 'trade' (made up of several 'deals' – Asset_1 and Asset_2 in the above) the costs and the risks to the market maker of taking on the position are less, so the bid–offer spread on the trade will be tighter than on a single unhedged asset. The trade in

Exercise 2 would generally be marked to a better overall price than the less-well-hedged asset. Similar arguments apply to other reserves mentioned below and later in the book.

If deals are marked to model there are uncertainties involved in arriving at the mid price. In the equity option example above, uncertainties arise from

i. The interpolation routine to interpolate for the maturity and the strike of the actual option held
ii. The uncertainty in the validity and accuracy of the model being used (other traders may use a different model). [This is not really an issue for the equity option example but might apply if we were pricing exotic options.]

Marking to model generally gives rise to additional reserves, which will depend on the product and model being used, and there may also be a general 'model reserve'. We shall discuss these in the context of credit derivatives products in Parts II and III.

Traders' P&L

Reserves may also apply to the calculation of traders' P&L – but not necessarily the same figures. The institution may take a cautious view of its books for a variety of reasons. On the other hand, reserves may be set against traders' P&L in order to avoid a situation where a trader can take a large mark-to-market profit (and bonus) on Day 1 then, subsequently, the deal turns out to be worth far less than anticipated.

Reserves have little impact on a portfolio with high turnover: a deal done on Day 1 with high reserves, and take off on Day 2, will release the reserves and the actual profit can be calculated on the basis of the buy and sell prices. Typically, a high turnover portfolio usually faces low reserves, and the case where reserves tend to be high is a buy-and-hold book. Faced with a certain level of reserves the trader may take the view that, if he puts a deal on, the level of reserves against that position may eat too far into the P&L accumulated so far. The deal will not be done.

Another problem in setting reserves is that some sort of approximation is made. This may imply that the reserves are low on one version of the trade and high on another, with the result that the trader deals only on one type of trade (where the reserves are too low).

Institutions take a variety of views regarding traders' P&L. At one extreme the traders' and the institution's reserves are the same; at the other extreme the trader may face an estimated bid–offer reserve only, and the institution takes the risk on other valuation estimates (and on the trader).

1.7 THE 'CREDIT CRUNCH' AND CORRELATION

A key element of the risk management process within a bank or an asset manager is understanding the risk on a portfolio of assets. In particular, how does asset class A (e.g. corporate bonds) perform when asset class B (e.g. equities) is falling? The answer to the question is often captured in the concept of asset correlation, but there is no single correlation that correctly captures the relationship and risk. During 'normal' times the correlation between asset classes may be low (say 20%) but during abnormal times (generally times of crisis of some form) the correlation rises sharply. Correlation itself is a variable and can indeed vary over a wide range. This aspect of correlation is generally not well captured in financial models and the problem is often tackled by looking at the tails of a value distribution (e.g. in VaR analysis). We shall see later that correlation is a key driver to the value of tranches of risk in CDO structures, and a large rise in correlation can have a devastating effect on the value of what in 'normal' times would be regarded as high-quality low-risk investments.

The credit crunch of 2008 was not a unique event. A credit crunch occurs when the banking system itself is compromised for some reason, interbank rates rise sharply relative to government rates and the banking system largely fails to function – lending to corporate borrowers is sharply reduced in a sudden change of risk appetite by the banks. This withdrawal of liquidity is rapidly felt throughout the financial system – for example, wealthy individuals often having to use their own financial resources to support companies in which they are closely involved rather than investing in peripheral activities – with the consequence that available cash for investment is sharply reduced. In such circumstances all asset prices fall as demand collapses – in other words, previously uncorrelated assets suddenly become highly correlated. Within the credit markets the crunch results in sharply higher spreads, and sharply higher actual default rates as bank lending to corporates is withdrawn, and has a dramatic impact on the pricing of structured credit as we shall see later.

CDO structures played a role in the credit crunch, the valuation aspects of which will become clear later in the book. It is important for the reader to realise the wide application of the term 'CDO' and the range of assets covered by the term. For example, a CDO referencing corporate credits may allow investors easy access to the corporate credit market. The impact of the new source of capital can drive the reward for risk (corporate credit spread) down but will have limited impact on the corporate's appetite for new borrowing. On the other hand, a CDO of mortgages can allow a bank to reduce its book of loans to individuals thus giving it scope to make additional lending, and the borrowers (individuals) are much more likely to take on additional loans, escalating the amount of debt in existence. This is a very real risk and has been one of the contributors to the size of the credit crunch starting in 2008.

1.8 PARTIES INVOLVED IN THE CREDIT MARKETS AND KEY TERMINOLOGY

Borrowers: Corporate and sovereign entities (through loans and bonds) and individuals (through mortgages, consumer loans, lease contracts, credit card debt).

Lenders: Generally the same institutions as above – banks, corporate entities, sovereigns, investment institutions and high-net-worth individuals.

Traders and hedging: Individuals within various institutions. Traders may be 'front book' traders whose role is to make a market (set bid and offer prices at which he or she is prepared to deal) and run a very limited risk generally offsetting deals done with a customer with other transactions ('hedges') done with other customers or the 'market'. 'Prop[rietary]' traders are taking a view of the market [direction] using their institution's funds. Such trades may be hedged but the meaning of the term 'hedge' may be very loose [for example, the trader may take the view that equities currently form a very good hedge for a bond position because of unusual circumstances at the time] but generally prop trades have a much higher risk profile than hedged trades.

Investors: Asset managers, pension and insurance funds, high-net-worth individuals. The investor is typically seeking risk reduction through diversification across different assets, asset classes, maturity, etc.

Economic equality: For example, a single-name CDS contract and an insurance contract on corporate debt may be 'economically equal' in the sense that the cashflows under the two contracts are equal. This does not mean that the two contracts are identical – in this case documentation may be written differently, as there may be requirements on the insured to make a claim under the contract, and the writers of the contract may be differently regulated.

Natural measure versus risk-neutral measure: The natural measure is often used by investors: What is this 'really' worth? What income do I anticipate I will receive? The risk-neutral measure is typically used by front-book traders: What is the cost to me of hedging this transaction so that I can lock in a known profit with minimal risk? The two measures are widely used in the 'valuation' of different assets but may provide widely different answers. [We shall return to this point several times in this book, and it is important to understand that the mathematical models we develop are often equally applicable under either measure – the difference being the data put into the model (or its '**calibration**') rather than the sums being done.]

Funded instruments versus CDSs: In the past the term CDS meant a single-name default swap, but is now more commonly used to mean a contract written like a single-name default swap (or interest rate swap) with periodic payments of premium and a contingent capital payment. A 'funded instrument' is more like a bond – the investor pays a large sum up front and receives larger periodic payments plus all, or a proportion of, the investment at maturity or default. The terms are often used to distinguish how tranches of risk of a CDO are documented and sold. In particular it is now common usage for the term CDS to mean any credit derivative (from a single-name risk to a complex non-standard structure referencing non-traded risks) which is documented as an unfunded trade.

2

Default and Recovery Data; Transition Matrices; Historical Pricing

2.1 RECOVERY: ULTIMATE AND MARKET-VALUE-BASED RECOVERY

2.1.1 Ultimate Recovery

In Chapter 1 we discussed recovery in the context of bonds and corporate default. The recovery described was that obtained on the ultimate wind-up of the company and distribution of residual assets. This is referred to as '**ultimate recovery**'.

Ultimate recovery occurs after the default event has taken place, after accountants have been appointed to wind the company up, and after assets have been sold. This process clearly takes a considerable amount of time. Table 2.1 shows the length of time this takes (for US defaults) and how much it varies, while Table 2.2 shows the mean and standard deviation of recovery rates for a range of debt seniorities.

It should be noted that the standard deviation of recovery is very high. Suppose recovery is equally likely to have any value between 0 and 100% (a uniform distribution[1]). Then the expected recovery level is 50% and the standard deviation turns out to be about 29%. The observed historical average recovery rate actually happens to be very close to that of a uniform distribution, but the standard deviation is even higher. Thus, if we assume that we have a large portfolio of debt, and that 100 of these names default, then on the basis of historical ultimate recoveries we would expect to get a recovery of around 50% on average for the 100 names. However, if we were to pick one of these names at random, its recovery would be more likely to be closer to 0 or 100% than to the average of 50%.

The historical data gives only very coarse information about the likely recovery level on any one specific company. Of course defaults are very rare compared with the number of firms in existence. The historical default information we have gives only a very rough idea what the recovery level for a particular firm might be. We shall see some more information in the following sections but, even then, we will find that we cannot make a very good guess at the recovery level for a particular firm and seniority of debt. This will turn out to have quite significant implications for the valuation and risk management of credit derivatives.

Who is interested in ultimate recovery data?

The manager of a book of non-assignable loans has a portfolio of buy-and-hold assets. If a name suffers a credit event then the manager has to follow the wind-up process through to ultimate recovery. Funds which invest in distressed or defaulted debt ('**vulture funds**') are also interested in ultimate recovery.

[1] See any standard text on probability and statistics: e.g. Billingsley (1995)

Table 2.1 Length of time in bankruptcy

Type	Count	Average	Median	Minimum	Maximum	Standard deviation
Ch 11	116	1.68	1.43	0.19	7.01	1.20
Prepackaged Ch 11	53	0.29	0.18	0.09	1.76	0.29
All	159	1.30	1.15	0.09	7.01	1.20

2.1.2 Market Recovery

The rating agencies have another definition of recovery: the price of traded debt in the marketplace one month after the 'default' event. This is clearly a very different concept from that of ultimate recovery. We shall refer to this definition of recovery as '**market recovery**' or 'one month post-default recovery' or simply 'recovery'.

Data for market recovery will clearly differ from ultimate recovery because they are different concepts (distribution amount versus market price of debt) and the data refers to very different stages in the company's wind-up process. The rating agencies also use a definition of the default event which is broader than the one that causes a wind-up of the company. For example, 'bankruptcy' constitutes default as far as the rating agencies are concerned, yet does not necessarily lead to a wind-up of the company. Failure to pay the coupon or maturity amount (perhaps because of an administrative error) is a default event in the rating agencies' eyes, but

Table 2.2 Ultimate recovery rates
(a) Average corporate debt recovery rates measured by ultimate recoveries, 1987–2008

	Emergence year			Default year		
Lien position	2008	2007	1987–2008	2008	2007	1987–2008
Loans	87.0%	96.6%	82.8%	61.2%	93.5%	82.8%
Bonds						
Sr. Secured	70.0%	85.5%	63.6%	51.1%	65.4%	63.6%
Sr. Unsecured	74.0%	60.0%	46.2%	76.8%	83.7%	46.2%
Sr. Subordinated	23.0%	62.9%	29.7%	9.3%	55.4%	29.7%
Subordinated	–	0.0%	28.9%	–	0.0%	28.9%
Jr. Subordinated	0.0%	50.0%	16.0%	–	–	16.0%

(b) Ultimate Recovery Rates and Standard Deviations

Ultimate recovery	Mean (%)	SD (%)	Number
All	51	37	954
Less than 1 year	65	37	241
More than 1 year	46	36	713

Source: Altman *et al.* (2004)

may not lead to a default event as far as bonds are concerned (payment may be made within the grace period). Thus data on ultimate and market recovery is based on non-identical sets of events.

Compared with ultimate recovery, market recovery data shows relationships of recoveries on different seniorities of debt. Ultimate recovery data will show only (at most) one seniority with a recovery different from 0 or 1. However, market prices of defaulted debt are estimates of what ultimate recovery rates will be. Instead of the inequality (1.4a) subject to the condition (1.4b), we find that (for a specific firm) higher seniorities have higher recoveries. In other words

$$1 \geq R_2 > R_1 > R_0 \geq 0 \tag{2.1}$$

is all we find. For example, post-defaulted (assignable) loans might be trading at 80, bonds at 50 and junior debt at 20, while ultimate recovery might eventually be 100, 65 and 0 respectively.

Table 2.3 shows market recovery rates for several seniorities of debt. Note that the standard deviation in market recovery rates is less than that of ultimate recovery data – and very substantially less for junior debt.

2.1.3 Recovery Rates and Industry Sector

Altman and Kishore (1996) have produced research showing that recovery rates depend not only on seniority but there is also a significant industry related effect. For example, utilities tend to have very high recovery rates whereas telecoms tend to be low. Table 2.4 is an extract of some of the recent results found by Moody's.

Table 2.3 Market recovery rates and seniority: Average corporate debt recovery rates measured by post-default trading prices

	Issuer-Weighted			Value-Weighted		
Lien position	2008	2007	1982–2008	2008	2007	1982–2008
Bank Loans						
Sr. Secured	63.4%	68.6%	69.9%	49.0%	78.3%	62.1%
Second Lien	40.4%	65.9%	50.4%	36.6%	65.8%	49.8%
Sr. Unsecured	29.8%	–	52.5%	22.6%	–	41.0%
Bonds						
Sr. Secured	58.0%	80.5%	52.3%	45.9%	81.7%	53.0%
Sr. Unsecured	33.8%	53.3%	36.4%	26.2%	56.9%	32.4%
Sr. Subordinated	23.0%	54.5%	31.7%	10.4%	67.7%	26.4%
Subordinated	23.6%	–	31.0%	7.3%	–	23.5%
Jr. Subordinated	–	–	24.0%	–	–	16.8%
Pref. Stock						
Trust Pref.	–	–	11.7%	–	–	13.0%
Non-trust Pref.	8.6%	–	21.6%	1.7%	–	13.1%

Table 2.4 Recovery and industry: Average recovery rates by industry category

	Issuer weighted	Mean recovery	Rate
Industry	2003	2002	1982–2003
Utility – Gas	48.0	54.6	51.5
Oil and oil services		44.1	44.5
Hospitality	64.5	60.0	42.5
Utility – Electric	5.3	39.8	41.4
Transport – Ocean	76.8	31.0	38.8
Media, broadcasting and cable	57.5	39.5	38.2
Transport – Surface		37.9	36.6
Finance and banking	18.8	25.6	36.3
Industrial	33.4	34.3	35.4
Retail	57.9	58.2	34.4
Transport – Air	22.6	24.9	34.3
Automotive	39.0	39.5	33.4
Healthcare	52.2	47.0	32.7
Consumer goods	54.0	22.8	32.5
Construction	22.5	23.0	31.9
Technology	9.4	36.7	29.5
Real estate		5.0	28.8
Steel	31.8	28.5	27.4
Telecommunications	45.9	21.4	23.2
Miscellaneous	69.5	46.5	39.5

2.1.4 Recovery and Default Rates and the Economic Cycle

We would expect default rates to rise during recessions: there is also evidence that recovery rates drop during the lows of the economic cycle. Figures 2.1 and 2.2 show the history of default rates and average recovery rates on bonds and its relationship to the economic cycle.

2.1.5 Modelling Recovery Rates

Historical market recovery data shows the following features, which are potentially useful in modelling recovery when analysing bonds and credit derivatives.

1. A strong relationship between average recovery rate (averaged over a large number of defaulted names) and seniority of the defaulted debt.
2. A large standard deviation in the recovery rate (looking at a specific seniority across a large number of names).
3. Some relationship between industry and recovery rate.
4. Some relationship between rating (investment grade versus high yield) and recovery rate (see Figure 2.3)
5. A dependence on the economic cycle.

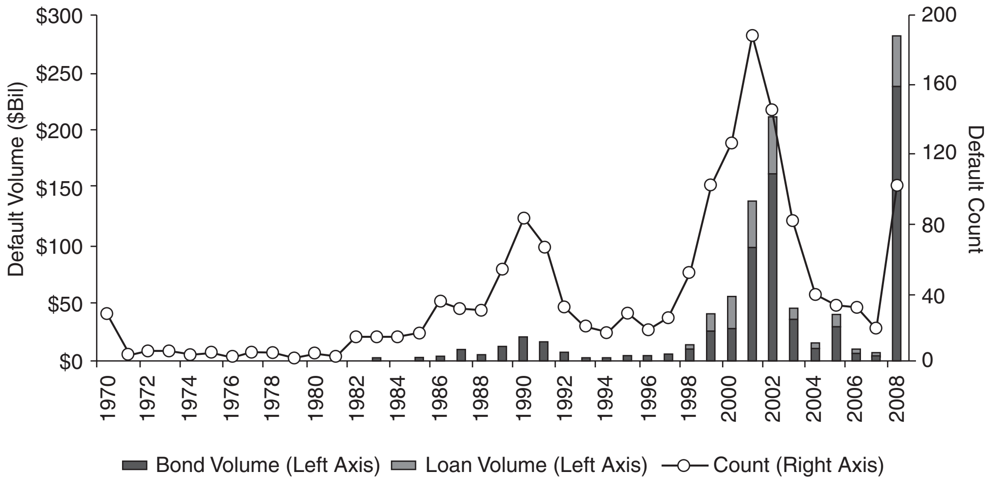

Figure 2.1 Default rates over time: Global Corporate Bond Default Counts and Dollar Volumes, 1970–2008
(© Moody's Investors Service, Inc. and/or its affiliates. Reprinted with permission. All Rights Reserved. Moody's Corporate Default and Recovery Rates, 1920–2008; 2009).

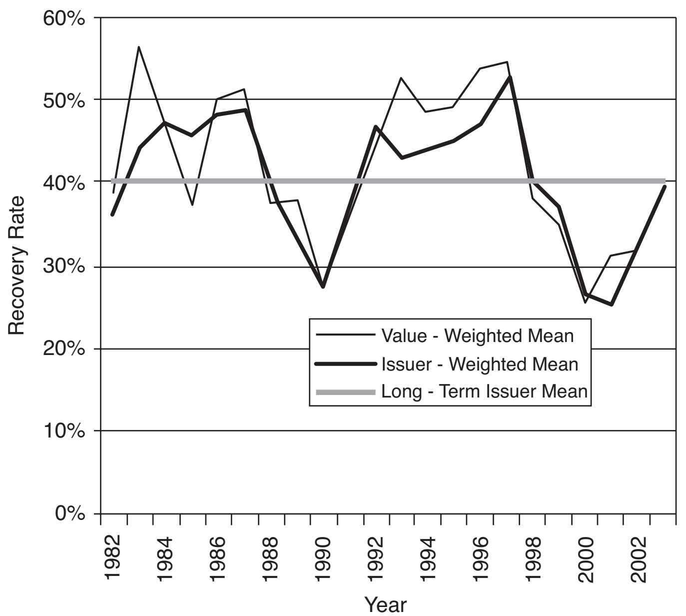

Figure 2.2 Recovery rates over time: Annual Average Senior Unsecured Bond Recovery Rates Exhibit Mean Reversion
(© Moody's Investors Service, Inc. and/or its affiliates. Reprinted with permission. All Rights Reserved. Moody's Default and Recovery rates of Corporate Bond Issuers Jan 2004).

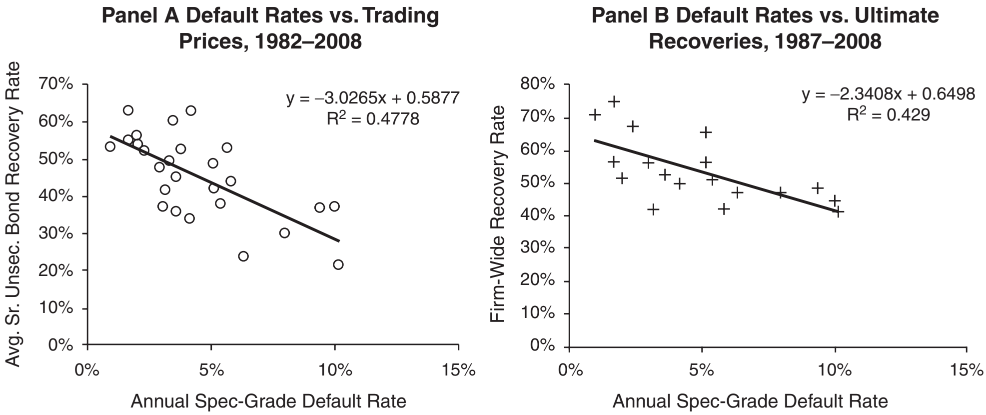

Figure 2.3 Recovery rate and rating (default rate): Correlation between Annual Issuer-Weighted Average Recovery Rates and Issuer-Weighted Average Default Rates
(© Moody's Investors Service, Inc. and/or its affiliates. Reprinted with permission. All Rights Reserved. Moody's Corporate Default and Recovery Rates, 1920–2008; Feb 2009).

We require a model of the recovery rate for any bond issued by a specific entity, not only to analyse that bond but also the credit derivatives related to it. The market typically uses the following simple model:

$$R(seniority = s, referenceEntity, rating) = const_{s,IGHY} \tag{2.2}$$

In other words, the recovery rate assumed is dependent on the seniority of the debt, s, and whether the name has an investment grade or sub-investment grade rating (*IGHY*), but otherwise independent of the reference entity and generally not explicitly time dependent.

Actual values assumed in (2.2) vary from firm to firm, but typical figures might be 10%, 40%, 70% for subordinated debt (senior secured) bonds and loans respectively on investment grade names, and 0%, 25%, 50% for high-yield names. Generally assumptions are still within 10% of these figures at the time of writing. Given that the impact of the economic cycle on recovery rates is slowly changing (Figure 2.1) then the assumed recovery represents an average over the life of deals we are valuing. Credit derivative deals (on a mature book) may have a life of less than 5 years – so the economic cycle has an impact on the recovery rate over the life of the deals, which may vary considerably over an economic cycle of roughly 10 years.

The figures quoted above are used in the valuation of 'vanilla' credit derivatives. We shall see later that these instruments have relatively little sensitivity to the recovery assumption, so the recovery assumption is not important for these instruments. We shall return to this topic when we discuss book management and 'digital' products. It is now [2009] much more common to make firm (and seniority) specific recovery assumptions, particularly for high-yield (wide spread) names. The assumed recovery for a specific firm and seniority may be very different from the assumed recovery for another with similar spread. This is also the case when actual defaults take place – the standard deviation of recovery is large and only the average recovery across certain classes (e.g. seniority) shows features of the class.

Table 2.5 Default risk and rating (Final column of Table 2.8)

Initial rating	Percent defaulted at year end
AAA	0.00
AA	0.01
A	0.05
BBB	0.27
BB	1.29
B	6.71
CCC	28.26

There is no difficulty in principle in implementing a time-dependent recovery assumption although this is rarely done in practice, and also needs careful controls.[2]

2.2 DEFAULT RATES: RATING AND OTHER FACTORS

Rating agencies provide historical data on defaults. It should be borne in mind that this data relates to the agencies's definition of default and is not necessarily directly relevant to either bond default risk (though the differences are likely to be small) or credit derivative trigger events.

Table 2.5 gives an example of historical default rates, and shows the key finding of the historical analysis which is that – as one would expect – default risk increases as rating decreases.

Historical default rates tend to be used by rating agencies and investors in the analysis of portfolios and credit structures. The historical data gives much more information than just default and recovery rates – which we discuss in the following section.

2.3 TRANSITION MATRICES

The change from being a non-defaulted firm to being in default is a 'change of state' and is – described as a transition. We can identify other changes of state – namely transitions from one rating at the start of a period to another (possibly the same, and possibly default) at the end of the period. The probabilities associated with these changes of state are given in a '**transition matrix**' or TM. Tables 2.6 and 2.7 give examples of two such matrices – the former a 'coarse' matrix where only broad rating bands (e.g. AA, A, etc.) are used, and the latter is a 'fine' matrix where individual notches are used (e.g. Aa1, Aa2, etc.).

For example, from Table 2.6 we see that there is an 84.05% chance that a BBB-rated name at the start of the year is also rated BBB at the end – a 6% chance that it moves into default, and a 4.34% chance that it gets upgraded to A. The 'NR' ('not rated') column refers to the chance that the name no longer has rated debt at the end of the year – for example, the only rated debt may have matured during the year. For convenience in subsequent calculations we include a row for defaulted names. We assume that once a name has defaulted, it is always in

[2] Time dependent recovery assumptions can easily lead to negative implied default rates. The assumptions made need to be carefully controlled to avoid this and intuitively sensible recovery assumptions may not be consistent with the shape of credit spreads and the requirement to have non-negative implied default rates.

Table 2.6 Coarse transition matrix

	AAA	AA	A	BBB	BB	B	CCC	D	NR
AAA	89.54	5.91	0.43	0.09	0.03	0.00	0.00	0.00	4.00
AA	0.60	87.96	6.76	0.51	0.05	0.09	0.02	0.01	4.00
A	0.06	2.10	88.09	5.05	0.43	0.17	0.04	0.05	4.00
BBB	0.03	0.23	4.34	84.05	4.14	0.72	0.23	0.26	6.00
BB	0.02	0.06	0.41	5.78	76.33	7.08	1.09	1.22	8.00
B	0.00	0.08	0.27	0.35	4.80	74.58	3.92	5.99	10.00
CCC	0.11	0.00	0.22	0.68	1.46	9.00	51.66	24.87	12.00

default.[3] Note that the sum of the figures in each row adds to 100% – every bond at the start of the year has to end up somewhere, either with some rating or in default or no longer rated. In contrast, the sum of the figures in the D column represents the chance that a defaulted name had defaulted the previous year (and is therefore still in default) or came from some other rating. The sum is clearly greater than 100%. Similarly the sum of other columns is less than 100%, reflecting the general drift to default.

If we wish to use the transition matrix to calculate (for example) the price a bond should stand at in order to give the buyer a fair compensation for historical default risk, then we need to make a reasoned estimate of what the transition matrix would look like if we knew how these 'NR' names would behave if they had been rated. One reasonable assumption would be that these names would have exactly the same experience had they been rated, as the other names in the sample. We shall use this assumption in what follows.[4]

Looking at the same row – debt rated BBB at the start of the year – we see that there is a 6% chance that the name no longer has a rating at the end of the year, and a 94% chance that the name is rated (including the possibility of default). We are assuming that the 94% of names is representative of the entire 100% of names, so if we increase the transition probabilities by a factor 100/94 for all those other states, we can ignore the 'NR' names. Table 2.8 shows the results of making this adjustment to the matrix in Table 2.6. The spreadsheet 'chapter 2.xls.' contains the worked example.

2.4 'MEASURES' AND TRANSITION MATRIX-BASED PRICING

We introduce two concepts that are crucial to the understanding of subsequent analysis. First, what we call the '**natural measure**' (also sometimes called '**actuarial pricing**' or, misleadingly, '**historical pricing**'). Suppose we want to know the 'worth' of a promise from a risky borrower to repay 1 EUR in a year's time. If the name were risk-free we would apply a discount factor (D) to the payment to get a current value. However, since the name is risky

[3] Of course this is not the case in reality – a company may go bankrupt (file for protection from creditors) and re-emerge from bankruptcy. But lacking information on how this re-emergence works we assume that the company stays in default. This is not a bad assumption since – even if the company emerges with no missed payments on debt – in practice the associated recovery value (i.e. the post default bond price) makes a reasoned guess on how that re-emergence (if any) might take place. Credit derivatives generally terminate on such events.

[4] Another argument might be that, since the bonds matured without default, the 'NR' names would actually have had a much better experience than the sample as a whole, with far fewer defaults.

Table 2.7 Fine transition matrix: Average one-year alphanumeric rating migration rates, 1983–2008

From/ To:	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca-C	Default	WR
Aaa	87.592	5.350	2.249	0.428	0.352	0.137	0.013	0.003	0.030	0.000	0.018	0.018	0.000	0.003	0.000	0.000	0.000	0.003	0.000	0.000	0.000	3.807
Aa1	2.526	77.611	7.472	6.321	0.943	0.342	0.067	0.134	0.047	0.010	0.000	0.000	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	4.523
Aa2	1.193	5.139	76.061	7.962	3.043	1.124	0.395	0.073	0.095	0.004	0.022	0.000	0.000	0.018	0.009	0.007	0.000	0.024	0.000	0.002	0.000	4.828
Aa3	0.213	1.555	4.568	76.256	8.042	2.556	0.677	0.194	0.172	0.073	0.012	0.031	0.016	0.016	0.000	0.000	0.000	0.004	0.000	0.000	0.037	5.576
A1	0.067	0.127	1.393	5.623	76.699	7.249	2.592	0.548	0.345	0.126	0.186	0.125	0.042	0.048	0.020	0.000	0.000	0.004	0.000	0.000	0.018	4.786
A2	0.083	0.036	0.256	1.163	5.332	76.592	7.484	2.740	0.821	0.351	0.172	0.102	0.121	0.041	0.035	0.005	0.025	0.030	0.010	0.000	0.025	4.576
A3	0.041	0.075	0.121	0.274	2.039	7.439	73.366	6.435	3.137	1.011	0.438	0.182	0.181	0.102	0.029	0.025	0.002	0.005	0.007	0.011	0.032	5.046
Baa1	0.033	0.064	0.091	0.121	0.303	2.041	7.222	72.629	7.054	2.875	0.813	0.399	0.318	0.403	0.060	0.034	0.060	0.034	0.005	0.018	0.132	5.292
Baa2	0.042	0.081	0.045	0.093	0.173	0.756	2.871	6.274	73.265	6.419	1.660	0.634	0.615	0.519	0.248	0.138	0.113	0.030	0.031	0.012	0.138	5.844
Baa3	0.053	0.021	0.032	0.048	0.133	0.247	0.628	2.912	8.649	70.021	5.146	2.469	1.122	0.798	0.372	0.269	0.205	0.117	0.173	0.056	0.288	6.242
Ba1	0.026	0.002	0.024	0.050	0.179	0.140	0.369	0.654	2.980	9.390	63.910	4.431	3.787	1.590	1.139	0.689	0.172	0.188	0.081	0.037	0.654	9.507
Ba2	0.000	0.000	0.029	0.000	0.029	0.088	0.071	0.322	0.803	3.172	8.681	63.023	6.452	2.804	2.300	1.073	0.221	0.179	0.125	0.096	0.702	9.828
Ba3	0.000	0.020	0.010	0.029	0.025	0.178	0.141	0.173	0.284	0.726	2.703	6.186	64.067	5.672	4.366	2.140	0.445	0.328	0.076	0.091	1.737	10.606
B1	0.029	0.015	0.017	0.008	0.042	0.088	0.084	0.056	0.131	0.262	0.488	2.657	6.378	64.162	6.553	3.860	1.104	0.575	0.296	0.366	2.368	10.461
B2	0.000	0.000	0.008	0.018	0.021	0.000	0.043	0.105	0.103	0.129	0.317	0.666	2.007	6.811	62.475	7.530	3.043	1.490	0.476	0.541	3.675	10.543
B3	0.000	0.006	0.046	0.000	0.010	0.024	0.078	0.046	0.056	0.114	0.106	0.256	0.654	2.747	6.545	59.098	5.281	3.738	1.101	1.085	7.318	11.690
Caa1	0.000	0.053	0.000	0.000	0.000	0.042	0.000	0.000	0.000	0.000	0.095	0.021	0.159	0.800	2.654	8.608	52.334	7.199	3.904	2.659	8.656	12.815
Caa2	0.000	0.000	0.000	0.000	0.000	0.015	0.000	0.000	0.074	0.280	0.125	0.103	0.530	0.876	1.575	4.223	7.424	49.032	3.730	4.415	15.481	12.118
Caa3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.521	0.174	0.130	0.000	0.195	1.302	1.888	2.323	6.360	42.197	5.513	22.900	16.497
Ca-C	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.085	0.000	0.255	0.212	0.870	1.528	1.570	3.756	4.371	38.892	30.066	18.396

Table 2.8 Coarse transition matrix adjusted for 'no longer rated' names

	AAA	**AA**	**A**	**BBB**	**BB**	**B**	**CCC**	**D**
AAA	93.27	6.16	0.45	0.09	0.03	0.00	0.00	0.00
AA	0.62	91.63	7.04	0.53	0.05	0.09	0.02	0.01
A	0.06	2.19	91.76	5.26	0.45	0.18	0.04	0.05
BBB	0.03	0.24	4.62	89.42	4.40	0.77	0.24	0.28
BB	0.02	0.07	0.45	6.28	82.97	7.70	1.19	1.33
B	0.00	0.09	0.30	0.39	5.33	82.87	4.36	6.66
CCC	0.13	0.00	0.25	0.77	1.66	10.23	58.70	28.26
D	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00

there is a chance (p, say) that the name defaults, in which case a recovery amount (R) is received (at the end of the year, say). In that case the promise is worth

$$W = ([1 - p] + p \cdot R) \cdot D \tag{2.3}$$

since the promised amount is paid with probability $1 - p$, and the recovery amount is paid with probability p. We may use various analyses – for example, historical data or detailed company analysis – to make an estimate of p and R. Such an approach is called the natural measure. We do not necessarily use unadjusted historical data – we are trying to make our best estimate of the real chance of default, and the real recovery in the event of default. These estimates are not necessarily the historical rates – we may explicitly adjust for the state of the economy, known information regarding the company, the state of the property market or other asset markets, etc.

The second concept is referred to as the 'risk-neutral measure'. This is no use for calculating what the price of the contract 'should be' but instead works from market data to calculate precisely the implied default probability and the implied recovery. Suppose the same borrower has also made a separate promise to pay 1 in a year, but nothing in the event of default. Suppose that both these promises are trading actively in the marketplace and that we can observe market prices of P_1 for the contract governed by (2.3), and P_2 for this second contract, which obeys the formula

$$W = [1 - p] \cdot D \tag{2.4}$$

We can calculate p from P_2 ($p = 1 - P_2/D$) and hence R from P_1 ($R = (P_1 - P_2)/(D - P_2)$). This is useful if there are other contracts with the same borrower of the same maturity but do not trade in the marketplace, and we want to calculate a 'fair' value for them.

It should be noted that the use of transition matrices does not prejudge the measure we intend to use. In fact we shall use both natural and implied TMs in following work.

We illustrate the technique for pricing using a simplified matrix. Suppose we just have two ratings A and B, plus default. We suppose the 3×3 transition matrix is given by Table 2.9.

Table 2.9 Illustrative TM for pricing examples

Rating	A	B	D
A	0.98	0.01	0.01
B	0.01	0.96	0.03
D	0	0	1

First, suppose we price 1-year zero-coupon bonds (whose claim amount is par) issued by the same entity which is rated A,[5] but of two seniorities – senior bonds with an assumed recovery of 50%, and junior bonds with an assumed recovery of zero. For simplicity we shall assume that interest rates are zero (discount factor = 1). From (2.3) we see that the senior bond has an expected value of

$$P_{\text{sen}} = 0.99 + 0.01 \times 0.5 = 0.995,$$

and

$$P_{\text{jun}} = 0.99.$$

Note incidentally that the senior bond gives a 0.5% – or approximately 50 bp – price appreciation over the year. The risk of default is 100 bp, and the loss given default $(1 - R)$ is 0.5 – so the rate of loss is 50 bp per annum, which equals the bond spread. Similarly, the junior bond gives a 100 bp return, and the risk of default is 100 bp, but the loss-given default is $1 - R = 1$ since $R = 0$.The rate of loss is again equal to the spread on the bond.

If we took a second entity rated B, then we would find prices of 0.985 (150 bp return = 3% × 0.5 LGD) for senior debt, and 0.97 (300 bp return = 3% × 1 LGD) for junior debt. Generally we find that the spread on the bond is equal to the rate of default times the loss-given default:

$$\text{Spread} = \text{Hazard rate} \times \text{Loss given default (LGD)} \tag{2.5}$$

Now suppose we have a 2-year bond of the 'A' issuer, and suppose the same transition matrix applies now and in the following year. At the end of Year 1 the (senior debt of the) issuer may still be rated A, or may be rated B, or the issuer may have gone into default. From each of the first two states there are similarly three possibilities – giving a total of 9 outcomes in all, each with an associated probability and payoff. Figure 2.4 summarises the possibilities, probabilities and payoffs.

We find that the value for 2-year senior debt is 0.9899 and junior debt is 0.9799.

Exercise 1

1. Repeat the above calculations for a B-rated 2-year bond.
2. Price a 5-year A-rated bond by the above method and 3 × 3 matrix.
3. Use the 'coarse' TM of Table 2.6 to price a 2-year BBB bond.
4. Use the 'fine' TM of Table 2.7 to price a 10-year Baa2 bond
5. Is (2.5) still true for 2-year bonds?

The spreadsheet 'chapter 2.xls' uses a coarse TM and a similar approach to calculate the fair spreads on par FRN bonds. Sheet 'TM data' contains in rows 16–23 the 8 × 8 TM derived from historical data in the rows above. The user can replace this matrix by any other 8 × 8 transition matrix (subject to row sum = 1 and row 23 remaining unchanged) whether historically based

[5] Actually debt is rated not the entity. We assume rating in the TM refer to the senior debt of the company. In this case the senior debt of the entity is rated A. The default probabilities and transition probabilities are given by this matrix and are now taken to refer to the company. The seniority of the debt we are pricing for that name only impacts the recovery assumptions we are using.

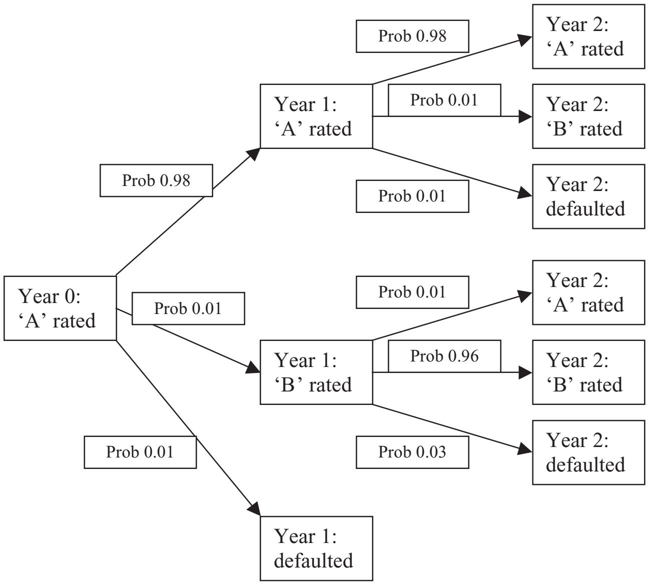

Figure 2.4 Two year evolution 'tree' of A-rated name, and associated bond payoffs.

or not. The sheet 'period TM probs' then calculates the probabilities of transition from each initial rating to each possible state at the end of Year *n*. This is equivalent to calculating the probabilities of arriving at each node on the tree (see Figure 2.2). The 'Spreads' sheet takes assumed recovery (cell C3), initial rating data (B5) and maturity (B6) to calculate a fair market spread (B9). B15–G21 calculates the spreads up to Year 5 for all initial ratings given the recovery assumption.

2.5 SPREAD JUMPS AND SPREAD VOLATILITY DERIVED FROM TRANSITION MATRICES

The transition matrix can be used to calculate more than just the historical fair-value spread, given the TM. If we have a 5-year bond, we can calculate the fair-value spread (if we assume that it is a BB bond with an assumed recovery of 40%, then the 'Spreads' sheet gives 104 bp). In one year's time there is a 0.03% chance ['TM data' cell B20] that the name will have migrated to AAA (at which point the 4-year bond would have a spread of 1 bp ['Spreads' cell E15]). Similarly we could find the probabilities of the other transitions and the forward spreads, and hence calculate the expected **forward spread** and the standard deviation (**volatility**) of the forward spread.

The sheet 'forward spreads and vol' performs similar calculations but for constant maturity instruments (more useful for default swaps) and calculates the forward spreads [C6–C12 for the instrument described in 'Spreads' B3–B9] and the standard deviation [E14] of the spread. B32–G38 performs the same calculations for all ratings and maturities with the recovery assumption of 'Spreads' B3 and expresses the volatility as a percentage of the initial spread.

Note that as rating changes, the spread jumps from one level to another. This is an important aspect of the credit market – spread movements in reality can be very large and very quick. Figure 22.5 illustrates this for some Telecoms 'spreads' (actually CDS premiums).

The 'jump process' implied by the transition matrix approach relatively realistically models an important feature of the credit markets. Rating (or, more generally, perceived quality) is certainly a driving factor behind spreads and spread changes. However, we often see spreads changing (on a daily basis or a secular change) where rating agencies are not changing the rating of individual names. We could say that the perceived quality is changing even though the rating is not, but 'perceived quality' has no other measure than spread. It is also very different from rating as there are more than seven levels of spread in the marketplace. Often the daily movements are small, although there may be a long-term drift in spreads which is not reflected in the rating changes. The spread volatility (cell E14) will not take into account volatility that arises from changing spreads when rating is constant, so E14 should be an underestimate of the spread volatility in absolute terms. Nevertheless, modelling and analysing historical data based around rating is a useful tool and we can make adjustments to reflect the higher volatility we would expect to see in actual spreads compared to historically implied spreads.

2.6 ADJUSTING TRANSITION MATRICES

Suppose we wish to replicate spreads currently seen on 5-year maturity bonds (averaged across a given rating) rather than use the historical TM and get a different spread from actual market levels. One way to achieve this is to increase the default probability for a given rating and multiply the non-default transition probabilities by a factor X so that the sum of the rows remains unity.[6] If $p(D)$ is the historical probability that the particular rating moves into default, and $p(i)$ that it moves to rating i, then we have

$$\sum_i p(i) + p(D) = 1 \tag{2.6}$$

If we increase the default probability by an amount x to $pp(x)$, then we can multiply the other transition probabilities by X, where

$$X = \frac{1 - p(D) - x}{\sum_i p(i)} \tag{2.7}$$

X is the ratio of the new survival probability to the old survival probability. 'Modified TM data' implements this with desired changes in L20–L26, and the following sheets repeat the calculations of the sheets described above. By adjusting L20–L26 it is possible to get 'Modified spreads' F15–F21 to agree with current market spreads rather than historical spreads.

It is possible to further modify the matrix in 'modified TM data' by additionally increasing the chances of transition from the existing rating to other ratings (but not 'default'). This can be done in many ways – one of which would be to choose the amount by which you wish to reduce the probability of the rating remaining unchanged, and increase the probabilities of transition to other ratings collectively by the same amount, distributing the additional amount by increasing these other probabilities by multiplying by a factor. If we reduce the probability

[6] In section 2.6 we are following the general approach developed by Jarrow et al (1997).

of the rating remaining unchanged from $pp(I)$ to $pp(I) - y$, where I is the initial rating, then we multiply the other (non-default) probabilities by Y, where

$$Y = \frac{1 - pp(D) - pp(I) + y}{\sum_{i \neq I} pp(i)} \tag{2.8}$$

Y is the ratio of the new probability that the name changes rating without defaulting, to the old one. 'Modified vols' calculates the new implied volatilities. Note that changes in default and in non-default transition probabilities are not independent, so a manual search for a solution is not easy.

Exercise 2 Implement the shift in transition probability described above. Find the shifts in default probabilities and the shifts in non-default transition probabilities required in order to generate the spreads given in the 'Modified spreads' sheet, and also to generate 100% spread volatility.

3

Asset Swaps and Asset Swap Spread; z-Spread

Asset swap deals are of interest to us for two reasons. First, the 'asset swap spread' is a widely used measure of credit risk in a general sense, and is also used as an alternative to talking about the price of a credit bond. Second, the asset swap contract itself is a derivative involving credit risk and, in some versions of the contract, embeds credit risk in a non-trivial way. For further details of asset swap products the reader should consult Das (2004), or Flavell (2002).

3.1 'PAR–PAR' ASSET SWAP CONTRACTS

3.1.1 Contract Description and Hedging

The contract we describe is referred to as a **'par–par' asset swap**. The investor pays par to an asset swap desk to gain exposure to a par notional amount of a particular fixed coupon bond issued by a chosen reference name. The asset swap deal pays the investor floating payments (of LIBOR [or some other reference rate] plus the '**asset swap spread**') until maturity or until the default of the reference name.

The investor receives the floating payments on pre-agreed dates – which may differ from the coupon payment dates and may be more frequent – and par at maturity. On default of the reference bond the investor receives a cash sum and the contract terminates. From the investor's point of view the asset swap deal converts a fixed coupon bond into an FRN initially costing par.

3.1.2 Hedging

The asset swap trading desk hedges the asset swap deal by entering two contracts. First, the desk pays market price for the reference bond (sometimes buying the bond from the investor). Typically the price differs from par. Second, the desk enters into a contract with an interest rate swap desk where the asset swap desk pays

(a) (on Day 1) the excess of par over the cost of the reference bond [which could be negative – i.e. a payment from the interest rate swap desk to the asset swap desk];
(b) (on bond coupon payment dates on the bond) the fixed coupons received on the bond

and, in return, receives from the interest rate swap desk

(a) (on asset swap coupon payment dates) the floating reference rate plus a fixed spread (the asset swap spread).

The asset swap deal and associated hedges are summarised in Figure 3.1.

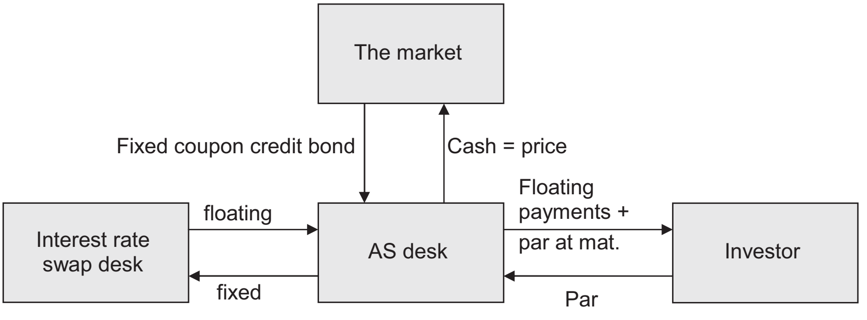

Figure 3.1 Asset Swap contract and hedges.

3.1.3 Default of the Reference Name

On default of the reference name the contract is unwound. The bond is sold for its post-default price (recovery), the interest rate swap is unwound – which may lead to a positive or negative cashflow depending on how interest rates have moved since inception. The net proceeds are returned to the investor, and the contract terminates.

The contract, as described above, is identical to an investment into par notional of the reference bond plus an interest rate swap as described above. It should be noted that, in this form, the investor has full default risk exposure to the reference bond – and maintains economic ownership of the bond.

3.2 ASSET SWAP SPREAD

The asset swap spread is the number found from a calculation which balances the values of the fixed and floating streams in the interest rate swap. There is no simple short-cut to this calculation. Figure 3.2 shows screen images for the asset swap calculation function on Bloomberg for a Carrefour 6.625 2013 bond.

The **key point** to note is that the asset swap spread results from two deals: a purchase of a risky bond and an interest rate swap contract which includes a cashflow on Day 1. The asset swap spread is only one measure of the credit risk on an asset, and we shall see alternative measures (maturity and *z*-spread) below and in Part II (CDS premium).

3.3 MATURITY AND *z*-SPREAD

By '**maturity spread**' we mean the difference between the yield on the bond and the yield off the swap curve interpolated to the maturity date of the bond. This is a quick and easy calculation. Sometimes 'spread' is measured relative to a different reference curve or to a specific reference bond – for example, a government bond. Alternatives are commonly used in defining the terms of a spread option contract.

The maturity spread is intuitively a measure of the risk on a bond. The riskier a bond, the less the market will be prepared to pay for it, hence the lower the price, the higher the yield and the higher the spread.

Rather than maturity spread, the '***z*-spread**' is more commonly used. In practice the two numbers are very close and will be identical if the interest rate curve is flat. Consider a

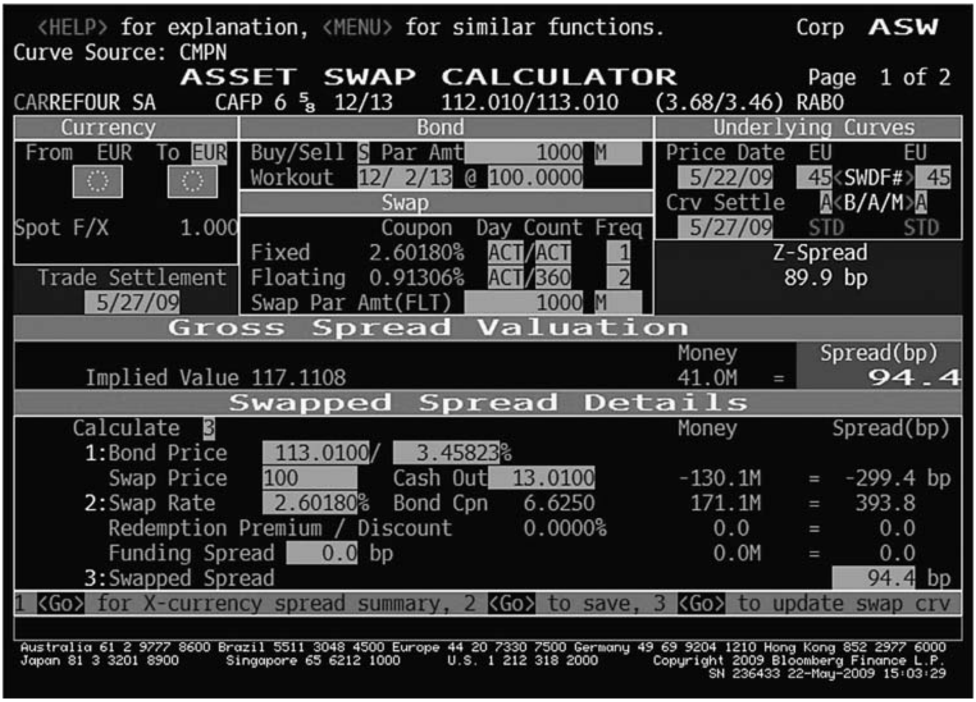

Figure 3.2 Asset swap calculation for Carrefour 6.625 2013 bond.
(Reprinted with permission from Bloomberg LP)

risky bond. If we discount the promised cashflows using discount factors (derived from the LIBOR/swap curve) then we arrive at a value V, which is (usually) different from the market price P – typically $V > P$ unless the bond is less risky than the banks contributing to the LIBOR rates, in which case discounting off the LIBOR/swap curve will give $V < P$. In the typical case, we need to discount off a more risky curve than the swap curve. Do the following: the discount factors correspond to zero coupon rates. Increase all these zero coupon rates by an amount z, and calculate the new discount factor corresponding to the shifted zero coupon curve. Now discount the promised bond cashflows using these reduced discount factors – the more z increases the more V reduces and we can find a z such that $V = P$.

> The z-spread is the amount by which the zero coupon swap curve has to be shifted up so that the market price of the bond equals the discounted value of the promised payments off this shifted curve.

If the reference name is very low risk, the z-spread may be negative.

Note that only the bond transaction (purchase of the bond at the market price) contributes to the calculation of z-spread (or maturity spread). These are ‘pure’ measures of the underlying risk associated with the bond, whereas the asset swap spread is dependent on two transactions – the bond purchase and an interest rate swap deal.

Figure 3.2 shows not only the asset swap spread but also the ‘z-spread’. In the case of the Carrefour bond we note that the z-spread is below the asset swap spread. Also note that, as the bond price is above par, the initial cashflow in the embedded interest rate deal in the asset

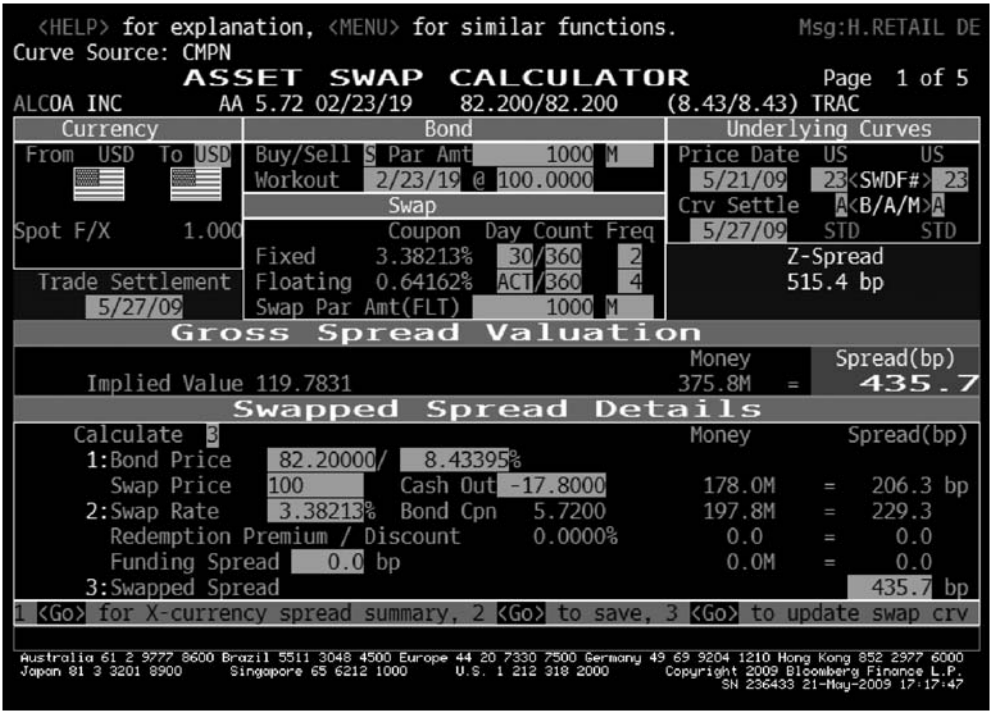

Figure 3.3 Asset swap and *z*-spread for Alcoa 2019 bond.
(Reprinted with permission from Bloomberg LP)

swap contract is from the interest rate swap desk – that is, money is being borrowed at LIBOR to invest in a more risky asset. It should not be a surprise that the asset swap spread is above the *z*-spread in this case[1] since the asset swap is a geared investment in the reference bond.

Figure 3.3 is also an asset swap calculation but now for a high-risk bond.

In this case the bond price is well below par – only part of the investor's cash is invested in a high-risk asset, the balance is invested in a low-risk interest rate swap. It should be no surprise that the asset swap spread is well below the *z*-spread as the asset swap is in this case a degeared investment in the bond.

If we consider non-par asset swaps – those where the market price for the underlying bond is paid – then the spread on this contract will be much closer to the *z*-spread (and if the interest rate curve is flat they should be equal).

3.4 CALLABLE ASSET SWAPS; 'PERFECT' ASSET SWAPS

Several variations of the par–par asset swap contract exist. We shall describe two and leave their analysis to the reader after Part II has been read.

[1] Where the price is 'close' to par, differences in timing of the fixed and floating cashflows and the shape of the interest rate curve can mean that the z-spread may be above or below the asset swap spread. The asset swap spread exceeds the z-spread if the bond is significantly above par, and is below the z-spread if the bond is significantly below par.

3.4.1 Callable Asset Swaps

The asset swap desk buys a bond as a hedge against an asset swap deal. If subsequently the z-spread on this bond falls (because the bond becomes less risky) then the price rises above par (say) and the desk could sell the bond, unwind the interest rate swap, and make more than par on its hedge. If the desk has the right to unwind the asset swap at par, then it can realise this profit. An asset swap with such an embedded option is called a **callable** asset swap.

After reading Part II, the reader will see that a callable asset swap can be analysed as a bond purchase, an interest rate swap, and an option driven by the z-spread on the bond (changes in interest rate curve level are assumed to be exactly hedged by the interest rate swap).

3.4.2 'Perfect' Asset Swaps

One drawback of asset swaps from the investor's point of view is the contingent loss arising on default of the reference name. On this event unwinding, the embedded interest rate swap contract may lead to a loss because of adverse interest rate moves. This threat can be removed by changing the terms of the asset swap: either by the asset swap desk meeting the unwind cost (or taking the profit) of the interest rate swap, or by the asset swap desk retaining any unwind loss but passing on any unwind profit.

In the former case the expected cost is small: if the interest rate curve is flat then the expected unwind cost at any future date is zero and, over time, losses arising from adverse interest rate moves will be offset by profits from favourable moves. Note that as the desk is typically a buyer of bonds (not of external asset swaps), it will have a considerable book of interest rate swaps where it is paying fixed, and will have a large contingent interest rate risk at any one time (interest rate moves have the same impact on the embedded interest rate swap deals on all the asset swap deals). It is difficult to hedge this risk directly – there is an exposure to the reference name(s) but the size of that exposure depends on interest rate levels. 'Double trigger' default swaps (see Parts II and IV) can, in principle, be used to hedge this risk.

Where the interest rate curve is not flat, there is an expected future unwind cost at any date in the future, and this cost can be measured using the techniques given in Part II.

In the latter product variation, expected value – even if the interest rate curve is flat – is not zero. The option can be seen as a swaption but contingent on the default of the reference name.

3.5 A BOND SPREAD MODEL

Investment grade credit debt is usually discussed in spread terms, which naturally leads to a 'spread model' of the bond's price. It is easiest to illustrate this model in terms of a z-spread, z, and the zero coupon curve underlying the LIBOR/swap curve, $r(t)$.[2] The bond's price, P, is then given in terms of the cashflow $c(t)$ at time t by the following:

$$P = \int_0^T c(t) \cdot \exp(-(r(t) + z) \cdot t) dt \tag{3.1}$$

[2] Here we are quoting r and z on a continuous basis.

Typically the cashflow is concentrated on specific coupon payment and maturity dates. The above description in integral terms is general, and includes 'lumpy' cashflows if the cashflow function c is a sum of Dirac delta functions. The integral description is mathematically simple and allows us to concentrate on the underlying features of the model and its implications. In contrast, a valid implementation must take into account details such as day count conventions, etc., and looks more like

$$P = \sum_{i=1}^{N} c_i \cdot \tilde{d}(t_i) + \tilde{d}(T) \tag{3.2}$$

where $\tilde{d}$ are discount factors off the swap curve shifted by the z-spread, N is the number of coupon payments, coupons c_i are at time t_i and maturity is at time T.

If we consider several bonds (of the same seniority) issued by the same entity, how are the z-spreads on these bonds related? Is there a term structure? Is there a coupon structure? How are z-spreads in different currencies related?

Description (3.1) relates the bond price to the promised cashflows, the discount curve and the z-spread for that bond. There is no explicit attempt to model default risk or recovery risk (or 'loss-given' default). The model is purely a translation of price into spread, and vice versa. Nor is there is any explicit attempt to model seniority differences. We would certainly expect senior and subordinated bonds (for example) to trade on different spreads, but the model (3.1) has no insight to offer on how much this difference should be.

We might guess that the z-spread curve for various issues has a term structure but no coupon structure. One implication is that, if we are able to observe three bonds of the same maturity but with different coupons, then (directly from (3.1)) we see that price differences must be proportional to coupon differences. Typically the relative paucity of bullet bond issues, together with idiosyncratic pricing in the bond market, make such implications difficult to test. To test the above, it is difficult to find instances of several bonds from the same issuer with the same maturity date and the same bond structure. However, when z-spreads become very high (i.e. the issuer is 'trading on a recovery basis') the above implication can be disproved.

Intuitively, if we believe that an issuer is likely to achieve a 30% recovery, then a bond priced at 70 (because it has a low coupon) has much lower risk [(70 – 30)/70 = 56% of notional] than a bond of similar maturity priced at 130 (because is has a high coupon) [where the risk is (130 – 30)/130 = 77% on notional]. We would expect the higher priced bond to stand on a higher z-spread.

The spread 'model' (3.1) can be related to the model we develop in Part II, where we price bonds using a default and recovery model. We find that, if recovery is assumed to be zero, then the spread model is a theoretically valid description of bond pricing, where the z-spread turns out to be the default rate (hazard rate) for the firm at time. Typically we would expect the default risk for a firm to be a function of time, so even though the basic form of the price model (3.1) is correct, we would wish to replace the constant z-spread by a function of time.

If recovery is more realistically anticipated to be above zero, then the spread model for pricing bonds cannot be reconciled with a realistic default and recovery model.

4
Liquidity, the Credit Pyramid and Market Data

4.1 BOND LIQUIDITY

We are used to looking at a government bond spread curve and seeing a sensible pattern – yields on bonds of different maturities follow a smooth curve (or surface) driven by maturity (and coupon, for example). A few bonds are special on repo and have yields that do not lie on the surface but are explained by this additional factor.

Credit bonds and loans generally do not trade in the same volumes as government bonds – they are less '**liquid**'. Figure 4.1 shows (maturity) spreads on British Telecom EUR large denomination issues (more than EUR500m) – large and supposedly liquid issues on one of the most heavily traded credit names. Even here the pattern of spreads is not as smooth as we would expect and the deviations cannot be ascribed to repo, seniority or other independently measurable factors. Such differences are usually ascribed to '**liquidity effects**' or '**illiquidity**'.

The situation is worse if we move away from the major borrowers. Many companies only have one or two bonds in issue. One may trade regularly and the other may trade rarely. Even if the maturities are similar, an investor would prefer the bond that trades regularly because it is then relatively easy to dispose of at a fair price, and there is also a transparent market which allows the valuation of this asset. How much cheaper (wider spread) should the other bond be? There is no straightforward answer to this question.

Bonds in issue are often more complex than the simple 'bullet' bond described above: one bond may be a convertible 5-year issue and may trade regularly; the other may be a 10-year callable issue and may trade rarely. Estimating a fair price for the callable, given the convertible spread, involves modelling (in some way) equity prices and volatility, interest rates and volatility, and default risk.

Suppose only one bond is in issue and it rarely trades. How can we estimate the bullet spread?

Looking at marks provided by credit traders may not give us the answer. Front book credit bond traders make markets in bonds and will attempt to make a market in illiquid issues and names. If possible, they will attempt to match a seller with a buyer on such issues. If the desk has to hold an issue on its books it will take account of the capital required to finance this position, the length of time the issue may be on the books, the return on capital required, and factor these items into the desk's bid price. Traders' mark-to-market levels are often driven by the last level at which an issue traded, and stale positions may not be repriced. Bond spread histories for such issues will usually show a 'step' graph pattern – periods of constant spreads with sudden jumps up or down when real deals occur.

4.2 THE CREDIT PYRAMID

The institutional asset portfolio of a commercial bank consists largely of bonds and loans. A large bank may have a portfolio of debt issued by 10 000 or more entities. This might

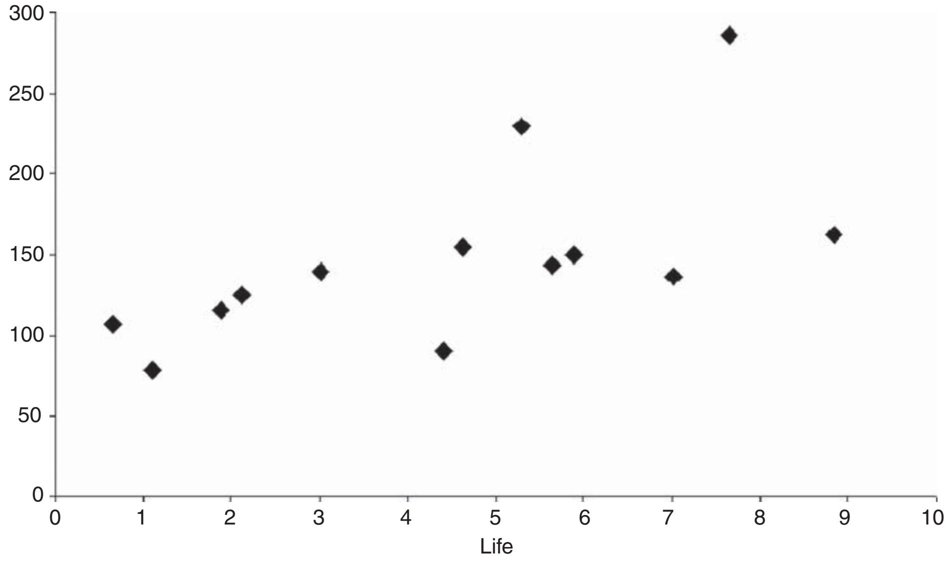

Figure 4.1 BT EUR bond spreads on large issues (greater than 500m; 2009 data).

consist of government debt issued by 20–50 countries, bonds and assignable loans issued by 2000 corporate (non-government) entities, and untraded debt issued by a further 8000 or more entities. Of the 2000 traded issues, perhaps 500 trade regularly in the bond market, perhaps another 500 trade irregularly in the bond and secondary loan market, and the other 1000 trade rarely. This pattern is illustrated in Figure 4.2.

Marking such a book to market would present considerable problems. Generally this is not a problem for 'buy and hold' investors such as commercial banks since they can report on a

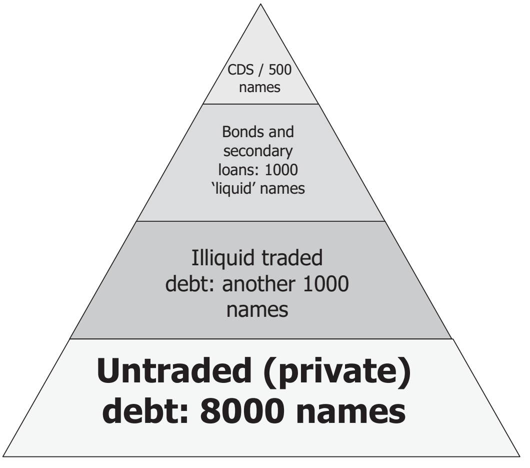

Figure 4.2 The Credit Pyramid.

historic accounting basis (book value, or cost plus accruals). But suppose the bank wishes to clear the economic risk associated with a large part of its portfolio off its books by issuing a derivative related to the reference entities. Investors in the derivative (who are synthetically buying the underlying risks) will need to be aware of the current spread associated with each of the names.

For 5% of the names we can look to an active market and get an accurate answer – at least in certain actively traded maturities. Other maturities present a problem. For the next 15% of the portfolio – the less actively traded names – there has not been an easy solution. The following section outlines several approaches to broadening the active market and to the provision of spread estimates at maturities or on names which are rarely traded.

The bulk of the portfolio does not trade. Information that is available is accounting information for the company which, if there is traded equity on the company, can be used to estimate default risk and spread (as is done by Moody'sKMV, and RiskMetric Group's CreditGrades™ (Finger, 2002), for example). Alternative approaches are to use the bank's internal rating for the entity, plus the bank's own history of borrowers ratings, defaults and recoveries. Internal ratings could be mapped to rating agencies ratings, and current average spreads per rating could be used as an estimate for the fair spread on the name (with some adjustment for seniority differences) – see section 4.4.

4.3 ENGINEERED AND SURVEY DATA

Several products have been developed to try to improve the amount and quality of spread data for the less liquid credits and maturities. Generally this falls into either

(1) 'survey data' where spreads are obtained as estimates rather than as actual trade or firm bid or offer prices from market participants, or
(2) 'engineered data' based on actual trades and quotes seen in the marketplace together with mathematical and statistical methods to fill the gaps in maturity or over time.

Some products combine both these methods. Note that internally a trader may provide the Product Control department with price estimates based on the trader's market feel – falling into the former category. The spreads estimated may be bond spread or default swap premium rates – the principles and methods used are nearly identical. We describe data products for default swap data below.

4.3.1 Survey Data

Banks and other institutions are asked to contribute estimates of spreads for certain maturities on a range of names. The list of names expands as the participants feel confident in estimating the spreads for the name concerned and as the market demands more names. The contributing individuals are effectively acting as market makers and trying to come up with a spread (bid and offer) which they would actually be prepared to deal on. As such, a good market maker will look at various sources of information – known deals, how the market has moved, how this company compares with other similar companies, ratings and changes, current company news, etc.

A criticism of the approach is that the spreads are only as good as the individuals' underlying estimates (although any sensible product offering will clean and rationalise the data it receives), and their methods are not transparent.

Table 4.1 A Mean and standard deviation

(a) 5-yr spread by rating (2002–2004)

Sector and rating	Mean (bp)	Standard deviation (%)
Bank AA	25	30
Bank A	50	20
Industrial AAA	15	80
Industrial AA	40	60
Industrial A	70	50
Industrial BBB	100	60
Industrial BB	200	60
Industrial B	300	50

(b) 5-yr spread by rating (based on Apr 2009 data)

Sector and rating	Mean (bp)
Bank AA	130
Bank A	370
Industrial AAA	33
Industrial AA	130
Industrial A	210
Industrial BBB	350
Industrial BB	750
Industrial B	1100

Two commercial sources of credit derivative data are Markit Group Limited (whose products are described in detail later) and CMA (a wholly owned subsidiary of CME Group).

4.3.2 Engineered Data

This approach uses actual trade, firm quote, or other data and will try to do two things.

1. Given certain points on the maturity spectrum, produce spreads for all other maturities. Typically this is done by fitting a mathematical curve to the known data where several maturities (say 2- and 5-year) are known or, if only one spread is known, by looking at the curve appropriate to similar names (for example, the same rating, or similar spreads) and using that shape to extrapolate the missing spreads.
2. Given a historic spread for the name, estimate where it is likely to be trading now, given data on the general movement of spreads over the intervening period. In order to assess this, appropriate 'tags' may be used – for example, similarly rated companies from the same industry may be used to get an estimate of spread movements appropriate to the particular name.

Such approaches are used by several data providers and internally within banks and other institutions.

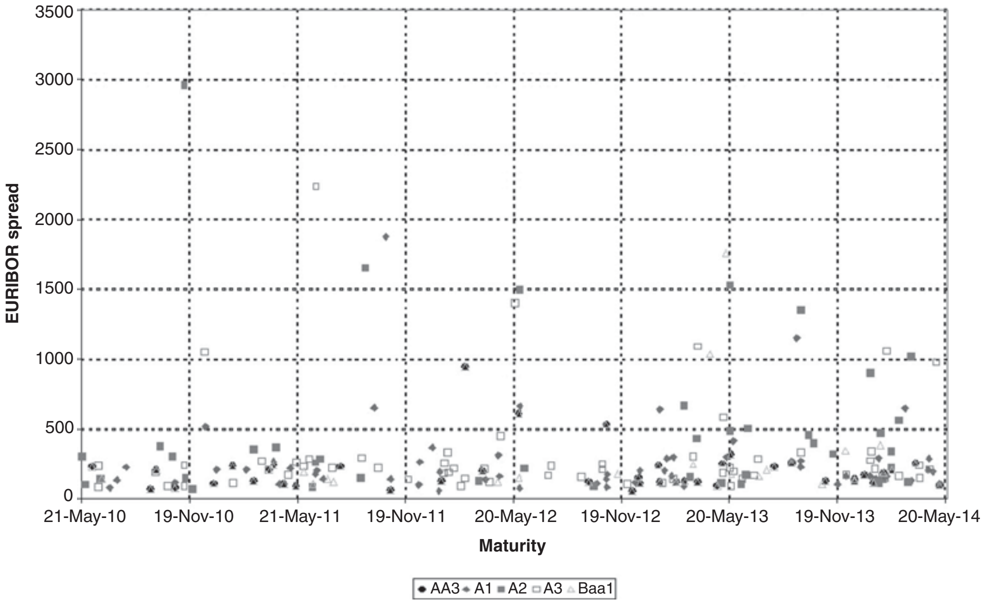

Figure 4.3 Bond spread by maturity and rating (2009 data).

4.4 SPREAD AND RATING

One method of estimating a fair spread for names that do not trade often, or never, is to use rating information. Of course where the name never trades it may not be rated by the public agencies, but a bank's internal rating – mapped to the equivalent rating agencies rating – can be used.

The starting point is to collect data for traded names and calculate the average spread and standard deviation – see Table 4.1 for examples of such an analysis (the results are purely illustrative and will vary over time as indicated).

It should be noted that spread is a very poor indicator of actual spread: a standard deviation of spreads within a single rating band of 60% being typical. Also, there will be considerable overlap between spreads on one rating and adjacent ratings. Introducing further tags (such as country and industry) generally does not improve the situation significantly. The exception is banks and other regulated financial entities, where the standard deviation is substantially less (around 20%).

Such a process of estimation is only appropriate for derivatives based on large portfolios of names (e.g. where a commercial bank wishes to obtain default protection on a large portfolio of loans).

Figure 4.3 shows spreads at a range of maturities for actively traded investment grade (by notch) names.

5

Traditional Counterparty Risk Management

5.1 VETTING

Counterparties risk arises from interest rate swap trades, credit default swap trades and many other deals. The first counterparty risk control measure is to accept as a possible counterparty to a deal only those entities which are (for example)

(a) regulated by a central bank or insurance regulator
(b) rated A (or possibly BBB) or better
(c) domiciled in certain countries.

Counterparties which do not meet these criteria will generally not be accepted as counterparties without other conditions being met – for example full collateral being posted or being accepted for certain types of trades or maturities only.

The intention of these restrictions is clearly to accept only 'low risk' counterparties. Even once a counterparty has been deemed an acceptable risk further measures described below are taken to control the risk on individual deals.

5.2 COLLATERALISATION AND NETTING

The second step (not always applied to all counterparties or to all products for a given counterparty) is to require that the counterparty 'collateralises' deals and signs a 'netting' agreement.

Collateralisation means that each deal is marked to market (and both sides have to agree on this value) and any change in the value from one day to the next is matched by a cashflow from one counterparty to the other. Thus a mature deal – for example, an interest rate swap or default swap that had an initial value of zero – may have moved to being a USD2m positive value for party A and the opposite for party B. Under the collateralisation agreement party B will have paid party A USD2m collateral (on which party B receives interest). If party B defaults, and the interest rate swap contract vanishes, then party A has USD2m capital which it can use to replace the deal with another counterparty.

To be effective, collateralisation also requires a netting agreement – in the event of default by party B the net value of all the trades with party A (some of which have a positive mark to market, others have a negative mark to market) only is a claim of one on the other.

Note that collateralisation does not eliminate the counterparty risk for two reasons.

1. Overnight change on the day the counterparty defaults is not reflected in the collateral posted until the next day. This risk may be positive or negative. Also, in practice, the failure of a counterparty may take some time to establish. A counterparty may fail to post collateral one day because of an administrative error. If the counterparty is failing it will probably try to maintain trading relationships as long as possible – and make excuses why collateral

has not been posted. In practice it takes between about 3 and 10 days to determine that a counterparty has failed and close a deal. A 10-day delay corresponds to roughly 20% of the underlying asset's annual volatility (being the square root of 10/260) and therefore represents a significant risk.
2. If the underlying contract is a credit derivative, there may be a correlation between the counterparty and the reference name(s) in the credit derivative [which is commonly the case for portfolio credit derivatives]. The default of the counterparty will be associated with a widening of the spread on the reference names and a substantial change in the value of the deal. The short-term move in the mark-to-market value may therefore be much more substantial than 20% of the annual volatility.

5.3 ADDITIONAL COUNTERPARTY REQUIREMENTS FOR CREDIT DERIVATIVE COUNTERPARTIES

Credit derivatives (and repo and asset swap deals) introduce a correlation risk between the counterparty and the reference name. Default of the counterparty may be related to a jump in the spread on the reference name(s), hence a step move in the value of the underlying deal when the counterparty defaults.

It is common to require that there is no correlation between the counterparty and the reference deal on credit derivatives, in addition to the other counterparty requirements described above. Assessing whether correlation exists between a reference entity and the counterparty is usually a subjective judgement and is typically based on some or all of

(1) region and industry overlaps
(2) direct and indirect business relationships
(3) equity correlation.

5.4 INTERNAL CAPITAL CHARGE

One way to control counterparty use is to assign a notional capital charge to the counterparty for each specific deal according to the perceived risk of both the counterparty and the deal. This initially requires the calculation of an appropriate measure of the risk on the deal – this might be the 'potential future exposure' (PFE) on the deal or the counterparty VaR for that deal. The more certain deals use up the allocated capital the fewer such deals are done in favour of less counterparty capital intensive deals. We shall see an example of PFE calculation for CDS in Part II.

6

Credit Portfolios and Portfolio Risk

In this chapter we introduce some further terminology applicable to credit risk measurement (VaR and CreditVaR), and we also look at some techniques used to assess the risk on credit portfolios. Detailed understanding of 'VaR' approaches are not required for this chapter – see Jorion (2000) or Grayling (1997) for more detailed information.

6.1 VaR AND COUNTERPARTYVaR

Consider an asset. We can calculate a forward value for that asset but are more interested in how far the value can fall below our expected value. More generally we are interested in the distribution of market values, and the low-value tail of this distribution. We shall refer to this concept as '**VaR**' – we shall assume that we measure this as the deviation of value below the expected value corresponding to a certain percentile of the distribution of the value. Our interest is in credit risky assets – the term '**creditVaR**' is sometimes used to refer to the risk to value arising from changing credit spreads (or defaults).

Additionally a trade may have counterparty risk – for example, an interest rate swap or a credit default swap has exposure to a counterparty. We can calculate the expected exposure to the counterparty at a forward date, but again our interest may be in how great this exposure to the counterparty could be. In this case we are interested in the high-value tail of the distribution of the asset value at the forward date. We refer to this concept as '**counterpartyVaR**'. For both measures we need to be able to produce a distribution of forward values of the underlying asset.

Typically we are not interested in the VaR or CounterpartyVaR numbers on a particular deal. We are much more interested in the VaR on the entire portfolio of assets, or the counterpartyVaR to a particular counterparty for all trades related to that counterparty. For example, suppose we own a 5-year ABC bond priced at par. Then this will have a certain VaR – say X. Suppose we also own an insurance contract on this specific bond covering any loss in value arising from the default event. This will also have a VaR – say Y – and the sum of the two VaR figures is $X + Y$. (Y will only equal X if the distribution of values is symmetric.) But considering the two deals together, if the bond falls in value the insurance will rise in value since the combination of the deals will always be worth par [see Part II]. In this case, the VaR of the pair of deals is therefore zero. We cannot add the VaR numbers for individual deals in order to get this: we have to simulate forward values for the entire portfolio in a consistent way, calculate the portfolio value on each simulation, then look at the distribution of portfolio values to find the VaR.

6.2 DISTRIBUTION OF FORWARD VALUES OF A CREDIT BOND

The approach described here and in section 6.4 is used by the RiskMetrics Group (Gupton *et al.*, 1997) and others for their CreditVaR calculations. The portfolio simulation approach is closely related to the Copula pricing model for CDO deals.

Suppose we need the distribution of values of a bond in one year's time. We shall replace the actual reference name by a 'generic name' – if the reference name's senior bond is rated A we shall replace the name by a generic name referred to by the rating A. We can calculate the expected forward price a bond of this name using the transition matrix as described in section 6.3. We shall do something extra here because we want to be able to price the bond in the presence of other reference entities whose transitions are correlated. For example, they might contain autos, telecoms, consumer good names, etc., and we would expect some significant relationship on how these name progress. We would expect autos to be subject to very similar economic influences and, if sales of autos drop significantly, all auto firms will be affected and all will tend to be downgraded with wider spreads. Similarly, consumer goods and autos will show some correlation because of general economic factors, while utilities and brewing will be much less correlated. So we wish to build in the idea of correlated transitions.

First, we shall create correlated transitions; then, once we have the forward rating for each entity, we use the forward rating and price the bond using the previous method. We have two fundamental choices – either to use the 'natural measure' (estimates of actual transition probabilities) or the 'risk-neutral measure' (one that replicates the bonds actual spread and spread volatility). If we choose the latter, then the TMs are calibrated to replicate the average spread by rating the bonds in the actual portfolio held – rather than market average spreads. Once we have chosen our preferred transition matrix, we can simulate the forward prices of the bond as follows.

Pick a random number x from a normal distribution.[1] Calculate the cumulative probability $\Phi(x)$ associated with x. Compare this probability with the probability from the transition matrix to which the A bond has migrated

1. AAA
2. AA or better
3. A or better
4. etc.

For example, suppose the one-year transition probabilities from A to AAA, AA, ... are 0.0009, 0.0225, 0.9043, 0.0548, 0.0073, 0.0026, 0.0001 and 0.0074. Then the probabilities that the A bond has migrated to AAA, AA or better, A or better, ... are 0.0009, 0.0234, 0.9277, 0.9826, 0.9899, 0.9925, 0.9926 and 1.0000. Suppose $x = 0$, then $\Phi(x) = 0.5$ and the name has migrated to A or better but not AA or better – in other words, it remains A. If $x = -1.5$, then $\Phi(x) = 0.067$ and the name has migrated to AA; if $x = 1.5$, then $\Phi(x) = 0.933$ and the name has migrated to BBB; if $x = 2.5$, then $\Phi(x) = 0.994$ and the name has defaulted. Figure 6.1 summarises the procedure.

The spreadsheet 'chapter 6.xls' 'from random n to rating' gives an example of this approach.

We can use this method to simulate the price of the bond at the end of the year (knowing the simulated rating and the forward life, we can calculate the spread using the TM approach described in section 6.3 and hence the price). Of course, if we are using the natural measure then the prices themselves are unrealistic – but typically we are looking at the distribution of price changes and we apply that change to the actual market price.

[1] There are a variety of ways of doing this. The method implemented in the spreadsheet 'Chapter 6 corr spreads.xls' follows 'Box-Muller' (Fishman (1995), Press *et al.* (2004)). Two random numbers from the uniform distribution are taken (a standard Excel function), one is converted to an exponential distribution, and then a simple transformation produces two independent numbers for the normal distribution. See the 'Normal Distribution' sheet.

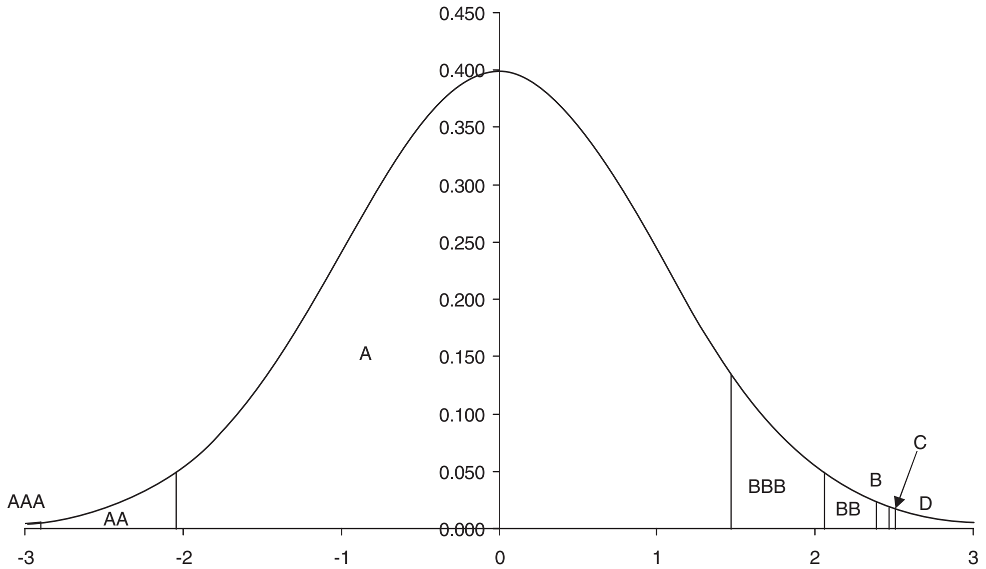

Figure 6.1 From random number to rating: Normal distribution and areas corresponding to rating transition probabilities.

If we use the risk-neutral measure we not only incorporate accurate bond pricing but also a volatility measure for bond spreads based on (for example) actual spread volatility rather than just volatility arising from rating transitions.

We can now use this simulated forward bond price distribution for the VaR calculation. Similar techniques can be used for credit derivatives – and we can use also this method to calculate the distribution of counterparty exposure on a credit default swap at a forward date.

If we need to simulate over an n-year period, we have two choices: either repeat the above method year by year; or calculate the n-year transition matrix and apply the above method once. The latter approach is usually adopted because of speed.

6.3 CORRELATION AND THE MULTI-FACTOR NORMAL (GAUSSIAN) DISTRIBUTION

We wish, somehow, to correlate the transitions and defaults of the names in the portfolio. This turns out to be trivial: if we can produce correlated numbers from the normal distribution, then following through the procedure in section 6.2 we get correlated transitions. But how do we produce correlated normal random numbers? There are various ways: a common and simple method is to take the correlation matrix R, and calculate the Cholesky decomposition, L (Press *et al.*, 2004). The Cholesky matrix is a triangular matrix with the following property:

$$R = L \cdot L^T \tag{6.1}$$

If x is a vector of uncorrelated uniform random numbers, then it turns out that Lx is a vector of random numbers with the correlations in R.

Example Suppose we have two names in the portfolio, and suppose that these have a 0.6 correlation. Then the above becomes

$$\begin{pmatrix} 1 & .6 \\ .6 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & b \end{pmatrix} \cdot \begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix} = \begin{pmatrix} 1 & a \\ a & a^2 + b^2 \end{pmatrix}$$

so $a = 0.6$ and $b = \sqrt{}(1 - 0.6^2) = 0.8$. So if we generate two independent uniform random numbers x_1 and x_2, then the two numbers x_1 and $0.6{\cdot}x_1 + 0.8{\cdot}x_2$ are normal random numbers with a 0.6 correlation. Generally if we take two (uncorrelated) random numbers x_1 and x_2 from a Normal distribution, then

$$\begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix} \cdot \begin{pmatrix} \mathrm{x1} \\ \mathrm{x2} \end{pmatrix} = \begin{pmatrix} \mathrm{x1} \\ \rho \cdot \mathrm{x1} + \sqrt{1-\rho^2 \cdot \mathrm{x2}} \end{pmatrix}$$

gives us two correlated Normal random numbers (with correlation ρ).

The spreadsheet 'chapter 6 corr spreads.xls', sheet 'corr fwd spreads' implements this approach for two names. The extension to many names is mathematically trivial but best handled in code rather than a spreadsheet. The process described above – generating correlated Normal random numbers using a correlation matrix and independent Normal random numbers – is referred to as the **Normal** or **Gaussian Copula** and is discussed in detail in Part III.

If we are looking at a very large portfolio – 500 or more reference names – the procedure described above for correlating the transitions can be replaced by something more efficient, using 'factor analysis' to simulate correlated spread movements and defaults (see Part III).

Note that symmetric matrices with unity on the diagonal, and off-axis elements between –1 and +1, are not always correlation matrices. For example,

$$\begin{pmatrix} 1 & .99 & .99 \\ .99 & 1 & .5 \\ .99 & .5 & 1 \end{pmatrix}$$

is not a correlation matrix. Imagine that the random variables are three coins which may land heads or tails. If A lands heads, then B is very likely to land heads, and C is also very likely to land heads. Likewise, if A lands tails, B and C are both very likely to land tails. So, in most experiments all three coins land with the same face showing. The correlation between B and C has to be high – it turns out that 0.5 is not possible. *The implication is that the correlation matrix cannot be set up arbitrarily, and cannot be stressed arbitrarily.*

6.4 CORRELATION AND THE CORRELATION MATRIX

As described in section 6.2, we replace names in the portfolio by generic names – the total number of names, N, is the same, we are just simplifying the pricing of each name. If we define an $N \times N$ correlation matrix, where there are N names in the portfolio, we choose N correlated normal random numbers and use them to calculate the forward rating. Once we know the forward rating, we calculate the forward price using the method of section 6.2.

The missing step so far is 'How do we obtain the correlation matrix required in the above, and what exactly does it mean?'

First think about the extreme case of 100% correlation – the x's we generate are identical for all names. Names that have the same rating at the start of the period will therefore have the same rating at the end. Names with different ratings at the start will not necessarily change ratings at the same time – the cumulative distribution profile of the transitions for a C name is quite different from that of an AAA name. If $x = 1$ then $\Phi(x) = 0.841$ the AAA name stays AAA but the C name moves into default. (See the cumulative probability table in 'modified TM data' of the 'chapter 6 corr spreads.xls' spreadsheet.)

Even for the simple case of 100% correlation, it is not intuitively clear what this correlation matrix means, or, more generally, how we can measure it directly. Observing the transitions of names over many years is not likely to lead to useful results since, in particular, default transitions are rare and a very large number of years would be required.

There are several approaches to correlation:

1. We could consider each name by name pair and calculate a correlation based on some data history. One approach commonly adopted is to use the equity price correlation matrix. The only justification for this is convenience – apart from those names that have no equity data, it is not clear that equity correlations have anything to do with correlated transitions.[2]
2. We introduced the idea in section 6.2 by describing the portfolio in terms of auto, consumer goods, and telecoms sectors. Here we are using industry as a '**tag**' to group companies together rather than considering a name-by-name correlation. A variety of tags are used in practice – for example, 'industry' alone, or 'industry, country, and high-yield/investment-grade'. Tag correlation is related to 'factor analysis', which is discussed in more detail in Part III.
3. Another approach is to choose correlations arbitrarily [usually driven by tags – e.g. auto with auto in the same region, 0.3; telecom / consumer same region, 0.2; etc.].
4. Yet another approach is to use some form of 'implied correlation' from market price data [see Part III].

In the context of VaR calculations, it is common to use correlations derived from equity data. The process of derivation of those correlations is typically not pairwise but based on a factor approach.

We also have to consider how to model normal ('low') correlation times and abnormal ('high') correlation times – such as occurs during a 'credit crunch'. A typical procedure is simply to stress the correlation number(s) being used to a higher level. Another approach is to use a '**regime change model**'. This is most easily understood if we explain it in terms of a simulation model. In time period 1 we randomly choose a correlation from one of two values – a low value (or set of values across different products) with a high probability or a high value with a low probability – and simulate price movements for this period. We then proceed to period 2 and repeat the process.

We shall revisit these topics in a different context in Part III.

[2] We shall discuss the Merton model and its relationship to this approach in Part III.

7

Introduction to Credit Derivatives

7.1 PRODUCTS AND USERS

Our purpose here is to give an outline of the credit derivative products traded, and to give some indication of how the market has developed. We shall define and examine these products in greater detail in Parts II to IV.

7.1.1 'Traditional' Credit Instruments

Fixed income derivative traders regard fixed income bonds as derivatives – they use the same techniques and models to value both cash and derivative instruments. Similarly, we shall regard credit bonds and loans as credit derivatives. We shall present models that cover these instruments as well as the more obvious credit derivatives.

The traditional credit market covers many more instruments than simply bullet debt. Callable or puttable debt follows the same pattern as for the non-credit market. Such bonds require a model of interest rates and credit (spread and default risk). In addition, convertible debt gives an option to exchange the bonds for a certain number of shares. These add a third dimension to the model requiring equity prices also to be modelled.

Commercial banks have been offering a variety of derivatives of varying complexity for nearly as long as they have been granting loans. A 'guarantee' or 'letter of credit' is economically the same as insurance on a specific debt, and is similar to a single name credit default swap. A 'facility' or 'standby facility' is an agreement to lend a certain amount of money at a fixed spread (over LIBOR, for example) for a predetermined term and within a given timescale. It is economically the same as a spread option on a bond.

We shall define and analyse the modern versions of these products, which trade in the investment banking and investment community generally. The commercial banking forms are often not priced in the same way, and are generally not traded but held to maturity or expiry, and do not require a mark-to-market value – unlike the traded equivalents.

7.1.2 'Single Name' Credit Derivatives

A (single name) **credit default swap** (**CDS**) pays a pre-agreed sum of money on a trigger (credit) event in return for defaulted debt of the reference entity. It is similar to insurance on the debt of the company – the main differences being that it is not an insurance policy and there is usually a range of deliverable debt. The CDS forms the core of the credit derivative business in terms of numbers of deals done.

Spread options arise in a variety of forms. A typical example is the right to sell a bond at a certain spread over a reference rate at a certain time in the future. Instead of a bond, the underlying instrument may be a default swap. Spread options rarely trade in naked form currently – however, they regularly arise embedded in other products such as 'callable asset swaps' and 'callable default swaps'. Callability or putability in bonds is usually more complicated, driven by both interest rate levels and spreads.

Total return swaps are also classed as credit derivatives, and typically arise when an investor (or bank) sells an asset to another investor (or bank) and simultaneously buys back all the cashflows and mark-to-market changes. Effectively the underlying traded asset is exchanged to an off-balance-sheet asset, which is economically equivalent. A total return swap is very similar to a repo trade.

7.1.3 Credit-Linked Notes

A **credit-linked note** (**CLN**) is a structured product in the form of a bond typically created by a bank. The note may terminate early, and repay less than par, on a trigger (credit) event of a reference entity or entities. The simplest example is where a single-name default swap is repackaged into a CLN, but the structure may be much more complicated than this – for example, being related to the risk on a mezzanine tranche of a CDO.

7.1.4 Portfolio Credit Derivatives

A trivial example is a '**basket**' of CDS. A single contract describes a collection of default swaps, but otherwise the two deals are identical. More often the deal is set up as an 'average basket' – subtly different from a collection of CDS in the way the premiums reduce as defaults occur. An average basket is actually a single tranche CDO. Examples are the iTRAXX and sub-index portfolios CDS.

***N*th to default baskets** are quite different products. The market operates mainly on 'first to default' baskets. A payment is made (in return for defaulted debt) on a trigger event for any one of a pre-agreed list of reference names – usually between 3 and 10. The contract then terminates. The *N*th to default basket is becoming a vanilla product among portfolio credit derivatives.

We shall use the term '**CDO**' or '**tranche of a CDO**' to mean protection of losses on a reference portfolio for losses after the first $x\%$ up to $y\%$. The product often occurs when a commercial bank seeks protection on its portfolio of loans. It may retain the first few percent of losses but seeks protection on the rest. Often the term CDO is used to refer to the entire structure – a reference portfolio (which may or may not exist as an actual portfolio) and tranches of protection which, in total, cover all the losses on the reference portfolio.

CDOs occur in many forms. Standard reference portfolios of CDSs have been created and are used to create CDOs with standard 'tranching' of protection. Examples are the iTraxx indices – the successor to TRAC-X and iBoxx indices – and CDX indices. The iTraxx Europe index relates to a standard portfolio of 125, and there are over 10 smaller indices (from 10 to 30 names) representing industry groups, sub- and senior CDSs, high-yield and other sets of names. Market makers trade standard portfolio structures based on this – generally a single tranche CDS (subject to a premium which is close to the average CDS premium when the portfolio started). In addition, they trade a standard indexing structure on the 125-name reference portfolio. The indices are described in greater detail in the Appendix.

Portfolio spread options have also recently been introduced. The presence of actively traded tranches of standard CDOs has opened up the possibility of trading spread and spread volatility on a standard benchmark. The portfolio spread option gives the buyer the right to buy (or sell) the underlying CDO tranche at a predetermined premium.

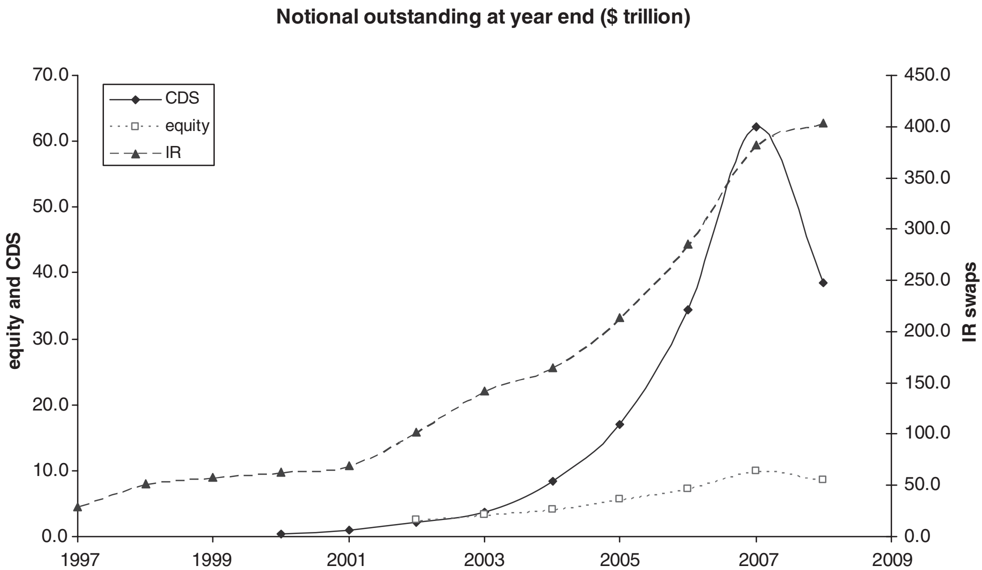

Figure 7.1 Growth in the credit derivatives market.
Data source: International Swaps and Derivatives Association (2004); CDS 2000 data – author's estimate.

7.2 MARKET PARTICIPANTS AND MARKET GROWTH

The modern credit derivatives market started to develop in the 1990s and saw rapid growth from the mid-1990s until the end of 2007. Figure 7.1 shows the historic growth of outstanding gross volume (longs and shorts) for the CDS market compared to the equity derivatives market and the much larger (by about 10-fold) interest rate swap and derivatives market.

Figure 7.2 shows the recent changes in CDO issuance. In this context, CDO excludes unfunded ('synthetic') CDO tranches but includes funded CDO structures, commercial and residential mortgage-backed securities and other asset-backed structures.

It should be borne in mind when looking at market data on credit derivatives that the underlying product is a structured product that is not traded through an exchange. There is no independent source of volume of transactions or size and type of deal, so data, particularly from the earlier years when the market was less transparent, has to be viewed with caution. Figures 7.1 and 7.2 also show quite different statistics – the former being notional outstanding, and the latter being notional amount issued. In addition, Figure 7.2 excludes the principal topic of the latter sections of the book – unfunded CDO tranches typically referring to corporate reference entity risk. In looking at such data we should remember that a variety of very different underlying risks (e.g. mortgages or corporate credit) are aggregated solely because the form of issuance is similar in that the risk is structured.

Figure 7.3 shows the breakdown of the market into products traded in terms of volume of transactions (notional traded) over the period from 1999 to 2006: single name default swaps form approximately 50% of the market. These products form the core of the CD business, are the vanilla trading products, and form the core of the credit instruments that are used to hedge

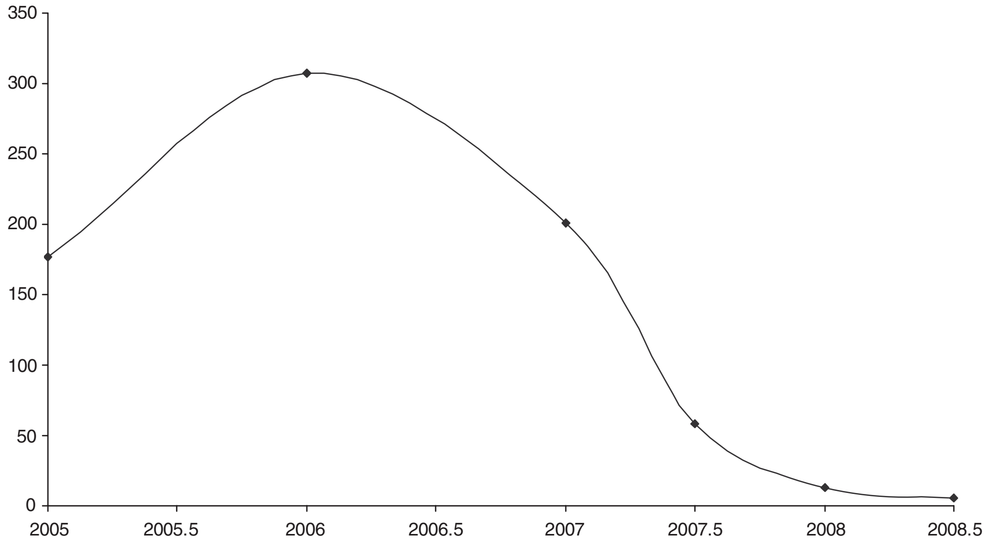

Figure 7.2 CDO issuance, (trillion \$).
Data source: Securities Industry and Financial Markets Association.

more complicated credit structures. Credit-linked notes are also largely embedded single name risks. Note that MBS and ABS products are not included in this survey.

Other single name products – total return swaps and spread options – form a very small percentage of the trades (approximately 10%). Note that embedded credit derivatives – in particular, spread options embedded in other products – are not captured by the survey.

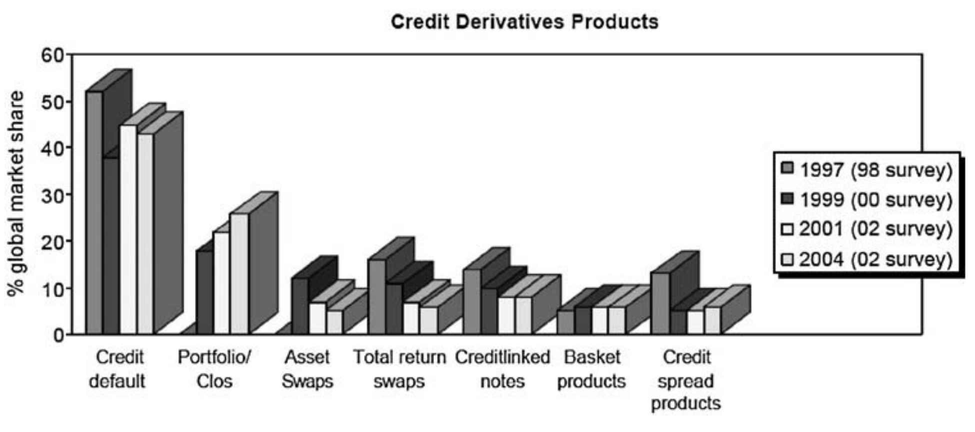

Figure 7.3 Market breakdown by product.
(This extract from the BBA Credit Derivatives Report 2003–2004 is reproduced with the permission of BBA Enterprises Ltd. All rights reserved. © BBA Enterprises Ltd. 2004.)

Portfolio products form about 35% of the total. By numbers of trades these are almost entirely first-to-default baskets that have come to be seen as vanilla portfolio products. CDO deals and related structures form a tiny proportion of the deals by numbers, but tend to be deals in large notional sizes – and entire CDO (all tranches of protection) is typically USD/EUR 500–1500m, and may be as large as USD/EUR 20bn.

By notional outstanding (Figure 7.3) we see a shift in emphasis from total return swaps. This reflects the typically short life (3 months or less) and the large notional size (often in billions) of the TRS deals.

Part II

Credit Default Swaps and other Single Name Products

8

Credit Default Swaps; Product Description and Simple Applications

8.1 CDS PRODUCT DEFINITION

8.1.1 Contract Description and Example

A (single name) credit default swap (CDS) is a bilateral, off-balance-sheet agreement between two counterparties, in which one party ('the writer') offers the other party ('the buyer') protection against a credit event[1] by a third party ('the reference name') for a specified period of time, in return for premium payment.

The reference entity in the transaction may be a corporate (including banking entity) or a sovereign. In the case of a sovereign reference entity the default risk typically refers to obligations issued in a currency other than that of the sovereign.

The 'buyer' and 'writer' are both referred to as 'counterparties'. Note again that 'buyer' and 'writer' are defined here in terms of protection rather than risk. As some traders and investors use the same terms to refer to buying or selling risk (the opposite position), it is obviously important to be clear which terminology you and counterparties are using.

The 'buyer' and the 'writer' are often banks, insurance companies, or hedge funds, but can be any corporate or sovereign entity in principle. The quality of the counterparty to the deal has risk and pricing implications (but note the contents of section 8.1.2).

We can see, before looking at details of the contract, that the deal is a portfolio deal – it is dependent on three parties. We shall examine the pricing impactions of this later – in most of Part II we shall be assuming that both the writer and the buyer are risk free.

CDS Cashflows

1. The contract (typically) pays par in return for 100 nominal of debt if the reference entity suffers a credit event before the maturity of the deal. Deliverable obligations of the reference name are specified in the documentation.
2. The buyer (typically) pays a premium quarterly in arrears (with a proportion up to the default date of the reference name in the event that default occurs before the maturity of the trade).

Example 1 On 2 January 2010, Barclays Bank buys 5-year protection (terminating on 20 March 2015) on Carrefour from Deutsche Bank, on a notional amount of EUR 10m, paying a premium of 100 bp p.a. (quarterly in arrears).

[1] For the moment think of a 'credit event' as 'default'. Section 8.3 defines the term more precisely – for typical CDSs, other events such as 'bankruptcy' and 'restructuring' can trigger the CDS payment.

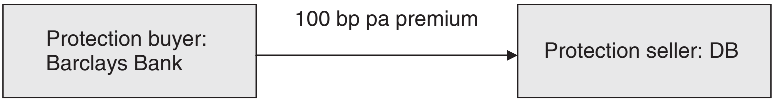

Figure 8.1 Vanilla CDS pre-default.

Pre-default (Figure 8.1): Barclays pays (usually on an actual/360 day count convention) a quarterly amount of (usually a little over) EUR 25 000 to Deutsche Bank (DB).

Post-default (Figure 8.2): Approximately one month after a 'credit event' Barclays delivers EUR 10m worth (based on notional) of debt of Carrefour in return for EUR 10m less the proportion of EUR 100 000 from the previous dividend date to the default date. Note that delivered debt may be YEN, USD or other currency debt, but the notional amount of debt is chosen to be the same as EUR 10m, taking the current exchange rate (on the day that notice of the bond to be delivered is given).

Note that the reference entity (Carrefour) is not a party to the deal in any form, although it is the occurrence of a credit event on Carrefour that triggers the capital payment from DB to Barclays.

The buyer of protection (Barclays) usually has the choice of what debt to deliver – within constraints such as G7 currencies (USD, CAD, JPY, EUR and G BP) and debt being less than 30 years maturity. This is called the **delivery option**. This option was originally introduced to reduce the risk of a squeeze developing on a deliverable issue. In the early stages of CDS trading there were examples of more protection being sold on a specific name and bond than there was deliverable debt to deliver into the contract. A credit event would cause that particular deliverable to rise to par. If the bond could be bought below par, the holders of protection with no debt to deliver could buy the bonds to deliver and obtain the capital gain on the CDS; holders of debt and CDS would have no incentive to sell debt below par; holders of bonds only would be bid until the price reached par.

8.1.2 Market CDS Quotes and Premium Payment

CDS quotes from brokers typically refer to inter-bank deals that are collateralised, and where the counterparties are rated A or better – termed '**professional counterparties**'. Deals on a certain reference entity, to the same maturity, would not necessarily be done at the same price if one or other of the counterparties is of a poorer quality (for example, an unrated corporate entity, such as a typical hedge fund) and/or no collateral agreement is in place.

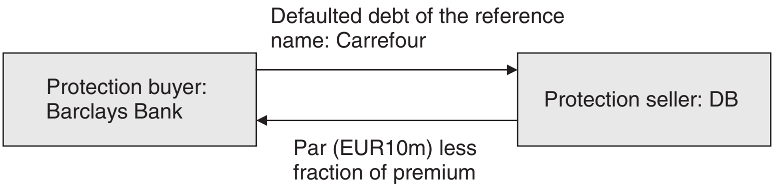

Figure 8.2 Vanilla CDS post-default.

Market CDS quotes also refer to a '**standard**' CDS contract as follows:

(1) payoff of par;
(2) physical delivery with the delivery option (i.e. there is no specified deliverable);
(3) premiums paid quarterly in arrears with a proportion to the default date on a actual/360 day count convention; or a combination of quarterly premiums and an '**up-front premium**';
(4) various forms of restructuring are quoted and, although CDS are normally on senior unsecured debt, subordinated CDSs are also are quoted.

CDS premiums are often referred to as '**spreads**', but this is misleading as they are not a spread to anything (the terminology arose because a CDS premium – expressed in bp per 100 nominal – is of similar magnitude to a bond spread).

Initially (until around 2002) if one traded for a settlement of 16 July 1999 on a 5-year CDS, then the maturity would be on 16 July 2004. If, some weeks later, the trader wished to hedge or unwind the contract, it would mean trading with a maturity that was an odd period – being a few weeks short of 5 years. In a move to increase liquidity, maturity dates were standardised to be 20 March, June, September or December. So a trade for settlement of 16 July 2005 on a 5-year CDS would have a maturity date of 20 September 2010 – i.e. it would be a 'long' 5-year contract initially. This had the advantage that for a period of up to 3 months everyone was trading the same maturity date – increasing the ease with which transactions could be unwound or hedged. In addition, premium payment dates were also standardised to be on the same quarter dates.

CDS premiums – expressed in bp per annum – would typically be up to 500 bp on investment grade names. For high-risk names the premium can rise to a large number of basis points – 2 000 bp (20% of notional per annum) not being uncommonly quoted. If the risk is exceedingly high then the premium could be 7,000 bp or theoretically even 100 000 bp per annum. In the latter case it implies paying 10 times the notional per annum – or 50 times the notional on a 5-year deal over the life of the deal if the reference entity did not default. Such high premiums met with protection buyer resistance and during 2009,[2] when spreads on many names had risen to very high levels, a combination of an up-front premium and a modest ongoing premium (of 100 or 500 bp) was introduced as the standard for USD quoted names. So a very high-risk name might have a 500 bp ongoing premium and a 70% of notional up-front premium. Note also that very low-risk names with a 100 bp ongoing premium could have a negative up-front premium.

8.1.3 Related Products

We classify any product which is broadly economically identical to a CDS as a CDS for the purpose of later discussions. In particular, if a CDS is written under insurance company documentation rather than ISDA documentation, and cashflow triggers are not materially different, then we regard this as a vanilla CDS.

'**Wrapped bonds**' have many of the features of a vanilla CDS contract. In the event of a credit event the writer (typically an insurance company) pays the promised cashflows on the bond rather than the original issuer. This is economically equivalent to paying the capital value of those cashflows at the time of default – a sum that will depend on interest rate levels at the

[2] This formed part of the CDS 'big bang' which introduced a variety of changes affecting the standard form of a CDS contract. These are discussed in greater detail in section 8.7 of this chapter.

time, as well as the life and coupon of the debt. Pricing and valuation are therefore different from vanilla CDS, though the modifications required are straightforward.

'**Average baskets**' refer to a portfolio of names and are documented in a very similar way to a single name CDS. Each name may have a notional amount and premium rate associated with it. On a credit event on one name the notional amount is paid in return for defaulted debt, the premium associated with that name ceases, and the basket continues with the remaining names. In this case an average basket is merely a portfolio of single name CDS contracts, the only difference being that a single piece of documentation is signed rather than many. Average baskets formed the predecessor to standardised portfolio credit derivatives – such as the products based on the iTraxx indices.

An alternative contract (a 'portfolio CDS') has a single premium rate which is reduced in proportion to the notional lost. In this case the basket is actually a single tranche CDO and behaves slightly differently from average baskets (see Part III). The iTraxx-based deals are mostly of this form.

Average baskets should not be confused with first-to-default baskets, as both types are sometimes unhelpfully referred to as 'baskets'.

'**Bank guarantees**' are traditional contracts granted by commercial banks to certain customers. Economically they are similar in form to CDS contracts. Typically they exist on the banking books and are not marked to market (or even 'priced' when they are sold in a way that an investment banker would recognise).

8.1.4 CDS on Loans: LCDS

The underlying reference obligation (see section 8.2.2) on a standard CDS may be senior or subordinated debt, or a loan. In the (rare) case where the reference or deliverable obligation is a bilateral loan, and deliverable loans are bullet instruments, then there is no material difference between this and any other CDS apart from the seniority of the debt. However, the typical case of a CDS where the deliverable obligations are loans has different features from a standard CDS and such a product is referred to as a LCDS. The reference obligation(s) are syndicated loans, and deliverable obligations are limited to such loans. The callability feature of the underlying obligations, together with the presence and impact of loan covenants, can mean that the underlying debt disappears and in such circumstances the LCDS may terminate. Practice is different in Europe and the USA: in Europe the LCDS terminates if the underlying loan is refinanced (called or terminating for other reasons and a new loan issued); in North America the contract can be terminated if no new debt is found within 30 days.

8.2 DOCUMENTATION

8.2.1 ISDA Documentation and Insurance Contract Differences

CDS deals are structured over-the-counter (OTC) products, as opposed to exchange-traded products, and as such are subject to their own documentation. This documentation specifies key items – maturity, premium, reference name, what constitutes a 'credit event', etc. – as well as technical details such as business calendars, legal jurisdiction, day-count convention. The banking community typically trades CDSs under an ISDA master agreement and ISDA standard documentation.[3] This documentation represents a format that can be varied to a greater or lesser extent (and as it has changed over time, many older contracts on the books of traders and investors have slightly different documentation). Again, the trading community prefers standardisation because it avoids 'documentation risk' in hedging. Much of the terminology in the ISDA documentation – including 'credit events' and 'restructuring' – are discussed in more detail below.

The general form of ISDA CDS documentation looks odd at first, as it closely resembles that of an interest rate swap (on which it is indeed based). Terms such as 'fixed rate payer' and 'floating rate payer' are not particularly helpful to an understanding of the product. The very name 'default swap' stems from the interest rate swap documentation that was 'borrowed' to document the first CDS deals, and from the analogy of paying a regular 'swap' premium (as opposed to a single up-front premium) in exchange for the protection on the reference entity's debt. At one time the term '**default option**' was used for deals where the CDS premium was paid up-front, although the term is no longer in common use.

Insurance Contracts and Documentation

A CDS is economically very like an insurance contract on debt issued by a company. The main differences between a CDS and an insurance policy are as follows:

1. An insurance policy calls itself an 'insurance' policy or contract and is written by an insurance company; a CDS is typically written under ISDA documentation and does not call itself an insurance contract (the writer may be a bank, insurance company or other institution).
2. An insurance contract requires the owner of the insurance to own the insured risk at the time a claim is made.
3. In addition, insurance companies (the writers of insurance policies) are regulated by an insurance regulator; banks are regulated by a central bank; and other bodies may not have a regulator other than generally through accounting and legal requirements.

Insurance documentation may look very different from ISDA documentation, and materially different conditions may apply – for example, replacement clauses for reference entities may exist in certain circumstances (e.g. if the reference entity rating drops below a certain level it can be replaced with another reference entity). Often, however, insurance companies will now write CDS contracts under ISDA documentation. The contract is then no longer an insurance contract but simply a CDS where the counterparty is an insurance company.[4]

What are the risk control implications of the following?

1. An insurance contract (written under ISDA) vs. a banking contract. (Minimal – the counterparty is strictly regulated.)
2. A CDS written by a hedge fund vs. a banking contract. (Unregulated counterparty, probably unrated.)

[3] ISDA documentation, including the Big Bang and Small Bang protocols, can be found on their website http://www.isda.org/protocol/index.html and http://www.isda.org/.

[4] The insurance company may have to go through the process of setting up an SPV to write the CDS, and itself write some form of 'guarantee' or insurance on the SPV.

3. A CDS written by a corporate vs. a banking contract. (As above, although large corporates typically have rated debt.)

Exercise 1 How would you control the risk in the above examples?

CDS Maturity, Premium, Trade and Effective Dates

Initially CDS deals were traded on a business day, with an effective date (the commencement of risk) several days later (now typically 1 day), and a maturity date at the business day anniversary of the effective date (say 5 years later). For example, a CDS traded on 5 February 2002 may have had an effective date of 12 February 2002 and a maturity date of 12 February 2007. This meant that a deal traded one day, and another offsetting 5-year deal (in the opposite direction) on the same name one day later, did not exactly match each other – there would be a day's unhedged exposure in 5 years' time (unless the second deal was deliberately done in a 'short' 5-year with a 12 February 2007 maturity). Maturities on new contracts were then fixed to be on 20 March, June, September and December, and coupon payments on the same dates. This meant that a new 5-year deal could have a life of anything from 5 to 5.25 years, and there is also a 3-month window during which trading on the current '5-year' contract will exactly match the maturity of other deals. This has been a major benefit to the trading community and has given an improvement in liquidity. Of course, deals under the old format persist on the books of traders and investors.

On a credit event a proportion of the current period's premium is paid. For example, if the premium is 500 bp and a credit event occurred on 10 May, then $51 \times 500/360$ bp would be paid by the buyer of protection ($51 = 11$ days from 20 March) $+ 30$ (April) $+ 10$).

Payoff and Premium Variations

It has been relatively common (around 5% of all trades) to see other premium schedules – semi-annual, annual, in-advance, without proportion (or refund). Single up-front (or at maturity) premiums also occurred in the past and partial up-front premiums are now the norm on USD deals. A proportion of the up-front premium may or may not be refunded (paid) on early default.

Payoff may be 'par plus accrued' on the delivered asset. This only usually happens when the deliverable itself is specified (see below), and is now very rare.

Deliverable Obligation

Occasionally one comes across a CDS where there is a specified deliverable – i.e. the buyer can deliver one and only one instrument. This is usually related to a commercial bank trade where the bank wants to obtain protection on a particular asset on its books. At one time regulatory treatment was unclear, and specifying the deliverable was an attempt to prove to an uncertain regulator that the protection did in fact cover the risky asset, and regulatory relief could then be obtained. It is now rare to see specified deliverables (see also 'reference obligation' below).

We shall address '**fixed recovery**' CDSs under 'digital' CDSs later.

Settlement – Cash or Physical

Cash settlement is an alternative possibility to physical settlement. On a credit event the writer pays par less the market value of defaulted debt of the reference name (less any accrued premium and subject to a minimum of zero). This typically involves a dealer poll to get independent valuations of defaulted debt of the appropriate seniority and taking an average price. The process has to be spelt out in detail in the CDS documentation, together with possible recourses for the other counterparty if they disagree. Usually the writer is the calculating agent.

Cash settlement is rare in a vanilla CDS contract. It is sometimes offered as an alternative if physical settlement is impossible for some reason (such as an inability to transfer debt because of a newly imposed legal restriction). It is more common in CLNs and portfolio CD products.

Convertibles

Convertibles may be delivered into a CDS contract if they meet the other criteria and are not explicitly excluded.

This was clarified[5] as a result of a case between Nomura and CSFB regarding Railtrack. Railtrack suffered a credit event (it was put into 'administration'). Bonds were not in default as no 'failure to pay' had occurred ('administration/Chapter 11' are credit events for CDS but do not constitute default for bonds). It was perceived that Railtrack would continue to meet its obligations on its bonds – and most bonds were high coupon – so the debt continued to trade over par. Holders of such bonds and of CDS protection would not deliver notice that the CDS had been triggered because delivery of above par debt would result in a loss. On the other hand, writers of protection would deliver such a notice but typically the buyer would fail to deliver since this would result in a loss on bonds over par (the choice not to deliver is always available). However, Nomura owned some convertible debt trading at a price of around 60 (because it carried a low coupon). By delivering into CDS contracts it owned, it could make a profit of 40.

This case points out an additional risk of writing CDS protection rather than owning a bond. A 'technical' credit event (where a trigger event occurs but bonds are continuing to trade around par) allows the delivery of low coupon debt into the CDS for a loss to the writer of par minus the market price of the debt.

8.2.2 Reference Obligations, 'Markit RED' and CreditIDs

Current documentation refers to a 'reference obligation'. The purpose of the reference obligation is to define the seniority of deliverable debt. If a credit event occurs then the buyer can deliver defaulted debt of the same seniority as the reference obligation or a higher seniority (for example, the CDS buyer could deliver transferable loans even if the CDS referenced senior unsecured debt). Some CDS deals written in the past under ISDA documentation do not have a reference obligation. In these cases the CDS refers to senior unsecured debt.

[5] The uncertainty centred around the non-deliverability of 'contingent debt' under the ISDA documentation. It was not clear whether convertibles were contingent or not, and the court case clarified this.

The aim of Markit RED[6] (Reference Entity Database) is to define a standard list of reference obligations for each reference entity that trades in the credit markets.[7] The aim is to help to avoid confusion and simplify the process of choosing an appropriate reference obligation of the correct seniority at the time the CDS is written. Any error in the choice of reference obligation may make the CDS legally unclear or may change the seniority from what was intended. The use of standard identifiers – the 'Markit RED 6 and 9 digit codes' – helps to avoid such errors.

Quoted from Markit [8] :

Markit RED provides confidence with market standard, scrubbed reference entity and obligation pairs.

- More than 3 000 scrubbed reference entity/obligation pairs.
- Audited, accurate legal reference entity names.
- Proprietary, tried and tested reference data scrubbing process.
- Unique, six-digit alpha-numeric Markit RED Codes to identify reference entities.
- Unique, nine-digit alpha-numeric Markit RED Codes to identify a reference entity/obligation pair.
- Complete re-scrubs at least annually or in the event of a corporate action.

The LCDS market introduces additional complications and has led to an expansion of the Markit RED Codes and the introduction of additional identifiers. Again quoting from Markit Group Limited public and private material:

It is important to note that due to the differences in geographical structure and syndicated lending practices, there are fundamental differences in the way in which LCDS is traded in North America and in Europe. In North America, LCDS is a means to carve out credit risk inherent on a reference entity. European LCDS is a means to buy and sell credit protection contained within a specific credit agreement issued for a borrower. Correspondingly, Markit introduced Credit IDs to better identify appropriate pricing for each European credit agreement for each credit lien. Credit ID is not a replacement for Markit RED LCDS; however, this is a complementary identifier to allow contributors to easily price ELCDS curves and to allow them to submit data with greater accuracy. The Credit Agreement-oriented CreditIDs are utilised for ELCDS rather than entity-based Markit Tickers to map a European LCDS curve. In addition, RED LCDS identifiers are included in the pricing report to ensure the highest degree of accuracy in the representation of credit curves for lien-ranked pricing and benchmarking purposes.

Markit RED has been further developed to support the growth in trading LCDS by providing transparent reference data. Markit RED verifies the reference entities, credit agreements and loan facilities used as reference obligations in the LCDS market.

In North America, Markit RED verifies and maintains all underlying syndicated secured loan reference data contained in the Syndicated Secured List (SSL). Reference entities are uniquely identified with 6-digit Markit RED Entity Codes. All identifiers are updated following a corporate event or refinancing. Markit works proactively with all major dealers of LCDS to ensure that the SSL provides updated information on most liquid names.

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[6] RED is operated by Markit

[7] In the process, Markit RED also identifies the precise legal reference entity – the issuer, or the guarantor, of the debt. For example 'Ford Motor Credit Company' and 'Ford Motor Company' are different legal entities: both trade in the credit derivatives markets with very different risk profiles.

[8] Information provided courtesy of Markit

In Europe, Markit RED has been established as the definitive standard for verification of LCDS reference data. The Markit RED service supports the latest non-cancellable ISDA documentation, providing transparency into refinancing and cancellation events through the Markit RED Continuity Procedures for European LCDS. Reference credit agreements are uniquely identified with 6-digit Markit RED Codes and the reference obligations of a common lien ranking are uniquely identified with 9-digit Markit RED Codes.

Markit RED identifiers for LCDS allow users to make a connection between synthetic and cash loan data. Additional benefits of Markit RED LCDS include:

- Markit RED LCDS provides regular, comprehensive monitoring of LCDS reference entities and obligations. Unique Markit RED Code identifiers allow subscribers to efficiently and accurately confirm LCDS trades and facilitate Straight Through Processing.
- Markit RED LCDS provides verified reference data for North American and European loan credit default swaps. By providing legally verified information in Europe, it greatly reduces subscribers' legal costs.
- Markit RED LCDS provides market transparency through managing LCDS Continuity (identification of one or more related Substitute Reference Obligations) in cases of refinancing, cancellation or when a substitute reference obligation is unclear.

Key Features

- Legally verified Reference Credit Agreement names and facility information for over 160 European Credit Agreements and over 580 market verified records of entity names and reference obligation details in North America.
- Corporate event and refinancing event notifications.
- Unique six-digit alpha-numeric Markit RED Codes to identify reference entities in North America and reference credit agreements in Europe for use in matching on DTCC.
- Additionally, unique nine-digit alpha-numeric Markit RED Codes to identify reference credit agreement and ranking pairs for Europe.
- Subscribers can request additions of new reference obligations to the Markit RED LCDS database saving legal costs and improving time to confirmation.
- Mapping of LCDS and Cash Loans pricing and associated reference data currently available for North America and Europe.
- All Markit RED data is available via website interface or automated XML downloads.

8.3 CREDIT TRIGGERS FOR CREDIT DERIVATIVES

8.3.1 Credit Events

Bonds are at risk to a 'default' event. What constitutes 'default' is specified in the issue document of the bond. Similarly, CDS documentation must specify what constitutes a 'credit event' – the event that triggers the capital payoff on the CDS. Like most in the market we shall usually refer to a CDS credit event as 'default' although it is wider than bond default.

The key credit events are:

1. Failure to pay. This usually has to be of a certain amount of money – the failure to settle a telephone bill on time will not usually meet the requirement!

2. Bankruptcy – Chapter 11 (US), 'In Administration' (UK) and similar events. This may occur without a 'failure to pay' (default), and without an ultimate debt restructuring (e.g. Railtrack in the UK).
3. Restructuring (see below).

8.3.2 Restructuring

'Restructuring' refers to the event in which the borrower rearranges the debt of the company, often to the detriment of the lenders but usually with their permission (the alternative – not giving permission – being worse). This may involve lowering the coupon, lengthening the debt, reducing the nominal, reducing the seniority, or otherwise replacing the debt with something of lower value. Sovereign borrowers typically restructure, and it is far from uncommon for corporate entities to restructure. Until around 2001 when Conseco agreed a restructuring with its bankers, it was normal to trade CDSs with 'original restructuring'. The precise restructuring clause included in the documentation has varied over time: the list below gives the current position and recent history.

1. No restructuring trigger (rare until 2009, then standard for US default swaps from April 2009).
2. 'Original restructuring' (USA until 2001, EU until 2003/2004).
3. 'Modified restructuring' (USA 2001–2009).
4. 'Modified modified restructuring' (EU, occasionally USA).

The detail of the following is not necessary for an understanding of the rest of the book, but is described for completeness.

'Original restructuring' did not have a materiality clause but required the following:

1. A reduction in the rate or amount of interest or the amount of scheduled interest accruals.
2. A reduction in the amount of principal or premium payable at maturity or scheduled termination date.
3. A postponement or other deferral of a date or dates for (a) payment or accrual of interest, or (b) payment of principal or premium.
4. A change in the ranking in priority of payment of any obligation causing the subordination of that obligation.
5. Any change in the currency or composition of any payment of interest or principal where this results directly or indirectly from a deterioration in the financial condition of the reference entity.

The Conseco case caused US market participants to question the clause. Conseco and its banks agreed to a restructuring of certain loans, including an extension of maturities and a commitment to pay down those loans early from the proceeds of asset sales. Holders of long-dated bonds (trading at very depressed prices) and of CDS protection, delivered such debt into CDS protection under the above restructuring clause. Participants realised that the CDS contained a significant 'cheapest to deliver option' triggered by restructuring, and this feature makes a CDS significantly different from a synthetic bond. (The same is true of 'administration/Chapter 11' without a failure to pay, but this is much rarer than restructuring.) New York dealers then started trading CDS with no restructuring trigger, but this caused regulatory problems in some areas and led to the introduction of the 'mod R' clause.

Modified restructuring sought to limit the cases when a restructuring could trigger a CDS contract and introduced 'restructuring maturity limitation'.

On the occurrence of a restructuring event only debt which has maturity on or before the earlier of

(a) 30 months after the restructuring date and
(b) the latest maturity of the restructured debt
is deliverable, but subject to any debt being deliverable that has a life less than or equal to that of the CDS itself.
The above do not apply if the seller triggers the CDS.

Generally this results in deliverable debt being of a similar maturity to the CDS (if this is longer than 30 months), and results in the CDS being closer to a proxy bond.

Other conditions introduced at the same time were:

(c) restructured facilities must have at least four creditors and require at least a two-thirds majority to approve,
(d) 'consent not-required debt' only is deliverable, or is generally transferable ('rule 144A reg S'),
(e) partial settlement is allowed.

Modified modified restructuring makes two changes to the above in a broadening of the scope of deliverable debt.

(a) *Restructured* debt with up to 60 months' maturity is deliverable (other deliverable debt remains at 30 months),
and
(b) the deliverable obligation must be *conditionally* transferable (consent may not be unreasonably withheld).

8.4 CDS APPLICATIONS AND ELEMENTARY STRATEGIES

The main application of single name CDSs is as a 'building block' for other more complex trades and as a liquid hedging tool. In the process, the vanilla CDS defines market levels. In this section we look at some basic applications of single name products in general terms, covering more detail (and more applications) in Chapters 10 to 14.

8.4.1 Single Names

Directional Trading/Relative Trades

If you take the view that spreads are going to narrow on BT, then you could buy BT debt. Alternatively, you could sell CDSs referencing BT – say 5-year at 200 bp. If you are right and spreads narrow to 150 bp then you could buy back the CDSs at 150 bp making 50 bp per annum for 5 years – a profit of a little under 2.5% of notional (we look at an exact calculation in Chapter 12).

Using a CDS rather than a bond does not require capital but does require that you are a 'professional counterparty' so that the buyer of protection will trade with you on an unfunded basis.

Alternatively, if you believe that BT will widen, a long protection position will make a similar profit if spreads subsequently widen to 250 bp and the protection could be sold for a 50 bp profit per annum. The alternative of shorting debt is difficult to achieve. To some extent, the CDS market developed as an alternative to an open repo market in credit debt. CDSs offer a simple practical means of taking a bearish position on credit debt.

Let us assume that you consider BT to be more attractive than Deutsche Telekom. Selling protection on BT makes money if spreads narrow, but loses money if spreads widen. The opposite position in Deutsche Telekom – buying protection to the same maturity – will largely offset this if spreads move together. Money will be made if – as you expect – spreads widen on Deutsche Telekom relative to BT.

Exercise 2 Suppose 5-year BT is trading at 150 bp while Deutsche Telekom is 200 bp. Work through the example where:

(a) spreads both widen by 50 bp;
(b) BT widens 20 bp but Deutsche Telekom widens by 70 bp.

Debt Hedging

An owner of 5-year senior unsecured France Telecom bonds could hedge the risk (in a general sense) by buying CDS protection. In the case of a credit event during the life of the CDS, the debt can be delivered into the CDS and par is received. The value of the bond (or its purchase price) may be 90, so this represents over-protection (equivalently, too much is being paid for the protection).

Prior to default, widening of the spread on FrTel bonds (leading to a fall in capital value) is likely to be accompanied by widening of the CDS spread (leading to a mark-to-market profit on the CDS). Therefore, some spread hedging occurs between the two. Precisely how much depends on the 'delta hedge ratio' and the 'basis risk', both of which we examine in detail in Chapter 10. The CDS therefore protects against not just the default event risk but also the mark-to-market change in value of the bond arising from spread change.

Suppose now that the FrTel asset is not a bond but a bank loan. It may be possible to arrange a CDS referencing loan at an appropriate cost, but liquid CDS contracts usually cover risk on less senior debt and are more expensive. Even so, loans – being more senior than bonds – are in principle deliverable into standard CDS contracts. But on a credit event the owner of the loan may have a problem delivering the loan, as it may be non-assignable. In this case the CDS owner will have to buy defaulted debt to deliver to receive par and a profit of par less the recovery on bonds. The loan will remain on the books until 'ultimate recovery' is received – which will probably be more than the recovery price paid for the debt delivered into the CDS. The portfolio manager can therefore protect non-assignable loans using vanilla CDSs, albeit at a cost.

8.4.2 Sector/Portfolio Trades

Just as CDSs can be used to reflect a 'bullish' or 'bearish' view[9] on an individual name, they can also be used to reflect an absolute (spreads falling or rising) or relative view of one sector

[9] Interpreted with respect to bond prices – i.e. 'bullish' means bond prices are rising, bond spreads are falling and hence CDS premiums are falling

versus another – e.g. bullish of autos versus telecoms. Some index-based CDS products may be appropriate – for example, the iTRAXX TMT (telecom, media and technology) and autos indices – but often the view has to be implemented by dealing in a portfolio of representative names.

If your view is that the motor industry looks attractive relative to telecoms, then again you could write protection on a portfolio of auto names and buy protection on a portfolio of telecom names in order to reduce the exposure to any single name. The usual minimum trading size is 5m per name. If you think five names is the minimum number to get acceptable diversification, then a 100 bp move corresponds to a profit or loss of around 1.25m (25m $\times$ 5-year $\times$ 1%). Clearly how the view is implemented depends on how large an adverse move you are prepared to withstand, and the size of your risk limits.

Exercise 3 Suppose your risk appetite is 100 000 (i.e. you are prepared to risk a 100 000 loss on a deal before cutting it and licking your wounds). You set a stop at a 20 bp adverse move. How many names could you incorporate in the 5-year mini-portfolio in order to implement the trade?

Hint: On 5m nominal, what loss does a 20 bp move correspond to?

An alternative approach would be to seek average basket deals, each referencing 5–10 names with a total exposure on each of around 20m. The average basket allows more names but with a smaller exposure per name because a single piece of documentation (for each basket) is used. The disadvantage is the limitation on the possibilities for unwinds. Average baskets have less appeal than single name products and offsetting trades are harder to achieve – liquidity is lower – often tying the buyer to go back to the original counterparty for an unwind.

Typically this style of trading is implemented by proprietary or hedge fund traders and as an overlay to the general hedging of positions by a trading book.

8.4.3 Income Generation

Certain funds sell options (for example, equity call options) as a means of enhancing return on a portfolio. Often this is done against positions actually held – a large rise in the share then results in a limited gain (on assets held) rather than an actual loss. CDSs can similarly be used to generate income by selling protection, but this would not normally have a natural hedge already in place. Such a trade often appeals to insurance companies or banks who are typically investors in risk, but so far investors have not generally traded in this way, preferring to build a portfolio of credit bonds instead.

The question then is how best to reduce risk on the portfolio (by diversification) and how to maximise a risk-adjusted return. This is generally handled by setting limits on individual exposures, country exposures, industry exposures, rating exposures, etc. Limits could be set in nominal terms, possibly by reference to risk-weighted exposure (nominal times default rate time life).

Exercise 4 The objective is to generate an income of 20m per annum subject to a ‘risk’ of the premium amount in any year. What portfolio make-up would you suggest?

Hint: The first problem is to clarify the question in the investor’s mind. This is best done by working through an example. Suppose we have a portfolio with an average spread of 100 bp – and a 2bn size. If this is a *managed* portfolio of BBB names, then the historical default rate may be 0.3% per annum (the one-year historical default rate, even though it is a 5-year portfolio). If there are 100 names in the portfolio, then we would expect 0.3 defaults, and probably[10] less than $0.3 + 2 \times 0.55 = 1.4$ defaults; so if recovery is 50% we would not anticipate a loss of more than 7m. Write a list of discussion points (e.g. spread and historical data; variability of rates over time; portfolios size and correlation; mark-to-market risk; ...).

8.4.4 Regulatory Capital Reduction

This example illustrates some of the basic principles of using CDSs to reduce regulatory capital. The example is not commonly implemented in this form – rather portfolio trades are done – and the example relates to the rather simpler Basel 88 regulations for clarity. Regulatory treatment under Basel II is discussed briefly in Chapter 19 with a more realistic example trade.

Suppose a bank owns a EUR 10m 5-year loan to a corporate entity generating 80 bp in spread. Currently the bank has to allocate 8% of notional as capital against this trade, which therefore generates a 10% return on capital.

The bank buys CDS protection on the reference entity at 60 bp – spreads may have narrowed since the loan was issued – for the same or slightly longer maturity then the loan (this is required in order to get any capital relief). The net income is now reduced to 20 bp. The regulatory capital depends on the counterparty to the CDS trade. If this counterparty is a G7 bank then the 8% requirement is reduced to 1.6%, so the return on capital is now increased to 12.5%. Capital which is released can be used to put in place new loans earning not just the spread but also fee income.

Economic Capital Reduction / Releasing Lines

The same deal could be done to reduce 'economic capital'. An institution has internal rules for assessing risk on any deal – loans and CDS etc. These rules may be rating and maturity dependent and will typically allocate an amount of capital to a deal which is used both to assess profitability – Return on Capital – and to restrict total risk. The regulatory capital example above applies equally to economic capital.

A commercial bank imposes limits on the exposure to a single client – a 'line'. Once a line is full no further lending can take place to that client. The option of selling existing loans so that new loans can take place may not be attractive. The bank may require the consent of the borrower to sell these loans, which may jeopardise the client relationship. BT may not be aware that a CDS deal referencing its debt has taken place – hence this offers a bank the chance to reduce the risk to BT on its banking books *synthetically* without risking any banking relationship that may exist between the bank and BT. The non-assignable loan stays on the books but, because CDS protection is bought, then the exposure (line) to BT is reduced during the life of the CDS, and new lending can take place.

[10] Think of the distribution of defaults in any year as a binomial distribution whose expected number of default is (number of names, n) times (default probability of a single name, q), with a standard deviation of root $(n \times q \times (1 - q))$. (See any book on probability – for example, Billingsley (1995).)

Other

We look in detail at examples of curve trading, recovery trading, and other applications from Chapter 10 onwards.

8.5 COUNTERPARTY RISK: PFE FOR CDS

The traditional approach to counterparty risk is covered in part I section 5, together with the question of correlation risk for CDS (and other deals).

Potential Future Exposure and Capital Allocation

A further step to prepare for counterparty risk is to reserve against it. Take a EUR 20m 5-year long CDS on a reference name subject to a contractual premium of 100 bp. The market spread yesterday was 200 bp and today it has moved to 210 bp. What is the current (approximate) M2M? What key factors affect the M2M in the future? At what time in the life of the deal does potential future exposure reach a maximum?

The current M2M is only the accrued premium if the premium on the deal is equal to the current 'fair' market premium. If spreads are static, M2M remains zero, but if spreads are volatile, it can become positive or negative. As life increases, the forward CDS spread becomes more uncertain but the life is shortening so the value of the premium difference is declining (see the spreadsheet 'chapter 8.xls').

If the premium on the trade is other than the current fair market premium, then there is the initial mark-to-market value to add on. This may be positive or negative (e.g. in the above example the premium on the deal is 110 bp from today's market level, increasing the exposure to the counterparty).

How would you allocate capital? One approach is to take the maximum PFE × default risk of the counterparty. If a cautious estimate of default risk is taken, this should reduce as the number of counterparties increases. A more sophisticated approach would be to look at counterpartyVaR (Figure 8.3).

8.6 CDS TRADING DESK

8.6.1 Mechanics of Transacting a CDS Deal

As a structured product a CDS transaction requires the verification and exchange of contracts, involving the legal department. For vanilla deals this has been streamlined. Initially (circa mid- to late- 1990s) 5–7 business days were allowed between agreeing the outline terms of the deal (**term-sheet**) and signing the documentation (**confirmation**). This has been reduced and the term-sheet is now no longer used for vanilla deals between professional counterparties, but a general form of contract is agreed between them before they begin dealing with each other. With that in place the particular deal contract is usually signed within one day of dealing.

A second important item to check is the quality of the counterparty and the counterparty/ reference entity correlation risk assessment – tasks performed by the credit risk department.

On the trade date: general terms are agreed with the counterparty (possibly via a broker) – reference entity, maturity, premium, etc. A term-sheet is produced for non-vanilla deals and passed to legal; the trade details are entered into risk management and reporting systems (as

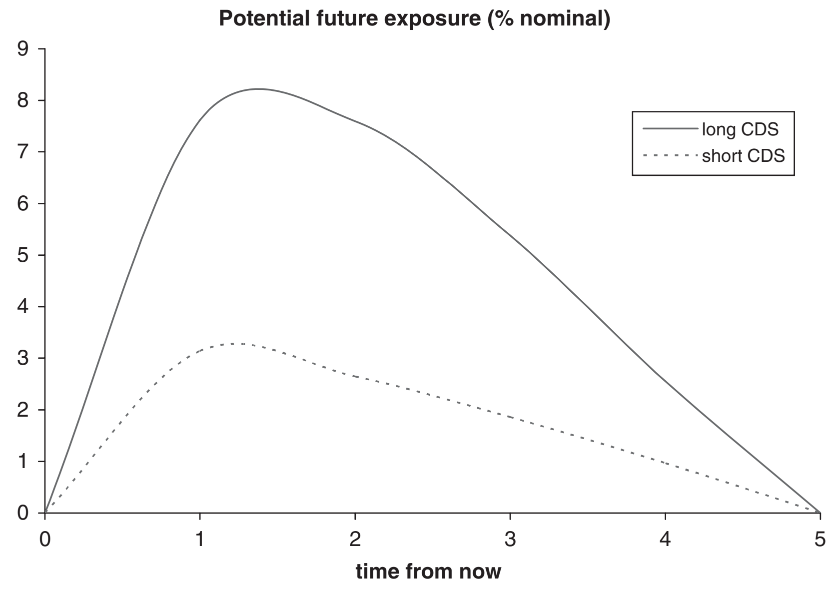

Figure 8.3 Potential future exposure on a 5-year CDS subject to 100 bp premium.

a ‘pending’ deal); credit risk approves lines for the counterparty and approves the correlation risk.

On the effective date (trade date + 1 days): The confirmation has been signed by both parties; the trade moves from ‘pending’ to ‘live’. The writer goes on risk and premiums start to accrue.

8.6.2 Trade Monitoring, Credit Events, Unwinds

Trade monitoring includes the following processes:

1. Update of market data for risk assessment.
2. Validation of and checks on market data against independent sources.
3. Generation of risk reports, P&L and all other reports.
4. Reconciliation of P&L changes with market moves.
5. Updating counterparty risk exposures; posting of collateral.
6. Monitoring for credit and rating events.

A credit event will spark a variety of actions. Deals will typically need to be rebooked, with updated deal information for credit event data – i.e. when the credit event occurred, what bond is delivered, the notional amount and currency, which notices were given and by which party, the amount of accrued premium, the final settlement amount (cash).

An ‘unwind’ will cause the original deal to stay on the books (to allow historical investigations), although the risk will be terminated. New data to be recorded includes the termination date, final payment, and calculation basis.

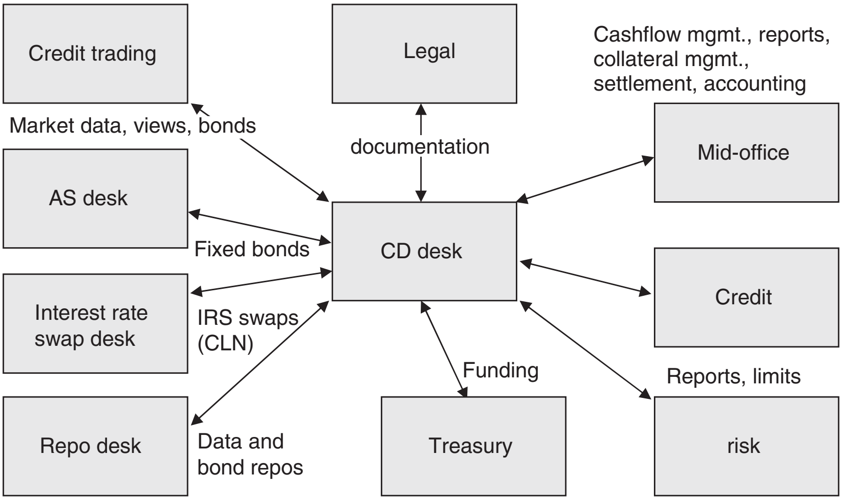

Figure 8.4 CDS dealing desk infrastructure.

8.6.3 CDS Desk Interactions and Organisation

The desk interacts with credit, repo and asset swap trading desks, the sales desk, treasury and interest rate swap desk (particularly where CLN deals are concerned), and with risk and mid-office departments who monitor the deal and cashflows arising from the deal. The interactions are summarised in Figure 8.4.

The 'desk' itself may consist of geographically separate sections whose primary responsibility is for a region (typically North America, Europe and Asia). As one market closes, responsibility for the book of that region passes to another desk – where there is limited trading in CDSs outside that desk's prime area of responsibility.

In addition, responsibility for investment grade reference entities and for 'high-yield' and 'emerging market' reference entities is usually split.

8.7 CDS CONTRACT AND CONVENTION CHANGES 2009

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8.7.1 Credit Derivatives: Review

The growth of the structured credit market was driven both by the desire of banks to move capital intensive assets off balance sheet and by fixed income investors hungry for credit-risky assets. One of the innovations that enabled this to happen was the development of the credit derivative market which, over the last 10 years, has transformed the trading of credit risk. Credit risk (risk of default) is essentially 'sticky': it is very hard to move because of legal restrictions, relationship issues, accounting obstacles, tax and operational challenges. Many of the most recent developments in the structured credit market have been possible because credit derivatives have removed some of these obstacles.

A credit derivative in its simplest form – the most common product is the credit default swap ('CDS') – transfers the credit risk of a debt instrument from one party to another. This process is often referred to as synthetic risk transfer. The effectiveness of the risk transfer is only as good as the legal contract between the two parties. For every party taking on the risk, there is another party laying it off. Should the risk-managing party actually own a debt instrument that needs to be hedged and the risk-taking party take on the risk with a view to holding it, then you have the simplest application of credit derivative usage. Since there is no requirement for the risk manager to actually hold any debt instruments, and the risk taker is not required to hold the risk to maturity, the two parties may just be trading the credit risk; this is the more common application of credit derivative usage.

A combination of risk management and trading ensures both a supply of synthetic credit risk and a well established secondary market. Should one party decide to bundle pools of this risk into a CDO and sell tranches to investors, the credit derivative is merely assisting in the functionality of the transaction. Securitisation of debt products has been around since long before the first credit derivatives were used; synthetic credit risk is merely another asset class within the underlying portfolio. If a CDO transaction fails due to poor structuring of the risk or actual losses within the portfolio, this does not mean that the credit derivative product is flawed: the flaw is most often with the users of the tool, not with the tool itself.

8.7.2 Overview of Recent Changes

The previous two years have seen dramatic changes in the CDS market, brought about by a combination of many factors that have emanated from the credit crunch and the subsequent economic and market crisis. The changes can be broadly divided in two: changes that have been applied globally to the contract itself, and changes to market conventions that have been introduced by geographic region. Many of these changes were implemented at the same time with the most significant being the so-called 'CDS Big Bang' in April 2009 and the 'CDS Small Bang' in July 2009.

The need for change stems from a combination of exponential growth in outstanding CDS contracts, combined with a belief that the infrastructure supporting this large sector of the financial markets is not suitably robust and contributed to heightened levels of systemic risk. The goal is same-day trade matching and centralised clearing to reduce both unmatched trades and redundant back-to-back trades.

8.7.3 Contract Changes

The main changes to the contract are (i) the 'hardwiring' of the auction-based system of settlement into the standard CDS contract; (ii) the introduction of a 'Determination Committee' to make decisions on whether significant trigger events have happened and, if applicable, to agree on key auction terms; and (iii) the introduction of new provisions for the effective date of a transaction.

In order to bring about the suggested CDS market changes and to enable old contracts to be bound by contractual and convention changes, the market was invited to sign agreements known as the Big Bang Protocol ('2009 ISDA Credit Derivatives Determinations Committees and Auction Settlement Protocol published 12 March 2009') and the Small Bang Protocol ('2009 ISDA Credit Derivatives Determinations Committees, Auction Settlement and Restructuring CDS Protocol published 14 July 2009'). This allowed changes to be applied both retroactively and to all future trades. Signing the agreements was achieved by the usual process of adherence: market participants send in a letter to ISDA stating that they want to adhere to a protocol. The closing date for adherence to the Big Bang Protocol was 7 April 2009, and for the Small Bang Protocol, 24 July 2009.

The Small Bang Protocol extended the geographic scope of some of the Big Bang Protocol changes and extended the auction hardwiring provisions implemented by the Big Bang Protocol to include Restructuring credit events. This had been specifically excluded because, following a Restructuring where Modified Restructuring or Modified Modified Restructuring is applicable, the settlement process applies differently to contracts with different maturities and depends on the party that triggers the credit event. The Small Bang Protocol addresses these issues and allows the settlement of Restructuring credit events to be brought within the auction framework.

The Determination Committee

In response to growing concern about the process of calling a Credit Event, the market has chosen to adopt a central decision-making body that can trigger both Credit Events and Succession Events. There is one Determination Committee (DC) in each of the following regions: The Americas, EMEA, Japan, Asia Ex-Japan and Australia/New Zealand. Each DC has the following responsibilities:

- Credit Event: The DC will determine whether an event has occurred, the type of event and the date it happened.
- Auction: The DC determines whether to hold one or more auctions, specifies the actual terms of the auction and can make amendments to the Auction Methodology.
- Auction Bidders: The DC approves requests from dealers that want to be Participating Bidders.
- Deliverable Obligations: The DC decides on acceptable Deliverable Obligations and any substitute Reference Obligations.
- Succession Event: The DC will determine whether an event has occurred.

Since the pronouncements of a DC will affect most participants in the CDS market, it is crucial that the make-up of the committee is seen to be broader than a handful of the largest dealers. To address this, the composition of the DC in each region is eight global dealers, two regional dealers, five buy-side members, two non-voting dealers, one non-voting buy-side member and ISDA as a non-voting secretary. To be a dealer member, the dealer must be a Participating Bidder in auctions and must adhere to the ‘Big Bang’ Protocol. The choice of dealers is based on trading volumes for transactions reported in DTCC’s Trade Information Warehouse. Buy-side members must have at least USD 1bn in assets under management, have CDS trade exposures of at least USD 1bn and be approved by at least one-third of the buy-side institutions meeting the previous two criteria. The buy-side members are randomly selected from this group and are in the DC for one year. The buy-side members in each DC must include at least one hedge fund and at least one unlevered asset manager.

To activate the DC process any member of ISDA can request the relevant DC to address a question about a Credit Event or Succession Event and at least one member of that DC must agree to consider the question. Each member is then required to vote on questions relating to an Event or to Deliverable or substitute Reference Obligations.

Credit and Succession Events

The DC process starts when a potential issue is presented to the committee by an ISDA member with the support or sponsorship of a DC member. If the issue relates to a Credit Event, the event must have happened within the last 60 days. For Succession Events this 'Lookback' period is 90 days. This represents an important change: under the old rules a Credit Event could be called at any time during the life of a CDS contract and up to 14 days after the final termination date. In theory, this left immaterial, legally defined but never-called Credit Events outstanding for many years, which presented both parties to the transaction with a risk management challenge, and the potential calling of a Credit Event many years later departed from broadly accepted market principles.

Once an issue has been formally logged with the DC, the Lookback clock stops ticking and nobody is affected by the length of time taken by the DC. The DC Secretary provides the committee with necessary information about the Reference Entity and transaction type and the DC then determines whether the event has occurred. If an 80% majority is not achieved on any question, it is referred to an external review panel (see below). If an event is deemed to have occurred, then the DC will further specify whether an auction is needed and what Deliverable Obligations are acceptable.

External Review

An external review panel is used where the DC cannot achieve a majority vote of 80%. The Review Committee (RC) is selected from an approved pool of independent firms and their decision is binding. If the DC only achieved a vote in the range of 50–60%, then two out of three of the RC can overturn the result; if the DC vote is in the range of 60–80% then three out of three reviewers is required. Members of the pool are nominated by members of the DC and confirmed by a majority vote. When the use of the RC is invoked, five reviewers are selected of which three are active and the other two are available as alternatives. If the DC selection is not unanimous, the DC Secretary (ISDA) then selects them at random. The external review process is allocated 10 business days, but this time can be extended with an 80% vote by the DC.

The Move to Auction-Based Settlement

For many years the standard CDS contract only allowed for Physical Settlement whereby, following a Credit Event, the Buyer of Protection delivered to the Seller bonds or loans that conformed to pre-agreed parameters such as currency, maturity and seniority. As the CDS market grew, so too did the concern that one day a Credit Event would occur and the notional of trades to be settled would exceed the notional of debt that could be delivered. This probably happened following the Argentina Credit Event in late 2001, but due to confusion as to when and indeed whether there had been a Credit Event, the Physical Settlement dates were spread out over many weeks allowing a relatively small number of bonds to be handed around the market, settling CDS contracts.

It definitely would have happened in 2005 when Collins & Aikman Products Co. filed for bankruptcy. As a member of an actively traded credit derivative index, the notional of contracts to be settled was large and, because of the mechanism of settling index-related Credit Events, settlement can only occur on one day. The market worked with ISDA to produce the 2005

CDS Index Protocol which was published on 26 May 2005. The protocol detailed an auction mechanism whereby a number of dealers (approximately 10) would voluntarily provide two-way prices on an agreed debt instrument, and following a complex mechanism of price selection and aggregation, arrive at a Final Price which would be used to cash settle all CDS contracts where the parties had adhered to the protocol.

It is worth noting that this form of Cash Settlement was a dramatic improvement over the previous mechanism for cash settlement involving a Dealer Poll. This mechanism (still present in older trades and particularly in credit-linked notes) only uses five dealers chosen by the parties to the contract at the time of the trade and has a much simpler price aggregation mechanism. Owing to flaws in this mechanism, the small number of potential price points and the frequent reluctance of the chosen dealers to give a price (they were never consulted and hence never agreed to take part in the poll), the final result was not considered reliable or accurate.

The Index Protocol was further refined each time a Credit Event occurred and, following the bankruptcy of Calpine Corporation in late 2005, was further modified for single name CDS contracts. However, the process of requiring market participants to sign up to a protocol following each Credit Event was not efficient, particularly as the recent economic downturn significantly increased the number of Credit Events. The next logical step therefore was to incorporate or ‘hardwire’ the protocol language into the CDS contract.

The Big and Small Bang protocols hardwired the auction methodology into standard CDS contracts, thereby leaving just the specific auction terms to be decided on by the DC following a Credit Event.

Auction Methodology

The Auction process involves two parts: Part 1 establishes both an approximate recovery value and the sum of the buy and sell orders from those participants wanting Physical Settlement. Part 2 allows the participants to cover these orders and in doing so establish the Final Price. The key components are as follows:

- Inside Market Midpoint (IMM): In Part 1 the participating dealers submit two-way prices on the Reference Obligation(s) stipulated by the Determinations Committee which also stipulates the maximum spread and quotation size. The prices are subjected to a complex aggregation process to produce the IMM.
- Physical Settlement Requests: In Part 1 participating dealers may submit buy or sell requests for the Deliverable Obligation(s) (as specified by the DC). These are netted to determine the Open Interest which is then bought or sold in Part 2.
- Adjustment Amount: If a participating dealer submits a price that is the wrong side of the IMM (for example, where a bid is higher than the IMM but the Open Interest is to sell), then the dealer pays a penalty based on the price differential and the quotation amount.
- www.creditfixings.com: Within 30 minutes of the end of Part 1, Markit publishes the IMM, Open Interest and Adjustment Amount on this website.
- Limit Order: In order to cover the Open Interest established in Part 1, Part 2 is open to dealers and investors who might want to buy or sell into the Open Interest. The relevant bids or offers used to calculate the IMM are automatically used as Limit Orders. Where the Open Interest is to buy, the lowest sell Limit Offers are used first; where the Open Interest is to sell, the highest buy Limit Offers are used first.

- Final Price: On completion of the Limit Order process the IMM is compared to the price of the last Limit Order used to cover the Open Interest. If this last price is between the IMM and the IMM plus or minus a preset cap amount (typically 1%) then this last price becomes the Final Price. If it is outside the range, then the Final Price is set at the IMM plus the cap amount if the final Limit Order is above the range, or minus the cap amount if it is below the range.

Auctions for Restructuring

When a Restructuring Credit Event occurs there are different restrictions on what can be delivered, depending in most cases on the domicile of the Reference Entity. For many years the market has used four different Restructuring clauses: Modified Restructuring (MR: generally used for North American and Australasian Corporates); Modified Modified Restructuring (MMR: generally used for European Corporates); Old Restructuring ('OR': generally used in Asia); and No Restructuring (XR: generally used by Loan Portfolio Managers in North America). MR and MMR apply if a Restructuring Credit Event is called by the Buyer and impose maturity limitations on the Deliverable Obligations used in Settlement. OR places no limit on maturity but typically a 30-year limit is imposed by way of the Deliverable Obligation Characteristics.

To allow Restructuring Credit Events to be settled by an Auction, CDS transactions need to be grouped into Maturity Buckets. This reduces the number of potential combinations of Deliverable Obligations that are deliverable into contracts with different maturities. Under this maturity bucketing, if the Buyer triggers a Restructuring Credit Event, then the Deliverable Obligations will be determined by reference to the Maturity Bucket into which the relevant CDS contract falls according to the remaining maturity of that CDS contract. This applies both to settlement in an auction and also to physical or cash settlement if no auction is held.

Following a Restructuring Credit Event the DC determines whether or not to hold an auction for each Maturity Bucket and determines the appropriate set of Deliverable Obligations for each Maturity Bucket.

Maturity Buckets

Since there are 10 different contract maturities and 4 maturity dates per year, there are in theory 40 different Deliverable Obligations to meet the maturity limitation of MR and MMR. Clearly the use of one Auction would not be sufficient, but running anything close to 40 auctions would not be practical. The market therefore decided to aggregate Deliverable Obligations into a small number of auctions and introduced Maturity Buckets of 2.5, 5, 7.5, 10, 12.5, 15, 20 and 30 years.

The Maturity Buckets replace the definitions of both Restructuring Maturity Limitation Date (RMLD) and Modified RMLD (MRMLD). Before an auction can be held, obligations and transactions need to be assigned to the appropriate bucket. A trade falls into a bucket if the maturity of the CDS is on or before the Maturity Bucket End Date. However, for this bucket to be used there has to be a Deliverable Obligation that (in addition to any standard limitations on Transferability and Multi-Holders) would also fit into the appropriate bucket. If there is no such Deliverable Obligation, then a shorter bucket must be used.

A Deliverable Obligation is a member of every Maturity Bucket where the bonds maturity date is less than the maturity limit of the bucket. Hence, a 4-year bond would be a member of the 5-year bucket and all of the longer Maturity Buckets. If it was the only deliverable that was available for a 10-year CDS, then it and the CDS would be placed in the 5-year bucket for settlement purposes. The use of this shorter Maturity Bucket is in keeping with the Rounding Down Convention, which states that where buckets are effectively identical, the shorter Maturity Bucket is used.

The decision of whether to hold an auction for a particular Maturity Bucket is made by the DC using information from the DTCC Trade Information Warehouse. An auction for a Maturity Bucket is automatically held if 300 or more transactions assigned to the bucket are triggered and five or more Dealer Members of DC are party to those transactions. If there is no automatic determination, then the DC votes as they would on other issues.

If no auction is held for a particular Maturity Bucket, then transactions in that bucket are subject to a Movement Option which can be exercised by either the Buyer or the Seller. The Buyer can move the transaction to the next earliest bucket for which there is an auction; the Seller can move it to the 30-year bucket, assuming that it has an auction. If both wish to exercise, then the Buyer has priority. The intention of the option is to allow the Buyer and the Seller to use the auction method wherever possible; if it is not exercised, then the transaction employs the Fallback Settlement Method. If the Buyer originally triggered the Credit Event, then the Fallback Settlement Method would still be subject to the Maturity Bucket limitations on what can be delivered.

Effective Date

The date when the Buyer and the Seller go on risk has for many years been set at trade date plus one calendar day. Hence a transaction done on a Friday would go live at 1 minute past midnight on the Saturday. Should either party choose to put on an offsetting transaction a few days later, there is the possibility that an event took place between the two dates resulting in a mis-match.

Under the new rules the Effective Date for Credit Events is set as Today minus 60 calendar days (90 for a Succession Event). The date rolls forward each day, so that at any point in time the Effective Date will still be the current date minus 60 or 90 days. Hence a transaction that is hedged one week later will, on the date of the second transaction, have an Effective Date equal to that of the hedge. All existing positions will therefore have the same Effective Date.

In order to bring the single name CDS market in line with the Index market, the former is having its Effective Date calculation changed, to be the same as for the single name market.

8.7.4 Convention Changes

The previous section focused on changes to the CDS contract which, due to their very nature, alter the extent of risk transfer in a CDS contract. Convention changes by contrast do not result in a legal change but in this case change the way that prices are quoted and therefore the way that contracts are traded.

Fixed Coupon

To bring CDS trading in line with CDS Index trading, effective initially in North America and later in Europe, single name CDS started to trade with a combination of a fixed rate coupon and an up-front payment. This is not to be confused with how the contract is quoted in pricing discussions where the market continues to either quote on the basis of a running spread or points up-front; however, once executed the cashflows now reflect a combination of fixed coupon and a variable initial payment to account for the difference between the market quotation and the contractual coupon.

An additional convention change relates to the first coupon payable. This is now paid in full regardless of the time between trade date and next coupon date. Any overpayment by the Buyer is captured by adjustment to the up-front payment.

Prior to the Big Bang Protocol, single name CDS were quoted using a par spread; only very high yielding credits were traded with a fixed coupon and an up-front payment. The market would change a name to points up-front once its credit spread had passed a certain threshold. As the economy deteriorated and more and more names moved to up-front trading, the North American market decided to formalise this trend with all contracts moving to a fixed coupon of either 100 or 500 bp. As a rule of thumb, investment grade name use 100 bp while non-investment grade use 500 bp. Europe's movement to fixed coupons started as a result of the Small Bank Protocol. The mechanism is the same, but Europe adopted more coupons: 25, 100, 500 and 1 000. In addition, the coupons of 300 and 750 were introduced to assist in the process of re-couponing old trades. At the time of writing, it is unclear whether these will survive as market convention coupons for new transactions.

Market participants should be indifferent to the size of the up-front payment relative to the running payment. From an operational perspective, however, the use of the fixed coupon is very beneficial: not only can new trades be processed faster but offsetting trades can be netted more easily.

Full Coupon

Since CDS coupons are payable on 20 March, June, September and December, there has to be some rule to determine how much coupon is paid for trade dates that fall between these coupon dates. The old rule involved the use of a 'short stub' period and a 'long stub' period, depending on whether the trade date was more or less than 30 days before a coupon date.

The new rule in both North America and Europe falls in line with the way that the CDS indices treat coupons: the buyer pays the full coupon on the next coupon date and, hence, on the trade date receives a payment from the seller equal to the accrued. This change, combined with the use of fixed coupons ensures that the only uncertain cashflow (other than those following a Credit Event) is when a trade is executed or terminated. The payment on the settlement date is a combination of an up-front payment to compensate for the fixed coupon being above or below the market spread and a payment of accrued interest.

9

Valuation and Risk: Basic Concepts and the Default and Recovery Model

Sections 9.2 and 9.3 develop a mathematical framework for the modelling of credit risk aimed at the valuation of CDSs and bonds. The model presented is largely the industry standard among the banking community. Section 9.4 generalises this model and shows under the circumstances in which the simpler model of 9.3 applies. In Chapter 8 we saw some of the major factors of relevance to the pricing of credit instruments. In this chapter we introduce a simple arbitrage model to help to identify the key driving factors in the valuation of CDSs, and to give a sound basis for the intuitive pricing of CDSs.

9.1 THE FUNDAMENTAL CREDIT ARBITRAGE – REPO COST

If BT debt is trading at 7% and risk-free (government) rates are 4%, what is the correct CDS premium for BT?

Let us assume that there is a 5-year FRN carrying a coupon (LIBOR plus a fixed spread S) such that the bond price is par. Let us also assume that default risk is constant on BT, so that the bond will always be priced at par, or at the recovery level in the default event. Suppose we buy a CDS on BT, to the same maturity as the bond, into which the bond – and that bond only – is deliverable and pays par plus accrued interest on the bond. In order to have a zero cost (self-financing) position we can lend the bond out on repo, receive cash collateral equal to the price of the bond, and pay the repo rate on the BT bond. We assume that the repo rate curve is flat (Figure 9.1).

Note that the position is insensitive to interest rate moves if the premium and interest payment dates coincide.

If default occurs, the bond is delivered into the CDS, receiving par plus accrued, which is used to unwind the repo transaction. If the accrued coupon equals the accrued repo cost and accrued CDS premium, there is no loss or gain and, therefore, no risk on the deal.

As the deal is risk free, there should be no profit or loss: the net cashflow is LIBOR plus the spread, S, less the repo rate, less the CDS premium. If the cashflow is zero, *the CDS premium, P, equals the spread on the (par FRN) bond, S, plus the amount special on repo*[1] *R.*

$$P = S + R \tag{9.1}$$

Typical investment grade credit debt trades non-special, so generally the CDS premium equals the par FRN bond spread: $P = S$. In particular, we see that 'risk-free' rates are irrelevant to the pricing of a CDS.

Generalising from constant default risk again has little impact if the spread curve is flat (we shall discuss this fully later).

[1] 'LIBOR less the repo rate' is referred to as 'the amount special (on repo)'.

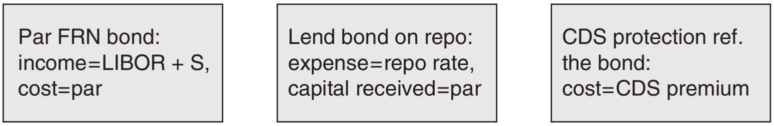

Figure 9.1 The fundamental credit arbitrage.

In reality (non-constant repo rate) there is a repo risk in the above. A term repo trade done at the outset has to be unwound on the default event at a cost that depends on the then repo rate. If the (theoretical) repo curve is flat, then the expected cost is zero, but if it is not flat there will be a minor difference in the above equation arising from the expected unwind cost on the default event. This will increase as the repo curve steepens and as we look at names with higher and higher default risk. If, instead of a term repo trade, a rolling overnight repo is then done, the cost is uncertain. The expected cost can be calculated in principle from the repo curve.[2]

The repo curve pricing impact is actually minor in the context of other assumptions we have made – the key ones being no delivery option and non-standard payoff – and we shall see later that, in reality, bonds and CDSs trade as separate markets with different but related spreads. The key point here is that the natural comparison is between the CDS premium and the (par FRN) bond spread to LIBOR. The argument applies more generally: the correct discounting curve for typical assets is the LIBOR curve (being the financing cost on repo-able assets).

We can set up a similar argument using fixed coupon par bonds instead of FRNs, and introducing a fixed/floating interest rate swap to eliminate interest rate risk. This gives the same result if the interest rate curve is flat, but introduces expected unwind costs similar to those incurred if the interest rate curve is not flat.

9.2 DEFAULT AND RECOVERY MODEL; CLAIM AMOUNT

In this section we discuss the formal description of the default and recovery process.

9.2.1 Claim Amount

The ‘claim amount’ is the amount that can be claimed and becomes immediately payable on default. It is specified in the issue document of the debt. Ultimate recovery is based on the claim amount: if there are insufficient funds to pay the claim in full, then a proportion of the claim amount is paid. In almost all cases of traded debt, the claim amount is par plus accrued interest (referred to as **claim amount of par**). For a few loan instruments the claim amount is the promised cashflows – economically equivalent to the capital value of those cashflows (and referred to as **claim amount of treasury**).

Claim of par plus accrued (in obvious notation):

$$C(t) = 1 + A(t) \tag{9.2}$$

[2] Of course this is theoretical – the repo market is not liquid, term repo deals are hard to find and, in particular, there is no repo curve data to allow these calculations to be performed in practice.

Often, particularly in CDS modelling, we refer to 'claim amount of par' to mean zero accrued:

$$C(t) = 1 \tag{9.3}$$

Claim of treasury: let the cashflow at time t be $c(t)$ and let $d(t)$ be the discount factors off the government curve. Assume that cashflows are paid periodically with n payments due before and including maturity at time T. Then

$$C(t) = \sum_{i=1}^{n} c(t_i) \cdot d(t_i) \tag{9.4}$$

9.2.2 Recovery Modelling

For corporate entities in the USA, UK and some other countries, the legal process is capable of enforcing equal recovery of the claim amount for all creditors with the same seniority of debt. In the event that the administrators or the company try to do something different, then creditors can take the company to court and enforce the distribution rules given in the documentation. For sovereigns no such process exists. In practice, although the documentation specifies what you can claim, there is no means of enforcing equal treatment of different claims. A good risk control measure would be to assess recovery on the basis of an alternative claim amount for sovereigns.

The simplest model is to assume that recovery is a fixed proportion of the claim amount for a given seniority of debt. Ultimate recovery is one of 100%, 0%, or some number between 0% and 100% for a particular seniority. But the settlement date of CDS contracts is long before ultimate recovery is known, and debt trades at a price that reflects the anticipated recovery level for that seniority. Therefore, around one month post-default junior, senior and loans will trade at prices between 0 and 100 – typically with junior debt at a low price (but not 0), loans at a high price (but not 100), and senior debt in between. Our recovery model could then be

$$R(\mathit{seniority}) = x(\mathit{seniority}) \tag{9.5}$$

where x is a constant for that seniority over all time (i.e. irrespective of whether default occurs in 2005 or 2015) and all reference entities. For example, we might (on the basis of broad historic data) assume that junior debt recovers 20%, senior debt recovers 50% and loans recover 70%. This assumption might apply no matter what the company is – any industry, any rating of debt. This is the most common form of recovery model at the time of writing.

A more realistic implementation (but one that is rarely used) is to take account of the industry group (for example, based on Altman and Kishore (1996)), and also investment grade/high-yield differences. This would make recovery a function of seniority, industry and rating. Our recovery model is then

$$R(\text{seniority, industry}) = x(\text{seniority, industry}) \tag{9.6}$$

where x is a constant for that seniority and industry over all time and for reference entities within that industry group.

Recovery modelling can be made more realistic. An investor with access to good fundamental company research could also try to estimate recovery levels on the basis of company accounts and other information to derive estimated recoveries specific to that company and seniority of debt. One could assume the time dependence of recovery – dependent on the economic cycle. This is rarely done in practice and rapidly leads to calibration problems as changing recovery over time has the effect of magnifying changes in the forward hazard rates and can rapidly lead to negative forward hazard rates. Also, the time dependence of recovery (because of the economic cycle) would naturally be associated with the time dependence of hazard rates. We shall see (in section 9.6) that this can lead to major complications over the modelling of CDSs if we want these dependencies to be reflected in a correlation of stochastic recoveries and hazard rates.

Recovery Uncertainty

Another way to make recovery modelling more realistic – which has little implication for CDS pricing but we shall see implications for portfolio products (particularly if we correlate the recovery rate with the numbers of defaults occurring) – is to reflect the uncertainty in our knowledge of the recovery rate for a specific company. For example, let us assume that the average recovery rate for senior debt is 40%. Then, on a large portfolio of names with senior debt we would, over time, expect to see the average recovery rate for those that default, to be 40%. But for any particular name the recovery may be anywhere between 0% and 100%, with a standard deviation (again an assumption based on historical data) of, say, 20%. We may need to investigate the impact of this uncertainty in our recovery knowledge. We can do this by performing only three sets of calculations. We can value the product with a low recovery assumption, with a high recovery assumption, and with an intermediate recovery assumption. Each of these recovery assumptions has a probability associated with it: the sum of these probabilities is 1; the expected recovery is 40%; and the standard deviation is 20%. We can choose the three recovery levels and then solve these equations to get the probabilities. In addition, certain constraints must be satisfied – all the probabilities have to be between 0 and 1, which acts as a constraint on our initial choice of recoveries.

Example (see 'chapter 9.xls') Suppose we take the three possible recovery levels to be 0% with probability p_0, 40% with probability p_{M}, and 80% with probability $1 - p_0 - p_{\mathrm{M}}$. Then the expected probability (assumed to be 40%) is

$$0 \times p_0 + 40 \times p_{\mathrm{M}} + 80 \times (1 - p_0 - p_{\mathrm{M}}) = 40$$

so $p_0 = (1 - p_{\mathrm{M}})/2$, and the variance (assumed to be 20%) is

$$(0 - 40)^2 \times p_0 + (40 - 40)^2 \times p_{\mathrm{M}} + (80 - 40)^2 \times (1 - p_0 - p_{\mathrm{M}}) = 20^2$$

from which we find that $p_{\mathrm{M}} = 0.75$. Hence we find that the probability of a 40% recovery is 75%, and of 0% or 80% recovery is 12.5%. We can take a more realistic view of CDS valuation by taking these three separate recovery assumptions, calculating the implied probabilities, and then weighting the resulting values by the associated probabilities. Generally this is not done (the results are not significantly different from the base assumption). However, we shall use this model when we look at first-to-default products, and also CDSs when we include counterparty risk, where there are some interesting conclusions.

Exercise 1 Why do we not assume that $R = 100\%$ as one of our three recovery assumptions?

9.2.3 Hazard (Default) Rate Model

A distinction is sometimes made between the **hazard rate** – the rate of default at any time assuming that the entity has survived up to that point – and **default rate** – the rate of default at a future time assuming only that the reference entity is alive now. We shall not make this distinction and use both these terms to mean hazard rate.

We model the hazard rate process as a 'Poisson' process (Billinglsley, 1995). This is a standard and widely used statistical model that describes the rate of goals scored at a football game and the time until a goal is scored, the rate of clients joining a queue and, in our case, the rate at which a company defaults and the time to default. The model can be used to describe multiple events – e.g. goals – but our application is simpler: we are only interested in the first occurrence of default for a company.

There are only a few results we need to know about the Poisson process. Let us suppose for the moment that the hazard rate (the rate of default at any time) is a constant, h. The probability that a default occurs in the next moment of time Δt is $h \times \Delta t$. The probability that the name survives from now (time = 0) to time t is

$$S(t) = \exp(-h \cdot t) \tag{9.7}$$

and the probability that it survives from now until that date and then defaults in that interval (the 'default rate' as defined in the first paragraph)

$$prob(def\_in\_t\_to\_t + \Delta t | alive\_now) = \exp(-h \cdot t) \cdot h \cdot \Delta t.$$

Any practical implementation requires us to make the hazard rate a function of time. (This is referred to as a 'time-changed' Poisson process.[3]) The survival probability becomes

$$S(t) = \exp\left(-\int_0^t h(x)dx\right) \tag{9.8}$$

(when $h(t) = h$, a constant, then the above formula reduces to the previous one). Similarly, the probability of default in time t to $t + \Delta t$ is

$$\exp\left(-\int_0^t h(x)dx\right) \cdot h(t) \cdot \Delta t \tag{9.9}$$

We can also allow the hazard rate itself to be a stochastic process. (We do this in section 9.4.)

Many of the practical pricing examples and illustrations in this book assume that the hazard rate is a constant. This is often sufficient to show the results we need. In many cases a hazard rate curve makes the sums more difficult but the point is unchanged: any practical implementation will have to implement a non-constant hazard rate curve.

9.2.4 Choice of Hazard Rate Function/Interpolation Process

The hazard rate function turns out to be very similar both in interpretation and mathematically to the forward (spot) interest rate, with the survival probability corresponding to a discount factor. When we perform pure interest rate calculations we have to choose a form of interest rate

[3] This corresponds to a change of variable in integration. In order to avoid our transformed time going backwards we clearly need the hazard rate to be positive at all times.

function when we bootstrap the yield curve in order to interpolate discount factors. We should think of the spot rate (hazard rate) function as a complicated way of interpolating between discount factors (CDS levels) at different maturities. We have similar types of choices for the hazard rate function – for example, we often choose a piecewise linear function for interest rates, equivalent to using a piecewise linear function for hazard rates.

Most examples in the book assume that the hazard rate is a constant, purely for simplicity of exposition. In practice we have to use more complex formula where the hazard rate is a function of time. When we do this it changes some very simple formulae into complex ones, and can have some serious implication for computing time. Any CDS calculation requires not only the evaluation of survival probabilities but also the evaluation of integrals of survival probabilities, discount factors and other functions (for example, see equation (9.13) later in this section). If we have to evaluate the survival probability numerically, then we shall have to perform numerical integrals involving numerical integrals, which will slow down the implementation process dramatically.

The following present some choices for the interpolation function: the first two interpolate the CDS curve, the remainder are hazard rate interpolation routines.

1. *Piecewise Constant on the CDS curve*. CDS levels are piecewise constant – simple but rather unrealistic, implying sudden jumps in the CDS premium as maturity changes and corresponding jumps in CDS valuations.
2. *Piecewise linear on the CDS curve*. Traders typically produce an entire curve of CDS premiums often based on a small number of maturities traded. A simple way to do this would be piecewise linearly interpolating between (or extrapolating from) known data. Traders generally do something rather more sophisticated – carrying around in their heads an appropriate curve shape for the name and general spread level and using this to generate the CDS premiums. The trader is performing on-the-fly data engineering – something that the quant can do mathematically but generally not as realistically as a trader on the basis of his market knowledge.
3. *Piecewise constant on the hazard rate*. Implying sudden jumps in the reference entity default rate as maturity changes.
4. *Piecewise linear on the hazard rate*. This is intuitively sensible and the integral (the survival probability) is also easy to calculate in closed form. There are, however, two criticisms. First, at the short and the long ends, extrapolation of the curve can lead to negative hazard rates or unrealistically high levels of hazard rate. The second objection is purely practical. Imagine we are calibrating our model to two CDS levels at 2 and 5 years. We have a linear curve: two unknowns and two simultaneous equations for the values of the CDSs, which is relatively slow to solve numerically.
5. *Piecewise linear on the hazard rate with constant short and long stubs*. This is intuitively sensible in the range from the shortest to the longest CDS (or other instruments) to which we are calibrating. It also has the advantage that
   (a) it eliminates the possibility of negative hazard rates at the short and long end and
   (b) at every stage of the calibrating process we only have to solve a single equation at a time – which is fast (whereas the piecewise linear approach requires us to solve a pair of non-linear equations in two unknowns for the first two maturities).
6. *Exponential form (tending to a limit at high maturities) on the hazard rate*. This can be useful when we want to perform a best fit (rather than an exact fit) to data and want to avoid kinks in the curve which a low-level piecewise linear model would introduce.

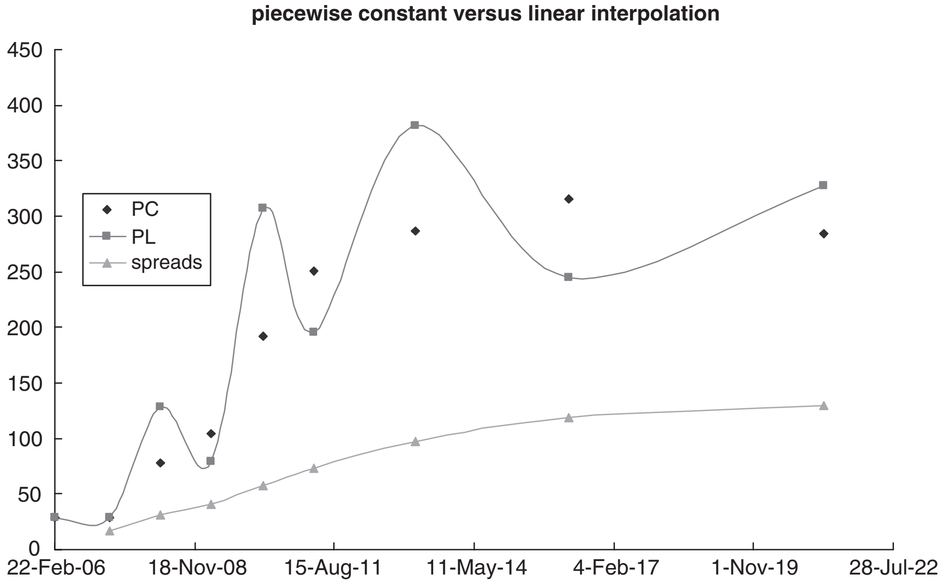

Figure 9.2 Piecewise linear versus piecewise constant hazard rate interpolation on imperfect data.

There are other interpolation formulae (a Cox–Ingersoll–Ross formula (Hull, 2002) is sometimes used). In practice, items 3 and 4 are the most common with piecewise constant on the hazard rate being the standard. The jumps in the hazard rate can be minimised by supplying a full curve of CDS premiums – hence the trader's need to engineer a curve of premiums – and has the advantage that minor irregularities in the CDS premiums lead to jumps in the hazard rate curve, but not wild swings, which can be the result of using piecewise linear interpolation with less than perfect data (see Figure 9.2).

For risk control purposes it may be useful to implement two interpolation functions in order to assess the impact of change between them. For example, items 3 and 4 will show very substantially different values on individual CDS deals.

Exercise 2 Consider a name where the 2-year CDS is 100 bp and the 5-year is 200 bp. What are the valuation implications of using a piecewise constant hazard rate (or CDS curve) over a linear rate?

9.3 DETERMINISTIC DEFAULT RATE MODEL

In this section we shall develop some simplified formulae for CDS valuation ignoring counterparty risk, which are of the same form as those generally used by the market. Section 9.4 will redevelop these in a much more general framework, and will show the assumptions under which the more complex formulae simplify to the approximation given here.

We also develop simplified formulae for bonds. These will be partially extended below and in section 9.4. In practice, the form shown here is acceptable although the hazard rate parameter for bonds turns out to be different from the hazard rate parameter for CDS.

In this section we assume that the hazard rate process in the Poisson default model is a deterministic function of time. This may seem odd, as anyone who has watched the credit markets knows that spreads and CDS premiums change day by day, and often by large amounts. Take the simplest case where the deterministic hazard rate is assumed to be constant. This is saying that tomorrow's spread will be the same as today's, but, of course, we know it will not be. Section 9.4 shows that, surprisingly, this does not matter for the purposes of valuation and risk calculations today. When we come in tomorrow and find that the spread has changed, we simply need to recalibrate our curve to the new spread and revalue.

We also assume that the recovery rate is a known constant and, for the moment, assume that only one specific bond is deliverable into the CDS. Also, for the moment, we shall assume that counterparties are risk free.

Notation:

Time now	t	
Hazard rate at time $s > t$	$\lambda(s)$	
Survival probability	$S(s) = \exp\left(-\int_t^s \lambda(x)dx\right)$	(equation 9.8)
Recovery rate	R	(equation 9.5)
Accrued on deliverable bond	$A(s)$	
Discount factors	$D(s)$	
Claim amount	$C(s)$ – given by equation (9.2)	
CDS premium per annum	p	
Maturity date of CDS deal	T	

(Note that both the survival probability and the discount factor have a dependence on the time now, t, which we have suppressed for simplicity of notation.)

9.3.1 CDS Valuation

In the event of default a physically settled CDS pays par in exchange for defaulted debt. It is easier to assume cash settlement in the development of the formulae but this is economically equivalent to (and therefore the same formulae apply to the valuation of) physically settled CDSs since the delivered debt has a market price of the recovery. The value of this payoff, if it occurs at time s, is

$$(1 - R \cdot C(s)) \cdot D(s) = (1 - R \cdot (1 + A(s))) \cdot D(s).$$

We see this has a maximum value if $A(s)$ is zero, and a minimum amount just before the payment of the coupon.

The probability that the reference entity defaults in the interval s to $s + \Delta s$ is given by equation (9.9). Hence the probability weighted value of the payoff is

$$(1 - R \cdot (1 + A(s))) \cdot D(s) \cdot S(s) \cdot \lambda(s) \cdot \Delta s.$$

Since t can be any moment between inception and maturity of the trade, the value of the contingent payoff (= **expected loss**) is

$$B(t) = \int_t^T \{1 - R \cdot [1 + A(s)]\} \cdot D(s) \cdot S(s) \cdot \lambda(s) ds \tag{9.10}$$

Turning now to the premium stream, we shall develop formulae for two cases. Case (a) where premiums are paid continuously (this leads to simple formulae which are easier to handle for theoretical analysis), and case (b), the standard situation, where the premium is paid quarterly in arrears with a proportion to the credit event date. Note that in both cases the total amount of premium paid is the same (whether default occurs or not). Also, if interest rates are zero, then the value of the premium payments is identical. In fact, assuming continuous premiums is a very good approximation to the standard case of quarterly in arrears with a proportion to the default date.

Take case (a) first. A premium payment due in the interval s to $s + \Delta s$ is $p \times \Delta s$, has a present value of $p \times \Delta s \times D(s)$, and will be paid if the name survives up to that time; so the probability weighted value is $p \times \Delta s \times D(s) \times S(s)$. This applies for all times up to maturity, so the value of the premium stream is

$$P(t) = p \int_t^T D(s) \cdot S(s) ds \tag{9.11}$$

Case (b) is notationally more difficult. Let the end quarter dates be at times s_i, and suppose there are n quarters up to the maturity date; so $i = 1$ up to n and s_0 is the premium due date on or prior to t. The value of these quarterly payments is the premium amount $(s_i - s_{i-1}) \times p/360$ discounted back by $D(s)$ multiplied by the probability that the payment is made – i.e. the chance that the name survives up to time s_i, $S(s_i)$. So the value of the quarterly payments made on the scheduled dates is:

$$p \sum_{i=1}^{n} \frac{s_i - s_{i-1}}{360} \cdot D(s_i) \cdot S(s_i).$$

We now have to take into account the proportionate premium paid if default occurs in the interval s to $s + \Delta s$ and assume that s is in the ith period. The amount of premium is $(s - s_{i-1}) \times p/360$, which is discounted back to get a value, and multiplied by the probability that default occurs in this interval. As this can occur for any time up to expiry, the value of the premium stream is given by

$$P(t) = p \sum_{i=1}^{n} \frac{s_i - s_{i-1}}{360} \cdot D(s_i) \cdot S(s_i) + \sum_{i=1}^{n} \int_{t_{i-1}}^{t_i} \frac{s - s_{i-1}}{360} \cdot D(s) \cdot S(s) \cdot \lambda(s) ds \tag{9.12}$$

Of course, there are many variations in the premium payment (frequency, in advance/arrears, with/without a proportion to default, etc.) giving rise to slightly different premium formulae, and potentially variations in day-count convention. We can see, by comparing (9.11) and (9.12) that continuous premiums are easier to handle notationally and shall use formula (9.11) for theoretical work from now on. Practical implementations will look more like (9.12) and the variations thereon.

The net value of a long protection CDS position on the books is therefore given by

$$V(t) = \int_t^T (1 - R \cdot (1 + A(s))) \cdot D(s) \cdot S(s) \cdot \lambda(s) ds - p \int_t^T D(s) \cdot S(s) ds \tag{9.13}$$

When the trade is fairly valued ($V = 0$) this gives a formula for the current fair-market premium, p_{M}. We can re-express (9.13) using this premium as

$$V(t) = (p_M - p) \int_t^T D(s) \cdot S(s) ds \tag{9.14}$$

The integral is the risky duration of the CDS. Periodic premiums will also have an accrued premium to be accounted for.

We shall examine some further consequences of (9.13).

9.3.2 Accrued Interest and the Delivery Option

Formula (9.13) refers to a (non-standard) CDS where the bond to be delivered is known in advance. Let us now assume that we do not know which bond will be delivered. A worst case for the writer is that, at every date, there is a bond available which has zero accrued. This gives the highest value to the long-protection position, and gives rise to an offer side premium taking into account the value of the delivery option related to the choice of a bond with low accrued. The formula becomes

$$V(t) = \int_t^T (1 - R) \cdot D(s) \cdot S(s) \cdot \lambda(s) ds - p \int_t^T D(s) \cdot S(s) ds \tag{9.15}$$

This is the standard form for valuation of a CDS contract where protection has been purchased – a standard CDS contract pays par, not par plus accrued – and effectively assumes a claim amount of par (without accrued) on the deliverable instrument. In cases where protection has been sold we simply treat the notional size as a negative amount, effectively reversing the sign of the equation.

Another extreme would be to assume that the only deliverable asset available at the time of default is a high coupon annual bond – with expected accrued (call it A) equal to half the annual coupon. In this case the payoff is $(1 - R(1 + A))$ as against $(1 - R)$ in the case of zero accrued. Let us take some numerical examples: if $R = 0$ then the payoffs are the same; if $R = 1$ the payoffs are also the same (CDS payoff has a cut-off at zero). If $R = 0.5$ and $A = 5\%$, the payoffs are $1 - 0.5 \times 1.05 = 0.475$ versus $1 - 0.5 = 0.5$, roughly a 5% difference. **So if we price a CDS assuming that a zero accrued bond is always available, then this may overestimate the value by up to around 5%**.

9.3.3 CDS Under Constant Hazard Rate

Assume now that the hazard rate is constant, $\lambda(s) = \lambda$. Then equation (9.15) becomes

$$\begin{aligned} V(t) &= \int_t^T (1-R)\cdot\lambda\cdot D(s)\cdot S(s)ds - p\int_t^T D(s)\cdot S(s)ds \\ &= ((1-R)\cdot\lambda - p)\int_t^T D(s)\cdot S(s)ds \end{aligned} \tag{9.16}$$

where the survival probability is now given by 9.7 (with $h = \lambda$). The fair value (ongoing – with zero up-front) premium for a CDS is when the inception value is zero, so

$$p = (1-R)\lambda \tag{9.17}$$

This equation is generally regarded as an approximation to the CDS premium (or, conversely, to the hazard rate given the market premium and recovery rate). We can see this result is exactly true at any level of CDS spread and under any interest rate environment as long as the hazard rate is a constant[4] (and premiums are continuous). It is a very good approximation for normal premiums.

9.3.4 Up-front Premiums

At inception equation (9.15) can be used to calculate the fair premium given the hazard rate curve or, inversely, to calculate the hazard rate levels given the fair premiums (see section 9.5). For the moment we assume that the hazard rate curve (and discount factors, etc.) are known. If there is zero up-front premium then at inception a fairly priced CDS will have a net value of zero. However, if there is an up-front premium then (9.15) will equal the amount of that premium at inception. After inception, (9.15) will in all cases equal the net value of the trade. In the case where there is no up-front premium, this value will initially be close to zero, but in the case where an up-front premium has been paid (purchasing protection), this value will start off as significantly positive but will be balanced by the up-front cash payment made.

9.3.5 Bond Valuation

The bond's 'dirty' price (including accrued) is the discounted value of probability weighted cashflows. These are in two parts: the promised cashflows, and the recovery in the event of default. We use the additional notation

Bond maturity date	M
Number of promised cashflows	m
Promised cashflow at time s	$c(s)$
Promised cashflow dates	s_i

[4] I am grateful to Charles Anderson for pointing this out to me.

Following through a similar process to the above we see that the price of the bond is given by

$$BP(t) = \sum_{i=1}^{m} c(s_i) \cdot D(s_i) \cdot S(s_i) + \int_t^M R \cdot (1 + A(s)) \cdot D(s) \cdot S(s) \cdot \lambda(s) ds \tag{9.18}$$

We shall examine some consequences of this formula.

9.3.6 Bond Price Under a Constant Hazard Rate

Assume now that the hazard rate is constant, $\lambda(s) = \lambda$. Then equation (9.18) becomes

$$\begin{aligned} BP(t) = & \sum_{i=1}^{m} c(s_i) \cdot D(s_i) \cdot \exp(-\lambda \cdot (s_i - t)) \\ & + \int_t^M R \cdot (1 + A(s)) \cdot D(s) \cdot \exp(-\lambda \cdot (s - t)) \cdot \lambda ds \end{aligned} \tag{9.19}$$

The first term is the sum of the discounted promised cashflows, **discounted off the interest rate curve shifted by the hazard rate**. In particular, let us assume that the forward interest rate is a constant r, i.e. $D(s) = \exp(-r(s - t))$. The first term is the sum of cashflows multiplied by $\exp(-(r + \lambda)(s - t))$.

If recovery is zero, then the bond price is only the first term and the bond price is the sum of the cashflows discounted off the interest rate curve shifted by the CDS premium rate (from (9.15)). The formula reduces to the simple bond spread model of Chapter 3 (equation 3.1), where the z-spread is the hazard rate.

Note that discounting of the interest rate curve shifted by 'the spread' (more correctly, the CDS premium) is only correct if recovery is zero. *Generally we must shift the interest rate curve by the hazard rate* and explicitly include a term for recovery.

9.3.7 Limiting Cases of the Bond Price

If the hazard rate is zero (the bond is risk free) then the second term of (9.18) is zero. The survival probabilities are unity and the formula becomes the standard (risk-free) bond pricing formula.

If the hazard rate becomes very large and $S(t)\lambda(t) \to \infty$ as $t \to 0$, then the survival probability to a future date becomes very low and the probability of default in the next moment of time approaches unity. In that case the recovery is paid imminently (and there are no other cashflows) so the bond price tends to the recovery. In this case bonds are said to be 'trading on a recovery basis'.

Note that if the hazard rate declines from this extremely high level, then default is almost certain within a finite interval but is deferred a little and is slightly discounted. Thus, *as the hazard rate reduces from very high levels, the bond price initially declines too.* Only as the hazard rate falls to levels where there is a significant survival probability does the bond price start to rise and pass the recovery rate. We see in section 9.5.3 that we can get multiple implied hazard rates for a given recovery assumption, if we calibrate to bonds.

9.3.8 Risky Zero Coupon Bonds

If the claim amount is par (as assumed in the above formulae) note that the zero coupon bond price also tends to recovery level as the hazard rate rises to high levels. Zero and low coupon bonds may have a claim amount given by a formula or a schedule of amounts rising from the issue price to par at maturity.

Exercise 3 Amend the above formulae for the case where the claim amount is, say, the discounted value of par, discounted at some fixed rate x.

Note that the ratio (risky zero coupon bond price)/(risk-free zero coupon bond price) does not have a simple form except when recovery is zero. Credit models which suppose the contrary will be at variance with the default and recovery model described here (and, more importantly, with market prices).

9.3.9 CDS and Bond Sensitivities

The valuation formulae allow us to calculate CDS and bond sensitivities to

(1) the default event
(2) the hazard rate curve or the calibrating data
(3) the interest rate curve.

We shall show some results for a couple of special cases.

First consider a CDS subject to a constant hazard rate and suppose that interest rates are also constant at r: the CDS value is given by

$$V(0) = ((1 - R) \cdot \lambda - p) \int_0^T D(s) \cdot S(s) \cdot ds = ((1 - R) \cdot \lambda - p) \int_0^T \exp(-(r + \lambda)s) ds$$

Evaluating the integral gives

$$V(t) = ((1 - R) \cdot \lambda - p) \frac{1 - \exp(-(r + \lambda)T)}{r + \lambda}. \tag{9.20}$$

The interest rate sensitivity is the derivative of the above w.r.t. r, the hazard rate sensitivity is the derivative w.r.t. λ, and the default event risk can be deduced from first principles or as the $\lambda \to \infty$ limit of the above (in either case it is $1 - R$). In practice the former two sensitivities are often calculated by shifting the interest or hazard rate, recalculating and taking the difference. The spreadsheet ‘chapter 9.xls’ worksheets ‘CDSsensitivity’ and ‘bond sensitivity’ shows some results.

Generally, interest rate sensitivities are small, and the hazard rate sensitivity is approximately

$$(1 - R) \cdot \frac{1 - \exp(-(r + \lambda)T)}{r + \lambda}$$

This can be interpreted as loss given default (LGD) times the **credit risky duration**. Note incidentally that the CDS hazard rate (and spread) sensitivity for different seniorities are in proportion to the loss given recoveries.

A bank will often wish to see sensitivity to a shift in the CDS curve of 1 bp. This will require recalibrating the implied hazard rate curve, then calculating the CDS values before and after the shift, taking the difference to get the CDS bp premium. The results will be very similar to the sensitivity to a shift of the hazard rate curve by $1/(1-R)$ bp.

If we look at a risky bond, assume that coupons of g per annum are continuous for simplicity, and the interest rate is r, then (9.18) becomes

$$BP(0) = \int_0^T g \cdot D(s) \cdot \exp(-\lambda \cdot s) ds + D(T) \cdot \exp(-\lambda \cdot T) + \int_0^M R \cdot D(s) \cdot \exp(-\lambda \cdot s) \cdot \lambda ds$$

Evaluating the integrals gives

$$BP(t) = \exp(-(r+\lambda)T) + (R\lambda + g)\frac{1-\exp(-(r+\lambda)T)}{r+\lambda} \tag{9.21}$$

Note that if the coupon is $r+\lambda(1-R)$ then the bond price is unity.

Exercise 4 Relate the above result to equation (9.17) and section 9.1.

Sensitivities can be obtained as above by differentiation or by stressing. Generally the interest rate sensitivity is significant, and we can see that, if recovery is zero, the above equation is symmetrical in interest and hazard rates and the sensitivities are thus equal.

Exercise 5

1. If the coupon is such that the bond price is par, show that the hazard rate sensitivity is equal and opposite to the CDS sensitivity.
2. Find the minimum bond price as a function of the hazard rate. Show this price is below the recovery rate if the interest rate is positive.

9.4 STOCHASTIC DEFAULT RATE MODEL; HAZARD AND PSEUDO-HAZARD RATES[5]

The development of the pricing formulae in the previous section assumed, rather unrealistically, that the hazard rate was a deterministic function of time. We show how to resolve this apparent problem in this section – at the cost of some rather more sophisticated mathematics although the treatment is kept as simple as possible. Knowledge of this section is not key to the rest of the book, although some results from this section are quoted elsewhere.

The generalisation we make is that hazard rates, recovery rates and spot interest rates are all stochastic variables with unspecified distributions. We shall also introduce risky counterparties

[5] I am grateful to Phil Hunt for conversations relevant to this section. Any errors and misunderstandings are my own.

into the valuation formulae. We redefine and add to our notation:

Reference entity stochastic hazard rate	$h(s)$
(and use h with a super-suffix for the writer and buyer hazard rate)	
Reference entity survival probability	$S(s) = \exp\left(-\int_t^s h(x)dx\right)$ (revised (9.8))
Survival probability (writer)	$S^w(s) = \exp\left(-\int_t^s h^w(x)dx\right)$
Survival probability (buyer)	$S^b(s) = \exp\left(-\int_t^s h^b(x)dx\right)$
Reference entity stochastic recovery rate	$R(s)$
Stochastic spot interest rate	$r(s)$
Discount factors	$D(s) = \exp\left(-\int_t^s r(x)dx\right)$

We assume that writer and buyer have zero recovery rates for claims on financial products.

We state that the value of a CDS deal is the expected value of the contingent payments. As in section 9.3 we obtain a very similar looking formula, bearing in mind that cashflows occur only if both counterparties to the deal survive:

$$V(t) = E\left(\int_t^T (1 - R(s)\right) \cdot D(s) \cdot h(s) \cdot S(s) \cdot S^b(s) \cdot S^w(s)ds$$
$$- p\int_t^T D(s) \cdot S(s) \cdot S^b(s) \cdot S^w(s)ds) \tag{9.22}$$

where E denotes the expectation operator (and we have taken the accrued on the deliverable to be zero for simplicity).

We have not been explicit about the stochastic processes behind these variables: for the moment the formula is very general.[6] Our interest at present is how to simplify the above formula. First recall that the expectation is itself an integral over a statistical distribution and it is possible to interchange the order on the integrations in equation (9.22) – in other words, we can take the expectation inside the time integral. We find expectations such as $E(D(s) \cdot S(s))$ and $E(R(s) \cdot D(s) \cdot S(s) \cdot h(s))$.

We would like to be able to take expectations of terms separately and multiply them together (for example, the expectation of the discount factors times the expected survival probability). We can only do this if the various terms are independent. We shall assume that

(1) interest rates and survival probabilities are independent
(2) interest rates and recovery rates are independent

[6] If 'value' is the market value then the statistical distributions are the 'risk-neutral' distributions and we are using the 'risk-neutral measure'.

(3) recovery rates and default rates are independent, and, if we include the counterparties, then we also need to assume that (4) the buyer, writer and reference entity survival (and default) are independent.

Note that we cannot assume that the survival product and hazard rate are independent:

$$E(S(s) \cdot h(s)) \neq E(S(s)) \cdot E(h(s)).$$

Assumption (1) is reasonable: although hazard rates are strongly correlated with the economic cycle, as are interest rates but only with a significant lag, so the correlation can reasonably be taken to be zero. Similarly, assumption (2) is reasonable. Assumption (3) is less reasonable; there appears to be a direct (negative) correlation between recovery rates and hazard rates. Nevertheless it is an assumption we need to make. Assumption (4) is also unreasonable as spreads (and hazard rates) tend to be positively correlated and we would anticipate that survival probabilities will be correlated. We therefore drop the counterparties from the equation by assuming they are risk free. We shall return to correct the valuation, including counterparty risk, in Part IV. We shall also test the significance of these approximations in section 9.7 below.

If we define

$$\begin{aligned}\overline{D}(t) &= E(D(t))\\ \overline{R}(t) &= E(R(t))\\ \overline{S}(t) &= E(S(t))\end{aligned}$$

then we can rewrite equation (9.22) as

$$V(t) = \int_t^T (1 - \overline{R}(s)) \cdot \overline{D}(s) \cdot E(S(s) \cdot h(s))ds - p \int_t^T \overline{D}(s) \cdot \overline{S}(s)ds \tag{9.23}$$

We cannot take the expectation of the two terms in the above separately because the survival probability is a function of the hazard rate. There is a correlation that we cannot reasonably ignore. But if we define λ by

$$\lambda(s) = -\frac{\partial \overline{S}(s)}{\partial s} \quad \text{or} \quad \overline{S}(s) = \exp\left(-\int_t^s \lambda(x)dx\right) \tag{9.24}$$

and note that

$$\begin{aligned}E(h(s) \cdot \exp\left(-\int_t^s h(x)dx)\right) &= -E\left(\frac{\partial}{\partial s}\exp\left(-\int_t^s h(x)dx\right)\right)\\ &= -\frac{\partial}{\partial s}E\left(\exp\left(-\int_t^s h(x)dx\right)\right) = -\frac{\partial}{\partial s}\overline{S}(s) = \lambda(s) \cdot \overline{S}(s)\end{aligned}$$

then equation (9.23) becomes

$$V(t) = \int_t^T (1 - \overline{R}(s)) \cdot \overline{D}(s) \cdot \overline{S}(s) \cdot \lambda(s)ds - p \int_t^T \overline{D}(s) \cdot \overline{S}(s)ds \tag{9.25}$$

We see that the valuation formula has the same form as the deterministic formula[7] (9.13) and the λ function (9.24) looks just like a deterministic hazard rate. We call λ the pseudo-hazard rate. The discount, recovery and survival factors are now interpreted as expectations, and the pseudo-hazard rate is a complicated function of other expectations. λ is neither a hazard rate nor an expectation of a hazard rate – it looks and feels like a hazard rate (in (9.25)) but it is not. Nevertheless, fitting the above equation to market data is exactly as before except that we are modelling the pseudo-hazard rate rather than the hazard rate. Nothing has actually changed in our modelling formulae for CDS (and for bonds) except that we now explicitly recognise that any of the variables are stochastic. Only the interpretation of the result has changed – the implied pseudo-hazard rate is different from the actual hazard rate and this has to be recognised in situations where volatility of hazard rate is relevant. In addition, we do not know the initial value of the hazard rate.

9.5 CALIBRATION TO MARKET DATA

9.5.1 Calibrating to CDSs and to Bonds

If we want a model to value credit derivatives it makes sense to calibrate to CDSs. The 'hazard rate' is then the 'implied' rate of credit events as appropriate to CDS documentation. On the other hand, if we wish to use the model to value bonds it would make more sense to calibrate to bonds and get an 'implied default rate'. We would expect the two rates to be different (see Chapter 11). Sometimes we do not have the luxury of the choice: there may be no CDS data for the name. For example, 'high-yield' entities (often young companies in a rapidly changing and competitive economic environment) will have perhaps one bond in issue only, and typically that bond will be callable (and possibly convertible). The name probably does not trade in the CDS market, but may figure in a CDO structure, so calibration of some form has to be performed.

The choice of calibration data has an impact on how we perform that calibration. For example:

1. Rich CDS data set (say 1-, 2-, 3-, 5-, 7-, 10-year maturities). We would choose six-parameter hazard rate function (say piecewise constant) so that we can replicate all the input data exactly.
2. Single CDS data point (say 5-year).[8] We would choose a one-parameter function (perhaps a constant hazard rate or, better, take the shape of the CDS curve from similar spread (and industry) names) to fit the CDS level exactly.
3. CDS and bond data. If several CDS maturities are available we would usually use these alone. However, we may have one CDS data point (say 5-year) and bond data at 2- and 10-year maturities. We may wish to replicate the CDS level and the curve shape information from bonds. We could fit a two-parameter hazard rate curve reproducing the CDS level and giving a best-fit to the slope of the hazard rate curve implied by the bond data.

[7] We have set the accrued to zero

[8] Typically in this case the trader (or another source) would engineer a rich curve of CDS quotes but there are circumstances where engineered data is not readily available and the risk control department for example may not want to rely on the trader's curve.

4. Bond data only (several bonds – say 5[9]). We may increase the bonds spread to estimate the CDS–bond basis (see Chapter 11), and fit a low-parameter model (say 2 or 3) in order to average out liquidity effects.
5. Single bullet bond. We would fit a one-parameter model to the bond price, then adjust the level to estimate the basis.
6. Callable bonds and syndicated loans (see section 9.5.6).

Note that 'calibration to bonds' does *not* mean taking the bonds spread and assuming this is the CDS spread, then calibrating to this notional CDS – as is commonly but erroneously done. It means using the bond price, and the bond price function (9.18) under the default and recovery model, to imply a hazard rate.

9.5.2 Implied Hazard Rates

'Implied' means derived from prices of financial products. Financial economists also call these 'expected' rates – in the sense of a statistical expectation. It is not necessarily a good estimate of future hazard rates ('anticipated' rates) in the same way that forward interest rates are a biased estimator of future interest rates. Prices for financial products contain an estimate of forward rates, plus the investor gets a reward for taking on the risk of uncertain forward rates. We use 'implied' rates in the pricing of complex derivatives because they reflect the cost of our hedge.

9.5.3 Calibrating to Bonds: Multiple Solutions for the Hazard Rate

We saw in section 9.3 that the bond price under the default and recovery model may be consistent with two hazard rates (if recovery is positive). This will typically correspond to a price marginally below the recovery rate. Additionally, given an assumed recovery rate, the floor for a bond price is generally slightly below this recovery rate. Figure 9.3 shows the situation for one bond under a constant hazard rate model. At prices slightly below the assumed recovery of 20, there are two hazard rates consistent with that price and recovery.

The situation becomes more complicated when we are calibrating to several bonds and have a multifactor hazard rate curve. Then we potentially have multidimensional surfaces of solutions to the hazard rate curve when prices are around the recovery assumption.

9.5.4 Calibrating to Bonds: Implied Recovery and Hazard Rates

An interesting and useful application of the price model to bonds is to derive information on implied recovery rates. The bond price sensitivity formula (derived from (9.18)) shows that, as the hazard rate increases, sensitivity to the hazard rate drops, and sensitivity to the recovery rate rises. This should be intuitively obvious – at very high hazard rates we have seen that the bond 'trades on a recovery' basis: changing the assumed recovery rate will have an almost

[9] In such circumstances (a rich bond universe) it is likely that CDS trade. However we may wish to explicitly model bonds separately from CDS in such circumstances in order to track the average basis.

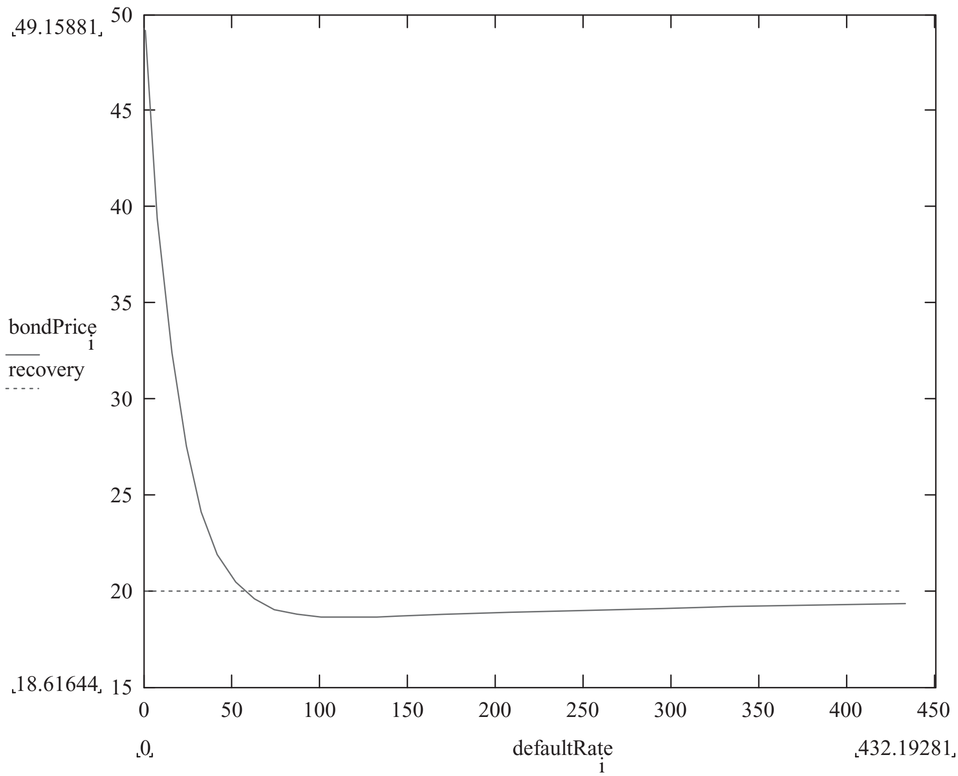

Figure 9.3 Bond price as a function of hazard rate for a given recovery (of 20%).

equal change in the bond price, but changing the hazard rate by 1 bp will have very little impact. The opposite is true when the hazard rate is very low: see Table 9.1 and Figure 9.2.

When spreads are 'high' it becomes possible to calibrate to several bonds using a simple hazard rate curve and treating recovery as a parameter to be fitted. Note that we can also

Table 9.1 Bond sensitivity to hazard and recovery rates at various bond prices (5-year bond with assumed recovery of zero)

Price sensitivity to changes in default and recovery			
Spread	Dirty price	1% change in RR	1% change in default rate
40	100.40	0.02	5.01 (5-year bond)
400	85.38	0.17	4.20
1 000	66.29	0.36	3.16
2 000	45.41	0.57	2.06
3 000	32.67	0.69	1.39
4 000	24.50	0.78	0.97
5 000	19.15	0.83	0.70
10 000	8.42	0.93	0.20
infinite	0.00	1.00	0.00

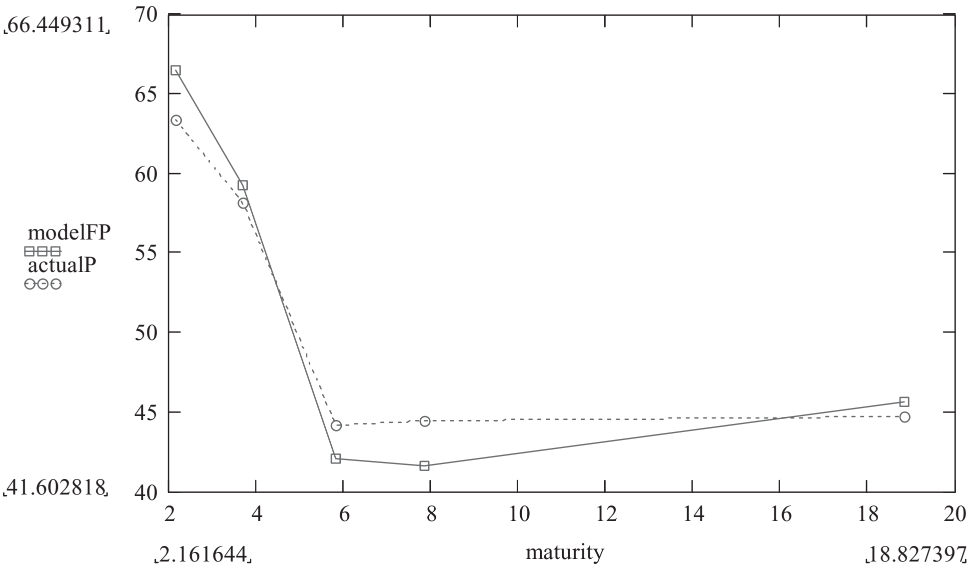

Figure 9.4 Zero recovery, two-factor hazard rate curve fitted to distressed debt. (Rms error = 2.1%).

partially calibrate to recovery when fitting to several CDSs of different seniorities. Figures 9.4 and 9.5 compare

(1) a linear hazard rate curve together with an assumed recovery of zero, and
(2) a constant hazard rate curve and an unknown recovery rate

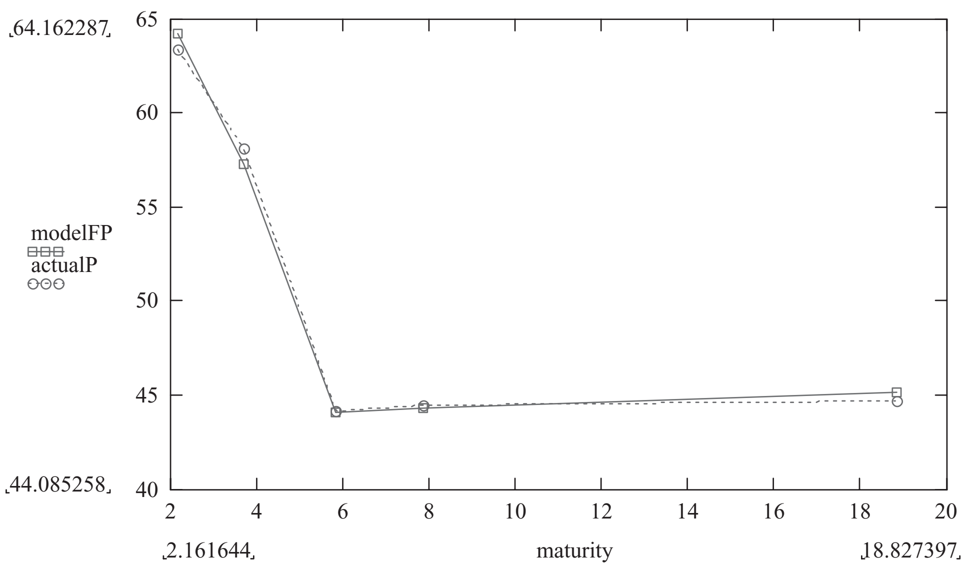

Figure 9.5 Constant hazard rate and recovery rate. (Implied recovery = 26%, rms error = 0.6%).

fitted to Russian bond data in 2000. In both cases a two parameter model is being fitted, and the model is a 'best fit' minimising the sum of square price errors. It turns out that the latter case – fitting to recovery and a single hazard rate – gives a markedly better fit than assuming recovery of zero and fitting a higher order hazard rate curve.

9.5.5 Implied Hazard Rate Curve and No-Arbitrage

The implied hazard rate should be positive at all times – the model should not allow for re-incarnation – and mathematically it is required for a time-changed Poisson process. Allowing negative hazard rates can produce problems in implementation (particularly under some simulation approaches). If negative hazard rates arise after calibrating to input data, then this indicates the presence of arbitrage opportunities. Negative hazard rates should always be detected and flagged by the CDS software – it not only indicated the presence of an arbitrage but the mathematical foundation of the credit model collapses and the results from any calculations using negative hazard rates will be suspect. In some cases the software itself may (reasonably) fail with such data. A simple example serves to illustrate corresponding inconsistencies at the CDS premium level.

Example Given a 3-year CDS of 50 bp and a 5-year CDS of 60 bp then there are constraints on where a 4-year CDS can trade. The default and recovery model interpolates a sensible value based on a hazard rate curve, and will probably come up with a number around 55 bp. The actual 4-year CDS may differ from this level but there are quite tight constraints on the no-arbitrage premium levels. If we enter 3-, 4- and 5-year CDS levels into a pricing model, then the calibration will alert us to an arbitrage if the implied hazard rate is negative at any point. Suppose the 4-year CDS traded at 35 bp: if we bought this contract we would pay a total of 140 bp over the 4 years, which is 10 bp less than if we buy the 3-year CDS. In other words, at this price the fourth year is not only free but the counterparty actually pays us to take on our risk!

Exercise 6 Show that, with the above 3- and 5-year spreads the 4-year CDS must be between about 37.5 bp and 75 bp to avoid an arbitrage.

9.5.6 Syndicated Loans

The market for syndicated loans has expanded considerably since 2000. Traditional (bilateral) loans are individually negotiated agreements and can vary enormously in their terms and conditions, particularly in the restrictions imposed on the borrower ('loan covenants'). Restrictions on the borrower can be in the form of certain financial ratios which must be maintained (e.g. percentages of different types of debt, total borrowing, issuance of equity, etc.) or can be commercial targets that need to be met (expenses to sales, sales growth, expenses growth, etc.). Failure to satisfy the covenants leads to a requirement to repay the loan in full – in practice typically leading to a renegotiation of the loan and refinancing on new terms. The syndicated loan contract has certain standardised features in (a) the form of the loan and (b) the covenants.

From the perspective of modelling, syndicated loans and callable bonds have the non-bullet nature of the debt as a common feature – syndicated loans being immediately callable by the borrower. This needs to be modelled in order to correctly imply hazard rates or, given hazard rates, to correctly value the debt or CDS on that debt (Loan CDS or 'LCDS'). However, modelling the callability poses several problems.

1. A random walk process is inappropriate for credit. We shall see later that the pricing of some credit instruments is inconsistent with a Wiener process model and realistic levels of volatility. A realistic model should use a jump process (such as implied by a ratings transition model), but there are implementation difficulties with such models and little readily available data to calibrate the model correctly.
2. For syndicated loans, bilateral loans, and most bonds the expenses of issuing the debt are substantial – often amounting to 2–5% of notional borrowed. A standard American option pricing model does not include expenses, and simply including the level of expenses will not capture the way the corporate treasurer actually behaves. A loan will not be refinanced if the refinancing merely saves the refinancing cost – typically the treasurer will look for a level of profit at least equal to the level of expenses.
3. The covenants mean that the loan may be terminated in different circumstances from the those handled by standard option pricing models – as the corporate performance deteriorates (and refinancing spread widens) a covenant may trigger the termination of the loan. Additionally, changed corporate plans or mergers may trigger the covenants and repayment – with refinancing spreads possibly wider or narrower.

A simple practical model is to ignore the presence of the covenants and implement a spread-tree model with a high level of volatility together with expenses plus profit margin converted into a spread hurdle. For example, if the refinancing costs are 2% and the loan is a 5-year loan potentially being refinanced after 3 years with a new 5-year loan, then it is unreasonable to view the refinancing cost as being spread over the remaining 2 years only – the company would probably be borrowing again if the loan reached original maturity. So the 5% refinancing cost plus, say, a 2% profit margin should be spread over the remaining 5 years and would correspond to, very roughly, a 140 bp spread hurdle. In other words, if the simulated spread on the reference entity is more than 140 bp below the initial spread (assuming a par issue price) then the loan is assumed to be called.

9.6 CDS DATA/SOURCES

9.6.1 Survey Data

The 'credit pyramid' in Chapter 4 outlines one of the major problems related to the credit and credit derivatives market – lack of good-quality data, both in terms of coverage and reliability, arising from the relative lack of liquidity in the markets. The 10 years up to 2007 saw a rapid growth in credit derivatives, with a concomitant increase in the amount of data available from market deals and via brokers. In the past, banks have relied on their own internal databases of CDS data collected from front offices and brokers. This is not sufficiently complete for the banks, and does little to help other institutions dealing less frequently in the CDS market.

As part of the process of supplying banks and other institutions with good independent market data to help in the pricing and monitoring of deals and risk, several data services have developed. Some examples are as follows.

1. Several credit derivatives brokers supply CDS (and other) data based on quotes and trades it has seen.

2. Markit, through their CDS pricing services, provide an independent source of pricing data covering investment grade and non-investment grade single name CDS. Data is sourced from sell-side contributors, including Markit's partner banks and other significant market-making institutions. Contributors provide data sourced primarily from their front-office (trading) desks via the systems that feed the firms' books of record.
3. CMA (part of the CME group) provides observed real-time indicative prices sourced from CMA's Buy-Side Data Consortium – 30 leading firms including Hedge Funds, Asset Managers and the buy-side desks of leading global Investment Banks.

Markit supplies[10] the following (as of June 2009).

- Daily credit pricing that is independently validated, accurate and robust.
- End of Day, regional closes and intraday CDS pricing.
- Rigorous data cleaning to ensure that only the highest quality data is used in forming composite prices.
- Data available at tier, maturity, currency and doc clause levels.
- Flexible access to data via multiple delivery channels, including internet, Markit Desktop and automated download.
- Over 20 000 CDS credit curves constructed daily (by entity, tier of credit, currency and documentation clause) across approximately 3 000 entities and tiers of debt. Historical data includes more than 8 years of daily CDS spreads where available.
- In addition, Markit provide data on industry and rating sector portfolios.

They also provide data services to meet the requirements of other asset classes, including bonds, loans, LCDS, and various structured finance instruments.

Figures 9.6 and 9.7 show the geographical and rating breakdown of the volume of data.

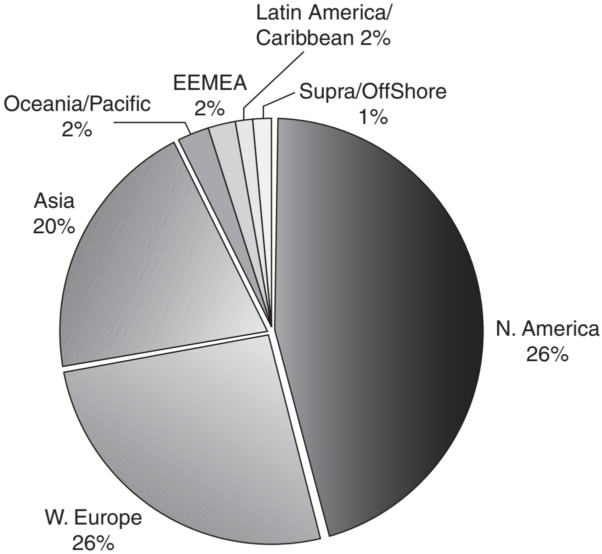

Figure 9.6 CDS data breakdown by region (*source:* Markit).

[10] Quoted from Markit marketing material

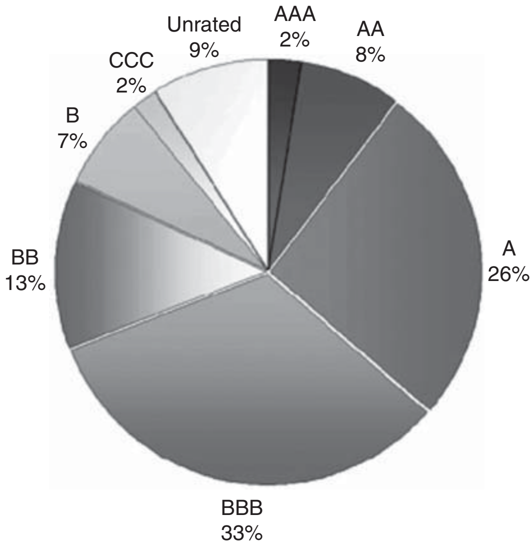

Figure 9.7 CDS data breakdown by rating (*source:* Markit).

9.6.2 Data Engineering

Data engineering is the process of taking poor-quality data and extracting 'best guess' estimates of the correct data – correcting for errors in the data – and of missing data. In the context of credit derivatives this can mean the following:

(a) Filling gaps in the time series
(b) Filling a full term structure of CDS quotes
(c) Producing CDS curves for names where no, or little, data is available.

Survey data in some cases requires data for each name covered and at each maturity. On the other hand, partial data may be received and data engineering techniques used to produce the data required. These various steps require different approaches.

Initial 'Quality' Test on the Data

The idea is to identify and reject bad or suspect data points. A standard approach for any time series is based on the change from yesterday compared with the market average. In the case of CDS spread – where a 200 bp name one day may be 1,000 bp the next – this can lead to many false rejections. CDS premiums exhibit jumps which are valid changes. The test described above might be amended and used as a tool to improve the data history by identifying sudden jumps which are quickly reversed. Even here correct data might be rejected.

Missing Time Points

Available data can be used to produce an index CDS premium. Indices may be very broad – all corporate investment grade names – with the advantage that index changes from day to day will be very little influenced by the appearance of new names in, or disappearance of old names from, the index. On the other hand, the index may be narrow – specific to US autos rate A – but then large spurious jumps from day-to-day will arise as names are added or removed from the index.

Once an appropriate compromise index has been produced, percentage index changes can be used to estimate today's CDS level for a name where no data is available, based on yesterday's level and the percentage index change (for the index appropriate to the name). Clearly such a process is an estimate that gets progressively worse as time goes on. The underlying assumption is that the name follows average market moves.

CDS Curve Generation

Yield curve modelling techniques can be applied to describe the shape of the CDS curve for those entities where CDS premiums at several maturities are seen. Fortunately CDS curves tend to follow similar patterns across most names – upward sloping and levelling off at long maturities for low- and medium-spread names, and downward sloping and levelling off, for high-spread names. The following functional form illustrates the approach and can be extended to give a more accurate description of a curve (Chaplin, 1998; Anderson *et al.*, 1996).

$$f(t) = A + B \cdot \exp(-\alpha \cdot t) \tag{9.26}$$

A and B are chosen to fit the curve while α can be chosen to reflect the overall curvature – defined in terms of the half-life (HL) of the levelling out of the curve:

$$\alpha = -\ln(0.5)/HL \tag{9.27}$$

A half-life that is proportional to the spread can give a good fit for most CDS curves from AAA up to B-rated names.

Once curves are calculated for different ratings or spread bands, an average curve can be selected – defining $\backslash A$ and $\backslash B$. Given a single CDS level for a name, the half-life is defined, and the $\backslash A$ parameter can be adjusted so that the curve passes through the known data point. The process can be refined to use earlier known curve data for the name.

Missing Names and Seniorities

A common problem is to price a CLO related to entities and/or debt which does not trade. Internal ratings can be used to map names with no market data to generic CDS curves for that rating. Adjustment to 'loan' seniority from 'senior unsecured' data is generally done on the basis of recovery assumptions.

9.7 MODEL ERRORS AND TESTS

We review some of the assumptions made in the process of building the model (9.18), and attempt to assess the significance of some of these.

9.7.1 Recovery Assumption

The impact of the assumed recovery rate, and of changing this assumption, on the implied hazard rate is huge. This will similarly have a huge impact on the value of digital CDSs (see

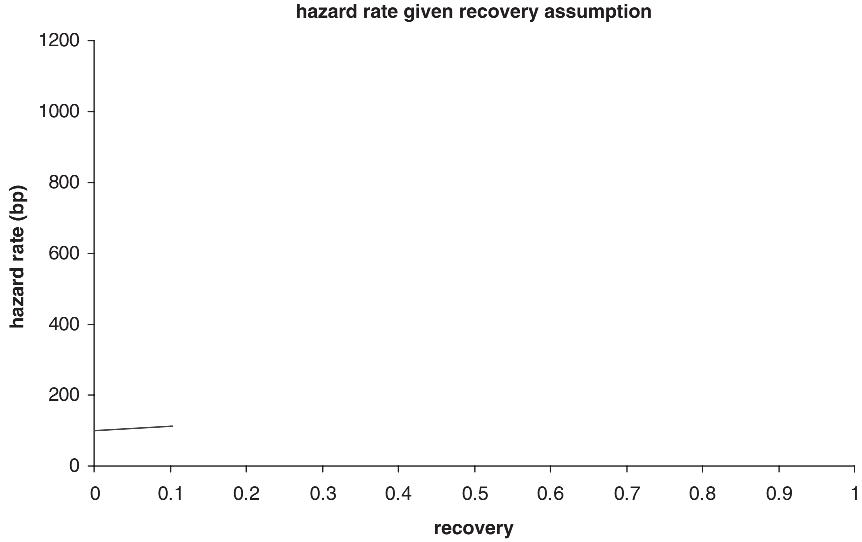

Figure 9.8 Implied default rate for a range of assumed recovery rates.

Chapter 14), but typically only a tiny impact on CDS values (the impact is zero on an at-market deal and greatest when the hazard rate is high). Figure 9.8 shows the implied default rate for a range of assumed recovery rates and a given CDS to which to calibrate.

9.7.2 Interest and Hazard Rate Correlation

This can be tested by setting up a model where both interest and hazard rates follow correlated diffusion processes. Let h be the hazard rate, r the interest rate, with drifts μ and standard deviations σ respectively, then define lognormal processes

$$\frac{dh}{h} = \mu_h \cdot dt + \sigma_h \cdot dz_h \tag{9.28}$$

$$\frac{dr}{r} = \mu_r \cdot dt + \sigma_r \cdot dz_r \tag{9.29}$$

where $E(dz_h \cdot dz_r) = \rho$ is the correlation between these variables.

We can use the above model to value CDSs of several maturities, and compare the results when 0 or 50% correlation is used. Figure 9.9 shows the results based on 1m simulations with a 10% interest rate volatility and 50% hazard rate volatility.

We conclude that – at least for investment grade names – interest rate and hazard rate correlation does not have a significant impact.

9.7.3 Reference Name and Counterparty Hazard Rate Correlation

The reader should note carefully the final paragraph of this section.

Assume that the buyer is risk free, but now include the writer risk as well as the reference entity risk, using a lognormal default rate model with correlated Wiener processes (similar to

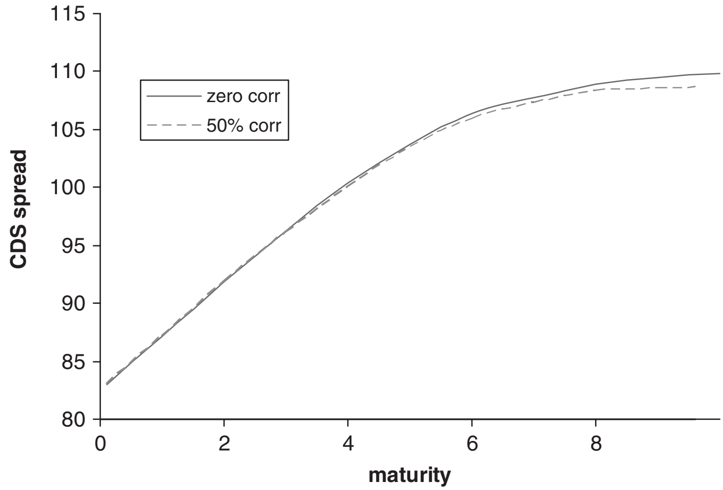

Figure 9.9 Impact of interest rate and hazard rate correlation on CDS pricing.

equation (9.26)). We choose two risks with nearly flat default rate curves – the reference name at 82 bp and the counterparty at 123 bp, and use this simulation model to evaluate the premium that should be paid for the reference entity CDS where there is a correlated risky counterparty.

Theory says that with zero correlation counterparty risk should have zero premium impact on continuously paid premiums – and this is what we see in Figure 9.10: the curves for a risk-free counterparty, and a risky uncorrelated counterparty, are well within the simulation error bars.

Default rate correlation has almost no discernible impact (well within bid–offer spreads) on CDS pricing for investment grade pairs and 5-year CDSs at spread correlations levels up to 95%, and less than 1% impact up to 60% correlation.

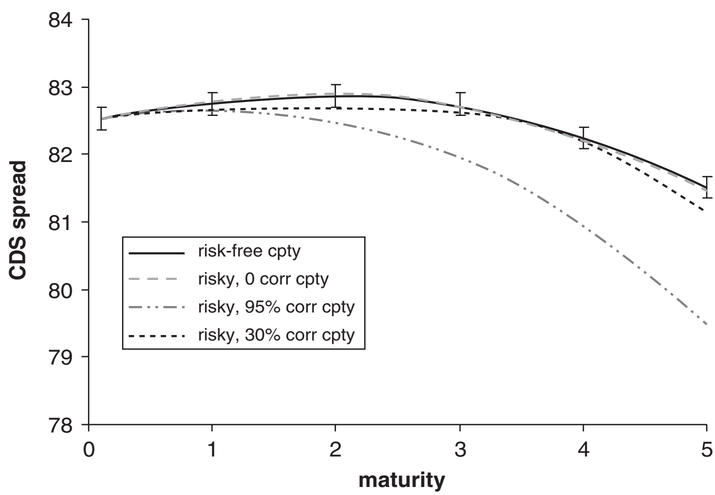

Figure 9.10 The impact of a risky counterparty on CDS pricing when spreads are correlated.

On the other hand, there is a discernible impact on very long-life (25-year) and high spread names (around 1 000 bp or more) (not shown). Although intuitively we expect correlated spreads to lead to higher risks that both names default in a certain time, the problem is that default is a sudden and surprising event. Widening of the spread on the remaining name does not significantly raise the possibility of a default within the remaining period of interest, at least not for investment grade names.

The above results should be treated with care. We have shown that if the reference name and counterparty have correlated spreads, *and those spreads follow diffusion processes*, then there is negligible impact in the pricing of CDS deals. However, the transition matrix, and experience in credit markets, tells us that spreads follow a more violent process than a diffusion. A jump process would be a better model. Part IV returns to this topic armed with a better model – and reaches quite different conclusions.

9.7.4 Interpolation Assumptions, and the Pseudo-Hazard Rate versus Stochastic Hazard Rate

Interpolation assumptions can clearly have a large impact on valuation results. Suppose a 2-year CDS on a reference name stands at 100 bp, and the 5-year CDS is at 150 bp. If we own a 3.5-year CDS and use a piecewise linear or piecewise constant hazard rate model, we find that the implied fair value CDS premium for the 3.5-year maturity is very close to 125 bp. On the other hand, if we use a piecewise constant spread assumption (not very sensible) we shall find the 3.5-year CDS fair value premium is 150 bp. The difference of 25 bp is a gross error but can easily be produced with a poor choice of interpolation function.

Other forms of hazard rate interpolation function (an exponential formula, for example) will generally lead to minor differences in CDS values. A simple reserve can be based on the use of an alternative interpolation routine and looking at valuation differences.

A more subtle question arises through the use of the pseudo-hazard rate model as opposed to a full stochastic model. Under the former model we have a near linear CDS curve if we assume a linear hazard rate curve. The latter model produces a sloping CDS curve if we have a stochastic hazard rate with a drift. How accurately does a linear pseudo-hazard rate curve fit the values arising from the stochastic model? Figure 9.11 shows the result of 1m CDS pricing simulations with 75% volatility.

Over a 5-year span there is an rms error of under a 1 bp. Over a 10-year span with a best-fit linear curve there is an rms error of under 3 bp (not shown).

9.8 CDS RISK FACTORS; RESERVES AND MODEL RISK

We summarise below the various risk factors we have discovered through the above analysis.

9.8.1 Captured and Hidden Risks

Risks Captured by our Model

1. Default event risk.
2. Default rate and recovery rate risks, now and forward.
3. Interest rate risks.
4. Repo rate term structure risks where bonds are valued or used for calibration or hedging.
5. Bond/CDS basis – if we use separately calibrated hazard rate curves for bonds and CDSs.

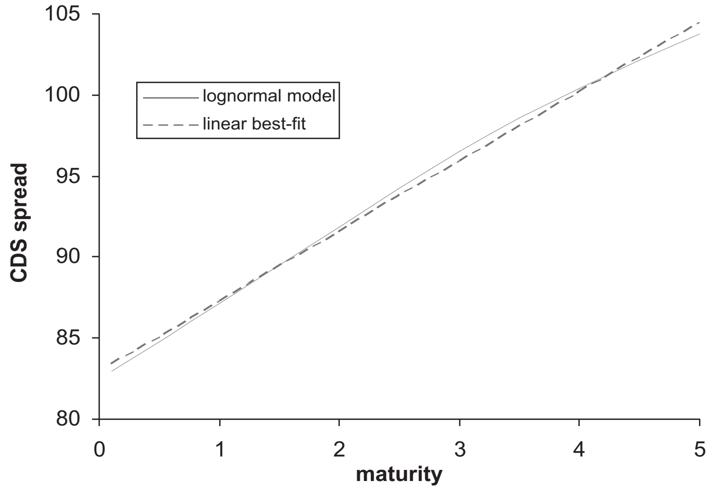

Figure 9.11 Linear pseudo-hazard rate versus stochastic hazard rate with drift.

Risks not captured by the model (hidden risks)

1. Correlation between interest rates and default rates.
2. Correlated counterparty risk.
3. Interpolation methods for default rate and interest rates.
4. Delivery option.
5. Liquidity.
6. Volatilities of CDS premiums, of interest rates, and of recovery.
7. Correlation between foreign exchange rates and hazard rates (relevant to cross-currency hedging).

9.8.2 Limits

Limits can be set on default event exposure to control unhedged event risk. These limits can be on a name, a class (US AA-rated corporates), a currency, or book(s). Digital (recovery rate) risk can be captured by changing recovery assumption, recalibrating, revaluing the books, and having limits on the difference. Term structure risks can be controlled by having limits on spread sensitivity in maturity buckets. *Repo risk:* What is the impact of repo risk and how does it affect hedges?

9.8.3 Reserves against Implementation Errors

Our model is a complicated interpolation procedure when we wish to value trades. Reserves are usually taken against each trade to reflect:

(a) 'errors' in the implementation of the model – for example, recovery assumptions or poor choice of hazard rate function;
(b) reserves for model risk – the hidden risks (see 9.8.4).

Typical examples of the former set of reserves are:

1. Bid–offer spread if the model is marked to mid.
2. Unknown recovery.
3. Interpolation.

Bid–Offer Reserve

When marking to model one can take fit to bid side data to get a bid, or fit to mids and make an adjustment (reserve) at the end. The bid–offer reserve or adjustment is one that should clearly be taken by both the trader and the institution: all other reserves arguably belong to the institution only.

In principle, bid–offer is name and maturity dependent. Some active names and maturities may have a near zero bid–offer spread; others may have a 30% spread. Increasingly traders seem to take a name-by-name approach to marking their books, explicitly estimating the bid–offer spread.

Marking on a deal-by-deal basis may overestimate the reserve. Well-hedged positions should be subject to a significantly narrower reserve (see Part I).

Recovery Reserve

Recovery is a key arbitrary number with almost no impact on some types of trades and a huge impact on others (digitals). A reserve per name can be calculated by shifting the recovery assumption to 0, or up 30% (with some practical upper limit), recalibrating and taking the worst change in the position value. This captures the large recovery risk in digital positions as well as the small recovery risk in vanilla trades.

Should the recovery reserve be netted across names?

In a diversified portfolio it can be argued that overestimates in one area will be compensated by underestimates in another. The mean will have less uncertainty. On the other hand, the mean recovery rate changes markedly over time. Even in non-recession years the mean moves 10% or so.

On the portfolio as a whole the impact of the reserve should be less than the sum of all the individual reserves. On the other hand, we know that the average recovery across a broad portfolio does vary over time – so a minimum recovery reserve on a portfolio can be set by shifting all recoveries simultaneously by, say, 10%.

Hazard Rate Interpolation Reserve

Pricing a 2-year CDS off a 5-year quote is subject to risk from the unknown shape of the curve – review the exercise at the end of section 9.2. The uncertainty increases as the deal moves further from the calibrating instruments. Clearly the interpolation reserve could be very large if inappropriate choices of hazard rate functions are taken.

Where active investment grade names are concerned, a rich data set and a linear interpolation curve will give rise to insignificant errors except at the short and long ends. One way to set an interpolation curve would be to use the form of the curve derived from engineered data and best-fit to the actual data for the name.

The reserve should be greater for inactive or atypical names (i.e. names where there is reason to believe that the risks are fundamentally different over time from other names of similar rating or industry). An extreme would be to base the reserve on a flat versus a generic curve shape. This would overestimate the reserves for a portfolio of such risks. A more reasonable reserve would be to compare the results using piecewise constant versus piecewise linear hazard rate interpolation.

Repo

This only matters where 'special' bonds are used for calibration. It does not apply to corporate debt except in very exceptional circumstances. The amount assumed can be stressed to zero or, up to an arbitrary amount, recalibrate and revalue. It is additive across the book.

Basis

The basis may be captured by separate calibration to bonds and CDSs resulting in a mark-to-market change as market levels change. The only reserve needed is to cover the error in its estimation, and this will already have been included primarily in the interpolation reserves.

9.8.4 Model Reserves

We can divide these into

(a) reserves for known simplifications in the model, and
(b) unknown deficiencies in our model.

The known deficiencies are listed in 9.8.1, and reserves can be implemented on the basis of the analysis of these deficiencies in section 9.7. We mention a few further points.

Claim amount reserve (for sovereigns)

Alternative claim amounts can be used in calibration and pricing to reflect the unknown reality of the recovery process for sovereigns.

Liquidity

This is partly captured by the bid–offer spread. Post-default liquidity is reflected in a potentially wide range of recovery rates for that name and seniority. An explicit reserve for this may be considered.

Unknown deficiencies tend to be handled by an arbitrary formula – perhaps a percentage of the expected loss, or a percentage of notional, depending on rating and maturity. Once a decision is made to trade a certain product by an institution, then model reserves would appear to be the responsibility of the institution rather than the trader.

10
CDS Deal Examples

We now look at a range of applications of the valuation formulae established above. The majority of these applications arise from trading or hedging strategies, and a thorough understanding of the analysis of these examples reveals the sources of risk in the deals and how best to control this risk.

Elementary Trading Strategies

At this point the reader should review the elementary trading strategies discussed in Chapter 8 in the light of the valuation and sensitivity results derived in Chapter 9. In particular, note that the value of a long protection position increases as the current (market) CDS premium increases – equation (9.12). Thus a bearish view (expecting CDS premiums – or bond spreads – to widen) of a particular name can be implemented by buying protection on that name. A bullish view can be reflected by selling protection or by buying bonds.

The following examples are divided into two parts: (a) purely CDS trades and (b) bond and CDS trades. A fuller analysis of CDS/bond (basis) trading is given in Chapter 11, some other CDS applications are covered in Chapter 12, and trading using credit-linked notes (CLNs) is covered in Chapter 13.

Risk Charts

We illustrate trades in this section with a 'risk chart' showing the spread sensitivity and the default event sensitivity. These are intended only to give a quick visual impression of the trade – they are not exact, nor do such charts in any sense replace a full risk report (which would include risks other than just these two). We shall assume 50% senior recovery, and 0% subordinated recovery where required. Positive spread risk means that we make money when spreads rise, and positive event risk means we make money on a credit event. For example, a bearish trade on FMC implemented by buying USD 10m protection in 5-year maturity is shown in Figure 10.1.

10.1 A CDS HEDGED AGAINST ANOTHER CDS

10.1.1 Cross-Currency Default Swap Pricing and Hedging

An investor wishing to take GMAC risk but wishing to invest in EUR has a limited choice as most debt is in USD or CAD and only around 2% is in EUR. A bank could create EUR GMAC debt synthetically by producing a credit note linked to GMAC in EUR (see below), effectively buying 5-year (say) GMAC protection in EUR. The bank could hedge this by selling protection to the same maturity on GMAC in USD. What price differences should there be, and what is the correct hedge, for long CDS protection in EUR hedged by short CDS protection in USD (or any other pair of currencies)?

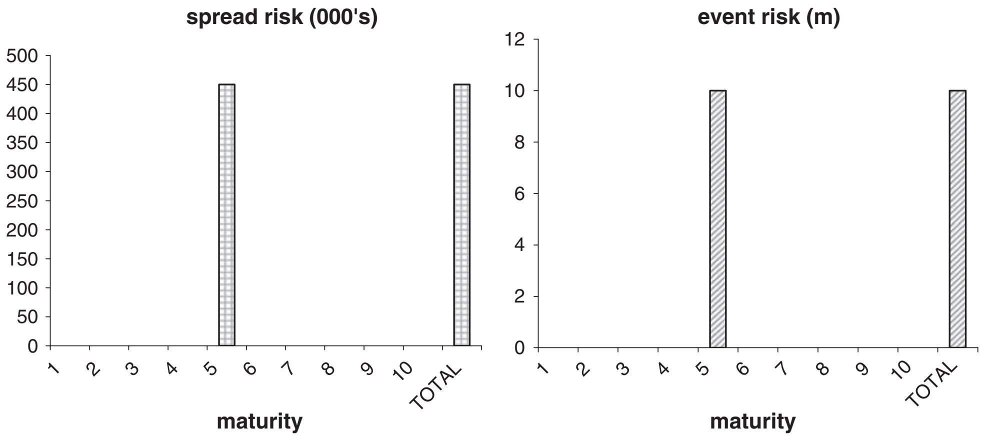

Figure 10.1 Risk chart for the example bearish trade.

Case A: Constant Hazard Rate/Constant Forward FX Rate

First assume that the hazard rate is constant. Equation (9.15) says that the CDS premium is the hazard rate for GMAC times the loss given default – there is no mention of currency or interest rates in this equation. The hazard rate gives the rate of default for the entity GMAC, and the recovery is the recovery rate on (senior unsecured) GMAC debt (all currencies of senior unsecured debt rank equally in default so have the same recovery). The implication is that the USD CDS premium and the EUR CDS premium should both be the same. In practice we can see the USD CDS premium in the market (say, 200 bp), we assume a recovery rate (say, 40%), and calculate a hazard rate: $200/0.6 = 333.3$ bp (3.333% p.a.). Given this hazard rate we can calculate that the CDS in any currency $= 333.3 \times 0.6 = 200$ bp.

So the theory (with a constant hazard rate) implies that the CDS premiums are identical in USD and EUR. If we buy 20m EUR protection at 200 bp, we should sell USD protection at 200 bp (although, of course, the bank will try to source protection cheaply and sell it expensively) but in what size?

If default occurs immediately we deliver 20m EUR of defaulted debt into the protection. In practice, on the long we would deliver EUR 20m worth of the cheapest USD or CAD debt (since this is pretty much all that is available) into the protection. We should make sure that we know what debt shall be delivered to us on the short protection hedge before we have to decide what debt we want to deliver[1] – we may be able to find something cheaper, but we cannot rely on that. Subject to this documentation requirement, we also want to make sure that the nominal amount delivered to us is the amount we need to deliver into the protection. If we are delivered USD debt then the amount is just the size of the USD protection we have written.[2] We want this amount to be financially equal to EUR 20m/EURUSD where EURUSD is the number of EUR per USD at the delivery notice date. We do not know what this amount

[1] It is the owner of protection who has the choice

[2] If we are delivered debt in another currency the nominal amount is determined by the current exchange rate (at the time the delivery notice is given stating what debt has been chosen): so is the size of the USD protection sold times the number of units per USD of the currency of the delivered debt.

will be but if the forward exchange rate is equal to the spot exchange rate (at all dates) then the expected exchange rate is just this rate. Suppose the rate is 1.25 then we need to sell USD 16m protection as a hedge. On default we get delivered 16m USD debt which, if the rate turns out to be 1.25, is EUR 20m worth of debt – the exact amount we need to deliver into the long protection. We receive EUR 20m cash which converts into USD 16m and this is paid to the owner of the USD protection we sold.

The actual exchange rate may be higher or lower than this but the expected value is 1.25. Even with volatile exchange rates the currency risk we are taking is small. A day after we put the EUR 20m trade on, hedged with USD 16m short protection, if the exchange rate changes to 1.20, then our EUR 20m protection is equivalent to USD 16.667m so we need to sell a further USD 0.667m of protection. But under the assumption of a constant hazard rate curve we sell the rehedge amount at the same premium of 200 bp. The total income on the new total hedge therefore continues to exactly match the outgo on the bought protection.

If hazard rates are stochastic we know (by assumption) that the expected spread is constant, so although rehedging will lead to profits or losses (i.e. surplus or deficit on the premium cashflows) the expected loss is zero. We can also see that there are risks to do with how the exchange rate and the currency rate might change together (FX and hazard rate correlation risk).

Case B: Constant Hazard Rate/Forward FX Rate Curve

If we now assume that the forward FX rate is time dependent (determined by the structure of the interest rate curves in the two economies) then the correct hedge amount to be short of at time t is EUR 20m/EURUSD(t) in order to cover the default event risk. If the EUR is rising then we will be selling more and more USD protection as time goes on. Note that at any moment we are paying out 200 bp on EUR 20m of protection = EUR 0.4m, and we shall be earning 200 bp on USD 20m/EURUSD(t) = USD 0.4m/EURUSD(t) = EUR 0.4m, so there is no income loss or gain as long as, under a constant hazard rate curve, we remain default event hedged.

Case C: Hazard Rate Curve / Constant Forward FX Rate

If the exchange rate is constant then the hedge ratio is fixed (apart from unexpected changes in the FX rate which may be either way). The two premium rates are set on Day 1, and again the two premiums will be the same. The premium is determined by equation (9.13) with $V = 0$, and because the forward FX rate is constant, this implies that the interest rate curves are identical, so that equation (9.14) is the same whether we are looking at USD or EUR – hence the CDS premiums are identical.

Unanticipated exchange rate moves will cause rehedging to be done at different CDS premiums from the initial premium (since the CDS is shorter and there is a hazard rate curve), giving a profit or loss. The expected value is again zero. This also applies to unanticipated hazard rate moves.

Case D: Hazard Rate Curve/Forward FX Rate Curve

Now let us assume that both the hazard rate and the FX rates are functions of time. In order to cover the default event risk, the hedge ratio at any time is still EUR 20m/EURUSD(t). But now the new hedges we put on are no longer earning 200 bp because the hazard rate curve is not constant so the forward CDS premiums will change – if the curve is upward sloping then the future USD CDS sales are at a higher CDS premium. These higher amounts will generate an excess income over the EUR cost. This implies that the CDS premium in EUR should be more than 200 bp. This is consistent with equation (9.13). Where the hazard rate is non-constant then a dependence on the interest rate curve enters in (9.13) and the existence of an FX rate curve also implies that the discount curves in EUR and USD are different.

First, we shall try to get a 'feel' for the impact of this on EUR CDS pricing and on our future hedges. Suppose EUR interest rates are 2% and USD rates are 3%, then this interest rate difference implies an FX curve with the exchange rate (EUR per USD) falling from spot 1.25 to 1.19 at 5-year maturity (EUR strengthening). The hedge sold rises from USD 16m to 16.8m. The CDS spread is changing but to first order the income is going to be around 200 bp. At the end of year 5 we need to sell an additional USD 800 000 protection, earning an additional USD 16 000. This income is in addition to the fixed USD 16m × × 200 bp = USD 320 000 p.a. on the initial hedge amount, and the total income is worth USD 336 000 × 1.19 = EUR 400 000, the same as our cost. We have to take into account the expected change in the hazard rate in order to see the second-order impact. Suppose the hazard rate is rising linearly from 0 to 667 bp in 5 years time (this has a current average of 333 bp and a CDS premium of 200 bp approximately). In the final year the average hazard rate is 600 bp, implying a CDS premium of roughly 360 bp. Thus, the fourth year produces a risk of 16.6m and an income of USD (320 000 + 6,400 × 3.6) × 1.19 = EUR 408 000. The excess income will average around EUR 4,000 per annum, or around 2 bp – in other words, we should be paying around 202 bp for EUR protection if 200 bp is correct for USD (this, of course, relies on the relative interest rates used in this example and the steep hazard rate curve assumed).

Let us look at the deal from the point of equation (9.13). The fair value premium at time t is given by

$$\frac{\int_t^T (1-R)\cdot D(s)\cdot S(s)\cdot \lambda(s)ds}{\int_t^T D(s)\cdot S(s)ds}$$

The only difference between USD and EUR is the lower discounting (by an assumed 1% p.a.) for the EUR. As this gives more weight to the higher hazard rates (at greater maturities), it should also give a higher premium for the EUR CDS – as we obtained above. An approximate calculation is given in the spreadsheet 'chapter 10.xls' worksheet 'FX and CDS curve' and indicates roughly 2 bp per annum less for the 5-year EUR CDS compared with the USD CDS. (Note that the significant error in the calculation of the USD CDS premium in the spreadsheet arises from the crude numerical integration. This error is similar in both USD and EUR calculations; the difference in the results is quite close to the accurate answer. Accurate calculations require much more accurate numerical integration: results from accurate calculations are quoted below in tables.

Table 10.1 shows the impact of different hazard and interest rate curves on CDS pricing in one currency relative to a base currency (with a flat 3% interest rate curve). The table incidentally shows the impact of quarterly premiums (in arrears with a proportion) compared with continuous premiums.

Table 10.1 Cross-currency pricing (relative to base currency with flat 3% interest rate curve); CDS paying quarterly in arrears with proportion to default date

CDS in base currency	Interest rates in second currency	CDS in second currency
200 bp flat; 40% recovery	3% flat	200 bp
	2% flat	199.7 bp (if premiums were continuous then the second currency CDS would also be 200 bp exactly)
100 bp at zero maturity, rising to 200 bp at 5 years	3% flat	200 bp
	2% flat	100 bp rising to 200.6 bp
	6% flat	100 bp rising to198.2 bp
	1.5% rising to 4% at 5 years	100 bp rising to 199.4 bp
1000 bp at zero maturity, falling to 500 bp at 5 years; 20% recovery	7% flat	1 000 bp falling to 519.8 bp

Other causes of currency-related premium differences

The example above related USD and EUR. For these two currencies – and for all major currency pairs – the relationship predicted above is observed in practice (Figure 10.2). This has not always been the case for bonds – in the mid-1990s substantial differences in bond spreads for JPY and USD debt of the same reference entity differed substantially. It was not possible to short the expensive bonds (and go long the cheap bonds to set up an arbitrage) and the CDS market did not exist in sufficient size to allow a synthetic trade to be set up. However, where an active two-way CDS market exists such differences should not persist.

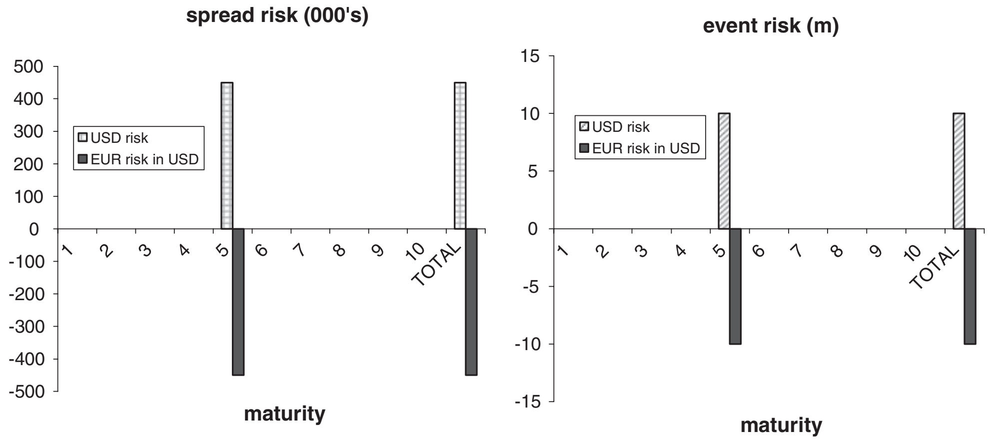

Figure 10.2 Risk chart for cross-currency trade hedged at the same maturity.

Hedging Based on Sensitivity Rather than Default Event

The examples above have discussed hedging in terms of default event hedging. Typically hedging will be done on a 'sensitivity' rather than 'event' basis for investment grade names. Results are similar, but as they are much more calculation intensive they are not given here. (For '**high yield**' = sub-investment grade names, hedging is more complicated and may be done on an event risk basis or a combination of event risk and sensitivity.)

Note on cross-currency 'book' risks

An investment grade trading or investment book of CDS deals which is net long protection in one currency and short in another, runs a risk whose expected value is zero (subject to the CDS being correctly priced). If forward FX exposure is unhedged and FX rates follow forward rates, then the book will be consistently selling (or buying) protection in one currency. Unexpected moves in either FX rates or all CDS levels will lead to an unanticipated profit or loss on FX rehedging on the net book position.

From a book perspective, net cross-currency positions should be limited to an amount with which the institution is comfortable. Examining a representative CDS book and stressing the exchange rate, CDS level, and interest rates in one economy, may indicate an unanticipated FX risk of (say) 0.2% of notional.[3] Naked exposure limits may be set at around 1% of gross notional – so if cross-currency hedging accounted for 10% of naked exposures, this would use up only around $10\% \times 0.2\% = 2\%$ of the naked exposure limit. Results depend on currencies, CDS levels, etc. Additionally an institution would wish to look at the unanticipated risk associated with the above – which will be significantly more restrictive.

10.1.2 Back-to-Back Trades, Default Event Hedges and Curve Trades

Typically a front book trader will find he has acquired or sold protection on a specific name (say BT) to a certain maturity (say 5 years). What risk should one hedge – default event, or spread (hazard rate) risk? There are various alternatives available to hedge emerging front book positions or to put on a deliberate risk position.

Back-to-Back Hedge

Where protection has been obtained cheaply (perhaps via a credit-linked note sold to a client), an exactly matching deal – same maturity date, size, currency and reference name – in the opposite direction removes default event, spread and interest rate risk. Only counterparty risk remains arising from the chance that the buyer of protection defaults after CDS premiums have narrowed.

Default Event Hedge/Spread Directional Trade

A purchase of JPY 1bn of 5-year protection could be hedged by selling the same amount of 3-year protection. If a credit event occurs within the 3 years, then debt delivered into

[3] Looking at Table 10.1, premium 'error' is (say) around 4 bp for the book or 0.2% of notional if they are 5-year deals.

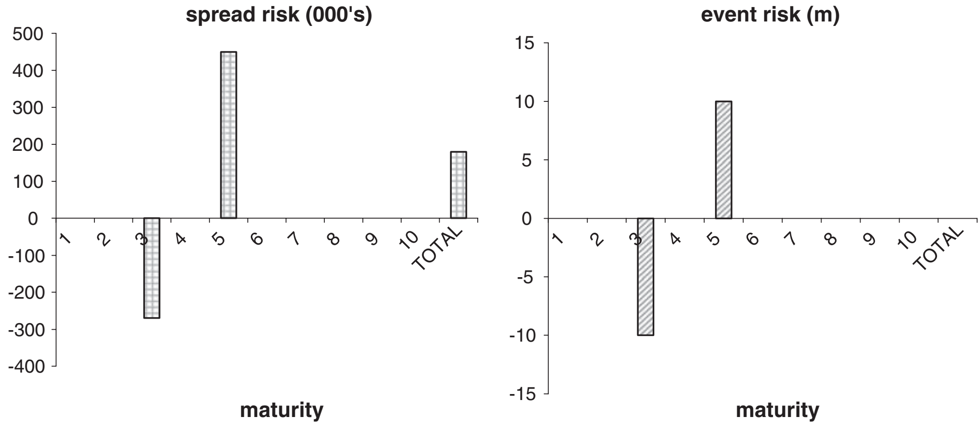

Figure 10.3 Risk chart for event hedged spread trade.

the protection sold, is in turn delivered into the protection bought (subject to documentation ensuring that sufficient notice is obtained on the debt to be delivered). If the 5-year protection costs 200 bp, then the 3-year may raise 150 bp. This leaves an income loss (‘**negative carry**’) of 50 bp per annum for 3 years, together with the final 2 years’ cost of the long protection. Of course, the expected forward protection on the remaining 2 years is around 275 bp, but there is no guarantee that the 2-year CDS will raise this in future. The risk reports will also show a significant spread risk in this case: as spreads rise, the long 5-year protection will make more money than the 3-year trade loses. This is a bearish trade on spreads while being default event neutral. (See Figure 10.3.)

Alternatively, if the name is highly risky, 5-year protection may cost 1,000 bp p.a. but 3-year protection may raise 1,400 bp – leaving a total income deficit of only 800 bp. The residual spread risk will be a small proportion of the risk arising from the 5-year deal alone.

CDS Spread Hedging Curve Trade

A longer CDS has a greater sensitivity to a shift in the hazard rate (or spread) curve. The spreadsheet ‘chapter 10.xls’ worksheet ‘spread hedging’ calculates the hedge ratios for a 5-year and a 3-year (and vice versa) CDS under a flat hazard rate curve. (The MathCad sheet ‘chapter 10 and chapter 20 hedge theory.mcd’ gives the mathematical formulae and also performs the calculations – the content of the MathCad file can also be read in the rtf version.) Long 5-year protection at 200 bp (say) could be hedged by selling approximately 160% of notional in 3 years at 150 bp raising 240 bp – covering the cost of the 5-year protection, leaving a **positive carry** of 40 bp. If no credit event occurs during these 3 years, 120 bp will have been earned, and protection can be sold in 3 years’ time for the remaining 2 years to earn more income. The hedge runs a significant default event risk during the first 3 years. In addition, the trade is exposed to changes in the shape of the curve. If 5-year spreads rise more than 3-year spreads, a profit is made – the trade is a **curve steepener**. (See Figure 10.4.)

Note that the hedge ratio (of 160% in this case) changes over time – in a year’s time it is more like 400%. The hedge is **dynamic**.

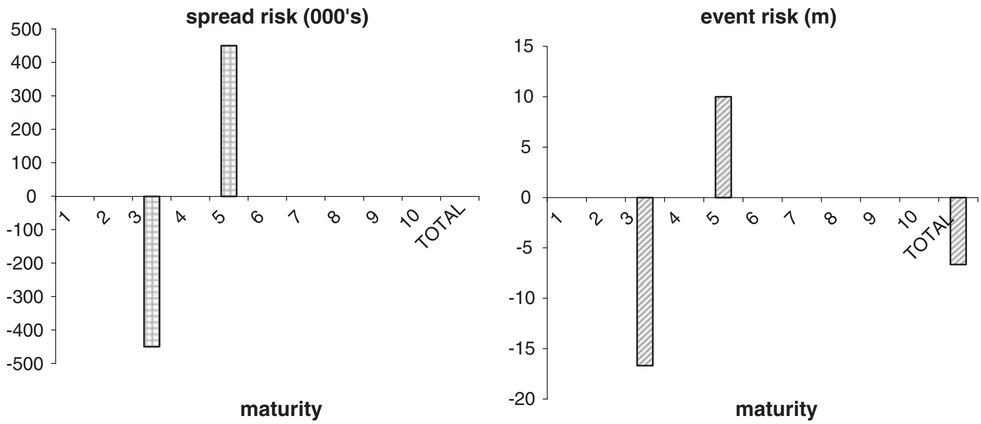

Figure 10.4 Risk chart for curve trade.

Consider for a moment an equity options trader who has written 3-month at-the-money (forward) call options on GBP 10m of Unilever PLC. The trader could delta hedge this position by buying 5m worth of Unilever shares (the option delta is 0.5). A small move in the price of the shares will see equal and offsetting changes in the value of the shares and the options. Yet a default event of Unilever will see the shares fall to almost zero – a loss of GBP 5m – which the option premium taken in will only be of the order of GBP 0.2m.

There are two main reasons that an equity trader generally does not worry about the default event. First, the positions are hedged by other options as much as possible, so *residual* risk to the share price is actually small. Extreme moves in share price are considered in reporting risk on the options book – although often not explicitly the default event. Another reason is that high-quality (investment grade) names are regarded as default risk remote.

The same philosophy generally extends to the CDS book – investment grade names are generally seen as default risk remote and a significant default event risk can be carried in each name as long spread risk is small. On the other hand, 'emerging market' names – where spreads are wide and default risk is substantial – will tend to have much tighter default event risk limits.

10.1.3 Hedging Both Credit Event and Spread Risk Simultaneously

Is it possible to hedge both default event and spread risk?

Typically a desk trading investment grade CDS will choose to hedge spread risk which will usually leave a net default event risk (and the desk loses money on credit events typically). The reader should note that it is theoretically possible to hedge both credit event risk and spread risk under certain assumptions (Figure 10.5). In principle, a long 5-year protection position in FMC could be hedged by short positions in both 3-year and 7-year CDSs. We require that the nominals are chosen so that the default event risk and the spread sensitivity are both zero. The two simultaneous equations can be solved to give the hedge nominals.

Suppose we own one nominal of FMC. Then we can sell notional N_1 of the 3-year CDS and N_2 of the 7-year CDS (we suppose, for simplicity, that the net value of all three CDS is

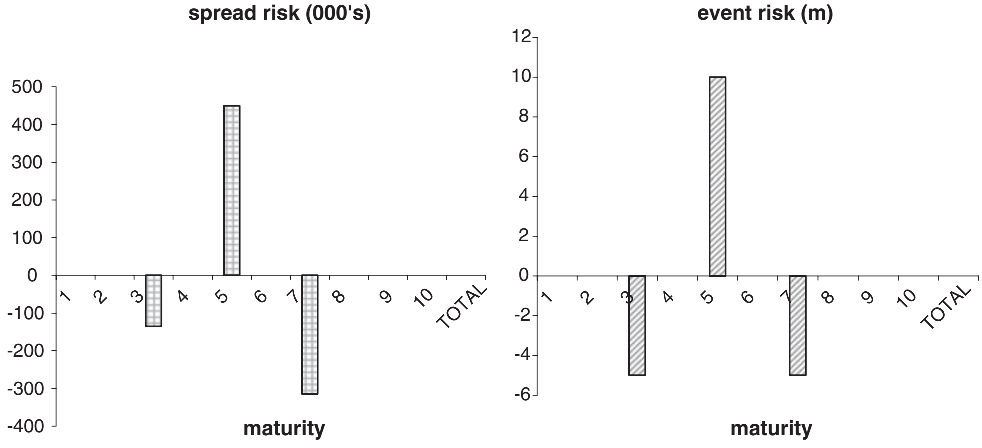

Figure 10.5 Risk chart for default and event hedge.

currently zero – i.e. they are fair market trades). We require:

$$N_1 \times (1 - R) + N_2 \times (1 - R) = (1 - R) \quad \text{default event losses are equal} \tag{10.1}$$

$$N_1 \times S_1 + N_2 \times S_2 = S_0 \quad \text{trade and hedge sensitivities (S) are equal} \tag{10.2}$$

This gives

$$N_1 = (S_2 - S_0)/(S_2 - S_1) \text{ and } N_2 = 1 - N_1. \tag{10.3}$$

If the CDS curve moves up or down in parallel, (10.2) ensures that there is no net profit or loss to first order. Also, on a credit event (and subject to the usual documentation conditions), the nominal delivered equals the amount we have to deliver (10.1). There is a solution to the equations as long as the sensitivities of the hedges are not equal (10.3).

Suppose we were to use a 1-year and a 10-year CDS as the hedges? Or suppose we are hedging a 2-year CDS with a 3-year and a 5-year, what problems do you foresee?

Equation (10.2) assumes that the CDS spread curve undergoes parallel moves. In reality this is not the case so hedging as above is flawed in practice. The example given – 5-year versus 3- and 7-year – is reasonable but the other examples will lead to large curve risk and large hedge notionals. Attempting to hedge slope risk with a third CDS will simply introduce curvature risk – 5-year spreads rise 10 bp, but 3-year rise 12 bp and 7-year rise 10 bp, whereas the slope hedge anticipates a 9 bp change. We could hedge this by adding a fourth CDS, but then there is 'twist' risk, and so on. The CDS curves are not as well arbitraged as the LIBOR curve, so shape changes do not necessarily follow a predictable pattern. On top of this, spread moves often include jumps – not infinitesimal moves.[4] Hedging both credit event and spread risk is not a practical possibility generally.

We shall revisit the topic under book hedging.

[4] Equation (10.1) ensures that we are hedged to the ultimate jump, but intermediate moves may be different from both extremes of (10.1) and (10.2).

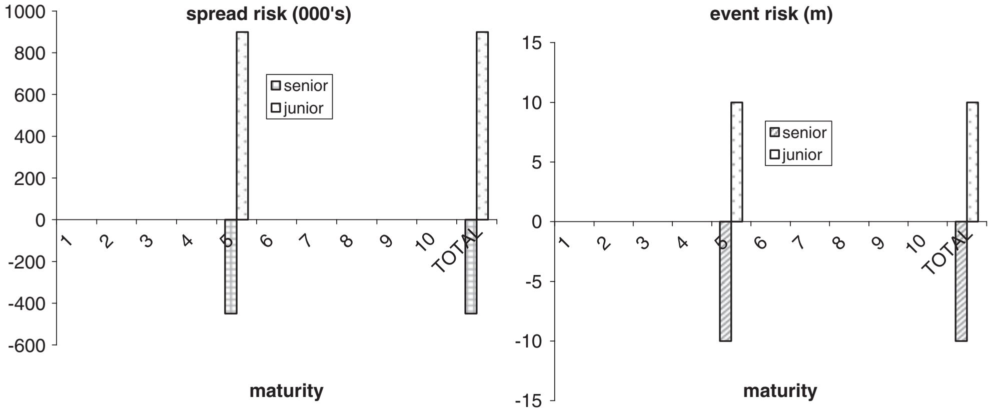

Figure 10.6 Risk chart for cross seniority trade.

10.1.4 Seniority Mismatch

Consider a reference entity where both senior and subordinated debt trades – typically such entities are financial entities (banks or insurance companies). A long protection position on junior (= subordinated) debt in EUR 10m against a short position in the same notional amount of senior debt can, at worst, lead to no loss in the default event. For example, if senior recovery[5] is 50% then junior recovery is less than this. A long junior CDS gets more than par 50% on a credit event and the sold senior CDS only has to pay par 50%. In the worst case, debt delivered into the short position can be delivered into the long position if necessary. It is more likely that we can sell the defaulted senior debt in the market at a higher price than it costs to buy junior debt to deliver into the long protection position (see Figure 10.6).

The same trade the other way round (long senior protection, short junior) can suffer a maximum loss of 10m. Junior debt may be trading at near zero post default, yet senior debt may be close to par.

As junior CDSs are more risky than senior CDSs, they carry a higher premium. Figure 10.7 shows some junior and senior CDSs premiums on financials (data from 2003). Based on around 50 names, subordinated CDSs typically traded at around twice the senior CDS premium. Why should this be?

Suppose the senior CDS trades at P_{sen} and the junior CDS at P_{sub}, and assume a recovery of R_{sub} on the junior CDS. Then (from equation (9.15)) the implied hazard rate for the entity is $P_{sub} / (1 - R_{sub})$. This hazard rate relates to the entity not the seniority, so the senior premium and recovery (again from (9.15)) are related by $(1 - R_{sen}) = (1 - R_{sub}) \times P_{sen} / P_{sub}$. If $P_{sen} / P_{sub} = 0.5$, then we get the relationship between junior and senior recovery rates shown in Table 10.2.

In particular, the ratio of CDS premiums is inconsistent with a senior recovery rate of less than 50%. Suppose (for example, based on fundamental research on the company) you think senior recovery is likely to be only 20%, and junior recovery is 0%. The market is being optimistic on senior risk so the senior CDS premium is too low. We should in this case buy senior protection and sell junior protection. Suppose we buy 10m senior protection at 50 bp,

[5] Remember that recovery here means 'one month post-default bond price'.

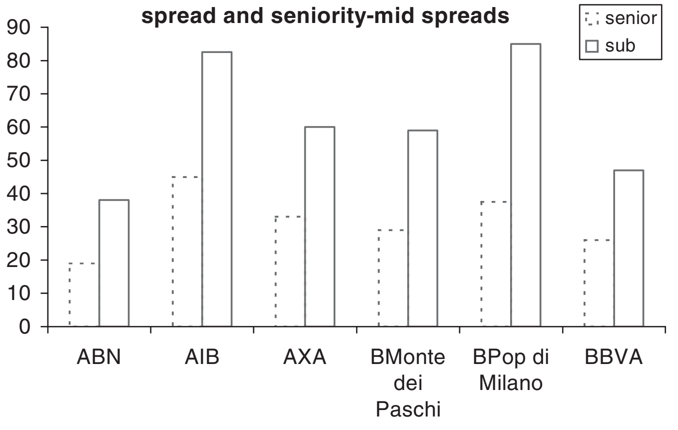

Figure 10.7 Senior and sub CDS premiums (sample financial data, 2003).

Table 10.2 Implied senior recovery rate when sub CDS trade at twice the junior CDS premium

Junior recovery assumption	Implied senior recovery rate
0	0.5
0.2	0.6
0.4	0.7
0.6	0.8
0.8	0.9

and sell $\times$ of junior protection (to the same maturity – typically 5-year) at 100 bp. On a credit event, if our recovery assumptions are right, we make $10\text{m} \times (1 - 0.2) = 8\text{m}$ on the senior protection, and lose $\times$ on the junior. We are default event hedged if $\times = 8\text{m}$. We have a positive cashflow of $8\text{m} \times 100\text{ bp} - 10\text{m} \times 50\text{ bp} = 30\,000$ p.a. (Such a trade may be capital intensive for certain institutions if the capital required to finance the positions is much greater on junior than on senior debt – and this may also be a hidden factor in the pricing of junior debt.)

10.1.5 Trade Level Hedging and Book Basis Hedging

The trades in sections 10.1.3 and 10.1.4 above are initiated by the trader based on a view of recovery rates for that name. The deals do not lead to any default event risk within the internal risk reports if the trader's recovery assumptions are also reflected in the risk reporting recovery assumptions. This is also true of spread (hazard rate) risk (equation (9.18)). If the risk reports are generated using different recovery assumptions from the trader's assumptions, then a spread risk will appear. This will be aggregated with other spread risks and, unless the trader is aware of the source of the net spread risk, that risk will automatically be hedged out, largely negating the intended effect of the trader's recovery rate hedge. It is therefore essential that recovery (and other) assumptions used for reporting be kept in line with the trader's assumptions or, alternatively, risk is not aggregated for certain trades. We shall discuss this topic in more detail in Chapter 18.

10.2 INTRODUCTION TO BOND HEDGING

10.2.1 Default Event Hedging

As loans may be non-assignable (they cannot be sold), CDSs represent a potentially attractive way to remove the economic risk[6] of the loan without selling the bond. If permission of the borrower is sought to sell the loan, this may damage the bank–client relationship, whereas a CDS deal is less conspicuous, and a portfolio CDS (CDO) deal is less judgemental about specific borrowers.

It may also be attractive to hedge bond positions by CDSs – perhaps as part of a term structure strategy, or to reflect a short-term view. Suppose you own USD 10m nominal of a 5-year FMC bond priced at par. What CDS would you use, and how much would you buy, to protect against the default event risk over the next year? Suppose the bond is priced at 80 or 120 – how would your answer differ?

If we buy USD 10m of a 2-year (or 10-year) CDS, then on default at any time in the next year we can deliver the bond and get USD 10m. We are fully event hedged in this case. If the bond is priced at 80, suppose that recovery on default is 50% and, for the moment, think in terms of a cash settled CDS. If we own a 10m CDS then, on the default event, we gain $10\text{m} - 5\text{m} = 5\text{m}$ on the CDS but only lose $8\text{m} - 5\text{m} = 3\text{m}$ on the bond. We need to buy only 6m notional of the CDS in order to be hedged. Alternatively, if the bond is priced at 120 we lose 7m on the default event and need to buy 14m of the CDS to be hedged. In fact the correct hedge ratio against the default event is given by

$$\text{CDS notional} = \text{Bond notional} \times (P - R)/(1 - R) \tag{10.4}$$

Unless the bond price is par, the correct hedge ratio requires an estimate of the recovery rate – hence generally *we cannot hedge the default event with certainty*.

Commercial bank loan portfolio managers generally do not mark their loans to market, so they stand on the books at the issue price – typically close to par. Also, the manager does not have to hedge the risk of the change in value of the loans. A notional-for-notional hedge is often appropriate for such portfolios. In addition, if it is required to reduce regulatory capital under current regulations, then the CDS life needs to be at least as long as the bond life.

Most market-traded CDSs are on senior debt (exceptions being financials where subordinate CDSs also actively trade). It may be necessary for a loan portfolio manager to buy CDSs on senior debt (rather than cheaper loan CDSs). On the default event the loan is potentially deliverable into the CDS (since it is more senior than senior bonds), although the portfolio manager would generally do better by buying defaulted bonds to deliver and keep the loans on the books to get the ultimate recovery.

[6] A loan plus CDS of matching maturity into which the loan is deliverable eliminates the default event risk (and also the mark-to-market risk) sufficiently well that the risk is seen to be no longer that of the borrower under the loan but that of the CDS counterparty. The economic risk to the borrower has gone, although the loan remains on the bank's books.

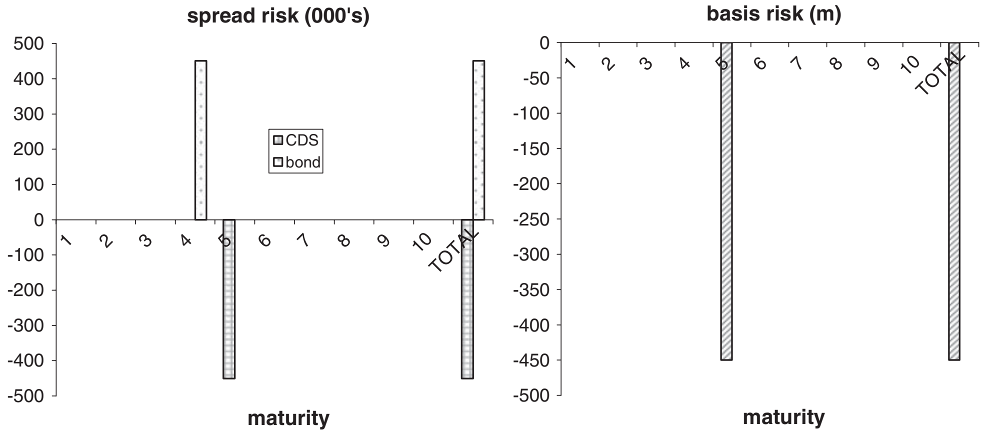

Figure 10.8 Risk chart bond and CDS spread hedge with basis risk.

10.2.2 Spread Hedging

If we assume that the CDS and the bond are priced off the same hazard rate curve, then we can use the sensitivity formulae of Chapter 9 to calculate the hedge ratio. For bonds priced at par, the bond sensitivity is almost exactly the same as the CDS sensitivity for a CDS of the same maturity – so a spread hedge (which is one-for-one) and a default event hedge are the same and we hedge both risks at the same time. If prices are away from par, then the hedge ratio needs to be calculated from the formulae.

In practice, bonds and CDSs do not move together – the difference is referred to as the basis. This is investigated further in Chapter 11.

Figure 10.8 shows a risk chart for a 4-year bond spread-hedged with a 5-year CDS. This leaves an event risk (not shown) and the more important 'basis risk'. A change in the CDS premium may be accompanied by a different change in the spread on a similar maturity bond – this difference is a change in the basis.

Basis risk is the same in magnitude as naked spread risk. Although we would expect the basis to be constrained, it can move very sharply in the short term. Limits on basis risk often impose tight constraints on basis trading by banks.

10.2.3 Convertible Bonds and Equity Risk

Convertible bond traders typically buy convertible bonds to strip out the equity conversion option and sell a matching option at a higher price. This leaves the trader with an exposure to the underlying credit over the life of the deal (typically there are several conversion dates when the bond and the risk will cease to exist, and the underlying bond itself may also have embedded calls and puts). Ideally the trader wishes to buy protection on the name, which 'knocks out' on conversion of the bond or on call or put. There then remains a pure interest rate risk which can be hedged with interest rate swaps and with swaptions (with some residual risks related to conversion dates and the contingent unwind cost on the default event).

CDS contracts with matching knock-out features do not trade – standard CDSs being bullet contracts. Typically the convertible desk will consider buying a bullet CDS to the final maturity date of the bond, but this may be too expensive. The question of pricing a matching structured CDS is left as an exercise. (*Hint:* Build a tree to evaluate the call and put possibilities. The equity price also needs to be modelled in a consistent way in principle – an approximation could be obtained by taking a lognormal equity price model to determine when and whether conversions occur. The dates and probabilities can be incorporated into the call and put tree, and the credit event risk can then be priced off this tree.)

10.3 HEDGE AND CREDIT EVENT EXAMPLES

We review some cases of 'defaults' from the perspective of a CDS protection owner, and also a bond and CDS owner, to see how the CDS and the hedging process worked in practice.

Sudden default

Figure 10.9 shows a (low coupon) bond price in the run up to the sudden default by Enron. Prior to October 2001 the price was stable; there was a lurch down and partial recovery as hints of problems appeared, followed by a sudden drop to a very low price towards the end of November when bankruptcy was announced over the weekend.

Exercise 1 Suppose in January 2001 you owned USD 10m of an Enron bond (say the low-priced bond in Figure 10.9) and USD 10m of 5-year CDS protection. How would the protection have worked over the credit event and how much would you have made? Suppose it was 1-year protection? How much notional would be required to hedge the position?

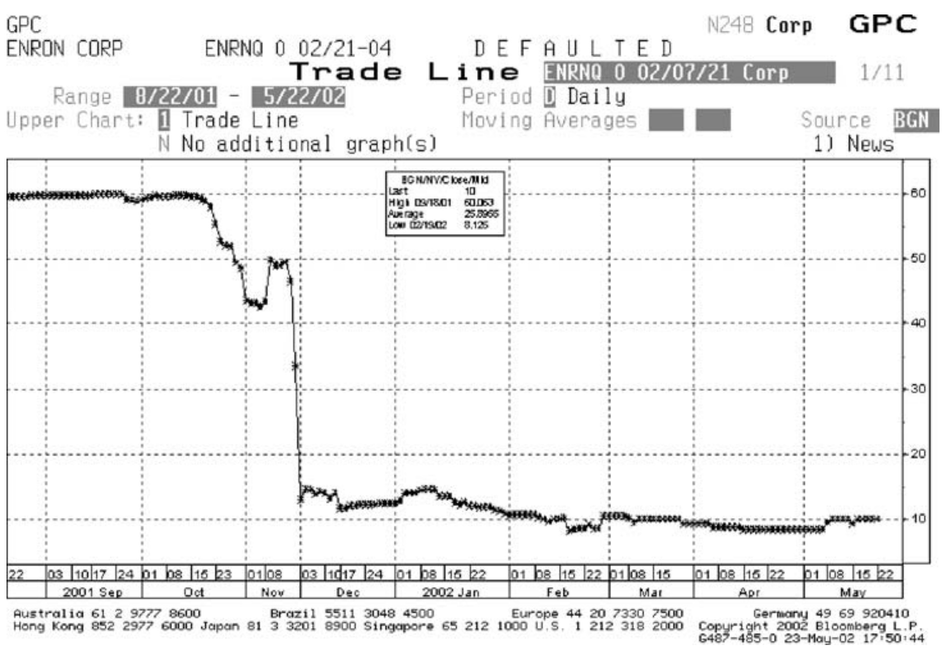

Figure 10.9 Sudden default: Enron.
(Reprinted with permission from Bloomberg LP)

Answer: The bond could be delivered into the (1-year or 5-year) CDS to obtain USD 10m; the bond position was worth around USD 6m at the beginning of October. The loss on the bond position if it was sold in the market would be around USD 5m, while a 10m holding would make around USD 9m – giving a USD 4m profit. With the benefit of hindsight a CDS position of about 5.5m notional would have been sufficient to protect the bond.

Protracted Spread Widening

The charts in Figure 10.10 show the gradual decline of Marconi shares to near zero over a period of years, accompanied by the gradual decline of an 8-year bond to a price of around 30. Even at the end of the chart period a credit event had not occurred (subsequently a restructuring was announced).

Exercise 2 Suppose in early 2001 you bought GBP 10m of the 9-year bond at a price of par, and simultaneously bought 10m 9-year CDS protection. How would the CDS have protected you during the period shown? Suppose it had been 10-year or 5-year protection? How much 5-year protection would you have needed to protect the value of your investment?

Figure 10.10 Slide into 'default': Marconi.
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Figure 10.10 *Continued*

Answer: Although no credit event had occurred over the period shown, the CDS would have provided value protection. At a price of 30, bonds were trading 'on a recovery basis' and imminent 'default' was expected. CDS contracts expected to cover the time horizon of the default would likewise have a capital value of par minus recover – around 70 points. All three maturity CDSs would have provided full capital protection. Initially the value of the 5-year CDS would have increased less quickly than the bond's value fell since it was of lower duration. At some point the effective duration of the 5-year CDS would actually have exceeded that of the longer CDS and the capital value of the 5-year CDS would have moved rapidly to 70 as bond price declines continued.

Technical Default

Figure 10.11 shows the price of a (high coupon) Railtrack bond over the period when it was announced that the company was being 'put into administration' (a credit event but not a default on debt). The bond price reacted but remained above par.

Exercise 3 (a) Suppose you owned the bond and an equal notional of CDS protection. How would it have protected you? (b) Suppose you owned a convertible priced at 60?

Answer: (a) It wouldn't unless you could find bonds below par to deliver. (b) The only below-par bonds were convertible – see section 2.1 in Chapter 2 for a discussion of this case.

Figure 10.11 Non-default credit event: Railtrack.
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Correlated Defaults

Figure 10.12 shows the rise in spread on Lehman followed by the rise in spreads on some other US investment banks shortly after the Lehman bankruptcy. Other US and European banks defaulted (Bear Stearns in the USA and many smaller banks) or sought massive amounts of government help to avoid bankruptcy (for example, Citigroup in the USA and RBS in the UK). The banking problems spread via a credit crunch to the rest of the economy and spreads on all credit rose dramatically.

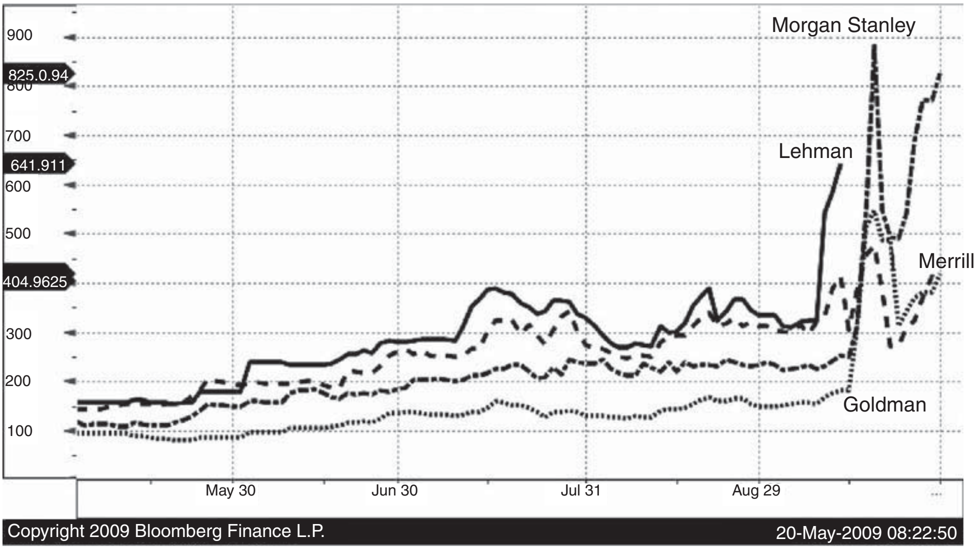

Figure 10.12 Correlated defaults: Lehman and other banks.
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11
CDS/Bond Basis Trading

If we wish to trade bonds against CDSs or vice versa (basis trading), or calculate VaR for a portfolio, then we need to understand how the (z-)spread on a bond relates to the CDS premium, and how a change in one relates to a change in the other. The default and recovery model has been developed with a single hazard rate driver for one entity. In this chapter we look both at the practical aspects of the basis, the factors driving the basis, and how the default and recovery model can be enhanced to model bonds and CDSs more realistically.

We define the 'basis' as the CDS premium expressed in basis points less the bond z-spread (for a specific bond and identical maturity CDS) (see 11.3 below). Basis is bond and maturity specific.

$$\text{Basis} = \text{CDS premium} - \text{Bond } z\text{-spread} \tag{11.1}$$

We saw in Chapter 9 that if a bond that

(a) has a single maturity (a bullet bond) (see 11.5)
(b) is an FRN
(c) is not, and never will be, special on repo at any date in the future (see 11.2 and 11.9)
(d) is priced at par (see 11.4)

and is hedged by a CDS that

(e) has the same maturity as the bond
(f) is triggered by exactly the same events that trigger default on the bond (see section 11.8)
(g) has only that bond as the deliverable instrument (see section 11.6)
(h) pays par plus accrued on that bond if a credit event occurs (see section 11.7)

then we can set up an arbitrage trade implying that the basis is zero. We see below that deviations from the above assumptions can introduce a positive, negative, constant or variable basis. In addition a basis may emerge related to new issuance or other market related factors – which we incorporate in a 'liquidity' element of the basis.

Under the above conditions a bond with a z-spread above the CDS premium presents an easily arbitraged trade (buy the bond on repo and buy the CDS) which generates a riskless profit. Factors giving rise to negative basis tend to quickly produce arbitrageurs, reducing the basis to a fair level. CDS spreads above bonds do not lead to a simple arbitrage – it is not easy to short issues via the repo market. (This is one reason that the CDS market developed.)

11.1 BOND VERSUS CDS: LIQUIDITY

Liquidity is the 'fudge factor' left when everything possible has been explained, and variable elements of the basis largely fall under this heading. However, one explanation for a change in the basis relates to market movements generally. If spreads start to move rapidly, CDSs are used as the liquid instruments in which to place bets on, or hedge, spread movements. The

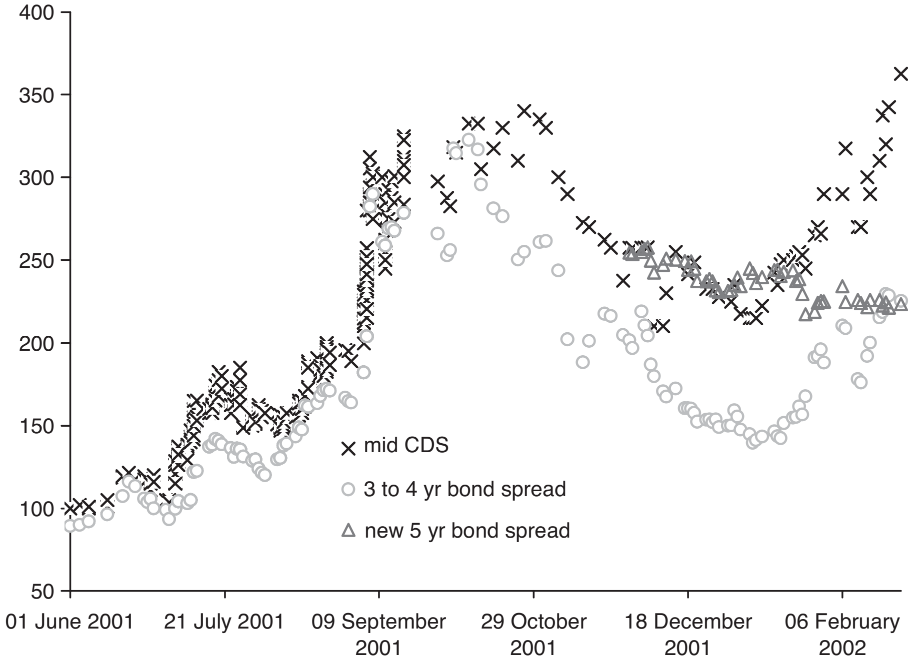

Figure 11.1 Alcatel 4- to 5-year bond spreads and 5-year CDSs.

CDS tends to react more quickly than bond spreads, so the basis initially increases as spreads rise, and decreases (possibly becoming negative) as spreads fall.

This factor is variable in magnitude and sign. There is a relationship to market movements and to new bond issues that are 'priced to go'. A bond priced cheaply in order to secure funds for the company will reduce the basis (and may make it negative) until the issue has been fully absorbed. Figure 11.1 shows CDS and similar maturity bond spreads, and variations in the basis can be seen particularly around the time of the new 5-year issue and shortly afterwards as spreads widen rapidly.

11.2 BOND REPO COST

We saw in Chapter 9 that (for the situation described above but where the bond is special on repo) the CDS premium should be the bond spread plus the amount special on repo. If the CDS premium remains fixed but the bond becomes special on repo, then the spread on the bond should decline.

For example: suppose the 5-year CDS is 500 bp, and the deliverable bond is not special on repo, then the *z*-spread should be 500 bp. Suppose the bond now becomes 200 bp special on repo.[1] The bond can now be lent out, against which cash collateral is received and on which interest at LIBOR minus the amount special has to be paid. The cash can be used to pay

[1] To the maturity of the trade.

off the loan taken out to buy the bond – saving LIBOR. Net income, therefore, increases by 200 bp, so the bond spread should drop to 300 bp.

It is rare for corporate (non-sovereign) debt to go special, but in 2002 5-year France Telecom CDSs were trading at 475 bp, whereas the comparable bond was at 290 bp – and special on repo – while other similar maturity debt was trading around 480 bp. A corresponding situation arose over Commerzbank data at around the same time. Sovereign (non-domestic) debt, on the other hand, often goes special on repo.

Repo is a bond specific element of the basis. It can be independently measured (although this requires access to a repo trading desk).

11.3 BOND SPREAD MEASUREMENT – *z*-SPREAD NOT ASSET SWAP SPREAD

We define the basis relative to the *z*-spread. Asset swap spread will show a different relationship to the CDS premium. For bond prices substantially away from par the asset swap spread will be further away from the CDS premium than the *z*-spread (see Part I).

11.4 BOND PRICE IMPACT

If we know that the recovery on a given entity is R, then debt priced at R is effectively risk free. If the entity defaults, we are going to get R back. Of course, the pull to maturity means that the price should rise above R later and there is a forward credit risk – only a 'perpetual' would not have a forward risk under a constant hazard rate. In contrast, highly priced debt (perhaps because it carries a high coupon) carries much greater risk. The spread on lowly priced debt should therefore be low, while that on highly priced debt should be relatively high.

Table 11.1 shows how the *z*-spread on bonds (with 2% and 5% coupons) relates to the CDS spread, at various levels of the CDS spread. Note that the coupon effect is considerable at high spread, with a 5% coupon bond trading at roughly 800 bp, while the CDS should be at 1 000 bp.

Above par, bonds should have a spread above the CDS and below par bonds should have a spread below the CDS (ignoring other factors).

Table 11.1 CDS and Bond spread: theoretical coupon effect (5-year, 50% recovery)

	5% coupon bond		2% coupon bond	
CDS premium	*z*-Spread	Basis	*z*-Spread	Basis
100	100.5	−0.5	94.5	5.5
200	197.3	2.7	185.7	14.3
300	288.3	11.7	272.2	27.8
400	376.0	24.0	353.9	46.1
500	459.3	40.7	431.0	69.0
600	538.1	61.9	503.0	97.0
700	612.6	87.4	571.6	128.4
800	682.9	117.1	635.3	164.7
900	749.0	151.0	694.8	205.2
1,000	811.2	188.8	750.3	249.7
2,000	1 323.8	676.2	1 155.6	844.4

Note, from equation (9.17), that if recovery is assumed to be zero then the z-spread is equal to the hazard rate (which in turn is equal to the CDS premium). In practice, because we do not know the recovery rate, we cannot be sure of the basis impact of the bond coupon.

Given a recovery assumption, we can fit the default and recovery model to the bond price (9.17) to get a bond-implied hazard rate. This then gives a bond-price-implied CDS premium (from (9.15)), and this can be compared with the actual CDS to get a *coupon-adjusted basis, given a recovery assumption.*

11.5 EMBEDDED OPTIONS IN BONDS AND LOANS

Callable bonds are treated in more detail in Chapter 15. We restrict our discussion here to the impact of embedded options on the basis.

Bullet bonds – bonds with a single maturity date – represent a large proportion of investment grade debt. Callable bonds (which may be redeemed early at the issuer's option) or puttable bonds (which may be sold back to the issuer at a predetermined price prior to the final maturity date) are also common. Syndicated loans are immediately callable by the borrower – the presence of an embedded option is the norm on such debt. These embedded options on fixed coupon bonds may be triggered by either interest rate or credit spread moves. Suppose that interest rate volatility is 10% (on a lognormal basis), with the yield level at 4%, then the basis point volatility is $0.1 \times 4\% = 40$ bp. Spread volatility varies with the reference name (and maturity, seniority, etc.). Table 11.2 shows some typical volatility levels based on individual CDS curves during the 2001–2003 period. (CDS data is used rather than bonds because of the constant maturity of the CDS and the better quality of the data.)

As the bond option is driven by the price which, in turn, is driven by the yield, we therefore need the yield volatility and a yield model in order to price the option. The yield volatility will depend on the LIBOR yield volatility, the spread volatility, and the correlation between the two. Table 11.3 shows some results at various spread and correlation levels.

(Calculations are in 'chapter 11.xls'.)

Table 11.2 CDS volatility: percent and bp (Figures are only indicative)

Spread level bp	% volatility	bp volatility
100	85	85
500	50	250

Table 11.3 Spread volatility, LIBOR/spread correlation and yield volatility

Spread	Spread vol	LIBOR/spread corr	Yield vol
100	85	−0.5	74
100	85	0	94
100	85	0.5	111
500	250	−0.5	233
500	250	0	253
500	250	0.5	272

We can see that the volatility of the bond yield is considerably higher than the volatility of LIBOR rates and, at higher spreads, the spread volatility dominates the LIBOR volatility. For a 5-year bond (duration of, say, 4) the price volatility of a par bond will be about 4% if the spread is 100 bp, and about 10% if the spread is 250 bp. The value of a 1-year at-the-money price option is about 40% of the price volatility – giving 1.6 points on a 100 bp spread name and 4 points on a 250 bp name. These price differences correspond to a spread change (= basis change) of about 1.6/4 = 40 bp and 4/4 = 100 bp respectively: the callable trading on a wider spread than a comparable bullet by these amounts. American options running to the maturity date of the bond will be even more valuable.

Floating rate bonds and loans (the latter typically being floating rather than fixed) are driven largely by the spread volatility. Loans additionally have early maturities triggered by breaches of covenants for which little historical data is readily available and no market-implied data. For both fixed and floating rate bonds the reader should again be cautious about applying a Wiener process model when the spread is fundamentally a jump process.

It is clear that embedded options will have a very substantial impact on the pricing of the bonds and on its basis to a similar maturity CDS. We briefly discuss models to estimate this basis in Chaper 15. It should be borne in mind that, although the implied volatility on the LIBOR (swap) rate is easily obtained, there is currently no source of implied volatility for the spread or CDS premium on a particular name. Given that spread volatility (and jump size and frequency) is typically the most important variable in driving the basis, then the uncertainty in that volatility means that basis adjustments for callables will, at best, be crude.

11.6 DELIVERY OPTION IN CDSs

If there is no delivery option (i.e. there is a specific deliverable) then the owner of protection may find that he has paid the premium for no protection. In the mid-1990s more default swap contracts had been written on specified deliverable instruments (of certain Korean bonds) than there was debt available to deliver. When a default occurred the price of the debt would rise to high levels as owners of protection tried to buy the debt to deliver into the CDS.

The delivery option reduces the risk of such a situation occurring again – and reduces the risk of bond squeezes being engineered. The tangible benefits of the option are two-fold:

1. Post-default trading may be at very different levels (within the same seniority) as certain investors are forced to dump bonds ('post-default illiquidity').
2. Since the CDS payoff is par, but bond prices should be recovery of par + accrued, a greater net gain is made by buying bonds with little accrued.

There is little data to asses the former. The large standard deviation of recovery rates we saw in Part I is largely driven by the recoveries across different names. Data giving the standard deviation of defaulted bond prices for a defaulted entity is generally not easily available, and many defaults are of names with only a single traded bond. Looking at a small number of specific defaulted names suggests that this arbitrage is generally relatively small, usually being within the (wide) bid–offer spread.

The latter factor is easier to understand. Suppose, no matter what the timing of the default, there is always a bond with zero accrued. The (theoretical) price in the market is then R, where

R is the recovery rate.[2] Suppose also that there is always a bond with 8% accrued. Then the theoretical price is $R \times 1.08$. If we owned such a bond we could sell it in the market, buy the zero accrued bond for R, deliver that bond into the CDS, and end up with $1 + R \times 0.8$. Assuming an average recovery of 50%, then the delivery option gives us 4% more than if we did not have the option.

Of course, this analysis will vary by name. For an entity with a rich universe of deliverable debt the accrued component of the delivery option may be worth 4–5% of the CDS premium. In other cases there may (at that time) be only one deliverable bond, so the option will have a low value – some value coming from the possibility that other debt may be issued in the future.

The delivery option in total may be worth around 5% of the CDS premium. This number will be relatively fixed over time but will vary according to name.

11.7 PAYOFF OF PAR

The CDS pays par on the occurrence of a credit event, whereas the risk on a par bond is actually par plus accrued. Given an assumed recovery rate R, then the risk on the bond is greater than that on the CDS by an amount that varies over time but whose average is $(1 + A/2 - R)/(1 - R)$ greater than that on the CDS. The CDS premium should therefore be less than the par bond spread by around 5% of the spread.

11.8 TRIGGER EVENT DIFFERENCES

Trigger events for default swaps are a superset of the default event for bonds. The default event is typically failure to pay, whereas credit events for CDS also include filing for protection from creditors and restructuring. On either of these events deliverable debt can be delivered into the CDS for par. Bonds will not necessarily be trading at low prices (e.g. Railtrack) but any debt that meets the deliverability requirements and is trading at a low price (perhaps because it carries a low coupon and has a long life) can be delivered in return for par.

CDS premiums should therefore exceed bond spreads. The greater the chance of a restructuring and the more low coupon debt that is issued by that reference entity, then the more the CDS premium should exceed par bond spreads.

It is difficult to estimate the financial impact of this trigger level difference. It will depend on the type of restructuring clause that is included, and will also depend on the reference entity. For example, implicit government support may make a default event unlikely, but bond spreads and CDSs will assess the impact of potential restructuring in different ways – the CDSs being driven by the event risk and the presence of low price bonds, while bonds would be driven by the event risk, and the outstanding maturity on those bonds.

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[2] Actually the situation is a little more complex than this. The ‘recovery rate’ is the one-month post-default bond price averaged over the debt for that name. Since defaulted bonds are quoted including accrued, R includes an average amount of accrued. The zero coupon bond should therefore have a price of R less the average accrued.

Some investment bank studies have suggested that the premium for the wider trigger event may be as much as 25%. In cases where CDSs have traded with different trigger events, the market seems to indicate that the widest trigger event set is worth about 10% more than the default event only as the trigger event.

11.9 EMBEDDED REPO OPTION

Imagine the situation described at the beginning of this chapter and if we could buy the bond on the same spread as the CDS, then we actually have a free option on the forward repo rate. If the bond goes special at any time in the future we can lend the bond out to earn an additional income of the amount special for as long as the bond remains special on repo.

Generally repo has not been a significant factor on corporate debt in the past – but this is not true for sovereign debt. Where there is a risk of a significantly special repo rate occurring in future, we would expect the CDS premium to exceed the bond spread by at least the value of this option on the repo rate.

Table 11.4 CDS/bond basis – factors and approximate magnitude

Factor	Description	Approximate magnitude (% CDS premium or bp) and sign (positive means CDS > z-spread)
Choice of spread	z-Spread is preferred to (par–par) asset swap spread	Nil for par bonds: Asset swap spread > z-spread above par; Asset swap spread < z-spread below par
Liquidity	1. Market direction and volatility related 2. New issue related 3. Other – fudge factor	Variable in magnitude and sign
Repo	Bond-specific amount special on repo	+ amount special in bp
Bond coupon	Coupon affects price affects risk affects spread	Nil at par: 200 bp when CDS = 1 000 bp
Embedded options	Bond calls and puts	Variable: explicitly model the option
Delivery option	Accrued effect at delivery; post-default liquidity effect	+ 5%
Trigger differences	Restructuring and other non-bond-default triggers	+10%
Par payoff	CDS pays par whereas bond risk is par plus accrued	−5%
Embedded repo option	Zero cost bond + CDS position allows the bond owner to enter a repo trade if the bond becomes special	+ 1% or less
Total	For bullet par bonds	+ 10% (5–30%) + amount special ± liquidity

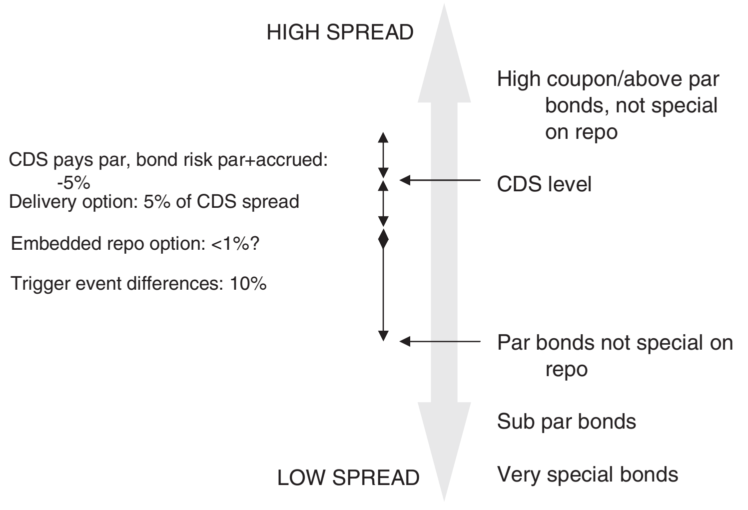

Figure 11.2 CDS/bond basis.

11.10 PUTTING IT ALL TOGETHER

Table 11.4 summarises the influences we have discussed in this chapter and gives approximate magnitudes. Note that the magnitudes are entity, bond and maturity specific, and change over time, so the figures are intended to be an indication of the rough magnitude only. Figure 11.2 summarises the relationships.

12

Forward CDS; Back-to-Back CDS, Mark to Market and CDS Unwind

12.1 FORWARD CDS

A forward start CDS is like a vanilla CDS except that the risk does not begin until the start date – say 2 years' time, and lasts until (say) 7 years' time. In the event of a default in the first two years the contract terminates (and there is no capital payment). No premium is paid or begins to accrue until the forward start date.

How would you hedge a EUR 10m position using vanilla CDSs?

The obvious choice is a long 10m 7-year protection position (at 350 bp say) and a short 10m 2-year one (at 250 bp). In the event of default in the first two years, we can then – as long as our documentation is correct – deliver the bond we get delivered on the 2-year short position into the 7-year long position, the forward CDS we have written knocks out, and we have no net capital gain or loss.

However, because the 2- and 7-year CDS premiums are different, the premium cashflows leave an unhedged risk arising from these premium differences, starting at zero and rising to a maximum of 200 bp after two years. After two years the premium on the forward CDS starts (at a higher rate than the 7-year CDS: say 407 bp) while the premium on the 2-year has ceased – a positive cashflow starts to reduce the accumulated negative cashflow from the first two years of the life of the deal. See Figure 12.1.

How does the exposure appear in the books? (See 'chapter 12.xls' sheet 'forward CDS'.)

Day 1: There is no value difference: the value of the fwd CDS and the 7-year and 2-year CDSs are all individually zero if the premium is 'at market'. (We look at sensitivities below.)

Day 2: Net premium is paid which appears in the P&L as a negative accrued. The value of the fwd CDS now exceeds the value of the hedges by an equal and opposite amount (other things being equal). The default event exposure will show a small negative – corresponding to the loss of the mark-to-market value of the CDS positions.

On Day 1, with the proposed hedge, the sensitivities do not sum to zero, so the trade – even on Day 1 – is not delta hedged. One way of obtaining a delta hedge on Day 1 is to sell only 93% of the 2-year CDS – even then the sensitivity gradually becomes more and more mismatched. (See 'chapter 12.xls'.)

We can see that there are problems with the proposed hedge. It does not hedge the capital risk (except on Day 1) and it does not hedge the spread risk of the deal (even on Day 1). In fact there is no practical hedge which eliminates either capital risk or spread risk at all times during the life of the deal. **Any hedge has to be dynamic**.

Typically, if the name is investment grade, then hedging is likely to be spread delta hedging. The sum of the sensitivities to a parallel shift in the hazard rate or spread curve is zero. This means that, initially, the hedge ratios are different from 100% in the shorter maturities (if the 100% hedges are implemented, a sensitivity mismatch is detected at the end of Day 1 and this

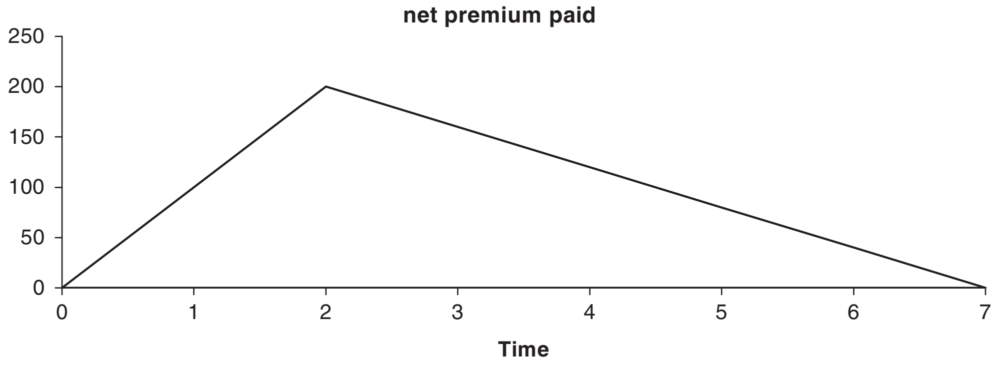

Figure 12.1 Net exposure on a long 7-year CDS, short 2-year CDS, and short '5-year 2 years forward' CDS position.

mismatch may then have to be hedged on Day 2). Day-by-day the sensitivities change by a small amount. Each day the trades are marked to market and the net sensitivity is calculated. The sensitivity mismatch arising from this deal may be well within the limits set for trading. Over time the sensitivity may exceed the limit and rehedging is required. In practice the deal will not be viewed on its own but in aggregate with many other deals, and a book-hedging policy will be followed. See Chapter 17.

12.2 MARK-TO-MARKET AND BACK-TO-BACK CDS

Suppose we sold a France Telecom 5-year CDS, receiving 300 bp for protection two years ago, and current 3-year CDS are at 100 bp on that name (with no up-front payment). What is the value of the deal?

In order to calculate the above we look at the fair market premium for a 3-year CDS, take our recovery assumption and calibrate the model, and use the implied hazard rate to value the cashflows under the deal, as described in chapter 9. (See 'chapter 12.xls' sheet 'valuation calc'.)

If we assume 50% recovery we find a value of +55 000 per 1m notional ('plus' because we are receiving 300 bp, whereas if we took out a new deal we would only receive 100 bp). The number makes approximate sense – the 200 bp excess premium for three years is worth a little less than $2\% \times 3 = 6\%$ of notional.

But suppose we had assumed 0% recovery – we find that the trade value is +55 800; and if we had assumed 90% recovery, the value is +49 000 (a range of roughly 10% of the value).

For non-investment grade names the valuation is even more uncertain. A name that has a 1 000 bp 3-year spread (otherwise the same trade details) has a value of −172 000 per 1m at 0% recovery, −150 000 at 50% recovery and only −70 000 at 90% recovery.

The value of the trades on our books is uncertain – very uncertain at an individual deal level for high-yield names. Looking at the (investment grade) book as a whole, the value of the book is uncertain to the extent that the recovery rate differs from the actual average recovery rate across the book. If we assume 50% recovery, it is plausible that this is in error by 20%. So the value of the investment grade vanilla CDS book is inaccurate to a few percent because of the unknown recovery rate.

Suppose (on the France Telecom deal above) we did a back-to-back deal with another counterparty in an attempt to lock in this profit. We write protection for three years for a premium of 100 bp. Ignoring the counterparty risk we have effectively secured a positive cashflow of 200 bp per annum for three years *contingent on the survival of France Telecom*. The value of this cashflow is uncertain. (It is the same value as the mark to market of the original deal since the 100 bp deal currently has zero value on any recovery rate assumption.) The back-to-back deal is a way of locking in a profit of favourable spread movements. How much value is locked in can only be determined precisely once the deal has terminated (through default of maturity) – a credit event tomorrow means that no profit is realised, but if France Telecom survives to the maturity of the trade, the maximum 600 bp is earned.

12.3 UNWIND CALCULATION; OFF-MARKET TRADE VALUATION AND HEDGING

An alternative to the back-to-back deal is an 'unwind'. In some cases there is a third option. Consider the following deal. Suppose you bought 5-year protection on Ford Motor Credit 2 years ago at 100 bp. The current 3-year CDS premium is 400 bp and you want to take profit. In this case we could buy a 3-year life bond to generate the excess spread income.

On this deal we could eliminate the risk to the reference name on a trade in one of the following ways:

1. Enter into a **back-to-back trade** with another counterparty. Earn 400 bp on sold protection and an excess of 300 bp over bought protection. If you ensure that notice of delivery on the written protection is before the notice you have to give on the bought protection, then there is no delivery risk. Again the present value of the premium excess is around 8.5%.
2. Go back to the original counterparty and ask for an '**unwind** price' – a sum that the counterparty pays to you to terminate the contract at the payment date.
3. Buy a 3-year bond. The bond gives a return equivalent to (approximately) 400 bp over the funding rate, and a 300 bp return over funding and default protection. A total of 900 bp of 9% of face value – with a present value of about 8.5%, say. In practice this is the least likely approach because of mismatching dates and risk.

In case 2, how much should the counterparty pay in order to effectively 'tear-up' the original CDS deal?

Looking at options 1 and 3 above, it is clear that the 900 bp is earned only if the reference entity (and the counterparties) survives the remaining 3 years. If default occurs tomorrow, there is no profit.

The calculation of the unwind price is exactly the same as the mark-to-market valuation calculation described above. (See the unwind calculation in 'chapter 12.xls'.) Given our recovery assumption (of, say, 20%) we calibrate to FMC and find that the implied hazard rate is 500 bp per annum. The current fair value premium is 400 bp whereas we are paying only 100 bp, so the value is the value of this premium excess over 3 years. We need to discount each payment at LIBOR, and we need to multiply by the survival probability. We expect to be offered 7.887% to unwind the trade.

The counterparty actually offers us just over 7%. Why?

Valuation and the unwind value is a recovery-dependent figure as we have seen. The counterparty is assuming a 70% recovery for FMC, which implies a much higher rate of default – hence a lower probability of receiving all the cashflows. Taking a 99% recovery assumption means that default is highly probable and the unwind price becomes a mere 0.74%. All answers are correct given the assumptions – the problem is that there is no determinant of the correct recovery assumption.

How can the trader argue for a better unwind value?

Some parties ask for the counterparty's recovery assumption each time a new trade is put on. In the above case the counterparty may have been using 40% at the time of the trade. Of course, the counterparty can reasonably argue that times have changed and so have its recovery assumptions, but this can easily be checked against recent deals done with that counterparty.

It became common prior to 2008 to standardise recovery rate assumptions to 40% on senior CDS across investment grade names. This, of course, did not necessarily make the results of a calculation correct but reduced argument between counterparties, particularly where there was a regular two-way flow of unwinds. From 2008, especially as spreads rose, the recovery assumption had more pricing impact and the market moved to different (name specific) recovery assumptions. Data was generally made available from the main data providers such as Markit and CMA.

A further argument can be used in special cases. Note that the value declines as the recovery assumption increases. Suppose the reference entity is Marconi which was bought at 250 bp some years ago, and in 2001 when spread had risen to 4000 bp you decided to unwind. Your calculation assumes a 30% recovery and suggests a value of 51.5% of notional. The counterparty offers just less than 1% of notional (arguing that it thinks recovery is going to be very high, and is using a 99% recovery assumption). In this case a much stronger argument can be used. Marconi debt is trading around 'on a recovery basis' of 30 (bond prices are similar irrespective of maturity) at this time. This is consistent with a recovery level of 30% and imminent default. Recoveries much above this level cannot be consistent with the bond prices and the default and recovery model. In addition, in this case the alternative of buying a 3-year bond at 30 is attractive – on the credit event the bond is delivered and par received, while if there is no default the high return is achieved.

P&L assigned to the trades is open to debate – it depends on the recovery assumption used. On any individual deal the actual profit achieved will exceed the expected P&L if the reference entity does not default (and losses will also be greater in magnitude). On the other hand, if the reference entity defaults early, a lower P&L will be achieved. The actual P&L on the book will agree with expectations if defaults are as expected and the recovery rates average out to be equal to the assumption used in the daily valuation of the deal.

12.4 'DOUBLE-TRIGGER CDS'

We discuss some further forms of this trade in more detail in Parts III and IV. For the moment we consider some of the simpler applications of this deal – where the only credit risk is the counterparty risk.

A large amount of banking business is in the form of interest rate swaps, cross currency swaps, forward FX deals, asset sales, etc., where there is a risk to the counterparty to the deal. On Day 1e the deal may have a mark to market of zero (the interest rate fixed and floating legs initially have the same capital value) but it may move away from this over time as interest rates rise, or the deal's initial value may be far from zero (e.g. an asset sale with delayed payment). On both types of deal there is (at least) a potential future exposure to the counterparty ('the buyer'): a default of the counterparty to the swap or asset sale leaves the seller with an exposure to the underlying which has to be unwound at market rates.

The seller therefore has a contingent exposure to some financial variables (e.g. interest or exchange rates) triggered by the default of a counterparty. The size of this exposure is neither constant nor a deterministic function of time. A CDS which pays an amount related to the value of some financial product on the default of a name is referred to as a 'double trigger' or 'variable exposure' CDS.

The valuation formula for such a product is similar to equation (9.13) – only the payoff function is different. We assume, for simplicity, that the premium is a single up-front payment: a better course in practice is to relate the premium to the then current exposure so that, as risk increases, so does premium income. $F(t)$ is the present value of the contingent reference asset (e.g. the forward swap unwind cost) at time t, and the survival and hazard rate functions refer to the reference entity (the counterparty to the deal). $F(t)$ may be positive or negative; and we have implicitly assumed that there is no correlation between the financial factors driving this valuation function and the hazard rate function for the reference name. Note also that the risk is digital.

$$V = \int_0^T F(x) \cdot S(x) \cdot \lambda(x) dx \tag{12.1}$$

Typically deals between professional counterparties are collateralised, so the forward exposure is only the mark-to-market change over a short period (7–14 days) while the counterparty failure is discovered. For collateralised deals the main concern is the possible correlation between the market moves and the risk of the counterparty defaulting (and the formula above is not suitable for this problem).

Double-trigger CDSs appeal to banks with uncollateralised counterparty exposures. Putting such a deal in place may give an unattractive level of disclosure about the bank's business to a potential competitor.

We shall return to this topic and the question of correlated counterparty risk in Part IV.

13 Credit-Linked Notes

13.1 CLN SET-UP; COUNTERPARTY OR COLLATERAL RISK

In a credit-linked note (CLN) the buyer of protection (seller of the note) transfers credit risk to an investor via an intermediate bond-issuing entity. This intermediary can be the buyer's own treasury or a third party's treasury using a **limited recourse note programme**, or via a **special purpose vehicle** (SPV).

The former route is conceptually simpler, and is cheaper to use once the note programme is set up. The purpose of the limited recourse note programme – a set of legal documentation and an issuing process – is to make it clear to investors and others that the failure of the note to provide the full cashflows (for example, on a credit event of the reference entity) does not constitute a default by the issuing bank.

It is helpful to think in terms of a simple CLN, where the embedded risk is exposure to (say) France Telecom. The note may mirror CDS terms and conditions – it may have a 5-year life, paying LIBOR plus (say) 250 bp (or it may be a fixed coupon note), but it will terminate early if France Telecom suffers a (CDS) credit event. In that case coupons on the note cease, and the investor receives the notional on the note less an amount determined by the issuing bank – CLNs are typically cash settled. The terms embedded in the note may (and often do) differ in detail from those of a CDS. For example, a specific bond may be referenced for the purposes of calculating the cash amount; accrued interest may not be paid up to the date of default; or a non-standard set of credit events may be used.

The process of issuance is for the bank to embed the terms of the note in the CLN issue document required under the issuance programme; the note is sold to investors (at a price which may be different from par); cash is received by the issuing department and deposited internally with the bank's internal treasury department.

The CLN is a funded instrument which offers the end investor (the buyer of risk) synthetic credit exposure to the reference entity (France Telecom), and gives the bank credit protection on the reference entity but no counterparty exposure.[1] The exposure to the reference entity is synthetic because the issuer is not the reference entity but the originating institution. It should be noted that, from the investor's point of view, the note gives the investor direct exposure not only to the reference credit but also to the issuing bank. Although the risks are simplified from the bank's point of view, they are actually more complicated from the investor's point of view (compared with direct investment in the reference entity's debt). The investor loses some or all of his investment on the first to default (within the life of the note) of either the reference entity or of the issuing bank. (We shall return to this viewpoint later.)

The CLN issued via a limited recourse programme is summarised in Figure 13.1.

An alternative issuing process is via an SPV. The SPV is a third party – usually independent of both the originating bank and the investor(s) – usually a company (corporate entity) often owned by a charity and established in an offshore financial centre. The process is somewhat

[1] If the price is below par there is still no counterparty exposure but the protection obtained is a variable amount and is less than the notional of the CLN.

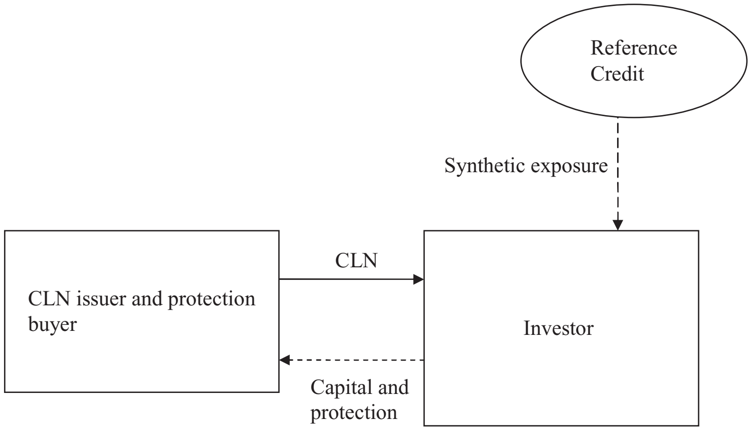

Figure 13.1 CLN via a limited recourse programme.

more complicated. For the note as described above, the SPV issues the note via its own issuance programme (not necessarily limited recourse programme) to the investor(s). Cash received is invested, typically in high grade and/or diversified assets (usually debt). In addition, the SPV enters into a contract with the issuing bank providing (say) 5-year CDS protection in return for a premium. The situation is summarised in Figure 13.2.

The bank now has the protection on France Telecom it requires, but obtained from a corporate (non-bank) entity – the SPV – to which it has counterparty risk. The bank is closely involved with the SPV (perhaps providing administrative or investment services) and as the SPV is well collateralised (often rated AAA), the counterparty risk is regarded as very small. Nevertheless, the bank indirectly carries both default risk on the collateral and some mark-to-market collateral risk (see below).

The investor now has synthetic exposure to the reference entity, to a portfolio of high-grade assets (but no direct exposure to the originating bank), to the SPV and to any third party obligations the SPV may have, including the CDS and any interest rate swap contracts with the originator.

The protection buyer eliminates **counterparty risk** to the investing institution – for example, it becomes possible to buy protection on Bank × from its parent company (which can be useful in restructuring certain subsidiary bank portfolios). Counterparty risk to the SPV and its underlying collateral is substituted.

The credit event under an SPV CLN (embedded risk on a single reference name)

On the reference entity credit event, or at maturity, collateral is liquidated. In the former case, collateral is sold to generate cash and, if there is sufficient, par less the recovery on the reference entity debt is paid to the original bank. Collateral risk is also borne by the investor – excess cash may be less than the recovery amount if the collateral was below par. The investor also bears collateral risk at maturity if there is no default event.

What is the risk that the collateral is insufficient to cover the loss on the reference entity (i.e. collateral is insufficient to meet the SPV's obligations under the default swap contract with the originating bank)?

The payoff required on the CDS is $100 - R$. Suppose the SPV collateral has value V at the time of the reference entity credit event. V may be less than par because of adverse market

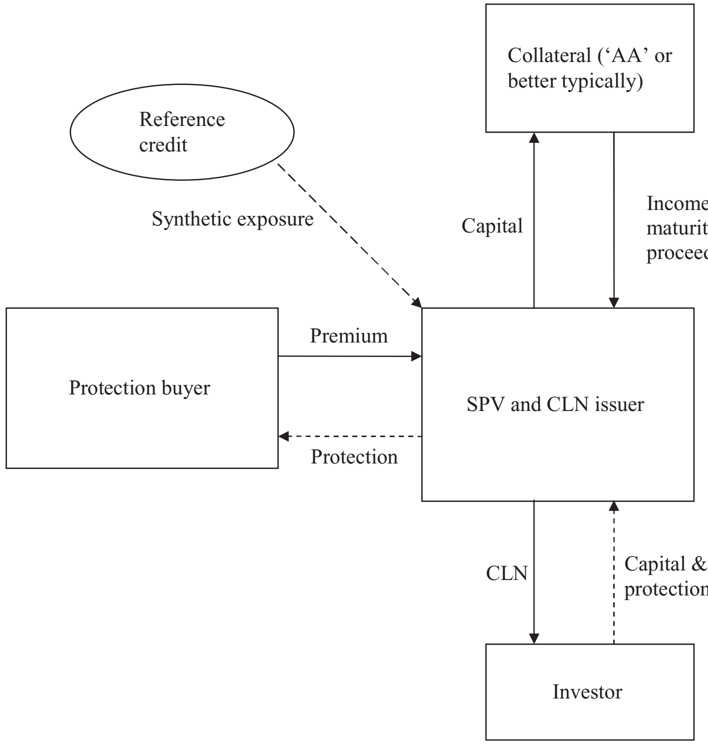

Figure 13.2 CLN via a special purpose vehicle (SPV).

moves on, and defaults in, the collateral pool. If $100 - R > V$, then there is insufficient funds to pay the amount promised on the CDS. This is the equivalent of an American put on the collateral at $100 - R$. If the collateral is short-life, diversified and high grade, and R is not close to zero, then this will have a negligible value that can generally be estimated using traditional options theory. We can model the collateral prices as a function of spreads and interest rates and we should introduce a distribution of recovery values, R. The risk is concentrated in the area of R near zero. Let us do a very crude back-of-the-envelope calculation. For SU debt approximately 10% of recoveries will be below 10%. The chance of high-quality collateral falling below 90 is small – party arising from default event risk. The risk of a loss of 10% may be 2% (including the, say, 5-year default risk on the collateral), giving a risk to the originating bank on the CDS of about $10\% \times 2\% = 20$ bp if a credit event occurs. The chance of a credit event on a 250 bp reference entity (assuming zero recovery) is roughly 12% over a 5-year trade. In other words, the bank would expect to pay below the market rate for the CDS protection it is buying by a figure of the order of $12\% \times 20$ bp $= 2$ bp.

13.2 EMBEDDED SWAPS AND OPTIONS

The CLN, particularly if issued via a SPV, may also include embedded interest rate or currency swaps. For example, if collateral is in both fixed and floating form, and in various currencies, swaps will be used to hedge out the interest rate and FX risk, but introduce *contingent* interest rate and FX risk. Typically contingent swap risk is carried by the investor – any loss on unwinding these swaps between the SPV and the bank on the occurrence of a credit event on the reference entity comes out of the collateral before any residual amount is paid to the investor.

If the reference risk is a bond with embedded options (callable or convertible, for example) then this risk will also be embedded in the CLN. Typically such notes would be issued to fund and hedge the convertible desk's activities and a back-to-back hedge by embedding the risks into the CLN is the most attractive route for the bank.

13.3 COSTS

A limited recourse programme is expensive to set up (of the order of EUR 50–100 000) but once established the marginal costs are low – EUR 10 000 or less. An SPV is set up for each deal and incurs a similar cost to those for a limited recourse programme. These costs are reflected in the terms of the CLN to the investor.

13.4 APPLICATIONS

We have already seen that the elimination of counterparty risk can be important, and here a CLN (or fully collateralised deal) is the obvious choice.

Large single name deals are more frequently done via a CLN. A bank may be seeking EUR 1bn protection on XYZ Company. Sourcing via CDSs is difficult (dealing size is normally EUR 10–25m), whereas the investor group for notes is wider and a larger sales force is available within the bank.

Investors may be seeking an exposure to a particular US reference name, but a risk denominated in EUR and to a specific maturity. The reference entity may have no EUR debt or no debt of the appropriate maturity, but a CLN allows a bank to manufacture synthetic XYZ debt to meet the investor's requirements. For example, this was the case for a while for Ford Motor Credit which at one time had little EUR debt. This then leaves the bank with a hedging problem that is easily handled on a book basis.

Where the bank is seeking protection via a CLN but is regarded as high risk, the investor is likely to prefer the use of an SPV to a note issued on the bank's own balance sheet. For example, if the bank borrows at 50 bp over LIBOR, this is a significant risk compared to other investment grade entities and may be unacceptable to the investor despite the additional spread received. In addition, the product becomes a significant 'first-to-default' risk and creates booking and monitoring problems.

The main applications of CLNs are in the structuring of more complex credit derivatives (see the example below) either for single reference names or for protection on losses between certain levels on a reference portfolio (a 'funded' tranche of a CDO).

There will be capital benefits or regulatory implications for the bank. Where a CDS protects a reference name the risk weighting (under Basle 88) is reduced to 20% (as long as the counterparty is an OECD bank). If a CLN is used, then cash resides in the bank rather than an SPV, and covers the risk weighting and more. If a SPV is used then there is no reduction of regulatory capital (the SPV is a corporate and gets 100% weighting) and no capital is raised.

13.5 CLN PRICING

13.5.1 Basic Pricing

Consider the following example: 5-year Fiat trades at 150 bp in the CDS market. The issuing bank's treasury borrows at LIBOR – 5 bp for 5-year USD or, alternatively, suitable collateral can be found paying LIBOR + 10 bp after swap costs. What spread can the bank offer on a USD 10m or 100m CLN via its limited recourse note programme, or via a SPV?

If the bank issues the CLN at par off its own book, then the hedge it puts in place and associated costs are as follows:

(a) Pay the set-up costs out of the capital received. The set up costs are (say) EUR 10 000 and this translates to about 2 bp per annum on the 10m note, or 0.2 on the 100m note.
(b) Write Fiat EUR CDS protection at 150 bp (we are ignoring bid/offer spreads)
(c) Deposit funds with its treasury department at LIBOR – 5 bp.
(d) If the CLN is fixed coupon then arrange an internal interest rate swap paying floating and receiving fixed.

The cashflow is matched between the hedges and the CLN if the CLN pays LIBOR plus 150 – 5 – 2 bp = 143 bp for the 10m note (144.8 bp for the 100m note). Additionally, the bank has a cross currency position to hedge and charge for – see Chapter 10.

If the bank issues via a SPV, then

(a) the set-up cost of the SPV correspond to 10 bp (1 bp) for the 10m (100m) note
(b) the CDS protection earns 150 bp as before
(c) the collateral spread is + 10 bp and
(d) the cost of the risk that the collateral is insufficient to cover the loss given default on the reference entity is approximately 2 bp,

giving a CLN spread to LIBOR of 148 bp and 157 bp respectively for the 10m and 100m notes.

13.5.2 CLN Pricing Model

From the bank's point of view the cashflows arising from the note are the promised payments subject to the reference entity survival plus payment of the recovery amount contingent on the credit event. The formula is identical to pricing the reference entity's debt (using a hazard rate appropriate to the CDS market if the CDS triggers are used).

From the investor's point of view the promised payments are contingent on the joint survival of the reference name and the issuing bank, and the recovery amount is contingent on the default of the reference entity and the survival of the bank.

From the investor's perspective this may present both booking and valuation problems. Although the CLN looks like a bond it holds two risks and booking the CLN as a bond on the underlying, ignoring the counterparty risk, is technically incorrect. Booking at least needs to record the existence of counterparty risk, and risk reporting should report this risk (at the very least the nominal exposure). Full valuation needs the methods of Part III, since the reference name and the issuing bank represent a correlated two-name portfolio in general. However, we can easily develop pricing results in the case of zero correlation.

There are two sets of cashflow under the note:

1. Promised cashflows – coupons and maturity amount. These are valued by discounting the cashflow (off the LIBOR curve) and multiplying by the probability of the joint survival of the reference entity and the counterparty to this date. If these two entities are uncorrelated then this is just the probability of reference name survival times the probability of counterparty name survival. We know these survival probabilities by calibrating to CDS data for these two names, given a recovery assumption.
2. Contingent cashflows:
   (a) The recovery on reference entity debt in the event of reference entity default before the counterparty
   (b) The recovery on the CLN in the event of counterparty default before the reference entity.

We can make several useful deductions:

(A) Let us first assume that recovery on both names is zero. Then item (2) is zero; the hazard rate is just the spread, and we can see that the CLN value is just the promised cashflows discounted at LIBOR plus the sum of the reference name and counterparty spreads.
(B) If we assume that only the counterparty recovery is zero, then recovery on the reference debt appears and again we are valuing the risky cashflow using the credit bond valuation formula (9.18) except that the LIBOR curve is shifted by the counterparty spread. The result is *approximately* the same as discounting the promised cashflows at LIBOR plus the sum of the reference name and counterparty spreads.
(C) If the counterparty recovery is also non-zero then a further recovery term appears.
(D) Generally we assume recovery on counterparty debt is non-zero but also assume that recovery on the CLN will be zero (it is not traded debt so it may be hard to find a buyer). In this case we can see that the CLN is *approximately* the same as discounting the promised cashflows at LIBOR plus the sum of the reference name spreads and counterparty *hazard rate*.

The final result should be noted.

The MathCad sheet 'chapter 22 N2D CDO CLN price and hedge.mcd' includes code for CLN valuation in the general correlated case (using the results of Part III), and uses this code to value a CLN with a correlated counterparty in addition to the uncorrelated case.

13.6 CAPITAL GUARANTEED NOTE

Any (credit-related) risk can be embedded into a CLN. A very simple CLN will embed a single vanilla CDS and it is easy to reflect the CDS settlement terms (physical) in a cash payoff on the CDS related to the recovery of the underlying. As a deal gets more and more complex, reflecting the 'recovery on reference debt' element of embedded CDS becomes more difficult to describe, and it becomes more common to embed a digital risk. In this section we look at one example of a more complex CLN where the embedded risk may be a single reference entity risk or a more complex instrument, and risk is embedded in digital form.

CLN Description

The CLN has a 20-year maturity, carries a fixed coupon of 6% and, for the first 5 years only, bears credit risk to a specific reference entity. If there is a credit event on the reference entity during this period, then coupons immediately cease (there may be a proportion paid to the credit event date). Par is returned at maturity whether or not a credit event has occurred.

Pricing and Valuation

We can write down the formula for the value of the CLN on Day 1 (assume the CLN is sold at par). We assume for simplicity that coupon g is paid continuously, we have

$$1 = g \int_0^5 v(t) \times S(t)dt + g \times S(5) \int_5^{20} v(t)dt + v(20) \tag{13.1}$$

where v is the discount factor and S the survival probability for the reference entity. The equation states that the market price is equal to

(1) the value of the coupon over the first 5 years weighted by the probability that the reference name survives to each payment date during that term;
(2) the value of the coupon over the 5 to 20 years contingent only on the fact that the reference name survives to the end of 5 years; and
(3) the value of par at maturity.

For a given reference entity and LIBOR curve we can solve this equation, knowing the survival probabilities and discount factors, to find the fair value coupon g on the CLN. Note that if the reference entity is risk free ($S(t) = 1$), then the coupon is the 20-year swap rate. The spreadsheet 'chapter 13.xls' implements the above formula for general risk and maturity periods. If the reference entity CDS is 200 bp, and the assumed recovery rate is 30%, then we find a coupon enhancement over the 20-year life of 50 bp. Figure 13.3 shows how the amount of protection bought under the note varies over its life, and shows the value of the final capital amount (assuming 4% interest rates).

Note that the enhanced coupon continues for 20 years in the event of no default, and that protection is in digital form.

Hedging

Per 100 nominal there is a digital protection amount falling from an initial value of just over 60 to a little over 50 just before the 5-year risk-termination date. At time t in the future ($t < 5$ years), the amount of the protection is just the value of the outstanding coupons that will not be paid following the credit event. This is

$$g \int_t^{20} v'(s)ds \tag{13.2}$$

where the prime indicates that the discount factors are those appropriate at time t (i.e. the forward discount factors).

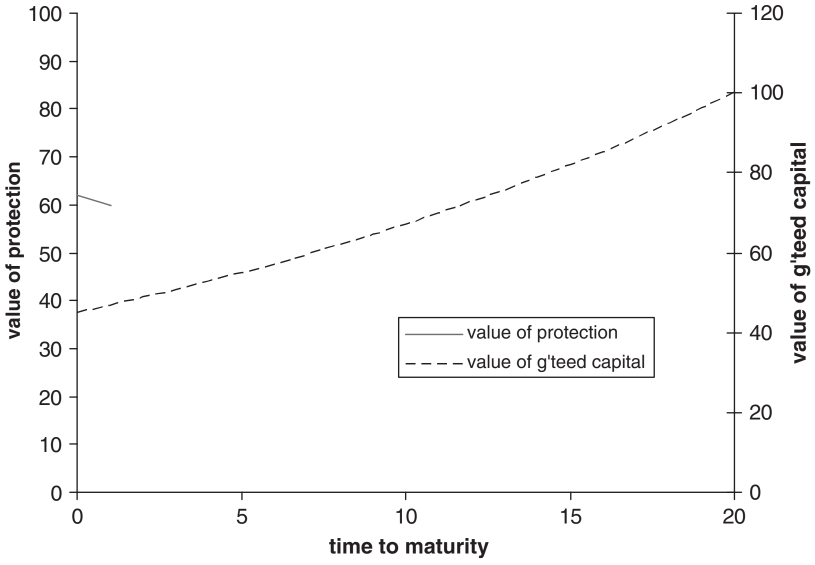

Figure 13.3 Value of protection and guaranteed capital amount.

For simplicity, we shall assume an average 55 digital protection (an accurate hedge calculation will take into account the exact protection, the exact life and the corresponding spreads available in the CDS market for the various maturities). Given the recovery assumption of (say) 30%, then the 55 digital protection can be hedged by selling vanilla protection on $55/(1 - 0.3) = 78.5$m per 100m CLN.

The trader can therefore enter into the following hedging deals once the CLN is sold:

1. Sell 78.5m 5-year vanilla protection on the reference entity at 200 bp, generating an income of 1.57m per annum for 5 years.
2. Enter into an interest rate swap with the IR swap desk, paying an initial capital amount of par, plus 1.57m quarterly for 5 years to receive par at maturity and a fixed coupon for 20 years.

On default of the reference name the fixed cashflows (excluding the maturity amount) have to be sold for a capital sum. This introduces a *contingent interest rate risk,* where a loss is suffered if interest rates rise. This capital sum has to settle par minus recovery on the 78.5m vanilla CDS sold – whose expected cost is $78.5 \times (1 - 0.7) = 55$m. This introduces a (large) *recovery rate risk*.

During the life of the deal interest rate moves (and possibly changing recovery rate assumptions) will cause a changing net exposure to the reference name, which will be rehedged.

Exercise 1

1. What is the impact on the coupon or reducing the final maturity (for a fixed 5-year risk period)?

2. What is the coupon impact of using a more risky reference entity?
3. If the CLN carries a floating coupon what impact is there on the initial hedges and on the risks?

The reference risk in the capital guaranteed CLN could be a first-to-default basket, the 'equity' protection of a synthetic CDO, a portfolio of CLNs, or any other credit risk.

14

Digital or 'Fixed Recovery' CDS

14.1 PRODUCT DESCRIPTION

A digital CDS is similar to a vanilla CDS except that the payment in the event of a credit event is not par in exchange for defaulted debt (or par less recovery), but par. Similarly, a fixed recovery CDS is a CDS where the payment on the occurrence of a credit event is par less the agreed recovery.

The difference between a digital (sometimes called a 'binary') CDS and a fixed recovery CDS is merely documentation. For example, consider a digital CDS in EUR 10m on Fiat, subject to a premium of 600 bp. A fixed recovery CDS on EUR 12.5m with an agreed recovery of 20% pays $12.5 \times 0.8 = 10$m in exactly the same circumstances and conditions as the digital. The correct premium for the fixed recovery CDS is $600 \times 10/12.5 = 480$ bp, giving the same monetary premium for the same monetary protection in both cases.

Digital products are not very important in their own right – very little trading takes place in a simple digital CDS. Generally a digital CDS does not form a natural hedge for many risks, as most risks are in the form of bond or loan assets with an unknown recovery or equity risk, where a full loss is taken on the credit event but the forward exposure is unknown. The key feature of a digital CDS in valuation and risk management terms is the strong recovery rate sensitivity of the value of these products.[1]

Digital risk arises in two main forms:

(a) Off-market vanilla CDS positions and non-standard premium streams have a small recovery sensitivity (which increases further the more the premium stream differs from the current fair value quarterly-in-arrears premium). In particular, a mature book of vanilla CDS positions will have a small digital risk.
(b) Structured credit products, particularly in CLN form, often have an embedded digital risk. Examples are the capital guaranteed note (section 13.6), a CLN with exposure to the 3–5% of losses on a reference pool, and counterparty exposure on an interest rate swap. We shall also see in Part III that portfolio products have a significant recovery rate risk.

A recovery stress on the individual reference name, together with recalibration and revaluation, reveals the level of digital risk in the books.

14.2 PRICING, HEDGING, VALUATION AND RISK CALCULATIONS

14.2.1 Simple Pricing

Given that a vanilla 5-year CDS is trading at 200 bp, where would you price a digital?

[1] This may seem counterintuitive. We know the price of a vanilla CDS from the market. The payoff is par minus the unknown recovery, but the chance of a credit event has to be inversely proportional to (par minus recovery) to justify the CDS premium. The digital CDS pays par but is driven by the same recovery dependent chance of default. Hence the digital premium is recovery dependent.

Our calibration for that name might be to assume a 50% recovery, leading to a 4% default rate. On the occurrence of a credit event the vanilla pays $100 - 50 = 50$ while the digital pays 100, so the digital should be subject to a premium of 400 bp. But suppose we had assumed that recovery is 0%. As the digital has the same payoff as the vanilla, the premium should be 200 bp. If the reference asset was a secured loan we might assume that recovery is 90%, and the digital then pays 10 times as much as the vanilla CDS and should be priced at 2000 bp.

There is no effective way of either knowing or hedging the digital risk (until such time that digital products trade actively). Faced with the prospect of doing a single isolated deal in a digital CDS, a trader would naturally take a cautious view of the recovery risk and make a wide bid/offer. On the other hand, if you regularly dealt in such instruments, and had a large diversified portfolio on the books, then you would be much more comfortable quoting close bid–offer spreads on digitals. This happens when constructing CLNs with an embedded digital risk.

14.2.2 Recovery Assumptions

In the case above we should note that there is a substantial difference between the status of the recovery rate assumptions for vanilla CDS and those for digital CDS products. We saw that a vanilla CDS has little valuation sensitivity to the recovery assumption (section 12.2) – so the actual recovery assumption is not very important. However, the simple example above shows that the fair value premium, and therefore the trade valuation, changes dramatically as we alter the recovery assumption.

Consider a portfolio where a 100m short in autos is hedged by a 100m long in utilities from the perspective of a credit event risk. Suppose we assume a 50% recovery across all senior secured names. If we also assume, for simplicity, that the CDS premiums are all 100 bp, then it follows that all names are subject to a hazard rate of 200 bp. But suppose the reality is that autos are subject to a 30% recovery rate (hence a 133 bp hazard rate), and that utilities are subject to 80% (hence a 500 bp hazard rate). If defaults are as expected (1.33 autos for every 5 utilities) then the results are as follows:

Vanilla CDS book: net loss $= 1.33 \times 0.7$ (autos) $- 5 \times 0.2$ (utilities) $= 0$
Digital CDS book: net loss $= 1.33 - 5 = 3.67$.

If we are trading digital risks we should put much more effort into making realistic recovery assumptions. (This does not mean that we have to trade 'digital' products as such – for example, portfolio credit products contain significant recovery (digital) risk.)

14.2.3 Valuation and Hedging

The valuation formula can be derived as in section 10.1 – we simply have to drop the recovery cashflows from equation (9.13). The value of a long digital protection is

$$V(t) = \int_t^T D(s) \cdot S(s) \cdot \lambda(s) ds - p \int_t^T D(s) \cdot S(s) ds \tag{14.1}$$

We can immediately see that if the hazard rate is a constant then the premium equals the hazard rate for a fair value trade.

Embedded digital protection in structured CLN products ultimately has to be hedged using vanilla CDSs. For simplicity, consider a single name digital CDS where the vanilla trades at 100 bp. If we assume zero recovery, then a long 10m digital risk is hedged by a short 10m CDS (the payoffs are both 10m), whereas if we assume 90% recovery, then 10m digital is hedged by a 100m vanilla position. In reality, on an investment grade portfolio, we shall typically hedge spread risk, not default event.

Exercise 1 Consider hedging digital CDSs from the spread sensitivity perspective. Compare the sensitivities derived from (14.1) and (10.1).

14.3 TRIGGER EVENT DIFFERENCES

In the mid- to late-1990s Enron attempted to develop a market for digital CDSs among corporate clients. They called these products 'bankruptcy swaps'; they were digital CDSs triggered by default or 'filing for protection from creditors'. They therefore had significant documentation differences compared to vanilla CDSs which would have further complicated hedging. (The market in bankruptcy swaps did not prove popular.)

Current embedded digital CDSs tend to reflect vanilla CDSs terms and conditions or, where differences occur, tend to favour the origination bank.[2]

[2] 'Documentation trading' – where essentially back-to-back trades are used to capture favourable documentation differences – used to be a common feature of the CDS market before standardisation of documentation took place. It is now largely confined to structured products in CLNs.

15
Spread Options, Callable/Puttable Bonds, Callable Asset Swaps, Callable Default Swaps

The biggest problem with spread option products is the lack of an open market and of any source of 'implied' spread volatility. This may be changing in the portfolio area – with the introduction of options in the iTraxx CDO tranches – but single name spread options remain a narrow market, and are largely restricted to embedded 'conservatively' priced options.

15.1 PRODUCT DEFINITIONS

15.1.1 Vanilla Spread Options and Variations

A credit-spread option is an option on the spread of a 'defaultable' bond (e.g. the yield on Fiat 5.75, May 2014) over a reference yield (e.g. the interpolated EURIBOR swap rate). At maturity (European), or at any time up to maturity (American), a credit-spread put (call) gives the buyer the right but not the obligation to sell (buy) the defaultable bond at the price implied by the strike-spread and the reference yield. For example, the bond may be trading at 150 bp currently, the option may be a European put exercisable in one year's time at a z-spread of 250 bp. The holder of the bond may also hedge the fixed coupons with an interest rate swap, thus interest rate moves have equal and opposite impacts on the bond and swap. If spreads on the bond rise to 400 bp in a year's time, then the bond plus interest rate swap value will decline below the initial price. Instead of selling the bond at a 400 bp spread, the option buyer can sell at 250 bp through the option, effectively making a profit of 150 bp relative to selling in the market.

Spread Option Cashflows

Suppose DKB sells a one-year European call to BoA on the Fiat bond at an exercise z-spread of 150 bp for a premium of 40 bp p.a. quarterly. We also suppose that BoA enters into a forward (starting in one year's time) interest rate swap related to the bond where BoA pays fixed, so that interest rate moves on the intended forward holding of the bond are hedged out.

If the spread in the market at expiry is above 150 bp, then it is cheaper to buy the bond in the market, but if the spread is, say, 100 bp the price calculated on a z-spread of 150 bp is less than the market price and the option will be exercised, see Figures 15.1 and 15.2.

Exercise 1

1. How much profit (approximately) is made in the above example?
2. Suppose it was a put option and spreads widened to 500 bp?

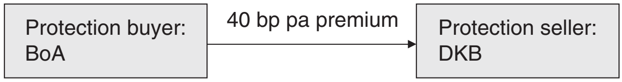

Figure 15.1 Spread option cashflows: pre-expiry.

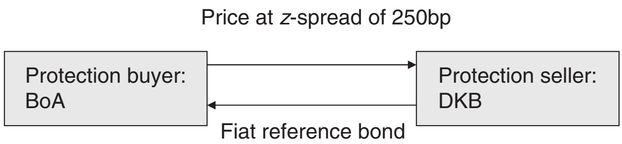

Figure 15.2 Spread option cashflows (if exercised): expiry.

Product Variations

- Premiums may be paid up-front or in swap form.
- The reference rate may be on a specific bond – generally a government issue. This introduces several new risks: the repo on the reference bond, and idiosyncratic price movement of the reference bond.
- The option may be a cash settlement – if exercised the writer pays the buyer the difference between the price of the underlying at the exercise spread, and the market price.
- A put may knock-out on default, in which case an owner of the bond and the option is left with a worthless option and a defaulted bond. Otherwise, the option is usually immediately exercisable on default.
- The option may be on a default swap instead of a bond. The call (put) option gives the buyer the right to sell (buy) the CDS at a predetermined spread. (Put–call terminology is used in a way consistent with the risk implication of an option on a bond.)

Exercise 2 Suppose you write a European call exercisable in 2 years' time on a (then) 5-year CDS on Toyota Motor Corp at 300 bp. The option is exercisable early in the event of default (defaulted debt is delivered for par). How would you hedge the credit event risk before the exercise date? What would the price difference be between a CDS call option without/with the knock-out?

15.1.2 Related Embedded Products

Callable Asset Swaps

The asset swap desk has the option to terminate the asset swap at par at any time prior to the maturity date. If the spread on the underlying asset has narrowed substantially, the reference bond (which is owned by the desk as a hedge against the asset swap) will have risen in price. Without having the call feature embedded in the asset swap, the desk does not benefit from the price rise of the bonds since the bond is economically owned by the asset swap counterparty through the asset swap. If the desk can terminate the asset swap at par, then the bond can be sold at market price (and the interest rate swap unwound) to yield a profit. (Note that a rise in interest rates will not significantly harm the profit arising from the spread narrowing, since adverse bond price movement arising from the interest rate risk is almost exactly offset by the interest rate swap held by the desk.)

Callable Default Swaps

A callable CDS is a vanilla CDS with the added feature that the buyer of protection has the right to terminate the CDS (at any time or at certain dates) on a narrowing of the fair market CDS premium below the trigger level for the option. This allows the buyer to terminate the CDS and replace it with a cheaper CDS.

A callable CDS and a callable asset swap are financially similar – the difference being the basis risk between the asset swap reference asset and the CDS on the reference name of the same maturity. Similarly, a spread put option on a bond is closely related to a puttable default swap.

A seller of a CDS with the option to terminate the swap if the spread widens beyond a certain level will do so if that level is breached. The termination could be followed up with a new sale at a wider spread or the name may be considered too risky (in which case any long CDS held as a hedge may be sold for a substantial profit). From the CDS buyer's point of view, the embedded option removes much of the benefit of the CDS – sudden default being protected – so the premium would have to be substantially below the vanilla CDS premium. An alternative is to allow the CDS premiums to ratchet up (driven by spread levels or ratings) rather than allow protection to terminate.

Commercial bank '**standby facilities**' or 'a **facility**': an agreement to lend a certain amount of money at an agreed spread over LIBOR at any time up to a specified data in the future. This is equivalent to the potential borrower owning a credit spread put option on its own debt. Usually the facility knocks out in the event of a default event.

15.1.3 Bond Price Options

A price option on a credit bond has the same format as a price option on a government bond. In the case of put options the question of the occurrence of a default before the option date has to be addressed. Price options are driven by both interest rate and spread movements.

A yield model is needed to value bond price options. This may be constructed by modelling the LIBOR curve and the reference entities hazard rate (or spread) curve, together with a correlation model, or may be driven by a one-factor model for the yield. Typically a Hull–White model is used or a binary tree implementation of a lognormal yield model. The tree approach discussed in section 9.2 (for the hazard rate) can trivially be applied to the yield.

The key problem in pricing all spread-related options is the unobserved implied volatility. For bond price options there is the additional problem of the yield/spread correlation required to calculate yield volatility.

Correlation Note

Note that the bond's yield and the LIBOR rate will have very different correlations according to the level of spread. At a low spread, the yield on the bond and the LIBOR rate are almost identical, so the correlation is close to 100%, but at very high spreads the yield is at a similar level to the spread, so the yield and the LIBOR rate correlation will tend to the spread/LIBOR correlation. Yield/LIBOR correlation therefore varies hugely with the level of spread. On the other hand, the correlation between the LIBOR rate and the reference entities spread would not appear to have any particular relationship to spread level and is regularly assumed to be zero. If it is necessary to have a two-factor model, LIBOR and spread with a constant correlation corresponds to a LIBOR and yield model with a complicated correlation structure – the former approach is preferable.

15.1.4 Applications

At one time – when CDS documentation was unclear and market participants were not sure that they could incontrovertibly establish that a credit event had occurred – deep out-of-the-money spread puts were used as an alternative to default swaps. As a reference entity went into default and the bond prices fell, the spread on the bond could always be calculated. If prices had fallen to very low levels, the option could be exercised.

What is the payoff difference of these two contracts in this case?

Take a 5-year FRN priced at par and paying a 100 bp spread. Suppose you own a spread option with a strike at 600 bp exercisable at any time up to one year from now. Assume 50% recovery and that default occurs tomorrow. The CDS would pay 100 − \R = 50. The FRN priced 500 bp away from current par value has an approximate price of 80, so the gain on the spread option is 80 − \R 30. Note that the deep out-of-the-money spread option protects the price of the bond at the exercise spread. A hedge ratio against default swaps is recovery dependent.

Commercial banks are active writers of spread options (standby facilities and irrevocable lending commitments) and in principle buying a spread put option would allow them to hedge their risk. In practice this almost never happens.

1. Commercial banks generally run such positions in a 'buy and hold' portfolio. Marking to market is not required and risk management is through portfolio diversification and 'knowing your client'.
2. Spread options remain conservatively priced – in other words hedging standby facilities with spread options is expensive compared with the commercial banks own assessment of risk (although this is usually not based on an options pricing model).

The largest application of such options is embedded into bonds, embedded into asset swaps and CDS. Ideally, spread options would allow the investor to hedge risks and take views on the market in a more flexible or tailored way. This will not happen until a more open market in spread volatility develops.

15.2 MODEL ALTERNATIVES AND A STOCHASTIC DEFAULT RATE MODEL FOR SPREAD OPTION PRICING

15.2.1 Model Approaches

The model described in Part I based around transition matrices (calibrated to the reference entity in question) is capable of calculation the forward prices of credit risky assets whether in default, or non-defaulted. A forward spread is only one of a handful of values, and the underlying hazard rate model is a pure jump process. It is driven by the Normal Copula, and has similarities to the portfolio pricing techniques described in Part III.

An alternative approach is to model the underlying hazard rate process as a stochastic process – following some sort of diffusion process – although we must bear in mind that this approach is fundamentally at variance with the way credit spreads actually behave. This will allow for an infinite range of forward spreads. Remember that the CDS pricing and calibration formulae discussed in Chapter 9 remain valid, even if the hazard rate is stochastic, as long as the hazard rate moves are uncorrelated with interest rate moves and other moves. However, the CDS calibrated hazard rate is not the true stochastic hazard rate but the pseudo-hazard rate valid for the CDS pricing only. (If we also wish to introduce an interest rate/hazard rate correlation then we must also rework our CDS and other pricing formulae.)

A discernible difference between these two alternative models arises, for example, if we consider a short-dated option. Suppose the underlying spread is 80 bp for a BBB-rated name. Under a diffusion model with a 100% volatility in one month's time, there is approximately a 0.001% chance that the spread will exceed 150 bp (about 3 standard deviations away), whereas under the TM approach there is around a 1.5% chance of a spread of 200 bp or more. (See the TM probabilities in 'chapter 2 and chapter 15.xls').

On the other hand, bank and insurance company names tend to have very low ratings volatility – a TM-based approach may overestimate the chance of a significant change in the spread.

The TM model has already been described in the context of simulated spreads for bonds in a portfolio – exactly the same process can be applied here. We now describe a simple stochastic hazard rate tree implementation of the second model alternative.

15.2.2 Hazard Rate Tree

A simple version of a tree model (see, for example, Jarrow and Turnbull, 1996) is described and implemented in the MathCad sheet 'chapter 15 stochastic hazard rate.mcd' (or view the rtf file).

We use a recombining binary tree for hazard rates (see Figure 15.3). For simplicity the hazard rate, λ, is assumed to be a constant (although the formulae can easily be amended for any expected hazard rate curve). The hazard rate can increase in any time interval to λu with probability C (conditional on no default), or to λd with probability $1 - C$. The logarithmic variance is assumed to be σ^2 per annum, which gives rise to a formula for u. We then calculate C in terms of u ($C = 1/(u + 1)$) and find u in terms of σ. The tree is a trivial recombining tree for the hazard rate conditional on no default. If q is the default probability in the time interval Δt, then the unconditional probability of an up move is $u \times (1 - q)$, and similarly for a down move.

We now have a three-legged tree. At each step the hazard rate may move up, down, or the reference entity may default. To price a bond (for example) we evolve the hazard rate tree to the maturity of the bond (where the price is 100 on the non-default nodes of the tree). We then work backwards ensuring that, at each step, the value of the bond is the expected value of the three optional one-step-forward bond prices discounted back, together with any coupon to be paid.

The same hazard rate tree and similar pricing formulae can be used to get the CDS premium. Note that, for bonds and CDSs, we have a process of trial and error in order to calibrate the hazard rate to produce the correct bond or CDS price. It is not accurate to use the pseudo-hazard rate obtained from CDS calibration.

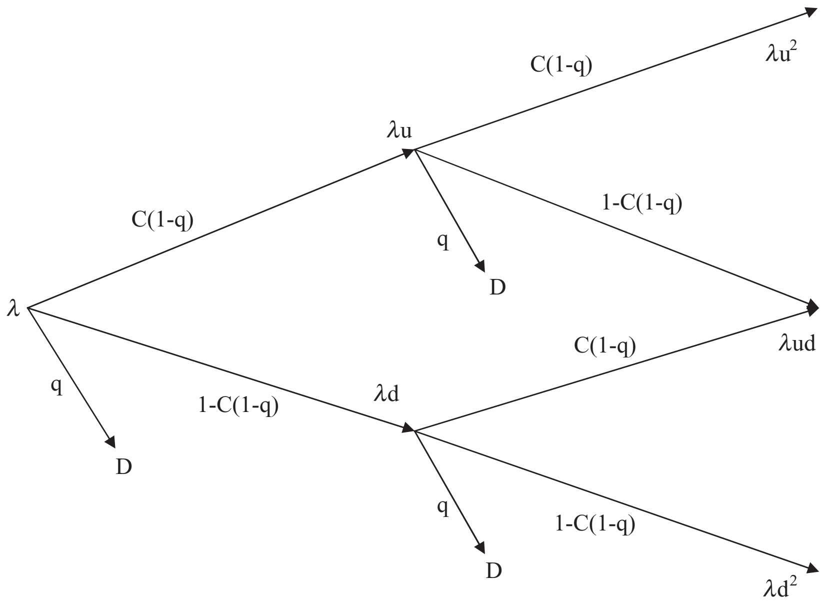

Figure 15.3 Hazard rate tree.

The tree now has bond, or CDS, values at each node on the tree. We can now use these to calculate the value of European or American call options on these underlyings.

15.2.3 Callable High Yield Bonds

The implementation referred to in section 15.2.2 actually calculates the option value for a price option on a callable high-yield bond. This follows the process described in Chapter 11 – the spread volatility (a guess based on historical volatility) is combined with the LIBOR volatility (from swaptions) and a correlation (assumed to be zero) to get a yield volatility for the bond. In this case the bond is a high-yield bond, and the volatility is dominated by the spread volatility. We assume that interest rates are a constant and model the stochastic hazard rate only in order to estimate the embedded option value.

15.3 SENSITIVITIES AND HEDGING

Spread (or yield) options have hazard rate, recovery and interest rate sensitivities as do CDS and, additionally, sensitivity to the hazard rate volatility. Since the mark-to-market levels are largely unknown, and since volatility is not a good measure of option pricing risk when the underlying process is a jump process, offsetting or hedging one deal with another is highly uncertain. Typically a risk control view of such a position would be to impose large reserves – reflecting uncertainty in value – and tight limits on outstanding positions. Buying options is usually handled by writing off the value of the premiums (and marking to intrinsic). The trading desk is then limited in the trades it can do involving selling options – hampering liquidity – but is able to buy cheap (low premium) options – hence the growth in embedded bought options.

The development in the standardised CDO market (see Part III) and options on CDO tranches may provide some help in assessing mark-to-market levels of volatility, although the conversion from implied volatility on portfolios and tranches of portfolios to individual name volatilities is not trivial.

16
Total Return Swaps

16.1 PRODUCT DEFINITION AND EXAMPLES

A total return swap (TRS) is an off-balance transaction in which the '**payer**' pays the '**receiver**' the total return on a reference asset (be it positive or negative) from the effective date to the 'fixing' date in return for a floating leg, usually LIBOR plus a spread. Cashflows – such as coupon payments – are usually paid to the receiver on the dates the reference asset makes those payments. A TRS may be arranged with several intermediate fixing dates before maturity. In this case usually the terms of the TRS may be the same as if each sub-period was a TRS in its own right done at fair market terms, effectively terminating and restarting a new TRS at each fixing date. We shall only consider a TRS with a single total return payment date.

The payer is often called the buyer of protection, as the TRS will compensate the payer for negative performance of the reference assets. If the asset declines in price, the payer pays a negative amount to – i.e. receives a payment from – the receiver. In particular if the asset defaults, a large payment is made by the receiver to the payer, and the contract terminates with a proportionate floating payment from the receiver.

Similarly, a receiver of the total return is often called a seller of protection, as the swap will expose him or her to negative performance of the reference assets.

On default of the underlying settlement can be cash or physical.

Counterparty risk is initially zero but increases as the reference asset moves away from its forward price curve. Counterparty risk may be positive or negative. Periodic cash-flow dates (short maturity) reduce counterparty risk.

Before expiry the reference asset cash-flows are passed to the receiver (Figure 16.1).

At expiry, default or on a fixing date the capital gain since the last fixing is paid, in return for the floating payment. The contract terminates at the expiry or default date. There may be an option to terminate at the fixing date – which is worthless if the TRS is remarked to current levels and terms at each fixing date (Figure 16.2).

What restrictions should there be on the reference asset?

There should be a readily available market price, and the ability to deal in the size of the contract at that price. For example the following would normally be acceptable.

1. US Treasury bond – typically up to USD several billion
2. Alcatel 5.875 2015– typically up to EUR 10m.
3. Alcatel equity – up to the typical dealing size
4. A forward delivery of oil – up to the typical dealing size

16.2 APPLICATIONS

Total return swaps were traditionally used between banks where one bank had exceeded its balance sheet limits and another bank had balance sheet available. A TRS allowed the former bank to remove the asset from its balance sheet while maintaining economic ownership of the bank. Usually the deals were of large size (USD1–2bn not being unusual), often on very liquid

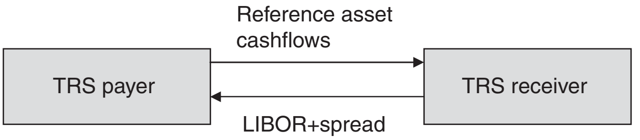

Figure 16.1 Pre-expiry: the reference asset cashflows are passed to the TRS receiver.

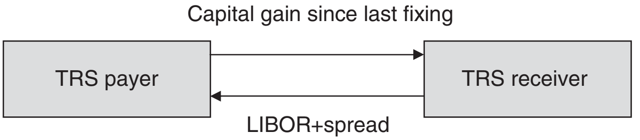

Figure 16.2 Expiry, default or fixing date.

assets (US Treasuries), often originated by one bank selling the asset to the other bank (buying it back at the end of the deal), and usually the TRS deal was for less than three months' life. From the perspective of the bank with available balance sheet, TRS are viewed as 'balance sheet rental' trades.

Recently TRSs have been used more by hedge funds. The TRS allows the fund to have economic exposure to the underlying asset without having to finance it. Alternatives include dealing in an economically similar but unfunded asset – such as writing a CDS. This presents problems for some hedge funds because, as unregulated and possibly also high risk corporate entities, they are unattractive counterparties for a bank buying protection – without substantial collateral being in place. Another alternative is buying the asset and financing it by doing a repo trade – we have seen that this is essentially what a TRS is.

A further application is to allow an investor to obtain economic ownership of an asset which he would otherwise not be allowed to buy. For example, assets within a certain country may only be legally owned by residents' of that country. A domestic bank could buy the assets and grant economic ownership (and funding) via a TRS.

16.3 HEDGING AND VALUATION

16.3.1 Pricing and Hedging

The payer may set up a hedge as follows.

1. Buy the reference asset at the market price P.
2. Enter into a repo trade with the asset as collateral for repo rate r for the term of the TRS. (This term repo may not be possible in practice and a rolling short-term repo is entered into, or the bank may finance the asset off its own balance sheet at its funding rate for the period.)

We can see that a TRS is the purchase of the asset and a funding trade performed one step removed. The TRS payer is taking risk on the funding rate rather than the end investor if indeed the investor were able to do the trade herself.

The payer then pays cashflows as follows.

1. Cashflows on the reference asset are paid to the 'receiver' as they are received.
2. At the fixing date, the reference asset is marked to bid, P_{fwd} (or sold if it is the expiry of the deal or the asset has defaulted), and the price gain $P_{\text{fwd}} - P$ paid to the receiver.
3. The repo rate is paid on the repo transaction based on the initial reference price.
4. LIBOR plus a spread, X, is received on the initial capital in the trade (P). (An alternative is to base the premium on the notional of the deal – this is just a rescaling exercise.)

The net cashflow on a single period TRS is LIBOR $+ X - r$. Thus a fair market price is achieved if X is minus the amount special on repo.

Even in the single period TRS there are risks. A term repo rate may not be available – and the bank will have to estimate the repo cost on a rolling basis. On the other hand if the bank is financing the trade off its own balance sheet then it will know this cost. Secondly, if a default event occurs on the reference asset, any term funding transaction will have to be unwound at prevailing rates.

A multi-period TRS will continue, but now the bank is financing not the initial price, P, but the fixing date price, P_{fwd}. The initial hedge could take into account the expected price change and the expected funding cost.

In practice the spread paid by the TRS payer will be above the fair value spread. A charge is made for access to the balance sheet, access to restricted markets, or access to funding which will be as large as the market will bear and will be chosen to generate a target return on capital for the payer.

16.3.2 Valuation

If we value the deal which expires at time t, where an asset cashflow c is expected at time t_0, then the value of the assets (bond, repo and TRS cashflows) is made up of the following.

(a) The bond, current value P_t and a loan of P
(b) The cashflow c is exactly matched by an outgoing cashflow, so has no valuation impact.
(c) A recovery arising from a credit event is exactly matched by an outgoing cashflow, so has no valuation impact.
(d) A credit event forces unwinding the repo trade at then current market rates. We would normally assume the repo follows the forward repo curve, so the expected unwind cost is zero.
(e) A payment of the total return amount at the maturity date if the name survives.
(f) The repo payment of the loan plus interest at the repo rate if the name survives.
(g) A receipt of LIBOR plus the spread on the initial capital if the name survives.

$$\begin{aligned} V = {} & P_t - P + (P_{\text{fwd}} - P)\cdot S(t)\cdot d(t) + P\cdot((1+r)^t - 1)\cdot S(t)\cdot d(t) \\ & - P\cdot((1+L+X)^t - 1)\cdot S(t)\cdot d(t) \end{aligned} \tag{16.1}$$

where $d(.)$ are discount factors off the LIBOR curve, $S(.)$ is the survival probability of the preference entity, L is the LIBOR rate and X is the spread.

Note that if we are performing the valuation at the outset then $P_t = P$ and the value is zero if $r = L + X$ as before.

17
Single Name Book Management

The aim of 'book' or 'portfolio' management is to understand and control the risks. Risk is related to potential changes in value, and these can be assessed in two different ways.

1. A '**level/duration/convexity**' approach. We may describe the full functionality of the value function by the values of all the derivatives of that function to all inputs. The first-order derivatives are variously called 'PV01', 'sensitivity', 'duration'. Second-order derivatives include 'convexity'. Book risk can be measured by the values of these derivatives – but typically we only look at derivatives up to second or third order. If a derivative is close to zero, then the book is hedged. On this approach a 1-year and a 20-year trade will both contribute risks to these measures (sensitivity, convexity, etc.) and will add to (or net against) each other. Note that risks beyond second order can be significant (in a 1-, 5-, 30-year CDS deal spread hedged and convexity hedged – 'twist' risk would be very large).
2. A '**bucketing**' approach. We look at sensitivities only (first-order changes only) but only aggregate risks on 'similar' trades (e.g. both 5-year BBB-rated US names). Typically a 1-year and a 20-year deal would not lead to an aggregated risk position.

We have already illustrated the bucketing approach in Chapter 10 with the simplified 'risk charts'. The subject of this chapter is how to control and manage risk in these CDS positions in practice.

17.1 RISK AGGREGATION

Credit derivatives books or portfolios within an investment bank or other institution are generally separated into main product categories – here we look at the 'single name book'; we look at the 'correlation (portfolio products) book' in Part III, and the 'counterparty risk book' in Part IV.

The single name CDS book contains all vanilla CDS contracts arising from market-making (client-driven) activities, proprietary trading or investment, and in-house hedging activities – for example hedges done between books (e.g. the, CDO traders will hedge residual name risk with the CDS traders). A separate single name (spread) volatility book will also be created to hedge volatility risk – residual single name spread risks will be passed down to the single name CDS book either explicitly via intra-desk deals, or implicitly at the risk level.

Managing the single name book requires the calculation of risks according to the pricing model of Chapter 9 (or the institutions preferred model). Typically these are interest rate risk (generally very small and we shall not discuss it further), spread or hazard rate delta risk, default event risk, recovery rate risk. Risk is calculated on an individual name basis. The trade is assigned to a bucket and a bucket may be defined as follows.

1. Reference entity
2. Maturity: For example 0–6 months, 6 months to 1 year, 1–2 years, 2–4 years, etc.
3. Geographical region: North America, Europe, emerging market, etc.

4. Currency.
5. Industry: for example, using rating agency industry groups.
6. Rating or spread range.
7. Seniority.
8. Others.

Buckets define a cell in a matrix into which trade sensitivities are posted. The natural next step is aggregation of risks. The risk charts in Chapter 10 illustrated the risks at the individual deal level, and an aggregation by maturity and across all maturities for certain types of deals. In practice, aggregation for a given name will be across all CDS (or bond) deals on that name.

An aggregation hierarchy might be as follows.

1. Name, currency, seniority and maturity.
2. Name, currency and maturity.
3. Name and maturity.
4. Name.
5. Industry and rating (by currency by seniority by maturity, and aggregates of these).
6. Region and rating (etc.).
7. Rating and maturity.
8. Rating.
9. All (e.g. a single spread sensitivity aggregated across all trades).

Risk limits are imposed on the various captured risks and certain buckets. In addition, reserves are held (i.e. capital is set aside) against risks not captured by the model but inherent in the deal. Generally the reserves are small on the single name book – the largest being bid-offer spread if this is not factored explicitly into the valuation process, another example is liquidity risk.

We briefly review some of the trades discussed earlier and highlight some further features of these deals and implications of the risk control approach.

Example 1: Naked Risk Risk is allocated to a bucket. Within that name this is the only risk, so the view (for the currency and seniority) of the deal is as in Figure 10.1. As we progress up the hierarchy – say to the industry, rating, currency, seniority and maturity bucket – we may find that the risk from this long protection deal is offset by other (effectively short protection) deals.

In other words, within the bucketing and risk limit approach adopted, naked exposure in different but similar names offset each other. Our rules allow us to hedge FMC with GMAC, for example.

Example 2: Cross-Currency Risk Figure 10.2 showed the position for a trade in a particular name, where long risk in one currency was offset by short risk in another. Within the bucketing approach we may be looking at an offset of short 5-year Fiat CDSs in USD offset by Fiat EUR protection in 3- and 4-year maturities, or long non-Fiat protection (e.g. other autos in USD).

Note also that the risk is dynamic, driven by interest rate moves. Cross-currency hedging can be controlled to a greater extent by having tighter limits on bucketed risk within a specific currency. Thus, if the autos A and BBB limits are EUR 40 000 per bp in 5-year euros, and a similar amount in USD, then cross-currency hedging is limited to 100m (100m $\times$ 0.0001 $\times$ duration of 4).

Example 3: Curve Risk Figure 10.4 shows the net risks for a curve trade on a specific name. Suppose the trades had been done (with the appropriate delta hedge ratios) in 1-year versus 10-year. What difference does this make?

The risk with curve trades is that spreads may not move in parallel but the curve tilts adversely. This risk is clearly greater in a 1/10 maturity trade compared with a 3/5 maturity trade. Clearly the bucketing picture for that name will be different with the risks being in very different parts of the maturity range. We can control the magnitude of these trades by putting tight limits on the exposure within each bucket, not at the specific name level but at the industry, rating, currency, and seniority level. Similar curve trades in various names will aggregate to large bucketed positions: the risk can be reduced by having curve steepeners on some names offsetting flatteners on others – or by having smaller positions.

Note that the hedges for a curve trade and the 'forward CDS' deal (Chapter 9) are dynamic. Fixed hedges will give rise to an emerging spread risk (at the total level, and changing spread risks in each maturity bucket for that name). Mismatches will appear at the individual name level and also in buckets as risks are aggregated.

A portfolio management approach allows:

(a) emerging risks on some deals to be automatically offset by emerging risks of the opposite sign on other deals, and
(b) reduction of risks by hedging aggregated emerging risk more frequently than is practical at the single name level.

Example 4: Cross-Seniority Risk; Digital Risk Figure 10.6 shows the position for a cross-seniority trade. Within the 'all seniorities' bucket, net spread risk does not know whether it came from senior, subordinated, loan or other exposure.

One way to pick up and control cross-currency hedging is through the default event risk measure. Based on assumed recoveries, the deals offset, but if we stress recoveries to 0% recovery and to 100% separately for different seniorities, we find a worst-case loss of zero on this deal. On the other hand, if the portions were reversed, the risk charts would be essentially the same, but the stress results would show a worst-case loss on the notional of the short subordinated protection (arising from senior recovery 100%, giving no gain on the long position, and junior recovery to 0%, giving a full loss on the short position).

Likewise, the aggregated spread risk bucket does not know whether the risk came from vanilla CDSs or digital risks. We can establish this by changing our recovery assumptions on all names, recalibrating, revaluing all the deals, and recalculating the bucket risks. Large movements in the bucket risk will reveal the presence of digital exposures from whatever source – including off-market CDSs, embedded digital products and cross-seniority trades.

17.2 CREDITVaR FOR CDSs

We saw in Chapter 6 how to use an (adjusted) transition matrix to simulate both default events and forward spreads. The transition matrix can be adjusted to replicate CDS premiums rather than bond spreads in the obvious way. The TM gives the default probability by year, hence we can calculate the value of a payoff contingent of the default event and a payoff contingent on survival (the premium stream).

The approach described there can be applied equally well to a CDS portfolio where we generate the capital proceeds arising from credit events before the time horizon, and the forward value of all CDS positions (aggregated by rating).

18

CDS and Simulation

Single name CDS products are easily handled by closed form models (such as those described in section 9.5). In 18.2 we briefly look at simulation applied to single name CDS products. This is not of great importance in its own right, but serves as a simple introduction to some of the simulation techniques of Parts III and IV, and enables us to understand some of the key concepts in a relatively simple context.

In addition, in section 18.3 we describe an interesting way to use the simulations to obtain not just product price but also – from a single simulation set – the hazard rate (spread) sensitivity of the CDS.

First we state a few more results from the Poisson process which are of relevance now and later.

18.1 THE POISSON MODEL AND DEFAULT TIMES

If the hazard rate is constant the expected time to default is

$$T = \int_0^\infty t \cdot \lambda \cdot \exp(-\lambda \cdot t)dt = \frac{1}{\lambda} \tag{18.1}$$

We can also show that the standard deviation of the default time is $1/\lambda$.

If we imagine that we can simulate default times (see section 18.2) then the average of these simulated times will tend to T, and the standard deviation will also tend to T.

Note also that, given our assumption that the hazard rate is constant, if, in H years (a given time horizon), the name has survived then the hazard rate is still λ – in other words, the forward expected default time at H is still T. Therefore, if we take all the simulated default times, and exclude those that are less than H, then the average of the remainder will still be T.

Results are more complicated if the hazard rate is a function of time. For many thought experiments the use of a constant hazard rate is sufficient.

18.2 VALUATION BY MONTE CARLO SIMULATION

For a given name and hazard rate we shall simulate the default time for that name. We also ignore counterparty risk, assume that premiums are payable continuously, and that the interest rate is also constant.

The process is as follows.

1. Pick a random number from any convenient distribution[1]
2. Find the cumulative probability associated with that number.

[1] In the case of a single name we have much more freedom in choice of random number generation. As this is also the case for more than one name, we need to be more careful over the choice of random number generator and how these numbers are used.

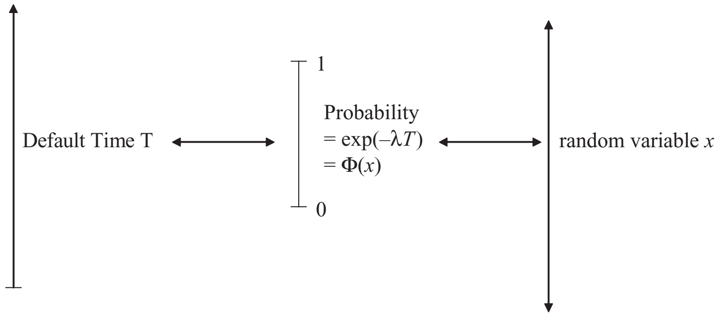

Figure 18.1 Normal random variable to default time.

3. Associate that probability with the probability of survival up to a certain time – the default time.
4. Given the default time t we know whether a capital payment (of 1-recovery) is made (if $t <$ maturity [M] of the CDS), and when the premium stream stops.

The simplest example is where we use a uniform distribution. See 'chapter 18.xls' worksheet 'CDS simulation' for an implementation of these examples. Pick a random x from [0,1]. Since the distribution is uniform the probability of getting a number of x or less is just x itself. So x is the probability of survival up to time t. We assume the hazard rate is constant[2] (i.e. there is no CDS term structure). Then we have

$$x = \exp(-\lambda t) \quad \text{hence} \quad t = \frac{-\ln(x)}{\lambda} \tag{18.2}$$

Alternatively, if we had chosen x from a Normal distribution then we would have to calculate the probability of seeing an x or less – using the cumulative Normal distribution – and our formula would become

$$t = \frac{-\ln(\Phi(x))}{\lambda} \tag{18.3}$$

Generally, under the Normal distribution but for a general hazard rate curve, we have

$$\Phi(x) = \exp\left(-\int_0^t \lambda(s)ds\right)$$

Techniques for solving the above equation for t, depend on the functional form chosen for the hazard rate. The process is illustrated in Figure 18.1.

We now value the cashflows for that simulation. Remember that we now know: (a) when the default takes place – each simulation is a realisation of the universe; (b) precisely what happens and when; (c) when the premium stream is terminated; and (d) when the capital payoff is made (if at all). Under one simulation everything is deterministic since all the uncertainty is

[2] In the case of a time dependent hazard rate the formula becomes an implicit one for the time t [where $\times$ equals the probability in equation (9.6)] and we have to find efficient numerical procedures to obtain the default time from the simulated random number.

Table 18.1 Simulated CDS premiums using different distributions.

Generating distribution	Mean	St Dev
Uniform	99.92	0.2
Normal	99.70	0.3
Exponential	99.97	0.1

reflected in the universe of all simulations, so valuation only requires discounting. If t is less than the CDS maturity then the benefits value is

$$B = (1 - R)\exp(-rt) \tag{18.4}$$

otherwise it is zero.

Similarly the value of premiums paid until the earlier of the default times and the CDS maturity, is

$$P = \frac{1 - \exp(-r \min(t, M))}{r}. \tag{18.5}$$

Now we repeat the process, adding to the value of benefits and premium stream for the value arising from this new simulation. We repeat many times – typically around one million simulations for a single name – and calculate the average (=expected) benefits value (and standard deviation), the average (=expected) premium value (and standard deviation), then finally the fair value premium (and standard deviation) or the net value of the trade (for an existing deal with known premium).

The process is very simple, requiring only a few lines of code to implement, and is illustrated in the 'chapter 18.xls' spreadsheet.

Table 18.1 gives the results of implementing the above process one million times for a single name with a hazard rate of 0.02 and a recovery of 50%. The correct CDS premium (for continuous premiums) is 100 bp.

Note that there is nothing special about using the Normal distribution in the case of simulation for a single name (or many uncorrelated names).

Exercise 1 Prove that under the simulation process described, the mean and standard deviations of the default time are the inverse of the hazard rate.

Take any (piecewise analytic) statistical distribution $f(x)$ with cumulative distribution $F(x)$. If we select x randomly then the default time is given by $t = \frac{-\ln(F(x))}{\lambda}$. The expected value of t is

$$E(t) = -\frac{1}{\lambda}\int_0^1 \ln(F(x))f(x)dx = -\frac{1}{\lambda}\int_\infty^0 y\exp(-y)dy = \frac{1}{\lambda}$$

where we have made a change in variable $y = \exp(-F(x))$. The integral on the right-hand side has been evaluated by integration by parts to give the required answer. Similarly, evaluating the expectation of the squares of the default time allows us to calculate the variance, and we get the required answer.

Notes on Credit Simulation

A. We simulate default time. This is very different from most simulation procedures in finance where a variable (for example, interest rates) is simulated along a time path. A single simulation produces a series of interest rates at different times – which are then used in product valuation. Simulation of default time – a single simulation – produces all relevant values for the product. This is much faster than path-based simulation but is only applicable to certain products. For example, it is not directly suitable for spread option products (we look at some modification in Part III).
B. A single simulation is a realisation of a particular universe. Producing many simulations is like looking at many possible universes or outcomes – only one of which actually occurs. Pricing based on the average of many simulations (also pricing off closed-form models) disguises the fact that the outcome may be very different from the expected result. This is an important point when we trade F2D protection hedged by a single name CDS. The realised P&L from the rehedging activity may be very different from expected.

18.3 SENSITIVITY

We could calculate the sensitivity of the CDS value to a change in the hazard rate (or spread) as follows.

1. Shift the hazard rate (by, say, 10 bp)
2. Recalculate all the default times (using the same set of random numbers used in the first set of simulations).
3. Recalculate the expected benefits value and the expected value of the premium stream, hence the expected value of the trade.
4. Subtract the value on this set of simulations from the value obtained on the first set, and divide by 10 to get the sensitivity per basis point (to an up-shift in hazard rate).

This is the process generally used when calculating sensitivities under complex CDO products. The uncertainty in the sensitivity is higher than the uncertainty in the valuation itself and we generally need to use more simulations when calculating sensitivities than just doing a valuation. Note that, if we had used a different set of random numbers in the shifted simulations, then the uncertainty would be even greater and we would have to use many more simulations to get a reliable estimate. It is good practice to reuse the same random numbers to generate the default times under the shifted hazard rate. Peculiarities of that random number sequence tend to cancel.

There is another way to calculate sensitivity which does not require recalculation of default times under a second simulation. We can extract for information from each simulated default time to produce the sensitivity as shown in Figure 19.2.

Note that if the hazard rate is shifted and default times are recalculated, all that happens is that every single default time in the valuation simulation run is shifted to an earlier default time by a factor depending on the hazard rate and the shift (equation (18.2) or (18.3)).

$$t^{shifted} = -\frac{\ln(F(x))}{\lambda + \Delta \cdot \lambda} = t\frac{1}{1+\Delta} \approx t - t \cdot \Delta \qquad (18.6)$$

For every default time that is shorter than the trade maturity, the payoff value increases because it is paid slightly earlier, and the premium stream is less because it is slightly shorter. For each such default, the increase in benefits value is just

$$\Delta B = B \cdot rt\Delta \tag{18.7}$$

and the change in the premium value is the loss in premium discounted by the time to default

$$\Delta P = -t \cdot \Delta \cdot \exp(-rt). \tag{18.8}$$

The two terms above give us the change in value arising from the shortening in the default time of those defaults that occurred before maturity. In addition, there are defaults that occur just after maturity that will shorten to just before maturity. All those default times between M and $M + M \cdot \Delta$ will now give rise to a benefits payment and an increase in the benefits value of

$$B_{\text{new}} = (1 - R)\exp(-rM) \tag{18.9}$$

and a reduction in premium whose value is

$$P_{\text{new}} = -\frac{M}{2}\Delta \cdot \exp(-rM). \tag{18.10}$$

We can evaluate these terms at the same time as we perform the simulation for value, and the sensitivity is then

$$S = \frac{\sum_{defaults} (B \cdot r \cdot t \cdot \Delta - t \cdot \Delta \cdot \exp(-rt)) + \sum_{nearDefaults} ((1 - R)\exp(-rM) - \frac{M}{2}\Delta \cdot \exp(-rM))}{sims \cdot \Delta \cdot \lambda} \tag{18.11}$$

Part III

Portfolio Products

19

Portfolio Product Types

In sections 19.1 to 19.4 we describe the core portfolio products. In addition to these products other types and variations are seen but – like securitisations and cashflow CDOs – rarely trade in the secondary markets. Cashflow CDOs and securitisations are included because of their important role in bringing certain assets to the market – in particular syndicated loans via LCDOs. We begin by discussing *n*th-to-default products which, although they are not major products, have many features in common with CDO structures and are often easier to test ideas on (both analytical and trading).

CDO deals that started in the late 1980s usually related to US high-yield reference names. Originally they were collateralised bond obligations (CBOs), introduced by Drexel Burham Lambert towards the end of the birth of the high-yield market. The subsequent high defaults in the bond market focused new efforts towards loans and collateralised loan obligations (CLOs), were introduced. Falling yields on loans meant that the funding arbitrage that drove CLOs would not work, so small high-yield bond buckets were introduced (usually 10% or less); the resulting structures with bonds and loans became known as 'collateralised debt obligations (CDOs)'. The CDO market was initially predominantly a US market, as US banks looked to focus efforts and origination and shift risk to institutions (insurance and pension funds). In Europe banks were more concerned about removing risk and optimising regulatory capital. This resulted in the birth of the synthetic market, and finally synthetic or partly synthetic structures on a much wider class of underlying reference risks. These risks were transferred to the CDO synthetically (via CDS-style contracts) rather than actually (bypassing ownership of the underlying asset). Modern terminology is to use 'CDO' as a generic name for these products and the term 'collateralised debt obligation' is now rarely used – many structures in fact cross boundaries between these old divisions and make them difficult to compartmentalise.

A CDO is best thought of as a product that provides protection on some or all of the 'tranches' of losses on a reference pool of risks (i.e. on instruments that define the magnitude of losses on the occurrence of a credit event). The terms 'tranched default basket' (TDB) or 'tranche of protection' are often used and provide a better description of the tranches of protection which form the underlying product. Occasionally a 'single tranche of a CDO' is misleadingly referred to as a 'single tranche CDO'. The term CDO may be used to refer to the entire structure of all tranches and the reference pool (and hedges), or often just refers to a single tranche of risk/protection – for example, protection on losses between USD 40m and USD 60m arising from a reference pool comprising 100 entities and a total notional of USD 1,000m.

CDOs can be considered in a number of ways. Cash CDOs are often compared to mini-banks; they raise capital usually for term; and invest in credit assets that yield more than the cost of funds. Like a bank they have senior debt, subordinated debt and equity. It was the discrepancy between capital required to support investment grade loans on balance sheet versus securitising them via a CLO that drove the creation of the synthetic market. Synthetic CDOs can be directly compared to the insurance–reinsurance market. A reference pool can be considered to be the whole risk. A protection seller takes on the risk for a premium and re-sells the excess risk above their capacity to a re-insurer, effectively tranching the risk, with (usually) decreasing risk premium. This analogy can be used to explain why so many insurance companies, both multiline and monoline, are involved in the CDS market.

It should be noted that a reference pool does not need to exist as an actual portfolio 'hedging' the CDO or *n*th-to-default basket. The term 'reference' pool correctly implies that the pool of risks does not actually need to exist as a portfolio somewhere, the term simply defines a set of events (survival or default) and defines the risk. The CDO is the protection on losses relating to that reference pool. Note also that there are usually no substitution rights on mergers of names in the reference pool of a CDO. However, if there is a concentration limit on the names and the merged entity violates that, then it may be required to be sold down.

19.1 *N*TH-TO-DEFAULT BASKETS

*N*th-to-default (N2D) products are small portfolio products both in terms of the number of reference names and the size of the deal. Trading volumes (in terms of numbers of deals) have been high compared to bespoke CDO deals but nominal exposures are typically less than EUR 25m. Trading is almost entirely in first-to-default products. These products provide a means of trading 'correlation' on small numbers of reference entities.

N2D products are modelled using the same mathematical framework as that for CDOs – only product cashflows differ – and form a useful tool for testing and understanding many features that are applicable to both sets of products.

19.1.1 First-to-Default Product Definition and Example

A first-to-default (F2D) swap is similar to a plain vanilla credit default swap. However, a first-to-default swap gives credit protection on a list of reference names instead of just one reference name, and will result in a payout after the first default only. The contract then terminates. (Figure 19.1.)

The typical contract pays par on the first credit event in return for debt issued by the reference name that suffered the credit event – 'physical settlement': the alternative 'cash settlement' may become more common following changes to the CDS market in 2009. Premiums are paid quarterly in arrears with a proportion up to the default date.

Documentation

Documentation follows CDS documentation. A reference obligation is given for each reference name; documentation is changed slightly to clarify the contingent payment made on the first

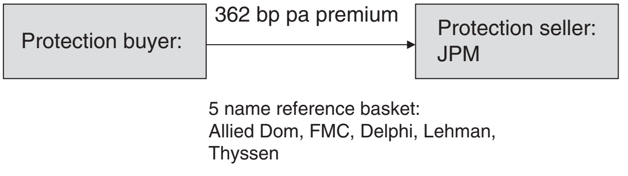

Figure 19.1 First-to-default basket: pre-credit event.

credit event, and termination of premium and future obligations on that event. The question of mergers is discussed below.

Premium Quotation Basis

It is usual to quote the premium as a percentage of the sum of the CDS premiums of the component names. CDS spreads may vary a little over a few hours or days while a potential trade is being discussed, but the F2D premium will remain close to a certain percentage (which varies according to the underlying reference names) of the sum of the premiums. When documentation is signed, the actual premium in bp per annum (not the percentage of the sum) appears.

Example Deutsche Bank (say) sells EUR 10m 5-year F2D protection on five reference entities – Allied Domecq plc, Fiat, Delphi Corp., Banco Santander, and ThyssenKrupp AG – in return for a fixed premium of 362 bp p.a. In the discussions prior to signing the deal the sum of CDS premiums on the individual names was around 600 bp, and a premium would have been indicated as (say) 60%. The buyer pays the 362 bp premium to Deutsche Bank quarterly in arrears until the earlier of the deal matures or a credit event on one of the names.

Suppose a credit event occurs on Delphi before the maturity of the trade (Figure 19.2). Deutsche Bank pays EUR 10m in return for delivery of defaulted debt of Delphi. The protection buyer pays a proportional premium up to the date of default. The contract then terminates and no further protection exists on the remaining four names.

Note: N2D baskets should not be confused with ‘average baskets’ (see Part II). Both of these products are sometimes confusingly referred to as ‘baskets’. An EUR 10m total exposure, equally weighted, average basket on the above names would initially cost 600 bp and, if the Delphi CDS premium was 140 bp, then on a Delphi credit event EUR 2m Delphi debt would be delivered and the average basket would continue in EUR 2m on each of the remaining names, subject to a reduced premium of 460 bp on the total EUR 8m.

19.1.2 Documentation and Takeovers

A new feature of portfolio trades is the merger or takeover risk. A CDS trade usually provides protection on the reference name or its successors (i.e. protection continues on the resulting entity after a merger or takeover – as defined by the ISDA ‘successor’ language or similar). A particular problem of portfolio products is that a merger of two of the names in the reference

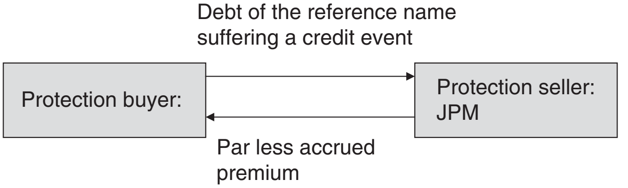

Figure 19.2 First-to-default basket: credit event.

pool results in a smaller basket (one fewer names). Current documentation treats mergers in one of two ways.

1. The documentation does not make any particular provision for the event of merger of two or more names in the basket. Note that a credit event of the successor entity may be assumed to be a credit event of both of the predecessor entities, so such an event would trigger a second-to-default contract at the same time as a first-to-default contract.
2. There may be replacement provision, which may operate along the following lines: one side suggests a list of potential replacement names and the other side selects from that list. The list may be restricted in rating, industry or spread – for example, any name on the list must have the same industry group as either of the merged entities, and the same rating as either of those entities had at the time the trade was done, or the suggested replacement names must have similar spreads to the merged entity.

The ISDA N2D confirmation gives the counterparties the choice of 'no replacement' or of 'with replacement'. Under the ISDA replacement provision, the replacement entity must be in the same industry group, and have a (bid) spread no greater than 110% of the spread on the name it is replacing (as determined by a dealer poll conducted by the protection buyer). The buyer must deliver a list of three or more eligible names to the seller within 10 business days of the successor event, and the seller then has 5 days to select the replacement entity.

What trading opportunities does a replacement clause open up?

On a merger event, the side that draws up the list has a contingent option (triggered by merger) to suggest names with relatively wide spread. The restrictions in the clause are designed to limit the value of this contingent option. In addition, the nature of the basket can be changed – the buyer of protection would like to increase the diversity[1] of the basket (which increases the value of the deal to the buyer) while the seller would like to reduce the diversity (which reduces the value of the deal).

What risk is present if there is no replacement clause?

The reduction of the number of names itself causes a drop in the value of the deal. Take the example in section 19.1.1, and suppose two names merged. The sum of CDS premiums for the reduced number of names is now (say) 480 bp, and the fair value premium for the basket drops to (say) 300 bp, so the buyer is now paying 62 bp above fair value. There is a further pricing impact: a four name basket is seen as different from a five name basket (there is less diversity) and is therefore priced rather differently. We shall see that the implied correlation on smaller baskets tends to be higher than on larger baskets, reducing their value to a buyer. The pricing correlation would rise, so the fair value premium would drop further as a percentage of the sum of spreads – hence, the value of the deal to the buyer will drop (on the above example this might give rise to a further drop in the fair value premium to 285 bp). A merger will therefore cause a significant downward jump in the value of the basket related directly to a reduction in the number of names and also to a change in the pricing correlation. The combined effect is that the F2D writer makes a mark-to-market profit and the buyer a loss, on merger events with 'no replacement'.

[1] For 'diversity' we can also read 'correlation'.

One of the leading market makers in this product argues that a 'no replacement' clause is preferable, primarily on the grounds that if replacement applies, and if a long basket is hedged with a short basket on the same names, then after a merger and replacement, the two baskets could refer to different portfolios. In addition, there is a 'consistency' argument: standard CDO contracts do not have a replacement clause although the greater number of names in a CDO compared to an F2D product means lower concentration and replacement is less important.

19.1.3 Second-(and Higher)-to-Default

An nth-to-default product is similar to an F2D except that nothing is paid on the ith credit event if $i < n$. If the basket reaches the nth credit event, then defaulted debt is delivered in return for the notional amount (less accrued premium) and the contract terminates.

Example Suppose, on the above reference pool, Deutsche Bank wrote EUR 10m second-to-default protection for 140 bp. The deal is unaffected by the first credit event (Delphi), except that it now effectively becomes a F2D on the remaining names. The premium continues at 140 bp on EUR 10m until maturity, or until a second credit event occurs. Suppose Fiat suffers a credit event after Delphi and before the maturity of the deal. Then the debt of Fiat is delivered to Deutsche Bank in return for par (less accrued premium) and the contract terminates.

Further product variations occur: for example, the first two to default from five and/or the inclusion of fixed recoveries.

19.1.4 Applications

In practice only F2D products trade frequently. Typical notional exposure is the same as (single name) CDS deals – around 10–20m, and 5-year deals are the most common. F2D baskets are regarded as the vanilla correlation trading product. They give a 'geared' exposure. In the above example the writer is taking on risk to reference names with an average spread of 120 bp, yet gets paid 362 bp.

Correlation Trading

The main application of F2D baskets is as a means of trading 'correlation' risk. The pricing of a F2D basket is dependent not only on the individual risks, but also on the way in which credit events on one name relate to credit events on another. We refer to the relationship as 'credit event correlation', 'credit correlation' or just 'correlation'. We would intuitively expect that correlation of hazard rates should be the source of the 'correlation' behind F2D products. We shall find in Chapter 22 that this is indeed the case, although for practical reasons, we choose to implement a model not by correlating hazard rates but by correlating default times.

We shall see how F2D can be used to trade correlation once we have developed a pricing model in the following chapters.

Mergers

F2D baskets can also be used to trade merger views (see section 19.1.2 above). Writing baskets, including potential merger candidates, and with a 'no replacement' clause, gives a mark-to-market profit on a merger event on the short protection trade. Deals can be hedged by buying protection 'with replacement'. If the premium differential is less than the jump in the premium expected on a merger (which can be calculated) times the probability of the merger (a guess) times a discount factor dependent on the timing of the merger, then an overall profit is made.

Standard Baskets

F2D products can include any names which the buyer and seller agree to trade. For a time certain standard baskets became popular and some market makers regularly quoted prices on such baskets. This allowed the investor to trade sector views combined with correlation views on relatively liquid (compared to bespoke) instruments. Trading in standard baskets has declined and market makers now no longer quote prices on such structures.

Exercise 1 A 5-year CLN in USD 10m, issued at par, has as its underlying risk the 'first to default' out of a list of seven names. If there is a credit event on any one of the names, the note terminates and the investor receives nothing. How would you hedge the CLN and, if the F2D costs 630 bp what spread would you expect to pay on the CLN? If the CLN is capital guaranteed with a 15-year life and a 5-year risk, what would the premium be?

In the remainder of this section we shall look at some special cases of correlation that will allow us to calculate some significant results and teach us a considerable amount about pricing generally.

19.2 'SYNTHETIC' CDOs

In this section we look at some standard indices and, standard tranching of these indices;, we explain the operation of synthetic CDOs; and look at variations of synthetic CDOs and some of their applications.

19.2.1 Standard Indices: Markit iTraxx and Markit CDX

Background and Terminology

Credit portfolio indices and standard portfolio CD products largely developed since around the year 2000. The Markit iTraxx and Markit CDX products outlined below, and described in further detail in the Appendix, were introduced in 2004. The indices are managed by Markit, and replace the TRAC-X (Morgan Stanley & JP Morgan) and iBoxx CDS indices that were set up and managed by groups of banks and other institutions. These latter indices were introduced in 2003, replacing earlier products such as Morgan Stanley's 'TRACERS', and rapidly became popular. The latest versions of these indices will undoubtedly change over time, but most of the general features described below are shared with their legacy products and will probably be inherited by successors.

The 'indices' are sets of reference entities chosen by Markit based on CDS trading volume data supplied by around 20 banking institutions – see the Appendix for details (and note that a recent minor change has been the introduction of standardised coupons). Trading on these indices, and on tranched CDO products based in these indices, takes place between counterparties (banks and other institutions) and does not involve Markit. There is the possibility that interbank transactions may clear in future through a settlement exchange.

The index product allows trading in relatively small size[2] yet gives exposure to a diverse portfolio of risks. As such it is a quick and low cost means of obtaining diversified exposure to spread risk. It also provides a clear benchmark of the performance of credit markets. An index provides a liquid, efficient, convenient and transparent means of trading general credit risk.

Composition of the Indices

The single name CDS market has strived to build up liquidity in as broad a range of underlying names as possible. The portfolio credit market has simultaneously developed CDOs based on portfolio structures. The indices managed by Markit are known as the Markit iTraxx (European and Asian names) and Markit CDX (North American names and managed Markit) are based on the most liquid names in the single name CDS market selected from a range of industry sectors. The main Markit CDX indices are divided into investment grade (CDX.IG) and high-yield (CDX.HY) indices. The current (July 2009) make-up of these portfolios are described in more detail in the Appendix. Key features are the following.

1. The indices are based on active names in the CDS market – market makers are committed to providing daily data to enable the calculation of the index (being the average CDS premium).
2. A variety of indices are maintained: from a diversified investment grade portfolio of 125 names (iTraxx and CDX.IG), a portfolio of 100 names (CDX.HY), to sector portfolios of 10–30 names, a 25 name subordinated financial portfolio, high premium and sub-investment grade portfolios. All names are equally weighted in each CDS-based index.
3. Reference portfolios are redefined every six months, are of standard 3-, 5-, 7- and 10-year maturities (to 20 June or December), and are known as different 'series'. Deals related to 'series 10', for example, may be rolled into 'series 11' (on then market terms) when the next series is introduced, or the 'series 10' deal may be held to maturity (with the original list of reference names, of course, remaining unchanged). To emphasise: the list of names in a specific series is static but a new series may replace (and typically does replace) some names with other names.
4. The average CDS premium $\times$ is calculated at inception and is defined to be the premium for the single tranche CDO (see below) based on that portfolio. Deals done on the portfolio will change hands with the payment of a capital sum (representing the off-market value of the CDO and a 'basis' – see below).

[2] If an investor deals in a 9m [3%, 6%] tranche then the reference pool is 300m, and the notional of each underlying reference entity is 300/125 = 2.4m. Since the tranche itself is a liquid-traded product, the counterparty hedges with that product rather than many small deals in the underlying reference names.

Single Tranche [0%, 100%] CDO Products

Tranche terminology: 'attachment' and 'detachment' points, 'thickness'

We define some terminology used to describe CDO tranches here since it is also useful in the context of risk on an entire portfolio. If we think of a pool of credit risks with a total exposure of EUR 1,000m, then initially losses are zero but as time goes on losses will rise, theoretically, all names could default with no recoveries, leading to a loss of the full 1bn notional. A portfolio of these risks is indeed subject to losses over this entire range. More generally a 'CDO tranche' may have an '**attachment point**' of a, and a '**detachment point**' of b, describes as an $[a, b]$ tranche, meaning that the writer of protection on this tranche starts to make payments once losses rise above $a\%$ of notional but stops once (if) they rise to $b\%$ – with a maximum payment of $(b-a)\%$. The '**thickness**' of the tranche is $(b-a)$. Attachment and detachment points, and the thickness, may also be referred to as notional amounts.

Protection on the entire portfolio of names can be thought of as a [0, 100] CDO tranche.

Average basket versus 'single tranche CDO'

Index-based CDS contracts are actually set up as single tranche CDOs rather than pure average baskets. An average basket (see Chapter 8) will have a premium rate associated with each reference name, and after default of that reference entity the total premium paid on the basket reduces by the premium rate associated with the defaulted name multiplied by the notional of that name. In practice index-based CDSs are [0, 100] tranches ('single tranche CDOs' covering the entire risk from 0% of losses to 100% of losses), and have a premium rate defined by the portfolio at the outset.[3] A default of a single entity does not affect the premium rate – the total premium falls by the rate applied to the notional for the entity defaulting. Thus the usual set-up for an index CDS is more correctly described as a CDO on the reference portfolio, where the CDO has a single tranche only. (Note that the term 'single tranche CDO' is often used to mean a '$[a, b]$ tranche' rather than a [0, 100] tranche.)

Funded and Unfunded Forms

All the portfolios trade on a single tranche basis, i.e. the trade gives protection on the entire portfolio covering all losses between 0 and 100% subject to a predefined premium rate levied on the remaining notional only. All names in a reference portfolio have equal weightings. The 'portfolio' defines the reference pool, and the initial premium on the single tranche CDO is also defined when the portfolio composition is first announced. Deals may be in unfunded (CDS) form, or funded.

Example (quoted from Markit) Counterparty buys EUR 10m Markit iTraxx Europe Exposure in Unfunded / CDS Form

[3] The premium is the average spread. Even at outset this is not the correct 'fair value' premium, but trading takes place with an exchange of capital up-front.

No Credit Event

- At issue of a given series, assume credit spread ('premium') of 45 bp.
- Market maker pays to counterparty 45 bp per annum quarterly on notional amount of €10m.
- With no credit cvents, the counterparty will continue to receive premium on original notional amount until maturity.

Credit Event – Physical Settlement

- At issue of a given series, assume credit spread ('premium') of 45 bp.
- Market maker pays to counterparty 45 bp per annum quarterly on notional amount of €10m.
- A credit event occurs on reference entity, for example, in year 3.
- Reference entity weighting is 0.8%.
- Counterparty pays to market maker (0.8% × 10 000 000) = €80 000, and market maker delivers to counterparty €80 000 nominal face value of deliverable obligations of the reference entity.
- Notional amount on which premium is paid reduces by 0.8% to 99.2% × 10 000 000 = €9 920 000.
- Post-credit event, counterparty receives premium of 45 bp on €9.92m until maturity subject to any further credit events.

When an index suffers a credit event when one of the reference entities itself suffers a credit event, this triggers a capital payment and a reduction in the outstanding notional on the junior tranche. The index then continues with one less name and the initial notional is reduced by the notional on that name.

The funded form would carry a EURIBOR + 45 bp spread and mirrors the above: on a cash-settled credit event the investor receives the recovery amount and the notional of the note is effectively reduced by the defaulting notional (0.8%).

Initial Premium

We can see intuitively that the 'correct' premium for a single tranche CDO should be slightly below that for an average basket. Generally the higher spread names are expected to default first, which will reduce the premium on the average basket faster than a [0, 100] tranche. The effect is not large because the constituents have broadly similar spreads. For a 100 name equally weighted 5-year index with average spread of 60 bp and spreads in the 20–100 bp range, the average basket premium is initially 60 bp and the [0, 100] tranche premium is 59 bp. A new tranche deal may carry a 60 bp premium but will change hands with an additional capital payment up front.

Secondary Trading

Since the premium on the (single) tranche is fixed, secondary trading takes place with the payment of an up-front premium.

Calculation of the capital value of index protection should be performed within a CDO valuation tool, to correctly take into account the premium waterfall. Suppose the deal is traded on a premium date 5 years before maturity, where the (fixed) premium on the index is 60 bp but, because average spreads have risen, the fair premium for the tranche has risen to 70 bp.

The approximate calculation is to discount the 10 bp excess quarterly premiums at interest rates (say 4%) plus the hazard rate as if the CDO were a single reference entity (i.e. assume a 40% recovery, so the hazard rate is 0.7/(1 – 0.4) = 1.17%, and we then discount at 5.17%) to give 0.44%. The capital value according to the Copula CDO pricing model may be 0.40%. Errors increase the more the diversity of spreads increases.

In addition to this difference between the average spread and the correct premium for the single tranche CDO, there will also be a 'basis' between where the product actually trades and the theoretically correct premium. Quoting from JP Morgan 'Credit Derivatives Strategy: Introducing DJ iTraxx Europe Series 1', June 2004:

As a note of caution, we have to emphasise that the DJ iTraxx index will not be trading exactly in line with the average of the single name CDS. Broadly there are two reasons for this. Firstly, there is a Basis to Theoretical between traded DJ iTraxx and the spread derived from the underlying CDS within it. Secondly there is the difference between the theoretical DJ iTraxx spread and the average spread level of the single name CDS. ... the basis [to theoretical] has traded in a range between –8 bp and +11 bp from July 2003 until today.

19.2.2 Index Options and Modelling Spread

The market for options on tranches of the iTraxx and CDX portfolios is less well developed and less active than that on the tranches themselves.

Product Description

- A limited range of maturity dates trade (the next and the following roll dates for the underlying portfolios).
- At-the-money forward strike only.
- European option.
- Up-front premium.
- Physical settlement (the buyer of the option is delivered a position in the underlying tranche; if the strike premium is different from the tranche premium then a capital sum is payable).
- If credit events on the underlying have already occurred then the protection can be immediately triggered leading to a capital payment or receipt.
- 'Call' is similar to a bond call – it is the right to buy risk/sell protection; similarly a 'put' is the right to buy protection; straddles also trade

Example Suppose the premium on the iTraxx [0, 100] is 51 bp mid. Assume the portfolio is a single name CDS with a gently upward-sloping CDS curve. We can calculate the forward CDS premium in 6 months' time – suppose this is 55 bp. Prices for the options might be

- Straddle: 50–70 cents
- Put: 30–40 cents
- Call: 20–30 cents

We can get a rough sanity check on these numbers as follows. At 55 bp the equivalent up-front premium (= expected loss) is about 55 bp $\times$ 4.5 = 2.5% where we assume that the 'duration' of the portfolio is 4.5. The volatility of the portfolio premium (and spreads) might be 50% per annum – for a 6-month option the volatility of the forward value of the up-front premium is $50\%/\sqrt{2} = 35\%$. An at-the-money option under the Black–Scholes model has a premium of

40% of the cash premium times the volatility for the period: $0.4 \times 2.5 \times 0.35 = 35$ cents (for either the call or the put). This is at the high end of the market quote, which implies that the volatility used was a little less than 50%.

Exercise 2 *What is the implied volatility on the above straddle?*

Typically these options are priced using the Black–Scholes model. We discuss the errors with this approach below.

Volatility

Volatility of the tranche premium is the key new variable driving the price of the option. This in turn is driven by the volatility of the CDS premiums on the reference names, conditional on no default, and on the distribution of defaults up to the exercise date.

If default times on the portfolio were uncorrelated, then we could correctly price an option on a tranche as follows:

1. Simulate defaults using the uncorrelated Normal Copula.
2. At the exercise date independently simulate forward spreads on non-defaulted names using an assumed spread distribution and volatility.
3. Valuation at the forward date can take into account defaults that occurred before the exercise date (the payoff under the portfolio protection can still be claimed), and the value of the portfolio protection on the remaining names.

The choice of the stochastic hazard rate process is completely open to us because all the reference entities are independent in the above situation. If, instead, we assumed that all the names were 100% correlated, then if one of the names (the highest spread name if all the recovery rates are equal) defaults before the exercise date, all the other default times are known precisely. In particular, the forward hazard rate process on the remaining names is known. In the general case where there is some default time correlation, we need a forward hazard rate process which is consistent with the defaults that have actually occurred, and with the remaining future defaults implied under the Normal Copula. The more defaults that have occurred on a particular simulation, the more likely it is that there will be several further defaults during the life of the deal (i.e. the higher will be the implied spread levels on those names at the exercise date). See section 19.3.1 for further discussion of this topic.

We need a hazard rate simulation process which is consistent with default time simulation model. There are currently two approaches:

1. Use an Archimedean Copula. It is then possible to build in a hazard rate process consistent with the Copula (but having the unrealistic implications of Archimedean Copulas) (Schönbucher and Schubert, 2001).
2. Use the (calibrated) transition matrix approach (described in Part I), replacing the portfolio by a portfolio of seven spread levels (effectively defining our own ratings).

An alternative is an extension of the outdated BET approach described in the first edition. For each industry (or other 'tag') replace the actual number of holdings by a smaller number of uncorrelated holdings which have the same hazard rate, recovery rate and notional size. Then use the multinomial distribution to simulate defaults and survival to the exercise date. For the surviving names we simulate forward pseudo-hazard rates (by industry) using any convenient distribution (say lognormal). We can then calculate the forward value of the underlying tranche or portfolio.

19.2.3 CDO Structures on Standard Indices

The iTraxx and CDX portfolios are used to create 'static synthetic CDOs' (the meaning of these terms will become clear in the following sections.) The key difference between these index products and 'bespoke'[4] ' static synthetic CDOs (described in the following section) is that the former actively trade in the secondary market, whereas the latter are typically primary trades only.

The motivation for developing CDO products based on an index is to meet the needs of traders and investors who require a portfolio hedge or a portfolio exposure. Standard tranche products are of major benefit to the professional institutions structuring credit solutions for clients. Where the bespoke client structure introduces correlation, the presence of an actively traded CDO gives a means of hedging the correlation risk embedded in the client structure using a liquid and relatively tightly priced standard instrument. These indices and products based on them are the key liquid CDO trading products and are the primary hedging vehicles for correlation and other risks arising from bespoke CDO transactions. They also provide the data source for 'implied correlation' data, transparent pricing information on implied correlation and its relationship to tranches for a relatively diverse portfolio.

Standard Tranched CDO Structures

The 125 name European (iTraxx) (Table 19.1) and North American (CDX.IG) and the 100 name CDX.HY portfolios are additionally traded with a non-trivial tranching structure – these individual tranches trade actively, and also certain super-tranches (e.g. the [3, 100]) constructed from them. The tranche attachment points are as follows:

1. iTraxx portfolio: these tranches are [0, 3], [3, 6], [6, 9], [9, 12], [12, 22] and, [22, 100].
2. CDX.IG portfolio: these tranches are [0, 3], [3, 7], [7, 10], [10, 15], [15, 30], and [30, 100].
3. CDX.HY portfolio: these tranches are [0, 10], [10, 15], [15, 25], [25, 35] and, [35, 100].

These CDO tranches, which are examples of static synthetic CDOs and operate in the same way as bespoke synthetic CDOs, are described fully in the following section.

From the point of view of a deal done on a specific reference portfolio (series) the CDO is a static synthetic deal. In 2 years' time when the deal has 3 years outstanding life, current

Table 19.1 Example indicative quotes on the iTraxx Europe Series 9 (July 2009)

5-year / tranche	Bid	Ask	Implied base correlation
[0, 3] + 500 bp p.a.	46%	46.5%	46.4%
[3, 6] + 500 bp p.a.	8.375%	8.875%	53.3%
[6, 9] + 500 bp p.a.	–3.75%	–3.25%	58.5%
[9, 12]	181 bp p.a.	185 bp p.a.	65.8%
[12, 22]	85 bp p.a.	87 bp p.a.	85.1%

[4] A deal based on a list of reference entities, and a tranching structure, chosen to suit the client.

average spreads may not be quoted by market makers (and currently there is no plan to quote such spreads). The then current spreads will refer only to the then current (5- and 10-year) series, and many traders and investors will roll their deals into the new series as it is announced.

Applications of Standard CDOs

The portfolio products (single tranche CDOs) can be used to go long or short protection (bearish or bullish of the underlying synthetic bonds). This can be used as a hedge, as an outright directional play, or as part of a sector trade using the smaller sector-based portfolios.

The tranched CDO products define fairvalue market levels in those CDO tranches. *Most importantly, an active market in CDO tranches allows banks to create CDO structures tailored to clients' needs and put an effective hedge in place*. We discuss this in more detail in section 22.3.

19.2.4 Bespoke Synthetic CDOs

Product Definition

A **reference pool** is a list of credit risky entities and related assets. The list defines the legal entities on which the risk is taken, and defines the asset, the size of the exposure, and how the loss is to be determined should a credit event occur.

The reference pool is **static** if, once it has been defined and a deal done on that reference pool, no additions or substitutions of reference entities or assets occur. Only credit events affect the pool, reducing the number of entities and assets covered.

A reference pool is **synthetic** if some or all of the instruments defining losses are non-cash assets. Examples include CDS or legal agreements transferring risk between one body (typically a commercial bank) and another (typically an SPV). Note that the CDO tranches defined on the iTraxx and CDX indices are static and synthetic – cashflows on the tranches are defined by reference to events on underlying reference entities as captured by CDS contracts on those entities. We shall look at a variety of other examples later.

For the purposes of this section, think of a **static synthetic reference pool** as a fixed list of CDS deals. Such a pool is typically long [or short] of risk only, but some bespoke deals may refer to a pool of mixed risks: long and short. Note that the reference pool is merely a list or reference entities and/or assets. It need not exist as a portfolio of assets owned by an investor.

A **tranche of protection** (also called a 'tranche', a 'CDO tranche', or – rather misleadingly and now rarely used – an '*n*th loss piece') on the reference pool with attachment and detachment points [a, b] reimburses losses between $a\%$ and $b\%$ on the total notional of the pool. The **notional size** of the tranche is the percentage size multiplied by the notional size of the reference pool. Note that the attachment and detachment points refer to losses, not to notional exposures.

Example 1 The [3%, 6%] tranche on a EUR 1bn reference pool of 125 reference entities covers losses from EUR 30m up to EUR 60m related to that reference pool. If there have been six defaults (on $6 \times$ EUR 8m = 48m notional) then the actual losses could be anywhere between 0 and 48m: if the loss is less than 30m, then no capital payment is made by the seller of protection on the tranche to the buyer, but if the loss is (say) 35m then a 5m payment is made.

The collection of all tranches (collectively covering 0–100% of losses), or any single tranche, is referred to as a **CDO**. A more precise terminology would be to consistently use ‘CDO tranche’ or ‘CDO structure’ but, in practice, context usually makes it clear what is meant. Many varieties of CDO, and many variations in the product, are described in this section.

The tranches of protection also go under a variety of different names which carry less information than knowing the attachment points.

1. The **first loss piece** (**FLP**) of a CDO absorbs the losses up to a predefined percentage of the sum of the notional on a portfolio of reference names. It is the [0, *b*] tranche of a CDO, also called the **equity piece**. The latter term has come into use because in some structures the originator of the deal ‘retains’ (i.e. writes protection on) the equity piece and also keeps all residual income after paying contractual costs. In this case the originator of the deal is more like an equity shareholder in that the return is very uncertain, rather than a bond holder.
2. A **mezzanine** or **second/third/. . . loss** piece starts to absorb losses once the first loss piece is exhausted. Mezzanine tranches may also be called ‘junior’ (more risky) or ‘senior’ (less risky).
3. The **senior** piece is the least risky [*a*, 100] tranche. If the senior tranche is rated AAA, and if the reference pool could suffer several defaults before the senior tranche loses its AAA status, then it is referred to as **super senior**. The tranche adjacent to the super senior tranche is then sometimes called the **senior** tranche.
4. Actual names used in the CDO documentation vary – synthetic static CDOs are increasingly giving just the attachment points. Cash deals and older transactions may give the tranches names – A, B, C, . . . being typical, albeit unimaginative, names.

Usually there are 3–7 tranches of protection covering the entire notional of the portfolio, though single tranche [0, 100] CDOs are now common (and often thought of, incorrectly, as average baskets). If a deal is a single tranche then the attachment points of the other tranches are irrelevant.

Figure 19.3 and Table 19.2 illustrate the above for a standardised tranching structure based on the Markit iTraxx portfolio. Every reference entity exposure is the same notional size – say 8m to give a total notional on 125 names of EUR 1bn. Note that the reference pool is an index portfolio managed by Markit while the CDO tranches are products traded by various banks and others based on the index portfolio.

The protection writer of the tranche receives a premium – this is covered in more detail in the following section. First we consider how credit events trigger cashflows.

Impact of Credit Events on the Junior Tranches

Suppose a deal is done on 9 July 2009 for a maturity of 20 December 2013, and credit events on reference names occur during the life of the deal. Each of the 125 reference names accounts for 0.8% of notional and suppose that recovery is 30% on the first, fourth, seventh, etc., default; 50% on the second, fifth, eighth; and 70% on the others, so credit events causes a loss of

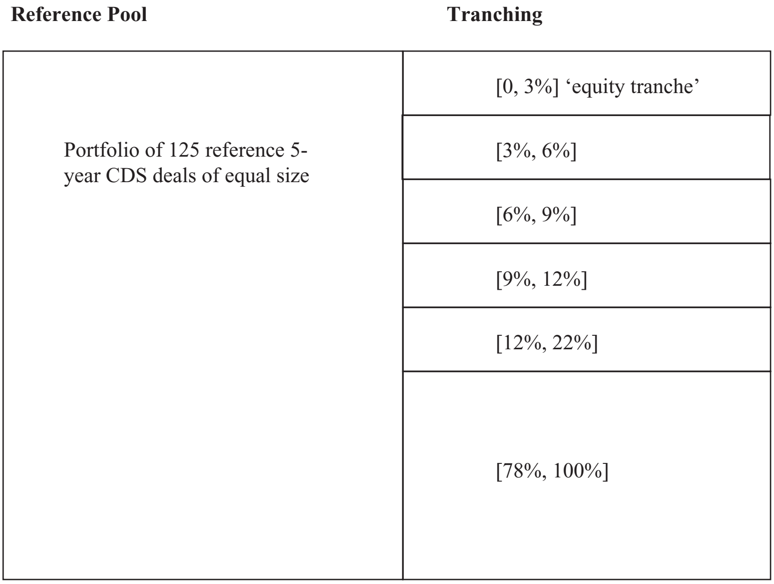

Figure 19.3 Standard tranching structure based on the iTraxx reference pool.

0.8% × (0.7, 0.5, 0.3) = (0.56%, 0.4%, 0.24%) on the first three defaults. The impact of the defaults on the junior tranches is given in Table 19.3.

Impact of Credit Events on the Senior Tranches

Example 2 Suppose the first-loss-piece of a CDO is 2%, and suppose a default amounting to 1% of notional (one name out of 100 in the reference pool) occurs and the loss to the FLP is 0.5%. The notional of this tranche reduces from 2% to 1.5%. The notional of the reference pool is 99% of the original amount. Suppose there is a single mezzanine tranche of 3%, and a senior tranche of 95%. After this first exposure the total protection may be 1.5% + 3% + 95% = 99.5%. The mezzanine piece still protects 3% of the original notional, and the senior piece also seems to protect 95% of the original portfolio. If the entire portfolio defaulted, with 50% recoveries, then the equity and mezzanine tranches would suffer a further 4.5% loss, and the senior piece would suffer the balance, (99% × 0.5) – 4.5% = 45%. The outstanding notional on the senior piece (50%) would remain even although there was no surviving portfolio to protect, and the senior piece would continue to receive premium (a reward for no risk at this stage).

Although some CDO senior tranches are documented as above, the current standard for synthetic CDOs is the following. For every credit event affecting the reference pool, recovery amounts are deducted from the notional exposure of the senior tranche. After the first default in the above example, the senior tranche reduces to 94.5% of notional and, if all names default, the senior tranche suffers losses of 45% (total portfolio loss less 2% + 3%), and has a remaining notional of zero (95% – 45% losses – 50% recoveries) = 0.

Table 19.2 Partial listing of the reference pool for the Markit iTraxx Europe Series 9 portfolio (July 2009)

	30-Jun–0-9	1Y	2Y	3Y	4Y	5Y
Accor SA	SEN	104.9	114.5	123.7	131.1	135.7
Aegon NV	SEN	201.1	200.8	200.5	200.5	200.5
Akzo Nobel NV	SEN	57.9	65.9	71.7	76.8	80.0
Allianz SE	SEN	66.4	69.3	73.6	77.8	80.3
Arcelor Finance SCA	SEN	25.1	82.7	88.0	102.4	112.0
Assicurazioni Generali	SEN	69.1	71.3	72.8	74.4	75.4
Auchan SA	SEN	29.6	38.4	47.2	54.2	58.5
Aviva Plc	SEN	193.5	198.0	202.6	207.4	210.3
Axa	SEN	108.3	112.8	115.8	118.4	120.0
BASF AG	SEN	57.1	61.0	64.6	67.6	69.5
BNP Paribas	SEN	49.0	54.8	60.7	66.1	69.4
BT Group Plc	SEN	123.0	142.8	150.3	162.2	169.5
Banca Monte dei Paschi di Siena SpA	SEN	79.1	80.9	82.4	83.9	84.8
Banco Bilbao Vizcaya Argentaria SA	SEN	78.7	83.3	86.7	89.9	91.8
Banco Espirito Santo SA	SEN	113.5	117.6	120.1	123.5	125.6
Banco Santander SA	SEN	75.4	79.2	80.6	85.0	87.7
Bank of Scotland	SEN	229.1	199.2	166.2	181.5	173.5
Barclays Bank Plc	SEN	124.0	128.7	131.5	134.2	135.9
Bayer AG	SEN	47.2	49.9	53.2	55.8	57.5
Bayerische Motoren Werke AG	SEN	123.0	140.5	155.1	165.9	172.5
Bertelsmann AG	SEN	187.6	205.5	212.3	217.8	221.2
British American Tobacco Plc	SEN	60.5	65.9	70.5	76.4	80.1
Casino Guichard Perrachon SA	SEN	83.5	95.1	107.3	118.1	127.9
Cadbury Schweppes Plc	SEN	21.9	27.8	30.0	32.9	34.6
Carrefour S.A.	SEN	35.9	39.4	43.2	47.8	50.6
Centrica Plc	SEN	50.4	53.9	57.1	59.5	61.1
Ciba Specialty Chemicals AG	SEN	24.8	26.2	28.3	29.9	30.8

Note that, since premium is usually related to notional outstanding, the former treatment of losses and recoveries is advantageous to the senior tranche writer (and therefore the premium should be lower).

Synthetic CDO Premium

Associated with the tranche's notional size is a premium. As the remaining notional reduces, the premium on the tranche also reduces. The term 'premium waterfall' is used later to describe a more complicated related feature for 'cashflow CDOs' – for synthetic deals the waterfall is very simple, being driven by the dates of credit events and other fixed parameters.

Premium is generally expressed as a percentage of (or number of basis points applied to) the notional of the tranche. In Example 2 above, the [2%, 5%] mezzanine piece is an initial USD 30m notional (absorbing losses from 20m to 50m) if the reference pool is USD 1bn of risks. If it was subject to an initial premium of 500 bp, the premium would initially be USD 1.5m. After 5 defaults with 50% losses (= 25m losses since each notional reference name exposure is 10m) the mezzanine tranche would reduce to 25m and the premium would remain at 500 bp on the outstanding notional, thus reducing to USD 1.25m.

Table 19.3 Impact of defaults on the two junior tranches

Credit event number	Loss	Outstanding protection on the [0%, 3%] tranche	Outstanding protection on the [3%, 6%] tranche	Outstanding protection on the [6%, 9%] tranche
0		3%	3%	3%
1	0.56%	2.44	3.00	3.00
2	0.40%	2.04	3.00	3.00
3	0.24%	1.80	3.00	3.00
4	0.56%	1.24	3.00	3.00
5	0.40%	0.84	3.00	3.00
6	0.24%	0.60	3.00	3.00
7	0.56%	0.04	3.00	3.00
8	0.40%	0.00	2.64	3.00
9	0.24%	0.00	2.40	3.00
10	0.56%	0.00	1.84	3.00
11	0.40%	0.00	1.44	3.00
12	0.24%	0.00	1.20	3.00
13	0.56%	0.00	0.64	3.00
14	0.40%	0.00	0.24	3.00
15	0.24%	0.00	0.00	3.00
16	0.56%	0.00	0.00	2.44
17	0.40%	0.00	0.00	2.04

Premiums are typically quarterly in arrears with a proportionate premium related to the losses occurring during the quarter and the credit event dates (similar to vanilla CDS). Similar variations in premium occur, with different frequencies being common. Occasionally premiums are fixed amounts independent of defaults (guaranteed to maturity, the equivalent of an up-front premium with no rebate).

On the standard iTraxx and CDX tranches, which trade in the secondary market, the premium fixed at the outset will not typically be equal to the current fair premium for that tranche, so there is additionally an 'up-front' premium (which may be positive or negative).

Exercise 3 Suppose a 5-year reference portfolio has 100 equally weighted reference names, each of which has the same 200 bp CDS premium, and we assume 50% recovery for all names. Approximately how many defaults would you expect to occur and what size tranche would be just wiped out by this number of defaults? How much variation in the number of defaults would you expect and what other tranches would be affected by this variation? Suppose we assumed that recovery was 0%, how much would this affect the results?

Hint: On a single name the hazard rate is given by (9.15) and the survival probability over 5 years is given by (9.6), with constant hazard rate. These are approximately, a 4% default rate and an 82% survival/18% default probability over 5 years. The expected number of defaults is $100 \times 18\% = 18$ defaults or 9% losses. If these defaults are independent, then the distribution of defaults is given by the binomial distribution and the standard deviation is $\sqrt{(100 \times 0.18 \times 0.82)} = 3.8$, so we would anticipate between roughly 10 and 26 defaults, or between 5% and 13% losses (taking two standard deviations). (Note that if we introduce correlation, then this has no impact on the expected number of defaults but changes the standard deviation.)

Exercise 4 Suppose the premium on the 1% equity tranche of a risky underlying 5-year portfolio is 200% (20 000 bp). How much premium is paid to that tranche over its life? What risks are there to both the buyer and the seller of protection, and how might the risks be lessened?

Hint: In one year twice the notional size of the tranche is paid in premium if there are no defaults. Consequently, we expect the equity piece to have a short expected life.

The equity tranche is subject to a variety of further premium variations. Often the premium is a combination of a fixed up-front amount (say 40% of notional) and an ongoing 'small' premium of, say, 500 bp. The premium on the junior tranches of the iTraxx/CDX-based CDOs have been of this form up to 2009, moving to this form for all tranches of the CDX-based deals from that year.

A further rare variation of equity premium is as follows. For each reference name there is an agreed premium: name A, 200 bp, name B, 20 bp, name C, 35 bp, name D, 60 bp, etc. The initial premium on the equity piece may be 10% of a 3% notional = EUR 3m on a EUR 1bn reference portfolio. On default of name A (the first to default) the reference portfolio suffers a notional loss of income of 200 bp × EUR 10m = EUR 200 000. The premium on the equity piece reduces by this amount to EUR 2.8m. Default of name B causes the premium to reduce further to EUR 2.78m, etc.

The above 'equity premium waterfall' typically reduces more slowly than the previously described percentage of outstanding notional. Theoretically the equity tranche may be wiped out, although the premium being paid is still a substantial proportion of the original amount. Usually the premium terminates when the tranche is exhausted.

Documentation; CDO Set Up

No standard format for CDO contracts exist other than for the iTraxx/CDX-based deals. Unlike a CDS contract, which is usually a few pages long, CDO documentation may be several hundred pages long (for each tranche). Documentation for synthetic static CDOs tends to be shortest – a little longer than a CDS contract – and usually appends a list of terms unique to the CDO structure.

Tranches of protection on a CDO may be bought or sold off a client's own balance sheet. An alternative is to set up the CDO structure through a Special Purpose Vehicle (SPV: see section 13.1, and below). The latter usually applies to collateralised structures, often cash-structures, and sometimes all tranches (or all but the equity piece) of protection are bought. We shall look at a variety of examples later.

19.2.5 Managed Synthetic CDOs: Compliance Tests (OC and IC Tests; WARF; Diversity Score) and Substitutions

In contrast to a static CDO, a **managed CDO** has a reference pool which is **dynamic** – it changes over time – and is managed by a credit portfolio manager. Reference pools are usually cash assets although they may also include synthetic risks (CDS, TRS) and long protection positions, but entirely synthetic managed CDOs also exist. Note that the ability to manage the CDO can be used as a partial solution to the ragged nature of the cash maturities – as a bond matures an alternative risk can be substituted.

Substitution Rules

Such structures require controls or rules under which reference risks might be changed. If the originating institution is the buyer of protection, and manages the pool, then it may be advantageous to put highly risky names into the reference pool. For example, the institution may have illiquid risky assets elsewhere on its books and wishes to obtain default protection. These substitution risks are handled in a variety of ways:

1. The originating institution and the portfolio manager usually take the first loss risk.
2. Substitution may be limited to replacing maturing bonds, or may allow the sale of one bond before maturity and the purchase of another.
3. Risk limits: the pool as a whole may have to maintain an average weighting better than a certain level, a spread below a certain level, or a WARF better than a certain level on a substitution.
4. Asset types may be constrained to bonds or loans or may have a maximum exposure to other types of risk (such as mortgage backed securities or tranches of other CDOs).
5. Concentration limits may apply: there may be maximum percentage exposures (by notional, or notional weighted by spread or WARF) on industry groups (e.g. a maximum of 10% by notional in any group); the portfolio may have to maintain a 'diversity score' of (for example) 40 (a measure of the lack of concentration in the portfolio – see below).
6. Price/spread limitations may also apply. In synthetic CDOs a maximum spread may be specified. In cash CDOs a minimum purchase price is often specified. This caused problems recently in the CLO market when loan prices dropped significantly below these levels.
7. Over-collateralisation, interest coverage or other tests may have to be passed for a proposed substitution to be allowed.
8. The manager may be required to remove assets if their rating falls below a certain level (e.g. Baa3).

Only if all the substitution rules are satisfied can the substitution be made.

Managed CDOs offer investors access to credit portfolio management. As such, one of the key deciding factors is the skill of the manager. An alternative application is to seek protection on a dynamic portfolio which already exists. For example, the credit trading desk of a bank holds long (and occasionally short) positions in a large portfolio of reference entities. Typically, risk is held for a short period – less than 3 months and often for much shorter periods. If a credit event occurs while the desk is long, it suffers a large loss. Using CDSs to protect risk would be difficult because of the turnover required and the bid–offer spread to be paid. A dynamic CDO offers an attractive solution where the trading desk could limit its losses (by retaining a small equity piece).

Note that a CDO managed under tight (or lax) constraints is by no means guaranteed to outperform a static CDO. The debate about the value of active management is as valid in the context of credit portfolios as well as in equity and other portfolios. It is generally agreed that credit managers are more effective at avoiding defaults in a portfolio than is a static portfolio, but by the time a credit is removed its price may have fallen to the recovery level, or even below, or the name ultimately may not default. The impact on portfolio performance of a requirement such as item 7 above, or of active management as a whole, is not necessarily positive. Active management tends to increase the volatility and dispersion of returns, i.e. less median returns, a higher number of good outcomes and a higher number of poor outcomes; 'fatter tails'.

Coverage Tests

The following are the most commonly applied portfolio tests. In principle any other test could be incorporated into a 'cashflow CDO' to trigger income diversion to a 'reserve' or 'collateral' account. An example of another test is the average spread on the portfolio – or the spread on any bond – compared to a predefined level.

Interest Coverage (IC) Test

The original form of the test came from the bond and loan market where it acted as a constraint on the issuance of debt by a company. The basic idea is to compare the collateral income to the outgo committed to that tranche and anything senior to it; the excess of income over servicing costs is called the '**excess spread**'.

Over-collateralisation (OC) Test

One definition is to take the ratio of the paramount of non-defaulted debt in the reference pool, plus expected recoveries on defaulted debt, to the tranche nominal for pari-pasu and senior tranches. The ratio has to exceed specified levels (greater than unity), which typically increase as we look at more senior tranches.

Variations in definition exist – for example, expected losses may be deducted from the numerator. Standard and Poor's have the concept of a Rated OC test in an attempt to provide a standardised measure. The main difference between an OC in a typical transaction is that it factors in not only non-defaulted assets and recoveries in the numerator but also expected losses and excess spread (where available). The aim of the ROC is to be able to compare OC ratios across asset classes – i.e. as a measure of leverage. S&P use a synthetic version of the ROC: the SROC is a standard measure for surveillance of synthetic CDOs. SROC is defined as [Portfolio – (SDR $\times$ Portfolio)(1 – WARR)] *divided by* [Aggregate portfolio amount *less* Credit enhancement *less* Any dynamic enhancement (excess spread)].

WARF test [Moody's]

The calculation of WARF (weighted average rating factor) is described below. It is a measure of the average rating on the portfolio: for example, a rating factor of 360 corresponds to Baa2 on 10-year bonds. The WARF falling below a predefined level on a portfolio test date may trigger the diversion of excess cashflow to the reserve account.

Rating factors are generated by dividing the expected loss for a rating category by the expected loss for the corresponding Aaa category of the same tenor. Thus, a Baa2 rating factor of 360 shows that a Baa2 expected loss is 360 times that of the Aaa. Most quoted rating factors relate to the 10-year tenor although it is technically possible to use any tenor. In earlier CDO transactions the remaining tenor was used to calculate the WARF. However, this resulted in breaches of the WARF as the multiple of expected loss increased as the time decreased, largely due to the marked decrease in the Aaa expected loss rates.

WARF Calculation

Ratings are replaced by numerical rating according to the maturity of the individual asset. The numerical rating is weighted by the notional holding of the asset, and the sum over all assets is then divided by the total notional:

$$WARF = \frac{\sum_{i=1}^{n} N_i \cdot nr_i}{\sum_{i=1}^{n} N_i}$$

where nr is the numerical rating taken from the table of Moody's 'idealised' default probabilities (see Table 19.4).

Diversity Score and Calculation

This uses a Moody's defined table to derive the 'diversity score' (D) for a portfolio. Apart from being used as a quantitative measure of a portfolio, the diversity score is a key step in Moody's 'binomial expansion technique' formerly used to rate CDO tranches and is still used to indicate the diversity or lack of correlation in a portfolio. The calculation steps are as follows:

1. Calculate the average notional exposure, A.
2. Calculate a weighting for each name, $w = \text{minimum}(1, \text{total\_notional}/A)$.
3. For each industry group (of the Moody's defined groups) calculate the weight W for that industry by summing the above w's for each name in that industry.
4. Using the lookup table (Table 19.5) go from the industry weight W to the industry diversity score d.
5. Sum the d's over all industries to get the portfolio diversity score, D.

The diversity score is currently only used in (new) leveraged loan transactions – tests or restrictions based around diversity score will be found in older CDOs. The Moody's Asset Correlation or MAC score is now (June 2009) used in new CDOs – based on correlated binomial methods (CBET) and Moodys CDOROM. The MAC is put into a CBET generator to come up with a correlated binomial distribution. This approach may be extended to leveraged loans in the near future by Moody's.

19.2.6 CDO Squared

If a reference entity in the reference pool of a CDO is a tranche of another CDO, then this CDO is often referred to as a 'CDO squared'. If all the reference entities are tranches of other CDOs then this CDO is generally called a CDO squared. The terminology 'CDO cubed' is occasionally used – where a reference entity in the CDO is a tranche of a CDO squared.

Table 19.4 Moody's idealised default probability

Moody's Rating	Year									
	1	2	3	4	5	6	7	8	9	10
Aaa	0.0001%	0.0002%	0.0007%	0.0018%	0.0029%	0.0040%	0.0052%	0.0066%	0.0082%	0.0100%
Aa1	0.0006%	0.0030%	0.0100%	0.0210%	0.0310%	0.0420%	0.0540%	0.0670%	0.0820%	0.1000%
Aa2	0.0014%	0.0080%	0.0260%	0.0470%	0.0680%	0.0890%	0.1110%	0.1350%	0.1640%	0.2000%
Aa3	0.0030%	0.0190%	0.0590%	0.1010%	0.1420%	0.1830%	0.2270%	0.2720%	0.3270%	0.4000%
A1	0.0058%	0.0370%	0.1170%	0.1890%	0.2610%	0.3300%	0.4060%	0.4800%	0.5730%	0.7000%
A2	0.0109%	0.0700%	0.2220%	0.3450%	0.4670%	0.5830%	0.7100%	0.8290%	0.9820%	1.2000%
A3	0.0389%	0.1500%	0.3600%	0.5400%	0.7300%	0.9100%	1.1100%	1.3000%	1.5200%	1.8000%
Baal	0.0900%	0.2800%	0.5600%	0.8300%	1.1000%	1.3700%	1.6700%	1.9700%	2.2700%	2.6000%
Baa2	0.1700%	0.4700%	0.8300%	1.2000%	1.5800%	1.9700%	2.4100%	2.8500%	3.2400%	3.6000%
Baa3	0.4200%	1.0500%	1.7100%	2.3800%	3.0500%	3.7000%	4.3300%	4.9700%	5.5700%	6.1000%
Ba1	0.8700%	2.0200%	3.1300%	4.2000%	5.2800%	6.2500%	7.0600%	7.8900%	8.6900%	9.4000%
Ba2	1.5600%	3.4700%	5.1800%	6.8000%	8.4100%	9.7700%	10.7000%	11.6600%	12.6500%	13.5000%
Ba3	2.8100%	5.5100%	7.8700%	9.7900%	11.8600%	13.4900%	14.6200%	15.7100%	16.7100%	17.6600%
B1	4.6800%	8.3800%	11.5800%	13.8500%	16.1200%	17.8900%	19.1300%	20.2300%	21.2400%	22.2000%
B2	7.1600%	11.6700%	15.5500%	18.1300%	20.7100%	22.6500%	24.0100%	25.1500%	26.2200%	27.2000%
B3	11.6200%	16.6100%	21.0300%	24.0400%	27.0500%	29.2000%	31.0000%	32.5800%	33.7800%	34.9000%
Caa1	17.3816%	23.2341%	28.6386%	32.4788%	36.3137%	38.9667%	41.3854%	43.6570%	45.6718%	47.7000%
Caa2	26.0000%	32.5000%	39.0000%	43.8800%	48.7500%	52.0000%	55.2500%	58.5000%	61.7500%	65.0000%
Caa3	50.9902%	57.0088%	62.4500%	66.2420%	69.8212%	72.1110%	74.3303%	76.4853%	78.5812%	80.7000%

Table 19.5 Industry diversity score (Moody's)

No. of firms in the same industry	Diversity score	No. of firms in the same industry	Diversity score
0	0.00	4.95	2.67
0.05	0.10	5.05	2.70
0.15	0.20	5.95	3.00
0.95	1.00	6.05	3.03
1.05	1.05	6.95	3.25
1.95	1.50	7.05	3.28
2.05	1.55	7.95	3.50
2.95	2.00	8.05	3.53
3.05	2.03	8.95	3.75
3.95	2.33	9.05	3.78
4.05	2.37	9.95	4.00
		10.05	4.01

There are no new cashflow features of such CDOs but they pose significant problems for valuation (on a risk neutral basis) and present complex correlation relationships. Underlying CDO tranches often contain the same names, but risk may be at different attachment points.

19.2.7 Funded (CLN) and Unfunded (CDS) Tranches

CDO Tranches; Funded and Unfunded Tranches; 'Buying' and 'Selling'

The reader should see a synthetic CDO tranche as a deal in its own right (which indeed it is) not requiring the existence of any further structure. There may be no other tranches of protection bought or sold – perhaps just a single mezzanine tranche on a reference portfolio. In addition, the reference pool of assets may not (and typically does not) exist as an actual portfolio held on someone's books. A typically 'CDO deal' amounts to only some of the tranches of protection being bought or sold. It is very rare for a reference portfolio of assets to be bought, and all tranches of protection to be bought from third parties.

A tranche of protection may be split into many transactions ('deals') all ranking *pari-passu* with each other. A deal may be set up in two forms:

1. A premium is paid in exchange for capital protection paid as, and when, a credit event occurs – known as a 'CDS style'.
2. The protection is embedded into a credit-linked note which carries a fixed or floating coupon (which includes the tranche premium) in return for the notional exposure of the deal (although sometimes CLNs are priced below par).

Recall that the sale of a CLN by bank × corresponds to the purchase of protection by bank X. Buying protection via CDS style deals may be accompanied by the sale of CLNs related to the same tranches, to achieve the same object.

'CDS style' deals related to a CDO are also referred to as **unfunded deals**, while CLN deals are **funded deals**.

Exercise 5 For a given reference pool of assets, what is the economic difference between the following?

1. A [3%, 6%] tranche related to a USD 1bn sized pool of which 15m is in CDS form and 15m is in CLN form, the two deals ranking *pari-passu*.
2. A [3%, 6%] tranche related to a USD 500m sized pool in CDS form, and a [3%, 6%] tranche related to a USD 500m sized pool in CLN form (the two deals are unrelated other than having the same reference pool).

Answer: None

'Buying' and 'Selling'

Often investors think in terms of CLN (funded) structures and, treating a CLN as a bond, talk in terms of 'buying a tranche' when what is meant is the purchase of a CLN with embedded CDO risk. For investors 'buying' means buying risk – corresponding to buying a bond – and therefore writing protection. For non-funded deals particularly, the meaning of 'buy' or 'sell' has to be clarified by adding 'risk' or 'protection'.

19.2.8 Relationship to *n*th-to-Default

We saw in section 19.1 that *n*th-to-default products are generally first-to-default for a notional amount of EUR /USD 10–20m on 2–10 reference names. In contrast, CDO deals may be for any tranche of protection (high risk to low risk), and typically refer to a reference pool of EUR/ USD 500m–5bn and between 50 and 500 reference risks.

Despite these large differences in deal characteristics, there is a close theoretical relationship between the two products. Consider a portfolio of N reference names – it does not matter what N is; we shall take $N = 10$ for illustration. Suppose each risk is of 10m, giving a total notional of 100m. Suppose all recoveries are 0%. The first default gives rise to a loss and would trigger an F2D contract (of, say, 10m size). It would also wipe out the capital protection granted by a 10% equity piece (and the premium on the equity piece would terminate). The second to default would trigger an S2D contract (again of 10m) and would wipe out a 10% junior mezzanine tranche.

Thus if recoveries are zero, an *n*th-to-default contract (in 100/N notional) is identical to an $[(n-1) \times 100/N - n \times 100/N]$ tranche of a CDO, for $n = 1, \ldots, N$, where N is the number of reference entities in the pool.

If recoveries are not zero, the 10m notional F2D pays par in exchange for defaulted debt – so effectively pays 5m if recovery is 50%, and the deal then terminates. If we make the equity piece 5m so that it terminates on the F2D, then there is a difference in the premium rates because the F2D is based on 10m while the equity piece is based on 5m. There is therefore still a close correspondence, but not an exact match, between the contracts if recoveries are not zero. This correspondence is useful for code testing.

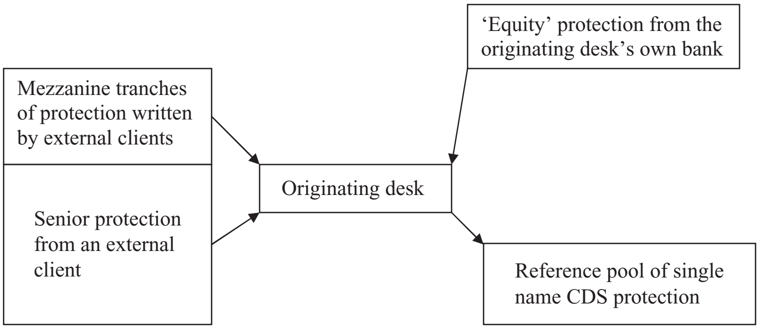

Figure 19.4 Static synthetic arbitrage CDO set-up.

19.2.9 Applications

Arbitrage CDOs

Synthetic static CDO structures have been used by investment banks since the late 1990s. Typically these were based on a reference portfolio of 100 investment grade names which are liquid in the CDS market and usually either entirely European or entirely North American names. The investment bank would typically buy protection on mezzanine tranches 'cheaply',[5] buy protection on the large senior tranche (in order to limit its own capital usage) at a 'fair price' or slightly above, and **retain** the equity risk – effectively obtaining protection on all tranches.[6] 'Retaining' the risk effectively means that the originating bank is writing protection on that tranche to the desk originating the deal. Typically it does not enter into a contract with itself to do so, but simply owns the risk by not buying protection. Often it will allocate internal capital to the full extent of the equity tranche, effectively paying a 100% notional up-front premium. Usually this is done by valuing the risk at notional in the bank's accounts. Some banks will value the risk using a model such as that described in Chapters 20 and 21.

The originating desk now owns all the tranches of protection and enters into a hedge where it sells protection on all the underlying names. The set-up is illustrated in Figure 19.4.

The desk receives premiums on the individual name CDS deals (or coupons on a bond portfolio), and pays premiums on the tranches. Any excess of the income on the CDS over the cost of protection is retained by the desk (i.e. the originating bank). If the equity tranche is not wiped out by losses, then any excess capital is released back to the bank at maturity. Such deals could generally be done to yield a profit and became known as 'arbitrage CDOs'.

Example Suppose the average spread on 100 names is 100 bp with 10m exposure each. The initial income is $1\text{bn} \times 1\% = 10\text{m}$ per annum. Tranching may be 3% (retained), 7% at 200 bp (average over several tranches), and 90% at 10 bp. The cost of the external tranches is $70\text{m} \times 2\% + 900\text{m} \times 0.1\% = 1.4\text{m} + 0.9\text{m} = 2.3\text{m}$, leaving an excess of 7.7m and an

[5] We shall define 'fair price' and what we mean by 'cheap' in Chapter 20.

[6] Some banks have favourable regulatory treatment, which means that they do not have to source protection externally. Those that do need to – and who find a potential writer of protection in the form of an insurance company – may face further complications. Capital reduction was only obtained if the protection was via an OECD bank, so it was often necessary to 'wash' the deal through an OECD bank with favourable regulatory treatment, which in turn dealt with the insurance company (often through an SPV).

exposure of 30m – an effective premium of just over 25%. The best case result would be no default, in which case the bank loses no capital and earns 7.7m × 5 years. The worst case is the immediate default of all names, in which case the 30m capital is lost.

Exercise 6 Calculate the expected profit using the CDO valuation software ('CDO pricer.xls').

Hint: (Read Chapters 20 to 22!) The expected profit is the net value of the assets held. The individual reference CDSs (traded at fair market premium) have zero net value. The tranches will have been dealt in at prices different from the fair market premium.

The question of hedging the structure above is discussed in the following chapters. The obvious (default event) hedge of 10m notional in each reference name (in the above example) may lead to a large (30% or so) mismatch in terms of spread risk depending on the amount and form of the premiums paid for the tranches – and in particular for the equity piece.

19.2.10 Portfolio Optimisation

A structurer wishes to design a deal (CDO tranche) meeting the objectives of a client which would typically include the following:

- Maturity and size.
- Constraints on the reference entities (maximum and minimum number, industry concentrations or exclusions, worst rating, geographical restrictions, etc.).
- Constraints on the tranching – e.g. minimum subordination level.
- Target rating for the tranche – e.g. BBB.
- Premium.

The structurer will want to select the reference pool to satisfy the constraints and to maximise the profit to the structuring group. This is a problem of 'optimisation subject to constraints' and such problems are common in business and engineering. Often the mathematics is simple ('linear optimisation') but in this case the interactions between the variables is non-linear and optimisation is far from simple. The procedure can be easily described:

1. Select a reference pool meeting all the constraints.
2. Find the subordination level that gives the desired rating. This requires use of the rating agencies' rating software which is currently (July 2009) freely available but as it operates via simulation, it is discrete, highly non-linear and relatively slow.
3. Calculate the fair market value for the tranche (all variables are now specified: reference pool, attachment and detachment point, premium, etc.).
4. Change the reference pool (and the subordination level) until a constraint-satisfying maximum profit CDO tranche is found.

In practice we need mathematical help in step 4 – the selection of a reference pool which increases the profit. It is impossible to go through all the potential combinations of (say) 100 names out of a universe of 2,000 and step 2 above may take a minute or so. Even with distributed processing random selection of reference pools it would take a prohibitively long time.

The following paragraphs, to the end of this section, give a description of the approach adopted by the major third-party CDO optimiser – Calypso's 'Galapagos Platform' – and is authored by Onur Cetin at the Galapagos Product Management team at Calypso.

Traditional optimisation methods rely on gradients, which basically involves making a best guess on the optimal step and direction to take in the search space in the vicinity of the current solution. This is a natural starting point, but unfortunately discrete problems like choosing a reference pool, coupled with simulations, makes this direction-finding hardly feasible.

Calypso's Galapagos Structurer software approaches this problem by looking at how nature deals with complex optimisations. Evolutionary algorithms – a smart blend of informed search and randomness that has no reliance on gradient calculation – can bring efficiency and flexibility to CDO optimisation problems. As a simulation of 'survival of the fittest', candidate portfolios evolve through generations by preserving helpful 'genes' – i.e. certain characteristics that drive the solutions towards the specified goals, while looking for innovation in the form of small random changes, or 'mutations'. This population-based approach can be implemented very effectively in a compute grid for distributed processing, so Galapagos has been a widely used decision support tool for structurers to quickly design and evaluate CDOs in a fast-moving market.

A natural extension of CDO portfolio optimisation is the active management of CDOs. The primary motivation for managed CDOs is that one can spot a deteriorating credit early and get out of the risk, with a loss much smaller than a potential default. Defending against downgrades/migration risk in this way is especially important for investors who have specific investment guidelines based on cost of capital/regulatory capital. Spotting the right opportunities to build subordination is also very critical in defending against idiosyncratic risks.

The CDO management problem starts with an existing reference pool to fine-tune rather than forming a CDO from scratch, so it is conceptually a simpler optimisation. But the process might require time-consuming iterations owing to the number of parties involved as well as complex constraints and covenants:

1. The manager decides to substitute a reference entity for another.
2. The manager runs a rating model to check if all criteria are met.
3. The dealer provides quotes and subordination adjustment.
4. The manager gets additional quotes and executes on best prices.
5. The exact level of subordination adjustment is determined.

Typical constraints include meeting several targets, such as:

- SROC
- Moody's metric
- Concentration levels and portfolio diversity
- 'Churn' limits (turnover limits)
- Loss triggers
- Other compliance tests
- Trading costs.

Calypso's Galapagos Manager aims to replace these cumbersome iterations in a typical CDO management process. While one addition and removal at a time seems to be a favoured strategy for managers to effectively leverage their credit-picking skills this is almost always suboptimal and a more effective defensive trading strategy can be employed by substituting several names at once using Galapagos. In this case, again employing evolutionary computation, many candidate portfolios go through substitutions (mutations) to adapt to sudden changes in the market or credit opinions. The platform essentially becomes a communication tool between a manager and dealers to continuously evaluate and re-optimise deals by integrating pricing models, rating models and data feeds.

The Galapagos platform has been used to 'restructure' CDOs as well, which typically involves changing the subordination and maturity of an existing deal along with a large number of substitutions to dramatically reshape the risk profile. This problem is not far from CDO management and gets complicated for such exotic trade types as CDO squared.

19.3 CASHFLOW CDOs

A '**cash**' or '**cashflow**' CDO is one where the reference portfolio is physical assets: bonds or loans or other financial assets. The reference portfolio actually exists as a 'risk' or 'hedging' pool – often the pool of assets exists and the CDO is created as a means of transferring all or part of the economic ownership. A key feature associated with cash CDOs is the ragged nature of bond maturities compared with the single maturity of the CDO. How is the excess (or deficit) of protection term to be handled in terms of premium paid and risk taken? Solutions include

(1) building up a cash reserve as bonds mature, and reducing the exposure for longer bond positions to maintain duration risk; and/or
(2) partial early maturity of the tranches with a reduction of notional and premium (**amortising principal**).

Another solution is a managed cash CDO which allows (at least) capital from maturities to be reinvested. An additional related problem is that cashflow (coupons and maturity amounts) received on the collateral pool, and liability payments (tranche payments plus taxes, fees and swap fees) are spread over a period of a year with payments concentrated on certain days. A solution here usually requires a cashflow account that has an initial sum (part of the overall collateral or a loan) and into which positive and negative cashflows are directed, or the use of a cashflow swap to smooth flows.

A further feature of cash CDOs, where the 'collateral' (reference) pool does not already exist as a real portfolio, is how to build up that pool as protection and capital (funding) is obtained. Typically such portfolios have a '**ramp-up**' period during which the assets are acquired. This may range from 3 months to over 1 year, depending on the type of collateral and its illiquidity.

Note that cash CDO structures will typically have embedded interest rate and FX swaps. Contingent risk on these arises from the unknown unwind value on the unexpected defaults of reference assets and the early unwind cost of these swaps is usually borne by the investor (writer of protection).

Cash CDOs primarily have application in the hedging of existing pools, or in allowing such pools of physical assets to be created to provide managed credit exposure for investors. Commercial banks or other hedgers could create a cash CDO on their own balance sheet related to their portfolio of commercial loans. Typically, however, commercial banks use synthetic structures partly because of their greater flexibility and they overcome issues that may arise around transferability (i.e. legality, borrower consents, etc.).

Apart from corporate (including bank) reference entity risk, one major class of cashflow CDOs is residential or commercial mortgage-backed securities (RMBS and CMBS). These have two key differences compared to corporate CDOs, namely:

1. The underlying assets are untraded (although they can be traded synthetically in a limited fashion through ABS CDS).
2. There is a very high correlation between the underlying assets – with high early repayment risk, risk of correlated defaults, and correlation between the underlying asset value of non-defaulted mortgages with the number of defaults taking place.
3. There are cashflow issues on the underlying assets with regards to available funds caps and the ability of the mezzanine tranches to 'pay-in-kind'.

19.3.1 Reference Pools

CDOs have been constructed with a variety of reference assets – not just bonds, loans and CDS but including the following:

(a) debt in any form – synthetic bonds or CDSs;
(b) contingent debt – including commercial bank facilities and guarantees;
(c) mortgage portfolios, credit card portfolios;
(d) lease contracts or trade receivables;
(e) equity;
(f) payments under contracts (particularly derivative contracts).

Terminology and valuation techniques will vary according to these variations. For example, a CDO of CDO tranches[7] requires (in principle) a 'look through' valuation to the assets underlying each CDO since a single reference name may appear in several reference pools. This can present logistical problems if the CDOs are dynamic. In addition, a reference CDO does not suffer a credit event – if funded it repays less than par at maturity – so 'credit event' calculations have to be reworked in terms of the outstanding notional of the CDO tranche.

19.3.2 Income and Capital Waterfalls: Reserve Accounts

The terminology '**cashflow CDO**' was coined for certain cash structures where income generated by the 'asset' (reference) pool was used in a certain way to service the CDO structure. The '**cashflow waterfall**' describes in detail how this income is used. This usually includes the use of 'quality' tests (such as income coverage or over-collateralisation tests described above) and diversion of income into a '**reserve** ([or over-collateralisation) **account**'. A cashflow waterfall can actually be applied to any type of CDO – the reference pool can include synthetic assets and need not exist as a real portfolio. The previously described premium rules for CDO tranches could be described by a trivial waterfall structure, but some synthetic CDOs have recently included a complex waterfall structure. The objective of a non-trivial waterfall structure and reserve account is to improve the quality (reduce the risk) of certain tranches of the CDO.

[7] Sometimes called a CDO squared or a CDO^2

A waterfall structure arises of necessity for deals backed by cash assets simply because it is those assets and those assets alone which generate the income to pay both the costs of running the structure and the premium on the tranches. Even if no tests (such as IC or OC tests) are included, reserve accounts and a payoff structure must still be defined – and this constitutes the waterfall.

The waterfall

Positive cashflow is credited to a collection account which typically also earns interest. Income (e.g. coupon payments on bonds) and capital (e.g. redemptions, sales and recoveries) are typically treated separately. The waterfall describes the priorities of application of funds from that account in any one period. Typically from highest priority to the lowest, the waterfall implements the following priorities (highest first):

1. Taxes (if applicable).
2. Trustee fees (if collateralised).
3. Payment under third party financial transactions – such as interest rate and FX swap payments (these typically match the cashflows on the bond assets so have to take priority).
4. External fees and expenses – taxes, rating agencies, legal, reporting and paying agents fees, etc.
5. Investment adviser's fee and structuring agent's fee.
6. Payments to tranches more senior than the equity tranche in order of priority.
7. Payments to reimburse tranche protection writers for losses they have already suffered.
8. On the failure of specified portfolio tests, the diversion of all remaining cash to a reserve account until the test is satisfied.
9. Excess cash is paid to the equity tranche holder.

The equity tranche holder is typically the originator of the deal, together with the collateral pool manager. The income they get from the deal is a combination of the high priority fee payments (item 3) and the low priority equity tranche payment (item 7).

Table 19.6 gives an example of how the initial annual waterfall on a cashflow CDO might operate. Income is shown net of funding costs and after the payment of interest and FX swaps (priority 1 in the waterfall). The portfolio is assumed to generate an average of 100 bp on a notional of 1bn initially. As credit events occur, income drops. Portfolio tests (item 6) may start

Table 19.6 Cashflow CDO – initial waterfall

Item	Income	Net cashflow
Asset cashflow after funding costs and after swap payments	1bn × 100 bp = 10m	10m
External fees	–0.05m	9.95m
Investment advisor's fee	–1m	7.95m
Structuring fee	–1m	
Super senior tranche	–850m × 10 bp = 0.85m	7.10m
Senior tranche	–70m × 25 bp = 0.175m	6.925m
Mezz B	–30m × 200 bp =	6.325m
Mezz A	–20m × 500 bp	5.325m
Reimbursement of tranche losses	0	5.325m
Diversion to reserve account	0	5.325m
Payment to equity tranche holders	5.325m	0

to fail and cashflow is diverted to the reserve account rather than paying equity. Further defaults may eliminate the equity tranche and defaults may take money from the reserve account or start to hit the senior mezzanine tranche. In principle, the CDO may reach a state where the equity tranche has been used up, no other tranches have been hit by losses (or have been hit but subsequently reimbursed), and the portfolio tests are passed. In that case the equity tranche holders continue to receive the excess cashflow.

Reserve or Collateral Account

The above tests, or other tests, may trigger diversion of excess cashflow into a 'reserve account' (also called an 'over-collateralisation' account). Funds in the reserve account may be released if defaults start to occur (the reserve account effectively acting as a new junior tranche to the extent that it holds any assets). Alternatively funds in the reserve account may be used to repay tranches early ('amortising principal') or 'defease' the senior swap if unfunded (i.e. reduce the senior swaps obligation by the reserved amount).

Note that CDO structures often have unique features and the management of these structures can pose significant control problems.

19.3.3 Funding and SPVs

An SPV may be used to house all or part of a CDO transaction. There are no differences in principle between this application of an SPV and that described for the creation of a simple CLN in Part II section 13.1. The SPV may contain the entire transaction – for example:

1. A synthetic CDO may be set up via an SPV. The SPV writes default swaps (the reference pool), and buys tranches of protection from the market. Tranches may be funded – and the SPV owns collateral to back these tranches – or unfunded, in which case there is counterparty risk.
2. A managed CDO may be set up in an SPV. Tranches of protection are sold by the SPV in funded form, and capital raised is used to buy the reference pool.

This is not always the case, however. We shall look at two examples below where the SPV buys and writes protection only on some tranches of the CDO.

A cash CDO (as discussed above) is one where the capital raised is used to purchase the risks referred to in the CDO. In contrast, a funded CDO is one where cash is raised and invested in collateral, but the CDO refers to other risks. This raises new management issues (e.g. collateral pool record keeping) and valuation issues.

Who takes the risk on the collateral pool with respect to losses and downgrades? There are two main approaches here.

1. A highly rated bond, typically issued for the SPV and consequently tailored to the SPV, is used with the investor taking the default and downgrade risk first and the counterparty taking it second. Additionally, the counterparty agrees typically to exchange par for any settlement (i.e. the investor does not take mark-to-market risk on settlement except for early termination).
2. The second approach is to use some form of financing trade (i.e. typically a repo or TRS), in which high-value collateral is pledged to the vehicle (usually in excess of the nominal),

in which the bank keeps the spread on the collateral and pays the interest rate index flat. In this instance it is subject to mark-to-market risk, but the bank typically takes the default and downgrade risk as collateral is ineligible if it is below a certain rating. This is potentially an additional source of profit for the originator.

To the extent that the investor is taking risk on the collateral, the collateral pool should be taken into account and modelled in the valuation of the CDO. To the extent that the originator is taking risk, the originator needs to factor this into the valuation models.

Example 1 This example is based on the BISTRO series of 'balance sheet' CDO transactions developed by JP Morgan. The SPV buys funded protection from the market on certain mezzanine tranches of a synthetic CDO – say the [2%, 8%] tranche. In turn it sells protection to the investment bank. In its turn the investment bank sells protection on a 2–100% tranche of the reference pool to a commercial bank. The commercial bank retains the first 2% of risk, and buys protection on the balance of the portfolio from an OECD bank in order to get regulatory capital relief. The investment bank may write the senior protection off its own books, or may buy it from a third party directly. Note that the risk on the reference pool of assets – commercial bank loans and facilities – is passed synthetically via the tranches of protection to ultimate investors through the investment bank plus (for the mezzanine tranches) through the SPV. In this case the role of the SPV is to hold collateral behind the funded notes remote from the investment bank. The transaction is summarised in Figure 19.5.

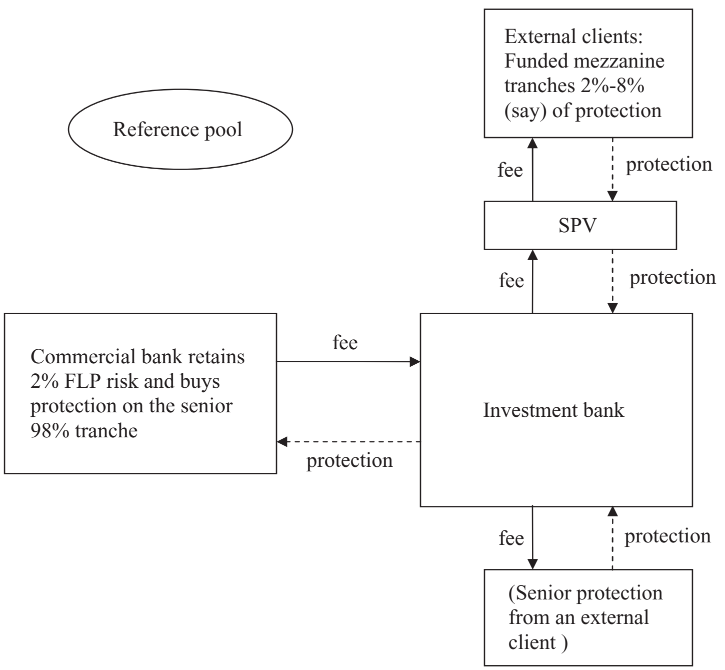

Figure 19.5 SPV holding the funded mezzanine tranches of a CDO.

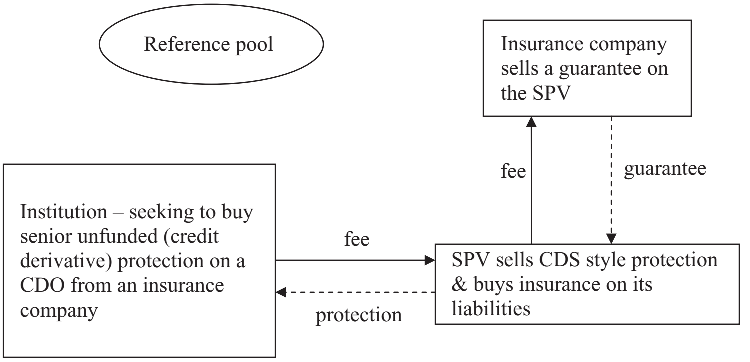

Figure 19.6 SPV holding an unfunded super senior tranche of a CDO.

Example 2 Protection on the 'super senior' tranche (meaning a tranche rated AAA with some significant margin before it would be downgraded) of a CDO is to be bought from a certain insurance company in standard credit derivatives form, yet the insurance company is unable to write credit derivatives although it can sell an insurance guarantee. An SPV is set up which buys a guarantee (on its liabilities) from the insurance company, and the SPV sell unfunded protection on the super senior tranche of the CDO, see Figure 19.6.

19.3.4 Balance Sheet CDOs

A balance sheet CDO is used to obtain capital relief (reducing economic risk or regulatory capital) on a pre-existing portfolio of credit risks. Typically the institution seeking protection is a commercial bank and the reference pool of assets contains non-assignable loans and a variety of contingent credit risks (including facilities and revolvers). Typically these are passed synthetically to an SPV which obtains protection in tranched form (or variations along the lines of Example 1 above). Passing the risks synthetically is quick, easy and flexible – allowing a wide range of complex risks and contingent risks to be transferred. (The problem in this case is less about the mechanics of the deal and more about correct valuation for products that do not trade in a secondary market.)

The calculation of regulatory capital has become significantly more complicated under Basel II than the previous regulations – both are described below. On the other hand, internal capital or economic risk is allocated by the bank according to the asset risks. The retained first loss piece is high risk and will be allocated the face amount of capital. The other risks are counterparty risks to the bank, and capital allocated will typically depend on the counterparty rating.

The following description to the end of this section has been contributed by Darren Smith (see section 19.6 for affiliation and disclaimer).

Under Basel II regulations there are three approaches to credit risk:

1. Standardised approach.
2. Foundation Internal Ratings-Based (F-IRB) approach.
3. Advanced Internal Ratings-Based (A-IRB) approach.

The standardised approach is basically an extended form of the Basel I accord in that standardised risk weights are attached to different asset classes but, unlike Basel I, the risk weights are banded by external ratings. There is some scope for using internal ratings for unrated assets but generally external ratings apply. This category is typically used by banks that are not large enough to support the infrastructure necessary for the internal ratings-based approaches.

The F-IRB approach allows banks to estimate their own probability of default from empirical data (regression models, etc.), but requires them to use the regulators' prescribed parameters, including loss given default in a model for estimating the risk-weighted assets. This is the so-called supervisory formula approach. Banks require approval from the regulator to utilise this approach.

In the A-IRB approach, banks are permitted to use their own proprietary models (subject to approval) to estimate RWA and all the prerequisite parameters (probability of default, loss given default and exposure at default). The supervisory formula is then used to estimate RWA, and capital required is a fixed percentage of the risk-weighted assets.

The supervisory formula is usually applicable to credit products (i.e. bonds, loans, lines of credit) for corporate obligations, but not structured finance products. An external rating is typically required.

Suppose a bank holds a portfolio of EUR 1bn, consisting of 100 borrowers at EUR 10m each. Under the former regulations (Basel I) the bank would be required to hold 8% capital against such a portfolio, i.e. the risk weight of the portfolio is 100%, giving risk-weighted assets of 1bn. It would only make sense to transfer risk in a balance sheet transaction if the risk weight of any retained pieces was less than 8%. For this to occur, the ratings on the loans would have to be at least investment grade and probably high BBB or single A. A capital structure could be obtained in which, say, the [6, 100] tranche was rated AAA, the [4.5, 6] tranche was AA, the [3.5, 4.5] was A, and the [2.5, 3.5] was BBB, with a [0, 2.5] retained equity tranche. If the rated notes could be sold, then the bank could reduce its capital from 8% to 2.5% or EUR 25m. However, if the underlying loans were rated, say, B, a [0, 21] equity piece would be required to get to a BBB attachment, much in excess of the current 8%, and making a balance sheet transaction unattractive.

This incentive to securitise higher-rated loans was an unintended consequence of Basel I, which, when being drafted, had not considered the effects of securitisation. Consequently, banks initially tried to increase return on capital by securitising the higher quality portions of their portfolio. This raised concerns of negative selection bias at the regulatory bodies and subsequently banks could not obtain the necessary approvals for such transactions.

Basel II attempted to address this by changing the risk weights progressively from the highest rated (less risky) to the lowest rated.

The standardised approach for Basel II, which is typically only adopted by smaller banks, varies the risk weights for assets based upon external ratings as well as obligor type. See Table 19.7.

Table 19.7 Standard approach (risk weights) for assets

Credit assessment	AAA to AA−	A+ to A−	BBB+ to BBB−	BB+ to B−	Below B−	Unrated
Sovereigns	0%	20%	50%	100%	150%	100%
Banks	20%	50%	50%	100%	150%	20%
Corporate claims	20%	50%	100%	100% to BB-, 150%	150%	100%

Table 19.8 Standard approach (risk weights) for securitisations

Credit assessment	AAA to AA−	A+ to A−	BBB+ to BBB−	BB+ to BB−	Below B+Or unrated
Securitisations	20%	50%	100%	350%	Deduction

In the **standardised approach**, an external rating needs to be obtained for any retained tranches (see Table 19.8). There is little or no incentive for a bank to securitise highly rated assets, as the risk weights for AAA to A– show. It is only at BBB and below is there any incentive to securitise.

Assume that the bank has a portfolio of leveraged loans rated B on average. The risk weight for such an asset would be 150%. Hence, on a portfolio of, say, EUR 300m, the risk-weighted assets would be 450m. Assuming an 8% capital requirement, the amount of capital to be held would be 36m. Assuming that the attachment point for an AA– is around 25%, then by retaining only the [25, 100] tranche the bank reduces its RWA to $0.2 \times 0.75 \times 300 = 45$m and its corresponding capital usage to 3.6m, a reduction of 90%. The capital released is 32.4m. However, the bank would pay premiums on the tranches.

For the above example, suppose that the portfolio paid EURIBOR + 300 bp, and that the retained tranche pays 50 bp. In addition, assume that the expected loss[8] on the portfolio was 1% per annum. Normally a bank would hope to deduct this expected loss from income. The risk-adjusted, or RAROC, would be 6/36 (i.e. 9 – 3)/36 = 16.66%. The bank would forgo EUR 7.5m in income for a new RAROC of 1.5/3.6 of 41.66%, bearing in mind that at least 3m of that income was the annualised expected loss.

Internal Ratings-Based Approaches and the Supervisory Formula

The internal ratings-based approaches allow banks, with their supervising regulator's approval, to use internal ratings to estimate risk-weighted assets and, accordingly, the capital allocated to them. Banks entitled to either of the internal ratings-based approaches can apply the supervisory formula to both the initial portfolio to determine the appropriate level of risk-weighted assets and appropriate allocation to any retained tranches.

The first step in utilising the supervisory formula is to estimate the 'Kirb', or the effective ratio of capital required to the notional of the portfolio. In Basel I, the Kirb was effectively the risk weight multiplied by 8%. In Basel II internal ratings models, Kirb is calculated for eligible assets from a formula. The formula for corporate, sovereign and bank exposures is given below.

Supervisory formula for corporate, sovereign and bank exposures

- Correlation: `R = 0.12 * (1 - EXP(-50 - PD)) / (1 - EXP(-50)) + 0.24 * [1 - (1 - EXP(-50 * PD))/(1 - EXP(-50))]`
- Maturity adjustment: `b = (0.11852 - 0.05478 * ln(PD))^2`
- Capital requirement: `K = [LGD * N[(1 - R)^-0.5 * G(PD) + (R/(1 - R))^0.5 * G(0.999)] - PD * LGD] * (1 - 1.5 * b)^-1 * (1 + M - 2.5) * b)`
- Risk-weighted assets: `RWA = K * 12.5 * EAD.`

[8] fundamental measure

where LGD is the loss given default, N is the cumulative normal distribution, G is the inverse cumulative normal distribution, EXP is the exponential function, and EAD is the exposure at default.

Risk-weighted assets can be derived from the Kirb by multiplying by 12.5, then by the exposures (typically the nominal exposures).

For example, assume a similar portfolio of B2-rated assets with a default probability of 20.71% at the maximum maturity of 5 years, and a loss given default of 35%. The Kirb would be 13.99% using the formula below. This implies a total risk weight for the assets of 174% or EUR 524.63m As long as any credit enhancement covers the Kirb plus some small add-on amounts, the risk weight for the senior retained piece will be floored at 7%, and the risk-weighted capital at 56 bp. So, by using an attachment of 16%, the risk weight for the senior tranche becomes 7%, and the RWA for the EUR 252m tranche becomes EUR 17.64m. Assuming an 8% capital requirement, the capital against for the selling bank is reduced from 42m to 1.4m, releasing 40.6m in capital.

Again the appropriate economics should make sense with regard to premiums paid.

As can be seen from above, Basel II incentives banks more to retain senior pieces from balance sheet transactions and sell junior pieces (buy protection).

The relevant formulae are summarised in Table 19.9.

19.3.5 Diversification and Risk Reduction Trades; Credit Bank

An insurance single name credit risk portfolio, developed on the basis of familiarity with the local market, may be geographically concentrated – for example, primarily Japanese. The company could write protection on a senior tranche of a portfolio of EU or North American credits (most simply using index products) to obtain geographical diversification at relatively low risk. The reverse trade might appeal to non-Japanese-based insurance companies.

Likewise, commercial bank loan portfolios tend to be concentrated both geographically and by industry. An EU bank could reduce its concentration risk at negligible cost by buying CDS protection on individual large exposures, and selling protection on an average basket of EU names where no exposure is currently held. Additionally, the bank could obtain protection in tranched form on its remaining risks (releasing lines, etc.) while writing protection on a portfolio again in tranched form. In practice, at present commercial banks rarely sell protection as it is not capital efficient and greater income can be obtained from existing clients (a large part of commercial bank income is in the form of fees rather than spread income).

Corporate treasury departments run substantial risks to trading counterparties – products or raw materials may be supplied to clients in large volumes for delayed settlement. The number of counterparties is usually relatively small and concentrated, and risks are typically rolling short-term risks (although may extend beyond 12 months in some cases). This leaves them doubly exposed – the loss of a client and the loss of a loan – in the event of the counterparty failure. One theoretical solution is the development of a 'credit bank' – a CDO-related to the reference pool of risks obtained synthetically from several corporate entities and other sources. The contributing corporates retain the first loss piece and an SPV obtains tranched protection on the remaining risks.

Table 19.9 Basel II formulas

	Description	Formula
K_{irb}	Ratio of IRB capital charge if the underlying pool were not securitised as a percentage of the notional of the portfolio	
T	Thickness: notional amount of the tranche	
L	Credit Enhancement Level: notional amount of all more junior tranches relative to the notional amount of the pool	
N	Effective number of exposures	$N = \frac{\left(\sum_i \text{EAD}_i\right)^2}{\sum_i \text{EAD}_i^2}$
LGD	Exposure weighted loss given default	$\text{LGD} = \frac{\sum_i \text{LGD}_i \cdot \text{EAD}_i}{\sum_i \text{EAD}_i}$
h	Probability that all exposures in the pool perform	$h = \left(1 - \frac{K_{irb}}{\text{LGD}}\right)^N$
c	Approximation of the mean parameter of the fitting function	$c = \frac{K_{irb}}{1-h}$
v	Variance of the conditional loss distribution	$v = K_{irb} \cdot \frac{(\text{LGD}-K_{irb})+1/4 \cdot (1-\text{LGD})}{N}$
f	Approximation of the variance of the fitting function	$f = \frac{v+K_{irb}^2}{1-h} - c^2 + \frac{(1-K_{irb}) \cdot K_{irb} - v}{(1-h) \cdot \tau}$
g	Precision parameter of the fitting function	$g = \frac{(1-c) \cdot c}{f} - 1$
a		$a = g \cdot c$
b		$b = g \cdot (1-c)$
d	Component of the supervisory add-on	$d = 1 - (1-h) \cdot (1 - \beta[K_{irb}; a, b])$
τ	Determined by the regulators. Tau is a supervisory parameter that captures the uncertainty in the loss prioritisation because the rules governing the cashflow waterfall are not known for a specific deal	$\tau = 1\,000$
w	Parameter that controls the rate of decay of the supervisory add-on. Determined by regulators.	$\omega = 20$
$K[L]$	Aggregate systematic (non-diversifiable) credit risk attributable to a first loss position up to and including L. General form shown.	$K[L] = (1-h) \cdot [(1 - \beta[L; a, b]) \cdot L + \beta[L; a+1, b] \cdot c]$
$S[L]$	Capital charge associated with systematic risk plus supervisory add-on. General form shown.	$S[L] = \left\{ \begin{array}{l} L, \text{ when } L \leq K_{irb} \\ K_{irb} + K[L] - K[K_{irb}] + \frac{d \cdot K_{irb}}{\omega}(1 - e^{\frac{\omega \cdot (K_{irb}-L)}{K_{irb}}}), \\ \text{when } L > K_{irb} \end{array} \right\}$
SFA capital charge	Supervisory Formula Approach capital requirement (where E is the pool's notional amount in dollars).	$k_{\text{SFA}} = E \cdot \max \left\{ \begin{array}{l} 0.0056 \cdot T \\ S[L+T] - S[L] \end{array} \right\}$

19.4 CREDIT SECURITISATIONS

Another way to think about a CDO tranche is as a source of capital for an enterprise. Typically, a company or bank will raise money in a variety of ways ranging from high-risk equity capital to low-risk borrowing from lenders or depositors. An entire CDO structure can be viewed in a similar way – capital being raised subject to a variety of risk levels. A credit securitisation is a means of raising capital to purchase some assets already in existence and owned by another entity. A securitisation usually looks like a cashflow CDO with a variety of tranches of risk being funded by the pool of assets being sold. Asset securitisations often falls into one of two categories, and some examples are the following:

Consumer credit securitisations include:

(a) residential mortgages (RMBS)
(b) credit cards
(c) auto loans/leases
(d) life settlements (life insurance contracts sold in the secondary market rather than being surrendered to the insurance company)
(e) personal loans
(f) student loans
(g) rental contracts (televisions and 'white goods').

These typically use an actuarial approach to estimating credit risk, as pools are typically very large (thousands of borrowers).

Commercial securitisations include:

(a) commercial mortgages (CMBS)
(b) trade finance
(c) SME loans
(d) lease transactions (equipment)
(e) aircraft and shipping finance).

These tend to use individually estimated parameters for credit risk and loss given default (i.e. internal rating estimates).

A feature of securitisations is that the underlying contracts do not trade in a secondary market – pricing off the hedge is not possible and some form of fundamental valuation is necessary. A prerequisite to being able to perform a fundamental valuation is that sufficient high-quality historical data should exist in order to be able to correctly model the economic impact of the development of the pool of instruments. We shall illustrate this by discussing a concrete example – life insurance policies – but also making reference to lease contracts.

Data Collection

Running a business related to life insurance (rental) contracts requires data to be held and maintained – contact details, contract details including renewal dates and any option dates, etc. An intended securitisation needs access to this historical data (which the business may simply throw away) and also needs the data to be reorganised into a different format. In the life insurance industry, collected data is not only used at the individual company level but is also collected by actuarial bodies acting for the profession as a whole. This is unusual, as lease contract information generally does not go beyond the originating company. The first step, therefore, is the creation of a historical database.

Data Items

It may be that the modelling of the future performance of individual contracts requires further data in addition to that collected for purely transactional purposes. This data will only become relevant as modelling of past performance is developed. Data collection, specification of the data required, and data modelling go hand in hand and the development of a data base usually evolves over a period of years. For example, life insurance modelling has been developed and enhanced over a period of around 200 years, and has therefore reached a relatively stable level. The key items recorded are contract type, age, sex, smoking status at the time of the policy, medical condition at the time of the policy, and date when the policy was taken out (known as a 'select date', which signifies – for 'normal' lives – that the individual differed from the general population at this date because he or she was accepted for a life insurance policy, indicating that his or her health condition was better than average). For rental contracts, likely candidates for relevant data include contract type, underlying product value, date the contract was taken out, geographical location, and contract options. A preliminary list is typically generated on the basis of available data and the product understanding of the business managers, which also, generally, indicates the type of mathematical model that is likely to be appropriate.

Data Modelling

The key component to being able to assign a (fundamental) value to the securitisation is being able to model the underlying contracts and cashflows generated from them. There are similarities here with the modelling of corporate entities – survival and mortality curves generally being the key components. The terminology itself is closely allied to the modelling of human survival. Note that here there are, or may be, some differences compared to the modelling of corporate default. 'Populations' of contracts may interact, in addition to being self-contained – the migration from one contract group to another, as well as simply surviving or terminating, may need to be modelled.

The process of building a mathematical model is not easily described and requires input from those who understand the business, mathematical modellers, and statisticians. A key component of the model development phase is calibrating the model with historical data and 'out of sample' testing (predicting a period beyond the historical data used for calibration but where data is available to test the accuracy of the prediction). This process generally reveals additional data that the business would be wise to start collecting.

It can be seen that securitisation for an unusual asset class is not a short process, and the overhead in developing the infrastructure to allow a valuation and sale is significant. Much of this infrastructure may be in place for business purposes, but often is not.

Rating a Securitisation

Once the groundwork is done and a viable securitisation model and structure has been developed, the proposed securitisation requires to be rated by investors. In the life products area, AMBest is the rating agency most commonly involved, but other securitisations are often rated by the credit rating agencies. The rating agencies are generally involved during the model development phase since the agencies will need to develop their own models and stress test the proposed securitisations.

19.5 RATING

The rating agencies provide a ratings service for some or all of the tranches of a CDO and other credit structures. The asset may be required to be rated by investors as part of a due diligence process and may be a prerequisite before they are able to buy a tranche. The rating agencies make it clear that the process of rating any sort of deal is not a purely mechanical one of entering data into a formula and getting the rating. This can be a problem for structurers – not being sure what the final rating will be until the rating agency provides it just before the product launch. Structuring a deal efficiently generally requires knowledge of the ratings that will be applied to tranches since this will influence the premium that has to be paid to investors (writers of protection). The structurer tries to minimise overall cost (optimisation) by adjusting tranche sizes and the reference portfolio (thereby changing the tranche ratings and affecting the premium that will have to be paid on the tranche). Often it is not minimum cost but where the investor interest lies.

However, the agencies have become more transparent and more formulaic (at least for static synthetic structures) – for example, S&P will supply its ‘CDO evaluator’ software to market participants. Generally the rating agencies’ methods are historically driven: ratings are used, together with historical default and recovery rates, as the basis for the determination of the expected losses on a tranche. Historically these ratings have driven the pricing of CDO tranches, and the cost of obtaining protection in tranched form has often been less than can be earned in a spread on the underlying portfolio (leading to ‘arbitrage CDOs’) and ‘value’ was generally found in the mezzanine area – i.e. from the structurer’s perspective, protection on mezzanine tranches can be bought cheaply.

One method used by Moody’s was the ‘binomial expansion technique’, which sought to replace the actual correlated pool by a smaller uncorrelated pool – which could be modelled easily and in closed form. The rating agencies have moved to a mathematical modelling process similar to that used by banks and investors, specifically using a correlated default time model based around the Normal Copula. The details of their models are not made public although structurers and others are allowed free access to the rating software to enable them to run the software to see what rating is indicated.

In outline, the current methodology takes individual default and survival curves based not on current spread information but on rating (transition matrix) derived default and survival curves. Recovery rates are based on historical data, and a correlation matrix is derived based on a variety of tags. Together with an interest rate curve this data is then used – subject to additional internal stresses – in a simulation implementation to derive an expected loss on a tranche, which is then converted to a rating.

19.6 ALTERNATIVE LEVERED CREDIT PORTFOLIO PRODUCTS

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The views and opinions expressed in this section are those of the author, who is writing in a purely personal capacity. No information or references to WestLB, its clients or transactions are included or inferred.

An alternative to standard CDO and correlation product are market value structures. In market value structures the risk is not only the ultimate default (or not) of the obligors in the reference portfolio but also the price appreciation/depreciation on the portfolio.

Market value structures do not rely on tranched credit correlations, although the prices of the portfolio will be somewhat correlated. Market value structures are typically employed on relatively liquid, low price/spread volatility financial instruments. In the past, these have included:

(a) Leveraged loans
(b) CDSs
(c) SIVs/Credit conduits (investments typically in highly rated ABS/RMBS and bank capital instruments)

The strategy is usually leveraged and leverage ratios typically vary from 5 to 15 times, although in distressed markets this has become more like 2 to 10 times. The risk taken on a leveraged market value structure is clearly that losses through price movements will effectively wipe out the investment. Several strategies are employed to vary leverage with price, two of which – CPPI and CPDO – have found favour in the credit markets in recent years.

19.6.1 CPPI

CPPI, or constant proportion portfolio insurance, is an investment strategy which first appeared in the 1980s for equity portfolios. Originally an equity index product – a product that is still commonly offered by retail banks to their clients – it has since been applied to other markets, including hedge funds, commodities and credit markets. Briefly it combines a low-risk or riskless zero coupon asset with an investment in a portfolio of a risky asset and cash deposit. In the more developed markets the leveraged investment is often offered as a fixed leverage option (call) on the underlying index, which may be additionally leveraged by lending against the market value of the call. However, in less-developed markets it is more often a dynamic leverage strategy based on the market value of the portfolio.

The initial invested amount is split between the zero coupon bond and the reserve amount. The reserve amount is invested into cash or cash equivalent investment (deposits, repo agreements, commercial paper, etc., or can be a coupon swap on the low-risk asset, i.e. a swap where the coupons on coupon-bearing asset are paid in return for an up-front amount) and a total return or performance swap on the risky asset. For credit-based CPPI, the total return swap is on a portfolio of credit defaults. Losses or gains on the total return swap are debited or credited to the cash account. The credit default swaps are usually on one or more of the standard indices (e.g. CDX) or tranches of indices, although bespoke credit managed portfolios can also be employed.

Unlike other leverage product, the investment is not tranched, other than the CPPI investor is clearly taking a junior position in a larger portfolio. Typically there is a maximum and a minimum leverage ratio linked to the present value of the portfolio. Interest earned on the reserve account and the premiums paid on the credit default swaps can be used to pay a coupon on the CPPI note, increase the reserve amount, pay for any senior protection or pay down the note. The minimum leverage amount is determined by the shift sensitivity ('PV01') of the portfolio, which in turn is linked to the coupon and duration of the swap. If the mark-to-market loss of the total return swap exceeds a 'low water mark', then the portfolio is liquidated and the investor receives only the principal at maturity. The dealer takes the risk, the so-called 'gap' risk, that the spread on the portfolio will 'jump' sufficiently to make the mark-to-market loss on the risky portfolio breach the low-water mark and exceed the total available reserve.

For example, assume an investment in a 10-year CPPI structure where the zero cost is 65% of the initial investment. For a 10-year investment, the duration is clearly 10 years for the zero, but the modified duration will be slightly less than 6.5 years if the cost is 65%, say 6.23. The initial reserve amount is therefore 35%. The size of the CDS portfolio or index is adjusted between the ranges, dependent on the size of the reserve account and the mark to market of the portfolio. Assume that the investment is one of the 5-year indices currently priced at 75 bp and that the index will roll every 6 months into the 'on-the-run' index. If we assume that the modified duration of the index is initially 4.4 years, then, if the initial leverage is 15, the initial gap risk taken by the dealer is $35/(4.4 \times 15)$, or a 53 bp jump in the index, or an increase of 70%.

If the spreads widen to, say, 20 bp, then the loss on the total return swap will be $0.2\% \times 4.4 \times 15 = 13.2\%$. The value of the reserve now becomes 21.8%. Assuming that the dealer is happy still with a 50 bp jump risk, then the leverage would become 9.35; however, if he wanted a 70% increase on the spread margin, then the available leverage would be 7.4.

These figures do not include any 'low water mark' to allow for friction and costs in liquidating the portfolio. This may be, for example, 5% of the portfolio value.

The reserve amount can be adjusted by

(a) the mark to market of the credit portfolio
(b) premiums paid on the portfolio
(c) interest earned on the cash account

For example, if the PV01 of the unlevered portfolio is –5 cents (i.e. a 0.01% change in the spread causes a –0.05% change in the price and, allowing for 5% for friction from the initial 35% to invest, leaves 30% of the investment to invest in the risky credit index, implying a 6% change in the spread (divided by the leverage) on the underlying portfolio before a loss is sustained. For instance, if the leverage was 25 times, this implies a 0.24% spread move on the underlying portfolio before the issuer is exposed to a loss. So the maximum leverage may be something like 30 times (if the portfolio PV is higher and/or the reserve fund grows).

The risk taken by the investor is:

(a) coupon or present value risk if the structure 'knocks out', i.e. the PV hits the low water mark and unwinds and is linked to the market value and/or credit default risk of the portfolio;
(b) credit risk on the issuer of the zero (potentially market risk also if the CPPI is subject to early termination for regulatory, legal or investor requests);
(c) counterparty credit risk on the performance swap counterparty;
(d) credit risk, if any, on cash investments.

The dealer or issuing bank predominantly takes the market value 'gap' risk.

The risk is usually priced and assessed using some form of jump diffusion model, e.g. Cox–Ingersoll–Ross or exponential Vasicek, linked to a state transition model in order to model different spreads, spread volatility, reversion rates and credit risk due to credit migration to enable the likely price evolution to be developed.

19.6.2 CPDO

As discussed above, CPPI structures the leverage moves proportionally to the present value of the portfolio, i.e. the higher the present value the higher the permitted leverage (up to a maximum). In contrast, a more recent structure was the constant proportion debt obligation (CPDO) in which the leverage moved inversely proportional to the present value of the portfolio.

The CPDO was a structure that found favour in 2006/2007 and was almost the completely opposite strategy to the CPPI. The structure again was a levered market value strategy based, typically, on one or more of the 'on-the-run' indices (iTraxx or CDX). There was no principal guarantee. The notes were initially rated by agencies at AAA/Aaa to both rated principal and promised rated coupons of up to LIBOR + 200 bp. The structure was typically a 10-year maturity based upon rolling the 5-year indices. Leverage was typically 5 to 20 times, depending on the present value of the CDS, but unlike the CPPI – whose leverage reduced when the present value of the portfolio reduced – the leverage on the CPDO increased when the present value of the portfolio went down (subject to a limit).

A typical CPDO structure was a 10-year structure on 5-year rolling indices (combined CDX and iTraxx), 10 times leverage with a range of leverage from, say, 5 to 15, paying a coupon of between LIBOR + 100 to 200 bp and rated AAA/Aaa by at least one agency. The initial investment is placed in a cash or cash equivalent instrument (e.g. Repo or TRS) and this pays the LIBOR/EURIBOR element of the coupon. Excess income (above 200 bp) from the fixed payments under the CDS, and positive gains from leverage changes and index rolls, are also paid into the cash account. Once the value of the deposit account and the present value of the CDS reach a target value equal to the principal plus the expected coupons (plus a margin for fees and expenses), the CPDO 'cashes in' and the CDS is terminated. Losses are deducted from the cash account, including credit losses through defaults, mark-to-market losses for changes in leverage and index rolls. If the value of the investment falls below a low water mark, typically 10%, then the deal 'cashes out', and the investor only receives the recovery value, if any, after termination costs.

The thinking behind a CPDO is that credit spreads are mean reverting. By keeping to the 'on-run' indices, liquidity premiums can be kept to a minimum and defaults reduced as credits that migrate below minimum rating or maximum spread tend to be excluded on the rolls. This brings in a small flaw to the mean reversion argument as high-volatility spreads will tend to be excluded on the roll to a new series, so losses are potentially locked in on these credits as there is no opportunity for it to recover. Additionally, rolling would not protect against credits that go to default within the period where the series is current.

Example Transaction

Assume a rated coupon of EURIBOR +2%, with a 10-year legal final and an initial duration of 7.5 years. The target price would initially be 115%. Assume that the index is 0.58% and the initial leverage is 10 times. The gap is 115% + 5% (margin for expenses, mark to market, etc.) *less* 100% *less* closing fees, or approximately 20%. The size of the gap determines the amount of leverage, which is given by a formula similar to the minimum of X and maxium leverage, where X = Gap/(SpreadPV × Multipler) and where the multipler is typically 0.75 (that is, initially [20%/(2.59% × 0.75)] =10.04 times, and the maximum leverage is between 15 and 20 times

Each payment date a total return swap will pay EURIBOR to the structure on the cash deposit and the structure will receive 5% (10 × 0.5%) of which 2% is paid out and the rest is reserved. Thus, on the first payment date the deposit duration would be, say, around 0.25 years shorter (assuming similar discount curves) and the deposit amount would have increased to

103%. Hence, gap would be 114.5% + 5% *less* 103%, or 16.5%. This would imply a leverage, assuming no change in spreads, on the CDS of 16.5%/(2.46% × 0.75) = 8.92 or 9.

As the cash account increases without mark to market the leverage decreases until the 'cash in' event occurs. Conversely, if there is a negative movement on the index (i.e. spreads widen), the gap will also widen. For example, if the spread widens to 70 bp from 58 bp on a 4.25-year duration, then a mark-to-market loss of around 5.1% is incurred (0.12% × 4.25 × 10).

The gap becomes approximately 22% and the leverage increases to 12 times. The knock-out level of, say, 10% would be reached if the spread on the index rose by, say, 140 bp. Unfortunately this is exactly what happened to some of the initial CPDO structures in late 2007 and during 2008.

The main issues with the CPDO were the pricing and rating methods. A similar pricing model as above is used to estimate future price evolutions. The main problems related to calibration and the credit crunch in general. Only 4 years' data was available to calibrate the parameters for the model, which led to a serious underestimation of the jump parameters, and consequently many of the CPDO structures failed as spreads blew out.

19.6.3 Advantages of Market Value CDS Products

One of the features particularly of index products has been their liquidity, even when other credit had become illiquid. Market value strategies, such as CPPI and CPDO, do not rely on correlation to determine prices (other than perhaps to calculate duration) and can be valued relatively easily. However, the valuation and rating methodologies are difficult to justify in terms of 'credit losses' – in particular, the rating methodology attempts to equate losses caused by defaults to losses through changes in market value. This is highly questionable as it is often comparing data gathered for 90 or more years on defaults to short-term price series collected over a few years for market value loss estimates.

CPPI is typically more suited to:

(a) retail/high-net-worth accounts that do not want to risk principal;
(b) diversified and concentrated investments (e.g. tranches on indices).

It has the advantage of principal protection, and can easily be rated to return of principal. However, the coupon can be less easily determined and may actually be zero and, therefore, difficult to rate. Consequently, CPPI is less interesting to banks and other institutions that rely on ratings to both principal and interest for capital determination purposes.

CPDOs – since they could be rated to both principal and interest – were targeted to financial institutions as part of a broad credit strategy and were prepared to risk capital.

The main differences are:

(a) typically based upon diversified indices (usually on-the-run 5-year iTraxx and/or CDX);
(b) indices are rolled to always be 'on-the run';
(c) principal and interest payments are rated;
(d) no principal protection, i.e. whole investment is at risk;
(e) credit risk can terminate early with a 'cash-in' event when PV of principal and future coupons has been reached;
(f) structure knocks out when a low water mark is reached, typically 10% of the initial PV with the investor only receiving recovery value.

20
The Normal Copula and Correlation

In this chapter we concentrate on the theoretical aspects of pricing models and, in particular, the 'Normal Copula'. In Chapter 21 we discuss application and some of the failings of the model and how to get round them.

The reader probably believes that the default correlation and correlated spreads have the same source so that correlating spreads is the most sensible model to develop. While this is true, it is impractical at present – we shall see later (from the pricing of F2D baskets using alternative models) that spreads have to be modelled as a jump process in order to replicate market prices with any sensible levels of spread volatility. Jump processes are difficult to handle – efficient methods have not yet been developed or accepted to incorporate jump processes in a way that can allow the models to be easily calibrated to market data and also lead to fast run times. As a consequence, the model that is in common usage is a correlated default time model.

20.1 DEFAULT TIME CORRELATION

We shall explain the concept of default time correlation by considering simulated default times.

20.1.1 Generating Correlated Default Times

First consider two names which have a default time correlation, ρ. Section 6.3 showed how to obtain the Cholesky decomposition from the correlation matrix and how to generate correlated random Normal numbers using this decomposition. Chapter 18 showed how to convert a normal random number into a default time given the hazard rate process (equation (18.3) if the hazard rate is a constant). The spreadsheets 'chapter 20.xls' and 'chapter 18.xls' produce correlated default times given CDS premium data and recovery assumptions (under the assumption that the hazard curve is constant), for any correlation.[1] Table 20.1 shows the first few simulated default times for uncorrelated ($\rho = 0$), 70% correlated, and 100% correlated default times.

20.1.2 Intuitive Understanding of Default Time Correlation

In Table 20.1 'name 2' has a higher hazard rate than 'name 1', therefore it has an expected default time that is before that of name 1. In the first pair of columns (zero correlation) in 8 of the 14 simulations name 1 actually defaults before name 2 – but as only 14 simulations are shown in the table no clear conclusion can be reached.

[1] Note that setting $\rho = 1$ does not give a correlation matrix (more on this later). We can let the correlation get close to unity but not actually be equal to unity. When we talk about 100% correlation we mean the correlations is very close to unity. (The simulation process can be extended to work with unit correlation using the factor method but not using the 'correlation matrix'.)

Table 20.1 Simulated correlated default times for 100 bp and 200 bp CDS names with recovery of 50%

#	Correlation = 0		Correlation = 0.7		Correlation = 1.0	
	Name 1	Name 2	Name 1	Name 2	Name 1	Name 2
1	24.5	29.8	30.4	9.3	37.0	18.5
2	1.6	9.7	14.5	11.0	127.2	63.6
3	14.1	37.8	3.1	4.1	3.6	1.8
4	9.9	1.9	10.1	4.1	36.0	18.0
5	47.8	25.6	20.3	15.3	1.7	0.8
6	3.1	20.4	112.5	63.8	75.3	37.6
7	30.4	3.3	67.3	26.1	7.8	3.9
8	5.9	6.1	21.3	7.9	156.2	78.1
9	6.1	43.1	193.9	16.2	45.6	22.8
10	75.8	26.7	43.3	26.7	31.8	15.9
11	63.3	20.9	29.2	14.8	44.8	22.4
12	12.7	35.6	10.5	5.8	17.7	8.8
13	1.7	23.0	24.6	26.3	70.5	35.2
14	76.3	15.4	5.8	4.4	59.3	29.7

Exercise 1 Extend the simulations to 10 000 and show that the expected default time of name 1 is indeed 50 years, and name 2 is 25 years. In particular, the average default time of name 1 is greater than that of name 2.

If we set the correlation to 70% then 12 of the 14 simulations show name 1 defaulting after name 2, and if we set the correlation to 100% then – on every simulation run – the default time of name 1 is twice that of name 2.

Exercise 2 Extend the simulations to 10 000 and show the following:

(a) At both 70% and 100% correlation the expected default time of name 1 is indeed 50 years, and name 2 is 25 years. In particular, the average default time of name 1 is greater than that of name 2.
(b) At the three correlation levels what proportion of name 1 defaults occurs after name 2?
(c) Using formula (18.3) explain the 100% correlation results.

We conclude that **changing the correlation has no impact on the expected default time of a particular name**, but does have something to do with how defaults of one name relate to defaults of another.

We can think of a single simulation as a roll of the dice which tells us the complete future of the universe (for these two names). We know what these two default times will be once we have cast our random numbers. If we think of many simulations we are actually thinking about many separate universes in which the two names default. By looking at the default time of name 1 and the default time of name 2 as random variables, we can calculate the correlation of the observed default times in the usual way.

Of course reality is totally different from the above. There is just the one universe in which name 1 and name 2 default and, even if we wait until they have defaulted we cannot calculate their default time correlation. Thus, the concept of default time correlation is mathematically valid, but is not directly observable. How then can we calculate the default time correlation of two names? We shall postpone this discussion until Chapter 22.

The extension of the above process to any number of names is mathematically trivial in principle.

Why do we use a model based around such an obscure and unobservable quantity?

We have already touched on alternative possible models – for example, correlated hazard rate processes, perhaps with jumps. The answer to the above question is that the model based around correlated default times is very quick to execute – even if we simulate default times – whereas a simulated hazard rate process requires simulating over time intervals, which is slow and also presents substantial problems (at least in terms of run-time) for the calibration of each single name CDS. In such a hazard rate simulation and calibration process it may take 24 hours to price a single CDO tranche, whereas the default time model can take one hundredth of a second (if a 'semi-closed form' approach is taken – see later). Against this enormous practical advantage of the Normal Copula default time model we are stuck with a model where we have no direct means of observing the key input.

20.1.3 Implications of 100% Default Time Correlation

We have seen above (Exercise 2(c)) that, at 100% correlation, the default times are in the precise relationship to the inverse of the hazard rates.

What happens to spreads and CDS levels during this process? If we look at the deal in, say, 5 years' time, what does the model tell us about the CDS premiums on the surviving names?

Take the 100% correlated column. If name 2 has defaulted but name 1 has not (simulation number 7 in Table 20.1) we know precisely when name 1 will default – i.e. in a further 3.9 years. The hazard rate function in this case has changed completely – the hazard rate is zero up to 3.9 years' time when it becomes infinite. We shall return to this point later.

Exercise 3 Using the simulated default times in columns G and H of Table 20.1, we can ask such questions as: 'if name 2 survives 5 years, what is the expected default time of name 1?' Working with hazard rates of 100 and 200 bp respectively, show that (columns Q to Y), if name 1 and name 2 have a positive default time correlation:

(1) the expected default time of name 1, conditional on name 2 surviving beyond (say) 5 years, is greater than 100 years;
(2) the expected default time of name 1, conditional on name 2 defaulting before 5 years, is less than 100 years;
(3) the expected default time of name 2, conditional on name 1 surviving beyond (say) 5 years, is greater than 50 years;
(4) the expected default time of name 2, conditional on name 1 defaulting before 5 years, is less than 50 years.

(Increase the number of simulations to ensure that you get significant results.)

What does this say about the forward (conditional) hazard rates (and spreads) of name 1 and name 2? The Copula model implies a correlated jump process for spreads, and pseudo-hazard rates and, implicitly, hazard rates.

20.1.4 N2D, Zero Correlation Case – Exact Pricing and Hedging Formulas

Special cases – where exact solutions are known – are of interest in their own right and are also useful as limiting cases to test more sophisticated models. These special cases allow us to perform tests on our model and its implementations. In addition, the zero and 100% correlation cases provide further insight on how correlation products work and also how our model treats them – with some surprising conclusions. Also, the 'semi-closed form' pricing approach for correlated portfolios can give very accurate answers in special cases – see chapter 22 and 'chapter 20 Normal Copula closed form N2D pricing.mcd'.

For readers who wish to skip the mathematics of this and the following section, we summarise the key results:

1. *Zero Correlation*
   (a) For a fair value F2D where the two names are uncorrelated and the hazard rates are constant, we see that the first-to-default premium is the sum of the premiums for the individual CDS contracts. See Exercise 4 below.
   (b) The F2D hedge ratios are less than 100% in each name. See Exercise 6 below.
   (c) The S2D premium is positive – so an F2D and an S2D product on two reference names costs more than the sum of the CDS premiums – and the hedge ratios are positive. See Exercise 7 below.
   (d) The sum of the hedge ratio for the F2D and the S2D is greater than 100% of the notional for each name in the basket. See Exercise 9 below.
2. 100% *Correlation*
   (a) The F2D contract is identical to a CDS on the higher hazard rate name, and the S2D is identical to a CDS on the lower hazard rate name.
   (b) Since recovery is unknown on specific names, the premium for the F2D may be anywhere between the lower spread name and the higher spread name. The sum of the N2D premiums remains equal to the sum of the CDS premiums.
   (c) The hedge ratios for the N2D contracts sum to 100% in each name.

Zero correlation

Consider the first to default product. For simplicity we assume the (pseudo-hazard rates are constant (i.e. there is no CDS term structure), and the interest rate is also constant. Remember that the CDS pseudo-hazard rate can be consistent with any underlying stochastic hazard rate process (in particular, including jumps) subject to certain constraints on various correlations, so the results in this and the following section are also valid in a jump process world. We also ignore counterparty risk at the moment. The formulas illustrate the case for a basket of two names, but the generalisation to many names is straightforward.

Since the names are uncorrelated, from basic probability theory we have the following results:

1. The joint survival probability of name A and name B is the product of the survival probability of A times the survival probability of B.

2. The probability (density) that A is the first to default at time t is the probability that A survives until t times the probability that B survives until t times the hazard rate of A.

The following formulas are implemented in 'chapter 10 and chapter 20 hedge theory.mcd'.

F2D Valuation

Assume that premiums are paid continuously (which is a good approximation to the typical premium schedule for 'quarterly in arrears with a proportion to the default date'). A real world implementation would of course take the details of premium payment and day-count conventions into account, but the differences are small and do not invalidate the key findings below.

A unit of premium $p.\Delta t$ is paid at time t if both A and B survive until t (with probability $S_A(t) \times S_B(t) = \exp(-\lambda_A t) \times \exp(-\lambda_B t)$). Its value is obtained by discounting at interest rate $r[D(t)] = \exp(-rt)$. The value of the premium stream up to the maturity T is therefore

$$P = p\int_0^T \exp(-(\lambda_A + \lambda_B + r)t)dt = \frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \tag{20.1}$$

The contingent benefits arise if both names survive until time t, then either A or B defaults at time t. This gives rise to a payment of (1 − recovery), and this payment is discounted at the interest rate r. Hence the value of the contingent benefits is

$$B = \int_0^T ((1 - R_A)\lambda_A \exp(-(\lambda_A + \lambda_B + r)t) + (1 - R_B)\lambda_B \exp(-(\lambda_A + \lambda_B + r)t))dt$$

$$B = ((1 - R_A)\lambda_A + (1 - R_B)\lambda_B)\frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \tag{20.2}$$

Recall that, under the assumptions we are making, the single name CDS premium is equal to hazard rate time loss given recovery (equation 9.17). Hence the net value of a long position in an uncorrelated F2D (equation (20.2) minus (20.1)) is

$$V = (CDS_A + CDS_B - p)\frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \tag{20.3}$$

For a fair value F2D where the names are uncorrelated, and where the hazard rate is a constant, we see the first to default premium is the sum of the premiums for the individual CDS contracts:

$$p = \mathrm{CDS}_A + \mathrm{CDS}_B.$$

Exercise 4 How can this make sense? We are paying as much for an F2D, which terminates at the maturity date or the default of either name, as we would for two individual CDSs which give the same protection up to the earlier default date yet give continued protection on the remaining name after the first defaults.

The answer is in our assumption that the (pseudo-)hazard rates are constant. When the first name defaults then, given our assumption, we can purchase protection on the remaining name at the same premium rate as if we had bought it at inception. Hence, F2D protection plus the potential future purchase of single name protection at a known price gives the same protection at the same cost as two single name CDS contracts.

This result clearly is not valid if there is a term structure to credit spreads for a single name – although the error is typically quite small.

Exercise 5 Given the above pricing result, how would you hedge an uncorrelated F2D basket?

The hedge ratios are **not** 100%. For a notional holding (long of protection, say) in an F2D basket we can hedge by selling notional amounts of CDSA and of CDSB (all trades are assumed to be done at a fair market price). For the moment we shall concentrate on hedging out hazard rate sensitivity (equivalently spread risk). In order to be hedged the portfolio must show no sensitivity to changes in the hazard rate (spread) on either name. We can calculate the sensitivity of the F2D from the value formula above, and compare it with the CDS sensitivities – this gives two equations for the notional amounts in the two CDS contracts. *The F2D hedge ratios turn out to be less than* 100% (even though – at zero correlation – the correct premium for the F2D is the sum of the CDS premiums).

Exercise 6 Why are F2D hedge ratios not 100% even at zero correlation?

The answer ultimately rests on the fact that the incidence of cashflows is different under an F2D compared with the portfolio of CDS contracts. If we were dealing with contracts where the premium is paid as a single up-front amount (given by the value of contingent benefits) we get the intuitive result that the F2D up-front premium is the sum of the CDS up-front premiums but we would still find that the F2D hedge ratios are not 100%.

If we view the F2D as two CDS contracts less a forward contingent purchase (at a known annual premium rate) of a CDS on the surviving name, then we can see immediately where the difference in sensitivity comes from. It is the contingent forward CDS deal that differentiates the F2D from the two CDS prior to the default of one name. We would therefore expect the F2D hedge ratios to decline as we look to higher spread names (because the contingent forward deal is more and more likely to be triggered). The reader can check this in the 'chapter 10 and chapter 20 hedge theory.mcd' sheet or by implementing the formulas in Excel.

S2D Valuation

Three cases give rise to a premium stream:

1. Both names survive to time $t <= T$.
2. Name A defaults at time $t_1 < t$ and B survives to $t < T$.
3. Name B defaults at time $t_1 < t$ and A survives to $t < T$.

This gives us the value of the premium stream p_2 for the second to default

$$P2 = p2 \cdot \left(\int_0^T \exp(-(\lambda_A + \lambda_B + r)t)dt + \int_0^T (1 - \exp(-\lambda_A t)) \cdot \exp(-(\lambda_B + r)t)dt + \int_0^T (1 - \exp(-\lambda_B t)) \cdot \exp(-(\lambda_A + r)t)dt \right) \tag{20.4}$$

We can expand and simplify this to get

$$P2 = p2 \cdot \left(\frac{1 - \exp(-(\lambda_B + r)T)}{\lambda_B + r} + \frac{1 - \exp(-(\lambda_A + r)T)}{\lambda_A + r} - \frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \right) \tag{20.5}$$

A capital payoff is made in one of two cases:

1. Name A defaults at time $t_1 < t$ and B defaults at $t < T$.
2. Name B defaults at time $t_1 < t$ and A defaults at $t < T$.

This gives the value for the contingent benefits of

$$B2 = \int_0^T (1 - \exp(-\lambda_A t)) \cdot \exp(-(\lambda_B + r)t)\lambda_B \cdot (1 - R_B)dt + \int_0^T (1 - \exp(-\lambda_B t)) \cdot \exp(-(\lambda_A + r)t)\lambda_A \cdot (1 - R_A)dt \tag{20.6}$$

Expanding, integrating and simplifying gives

$$B2 = CDS_B \cdot \frac{1 - \exp(-(\lambda_B + r)T)}{\lambda_B + r} + CDS_A \cdot \frac{1 - \exp(-(\lambda_A + r)T)}{\lambda_A + r} - (CDS_A + CDS_B)\frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \tag{20.7}$$

So the capital value of a long protection S2D deal in these circumstances is

$$V2 = (CDS_B - p2) \cdot \frac{1 - \exp(-(\lambda_B + r)T)}{\lambda_B + r} + (CDS_A - p2) \cdot \frac{1 - \exp(-(\lambda_A + r)T)}{\lambda_A + r} - (CDS_A + CDS_B - p2)\frac{1 - \exp(-(\lambda_A + \lambda_B + r)T)}{\lambda_A + \lambda_B + r} \tag{20.8}$$

The fair value new CDS premium is given by finding the p_2 that makes the trade value zero. (Examples are given in the MathCad example sheet.)

Exercise 7 The above formula gives a non-zero S2D premium. Clearly the S2D has some value so has a positive (non-zero) premium no matter how remote the probability that both names default. Consider two portfolios (of the same maturity):

1. A CDS on name A and a CDS on name B; both names are subject to 100 bp premiums.
2. A F2D on names A and B at 200 bp, and a S2D on names A and B at 8.8 bp.

In the zero correlation case, if I buy first-to-default and second-to-default protection I am paying more than the sum of the two CDS premiums. Both portfolios – FSD plus S2D or CDSA plus CDSB – give the same capital protection. Why should the F2D plus S2D portfolio cost more?

It is true that the two portfolios have the same capital (contingent payoff) flows in all cases – whether there are no defaults, whether A defaults and B survives, whether B defaults and A survives, or whether both default before the maturity date. But the premium flows are different. We have seen that the F2S premium is a high number (= the sum of the CDSs at zero correlation); and the S2D premium is typically a low number – much less than the lower of the CDS premiums in the case of zero correlation. Imagine for a moment that the F2D plus S2D premium is the same as the sum of the CDS premiums. If no defaults occur the same premium is paid on both portfolios. But in any other event, after the first name defaults, less premium is paid on the S2D contract than on the remaining CDSs (whichever name survives). Hence the premium in the case where both names survive has to exceed the sum of the CDS premiums. (This argument is true at any correlation except 100%.)

Exercise 8 Prove that if the interest rate is 4% and recovery is assumed to be 50%, then the S2D premium above is 8.8 bp.

Exercise 9 Do the F2D plus S2D hedge ratios add to 100%?

No. At this stage the result probably does not surprise you. For 1 000 bp and 2 000 bp names and a 5-year F2D and S2D together, we find the hedge ratios are approximately 105% in the first name and 120% in the second. At 100 bp and 200 bp we find 103% and 107%.

20.1.5 N2D, 100% Correlation – Exact Pricing and Hedging Formulas

We shall use the default time simulation process of Chapter 18. We simulate 100% correlated defaults, and then value various portfolio products. What does this mean?

We saw in Chapter 18 that we could obtain a default time from a random number (equation (18.3) reproduced below). If default times are 100% correlated, then this means that we use the same random number to generate the default times for names A, B, ... – a single random number × generates the default times for all the names i in the portfolio:

$$t_i = \frac{-\ln(\Phi(x))}{\lambda_i}$$

Note that if name A has a 1% hazard rate, and name B has a 2% hazard rate, then name A defaults after twice the time it takes B to default. For example, if the randomly generated probability is 0.015, then A defaults after $-\ln(0.15)/0.01 = 420$ years, while B defaults after $-\ln(0.15)/0.02 = 210$ years.[2]

Suppose A has a higher hazard rate (lower survival probability and shorter simulated default time) than B. Then the probability that A and B survive to time t is just the probability that A survives to time t. If we are looking at F2D and S2D contracts, then the F2D is triggered if and only if the CDSA is triggered, and the S2D is triggered if and only if the CDSB is triggered. Thus the F2D *is* CDSA and therefore has the same premium and is exactly hedged by a 100% holding in name A, and the S2D *is* CDSB.

We can also follow through the probability formulas to calculate the premiums and hedge ratios explicitly – this is done in the 'chapter 10 and chapter 20 hedge theory.mcd' sheets.

In this case the sum of the basket premiums is equal to (and not greater than) the sum of the CDS premiums.

[2] This may make no intuitive sense and/or the model may seem to be non-sensical. Nevertheless this is a result of the mathematical model. The question of whether this is realistic model or not is discussed later.

20.1.6 N2D, Recovery Uncertainty

There is one oversimplification we have made which is particularly relevant at 100% correlation. Consider the following:

Exercise 10 At 100% correlation, calculate the F2D and S2D for a two name basket where name A has 100 bp CDS, and name B has 200 bp CDS. When is the F2D premium 100 bp, and the S2D premium 200 bp? Is an F2D premium of 184 bp possible (maintaining the 100% correlation assumption)?

The F2D premium may in fact be anywhere between 100 bp and 200 bp and then the S2D is 300 bp minus the F2D premium.

If we assume that both names have 40% recovery, then the hazard rates (from equation (9.17)) are 0.01/0.6 = 0.01667 and 0.02/0.6 = 0.03333. Name B has the higher hazard rate and therefore defaults first so F2D premium = 200 bp, S2D premium = 100 bp. But suppose B is junior debt, while A is a secured bank loan: we might assume zero recovery for B, but 80% recovery for A. This gives a hazard rate of 0.02/1 = 0.02 for B and 0.01/0.2 = 0.05 for A. Name A has the higher default rate and defaults first, so the F2D premium = 100 bp and the S2D premium = 200 bp.

However, what we have so far is a modelling oversimplification. We are assuming a recovery rate and deriving an implied hazard rate from CDS data. In the case of CDS pricing (section 9.4) we saw that this was possible and reasonable as long as the recovery and hazard rates are independent – and then the recovery rate is just the (statistically) expected recovery rate. But now we are valuing more complicated products and the presence of recovery uncertainty should be considered.

We implement the recovery uncertainty model of section 9.3. Imagine the following. Both A and B have an expected recovery of 40%, but we have an uncertainty (standard deviation) of 30%. We can model this by assuming 0, 40% or 90% possible recovery rates with probabilities p_0, p_{Mean} and $1 - p_0 - p_{\text{Mean}}$. We have two equations for the mean and standard deviation to solve in two unknowns:

$$0.4 = p_{Mean} \cdot 0.4 + (1 - p_0 - p_{Mean}) \cdot 0.9$$

and

$$0.3^2 = 0.4^2 \cdot p_0 + 0.5^2 \cdot (1 - p_0 - p_{Mean})$$

We find $p_0 = 0.25$ and $p_{\text{Mean}} = 0.55$ (and $p_{90} = 0.20$).

If we assume recoveries on A and B are independent then we can perform 9 different pricings and get the results shown in Table 20.2.

The expected premium for the F2D and S2D respectively are therefore

$$\begin{aligned}200 &\times (0.0625 + 0.1375 + 0.1375 + 0.3025 + 0.05 + 0.11 + 0.04)\\ &+ 100 \times (0.05 + 0.11) = 184,\end{aligned}$$

$$\begin{aligned}100 &\times (0.0625 + 0.1375 + 0.1375 + 0.3025 + 0.05 + 0.11 + 0.04)\\ &+ 200 \times (0.05 + 0.11) = 116.\end{aligned}$$

Note that the sum is unchanged, but hedge ratios for the F2D are 84% for name B and 16% for name A.

Table 20.2 F2D and S2D pricing under recovery uncertainty (100% correlation)

The CDS on name A is assumed to be 100 bp, and on name B is 200 bp.

Assumed recovery A	Implied hazard rate A	Assumed recovery B	Implied hazard rate B	F2D premium	S2D premium	Probability
0.0	100 bp	0.0	200 bp	200 bp	100 bp	$0.25 \times 0.25 = 0.0625$
0.4	167 bp	0.0	200 bp	200 bp	100 bp	$0.55 \times 0.25 = 0.1375$
0.9	1000 bp	0.0	200 bp	100 bp	200 bp	$0.2 \times 0.25 = 0.05$
0.0	100 bp	0.4	333 bp	200 bp	100 bp	$0.25 \times 0.55 = 0.1375$
0.4	167 bp	0.4	333 bp	200 bp	100 bp	$0.55 \times 0.55 = 0.3025$
0.9	1000 bp	0.4	333 bp	100 bp	200 bp	$0.2 \times 0.55 = 0.11$
0.0	100 bp	0.9	2000 bp	200 bp	100 bp	$0.25 \times 0.2 = 0.05$
0.4	167 bp	0.9	2000 bp	200 bp	100 bp	$0.55 \times 0.2 = 0.11$
0.9	1000 bp	0.9	2000 bp	200 bp	100 bp	$0.2 \times 0.2 = 0.04$

We can see that different assumed recovery rates for different names, or uncertainty over the recovery rate, means that the premium for an F2D under 100% correlation may be less than the highest individual CDS premium.

Extending this method to higher dimensionality products rapidly becomes unmanageable. For a three name basket we need 27 valuations, four names requires 81 valuations, and 12 names requires over half a million valuations. A practical solution is to perform a small number of simulations for some randomly chosen combinations of recoveries (the random choices being driven by the probability of that combination of recoveries).

Exercise 11 Show that recovery uncertainty has no impact when correlation is zero.

20.2 NORMAL COPULA

The process described above, taking a correlation matrix, generating independent Normal random numbers and converting them into correlated random numbers, then producing correlated default times, is an application of the 'Normal (or Gaussian) Copula'. The Normal Copula model is the industry standard valuation procedure among investment banks and others for the valuation of portfolio credit derivatives − *n*th-to-default baskets, CDOs and default swaps with correlated counterparty risk. We describe Copula methods more generally later – here it merely carries the implication that the Normal distribution is being used to produce correlated default times when the only information we can observe is the (unconditional) spread curve for individual names. The Normal Copula is a very simple and a rather special case of Copulas. Note that even though the same mathematical model – the Normal Copula – is used, it is often applied in different ways by investment banks marking to model (risk-neutral pricing) compared to investors (who often perform a fundamental valuation).

The following summarises the method bringing together what we have seen in Chapter 18 in the context of multiple assets and introducing correlations.

20.2.1 Generating Correlated Default Times under the Normal Copula

The description below is in the context of simulated default times with a general correlation matrix and immediately lends itself to simulation pricing (described in sections 20.2.3 and 20.2.4). The same underlying process can – in certain circumstances – be implemented far more quickly semi-analytically (described in section 20.2.5) The process is an extension of that described for a single name CDS – we just have more names and an additional step of correlating the Normal random numbers. The process is as follows.

1. Given a correlation matrix R we find the Cholesky[3] decomposition L such that

   $$R = L \cdot L^T$$

2. Pick n independent random numbers x_i from a Normal distribution.
3. Multiply the Cholesky matrix by the vector of n independent random numbers – this gives n correlated random numbers z_i from the Normal distribution.

   $$z_k = \sum_{j=1}^{n} L_{k,j} \cdot x_j$$

4. Find the cumulative probability associated with each number $-\Phi(z_i)$.
5. Associate that probability with the probability of survival up to a certain time – and get the default time – for each name. $t = \frac{-\ln(\Phi(z_i))}{\lambda}$ in the case where the hazard rate is a constant.
6. Given the default times $\{t_i\}$ we can calculate the cashflows under the N2D and static CDO tranche products (with a complex waterfall) and value these cashflows.

We need to know no more about the Normal Copula than we have seen already – namely

(a) how to generate correlated Normal numbers and
(b) equating the nth name survival probability to the cumulative probability associated with the nth random number.

We shall return to the topic of other Copulas in Chapter 23, and put the Normal Copula in the context of other approaches. It was first used (although not referred to by the name 'Normal Copula') – as described in Part I – by RiskMetrics Group (Gupton *et al.*, 1997) in their credit risk analysis. David Li (2000) first identified and used the Normal Copula to generate default times and price correlation products, other Copulas have been used by various researchers including Schönbucher and Schubert (2001). A review of the Copula approach in the context of other models is given in Finger (2000).

Default Time versus Rating Simulation

Note that the default time approach using the Normal Copula described above, and the approach for simulating forward ratings using the Normal Copula, described in Chapter 6, are related but not identical. If we choose a time horizon for a portfolio (say one year), and simulate forward ratings for a portfolio, then some names will default but most will survive and migrate to another (or stay the same) rating. Consider again two names where the default time correlation is 100%. On those default time simulations where one name has defaulted we know when the

[3] There are alternative methods such as 'singular value decomposition'. Experience performing over 1000 different CDO valuations has shown the Cholesky decomposition to be stable and accurate.

other name will default. But in the rating simulation[4] a wide spread name might move into default, but a 100% correlated AAA name would probably remain AAA (*exercise*: review section 6.2 and think through the process of rating simulation in the context of a CCC and a AAA name). What is the AAA name's expected default time at our time horizon? It is certainly not determined by the rating simulation approach, in contrast to the default time simulation approach. We conclude that, although they use the same methodology, the rating simulation and the default time simulation approaches are different and will produce different answers.

20.2.2 Correlation Types: Pairwise, Tag, Tranche/Compound and Base Correlation; Factor Correlation

The reader will have already asked: Where does the correlation matrix come from? We shall not answer the question here but prepare some of the ground by defining different types of matrix.

- *Pairwise Correlation Matrix*
  A correlation matrix in which the numbers in the lower left (or upper right) diagonal are generally different from each other is referred to a pairwise correlation matrix.
- *Tag Correlation Matrix*
  A simple case of the above may be where the correlation between name A and name B is defined by the certain characteristics of A and B (known as tags). For example we may use industry and investment grade / sub-investment grade as tags and define the correlation between A and B to be 0.2 if they share the same industry and are both investment grade or both sub-investment grade; or 0.1 if they share the same industry or grade status (but not both); or 0 if both tags differ.

  This rule will generate a matrix with only three numbers (at most) appearing off the diagonal. Note that is may or may not be a correlation matrix, and we shall return to this point later.
- *Tranche/Compound Correlation*
  If the matrix contains only one value off the diagonal (e.g. 0.2) and is used to price a specific tranche (e.g. the [3, 6] tranche of an iTraxx CDO) then it is referred to as tranche or compound correlation. A different tranche may be priced with a different matrix – for example, the [6, 9] tranche might be priced with a correlation of 0.25.
- *Base Correlation*
  Base correlation is tranche correlation for a tranche with attachment point of zero. For example the [0, 3] tranche may be priced with a correlation of 0.15 and the [0, 6] tranche with a correlation of 0.2. We shall see later that these two base correlations can be used to value the [3, 6] tranche.
- *Factor Correlation*
  The use of factor correlation is a means of implementing a correlation matrix where there is a single value off diagonal – so can be used for base or tranche correlations and can be applied to either simulation or semi-closed form pricing as we shall see later.

The random number generation in the Monte Carlo method can be simplified to the direct generation of correlated random numbers via

$$Z_i = \sqrt{\rho} \cdot F + \sqrt{(1-\rho)} \cdot X_i \tag{20.9}$$

[4] The transition matrix can be calibrated to the two names to give the correct spreads.

where Z denotes the correlated (Normal) random variables, F is the common factor and $\times$ denotes the uncorrelated variables. (Show that this gives a correlation of ρ between the Z's.)

Factor correlation also forms a core element in the semi-closed form approach described in section 20.2.5.

20.2.3 Simulation Pricing

The process of calculation product values for nth-to-default and CDO tranches is mathematically trivial given a set of simulated correlated default times. In certain circumstances there are alternative (semi-closed form) solution methods (discussed in more detail in section 20.2.5). The simulation technique is easy to understand and to apply. It will handle a general correlation matrix whereas semi-closed form implementations will not, and will handle many 'tag' correlation matrices (see section 20.2) more quickly than the semi-closed form approach. The valuation formulas under simulation can also be trivially extended to handle many cashflow CDOs with OC, IC and other tests, and handle the operation of a reserve account or tranche amortisation rules.

The key point to understand about valuation using simulated default times is that default time uncertainty, and correlation between names, is handled by the simulation process. Within a given simulation the survival probability of a name up to its default time is unity, and zero after the default time. Each set of simulated default times for the names in the portfolio is a realisation of the universe – there is now no uncertainty left. The default times generate a certain set of cashflows, and all we have to do is value those cashflows using the LIBOR-based discount factors.

We illustrate valuation of both products by example.

N2D baskets

We illustrate the process for a two name basket and a first-to-default contract – larger baskets and higher N2D contracts are handled by a trivial extension of the procedure. Suppose the product maturity is N years, and suppose we assume recoveries of R_1 and R_2 for name 1 and name 2 respectively. During the simulations we keep track of the value of capital payments (the 'contingent benefits stream', B, also called 'the capital value' or 'expected loss'), and the value of premium payments (paid at a rate of 1 bp per annum – the 'premium stream', P). It is also useful to keep sum-of-square terms to allow us to estimate the standard deviation of the expected value we calculate. In section 3.5 we show how to extend the process to give spread sensitivities.

Simulation No. 1

Let the simulated default times be T_1 and T_2 years, then there are three possible cases[5]:

1. $T_1 <= N$ and $T_2 > T_1$: generating a capital flow of $1 - R_1$ at time T_1, and premiums ceasing at time T_1.
2. $T_2 <= N$ and $T_1 > T_2$: generating a capital flow of $1 - R_2$ at time T_2, and premiums ceasing at time T_2.
3. $T_1 > N$ and $T_2 > N$: generating no capital flow and premiums ceasing at time N.

[5] Default times are simulated to around 16 decimal places – nanosecond accuracy. We ignore the possibility $T_1 = T_2$.

The valuation formulas will depend on the yield curve, day-count and other conventions. We shall take the simplest case of a constant zero coupon rate, r, and ignore day-count conventions. Then the cashflows increment the values of the contingent benefits, B, by

1. $(1 - R_1) \times \exp(-r \times T_1)$
2. $(1 - R_2) \times \exp(-r \times T_2)$
3. 0

and increment the value of the premium stream, P, by

1. $(1 - \exp(-r \times T_1))/r$
2. $(1 - \exp(-r \times T_2))/r$
3. $(1 - \exp(-r \times N))/r$

respectively.

We repeat the process many times, using a new pair of simulated default times for each repetition. Finally we can calculate the expected value of the benefits as B/n, and of the premium stream (per bp of premium) as P/n, where n is the number of simulations. The simulated product value is

$$V = \frac{B}{n} - x\frac{P}{n}$$

where $\times$ is the actual premium paid on the deal. Alternatively, for a pricing, we can set $V = 0$ and calculate the fair value premium.

CDO Tranches

To value a CDO tranche we need only the simulated default times (and discount curve). The only difference between N2D and CDO products are the cashflows generated by credit events. Consider a tranche with attachment points $[a, b]$, life T, and a reference pool of N names. Given a set of default times for the names in the reference pool these are first sorted into increasing order. We suppose a reference entity j has notional exposure N_j in the pool and defaults at time t_j with recovery R_j, and define two functions of time:

1. The initial notional of the tranche minus losses to date, $E(t)$, and the notional of the $[0, a]$ base tranche $E_0(t)$. We know $E(0) = b - a$ in percentage terms. For each default the loss $N_j \times (1 - R_j)$ reduces E_0 by the amount of the loss or to zero. If there is any excess of the loss over the reduction in the junior tranche then E reduces by this excess. When the $[0, a]$ base tranche is exhausted all defaults lead to an excess equal to the loss and reduce E by this loss.
2. The notional for premium calculation, $N(t)$. If the tranche is not the senior tranche then $N(t) = E(t)$. For the senior tranche only, the notional for premium calculation typically follows
   (a) $N(t_j+) = N(t_j-) - N_j$ if the loss hits the senior tranche (the tranche notional drops by loss plus recovery) and
   (b) $N(t_j+) = N(t_j-) - N_{j\times} R_j$ if the loss hits junior tranches only (the tranche notional drops by recovery), and the obvious extension if the loss happens to hit both junior and senior tranches).

The values of the contingent benefits, and of a unit of premium stream, on the tranche are then given by

$$B = \sum_{i=1}^{N} if(t_i < T, N_i \cdot (1 - R_i) \cdot \exp(-r \cdot t_i), 0) \tag{20.10}$$

$$P = \int_0^T N(t) \cdot \exp(-r \cdot t) dt \tag{20.11}$$

and the deal value (**net value**) is

$$V = B - pP \tag{20.12}$$

where p is the actual premium being paid (and $V = 0$ for a fair value deal).

Exercise 12 What tests could you apply to check consistency of the simulation program across products? What extremes would you test?

Some possibilities are:

(a) N2D and CDO with one name should relate to the underlying CDO.
(b) N2D with two names and correlation of 0 and 100% has known values.
(c) N2D and CDO with particular tranches and zero recovery assumption should generate the same fair value premiums.
(d) At very high spreads and zero recovery, all tranches will default. If interest rates are zero then the benefits values will tend to the notionals.
(e) Select a set of simulated default times and check the program cashflows are correct.
(f) Check the pricing of CDO tranches against market-traded index products (iTraxx or CDX tranches).

More Complex CDO Structures

We shall look at the valuation of CDO squared and cashflow CDOs in Chapter 22.

The Number of Simulations

In practice it is best to do a few 'housekeeping' activities:

(a) It is often useful to generate random numbers using the same seed, so that different valuations for the same product with minor changes, can be compared.
(b) If we calculate the sum of square values for the contingent benefits and premium streams, then we can estimate not just the mean but also the standard deviation of the results, and hence get a standard deviation of the resulting deal value or fair value premium.

If we increase the number of simulations n by a factor of 100, then the standard error of the result falls to about 1/10th of its previous value. The number of simulations we need to perform will depend on the accuracy required. Typically for a two name basket 1 million simulations is usually sufficiently accurate for valuation, and for a 100 name basket 20 000 simulations is sufficient (roughly 2 million simulated defaults in both cases). For CDO tranches – and for

N2D above F2D – there is an addition influence coming from the seniority of the tranche. We find that the absolute error of the calculation decreases as tranche seniority increases, but the percentage error increases.

20.2.4 Variance Reduction Techniques

The process described is known as 'crude' Monte Carlo. There are well-documented techniques for reducing the number of simulations required (hence speeding up calculations) (Jäckel, 2003). Such methods may be useful in a production environment. Crude Monte Carlo has the advantage of simplicity and easy estimation of errors. In addition crude Monte Carlo will converge on the correct answer (given a correct implementation) and can serve as a useful check on other short-cuts including semi-closed form pricing methods which generally make approximations. We mention three variance reduction techniques that can be useful.

1. *Control Variate:* Simulations are used to generate the value V_1 of the actual product and a similar product V_2 where an accurate answer V_{A2} is known. The error $V_{A2} - V_2$ can then be used to adjust V_1 to obtain a more accurate estimate. This can be used if the correlation matrix is driven by single number (so we can use semi-closed form solutions) and
   (a) we are comparing a CDO which has a complex waterfall with a CDO which has a trivial waterfall, or
   (b) when we want to value a CDO using a complex correlation matrix
2. *Low Discrepancy Sequences* (Jäckel, 2003): Sobol sequences are widely used but are known to have problems in high (around 15 or more) numbers of dimensions.
3. *Importance Sampling:* For example, in the F2D simulation process for investment grade names, relatively few simulations generate a capital payoff. Importance Sampling is a technique for changing the default distribution to generate more defaults – hence use fewer simulations to get an accurate answer – but make adjustments to get back to an answer appropriate to the actual distribution. Joshi and Kainth (2003) describe the application for N2D products.

The fourth method of speeding up calculation is a hardware solution. Using distributed processing and perhaps 100–1,000 idle (during overnight processing) PCs the calculation time can be reduced by a similar factor.

It should always be remembered that the best way to speed up any Monte Carlo implementation is to avoid unnecessary calculations and other 'tricks' For example, there is no point in calculating the default time if it is greater than the life of the deal. So at the start of the pricing routine, calculate for each name what random number corresponds to the maturity date, and if the simulation generates a random number the wrong side of this, then there is no need to go through the slow conversion to a default time – simply take maturity plus a margin. By using this and similar trick the calculation time for a simulation pricing on a 125 name investment grade pool can be brought down to a matter of seconds. In addition, if sensitivities are needed repeated pricing and taking differences will not yield accurate answers in a reasonable time – instead methods such as those outlined in section 18.3 should be used.

20.2.5 Semi-Closed Form (SCF) Pricing

This implementation has become standard among the banks and traders in portfolio credit derivatives for the purpose of CDO and F2D pricing. It can produce fast valuation and vast calculation of sensitivities. It is important to understand the following points.

1. The implementation is an alternative means of performing calculations using the same Normal Copula model described and illustrated above with a Monte Carlo simulation implementation. In certain circumstances (where the SCF approach is not making significant approximations) the two implementations should produce the same answers.
2. In general, the SCF implementation is an approximation. The approximation can be made more accurate at a cost of increased run time. Generally we can use Monte Carlo to check the accuracy of valuations.
3. Several approaches are used: Laurent and Gregory (2003), Andersen *et al*. (2003), Mina and Stern (2003). We shall describe the Andersen approach in detail below.
4. SCF can be 100–1,000 times faster than the simulation methods, and the key advantage is that this allows the rapid and accurate (in certain circumstances) calculation of spread and recovery hedge ratios for each underlying reference entity.

The idea behind the Andersen approach is to price an n name reference structure from an $(n - 1)$ name structure. The method is implemented as follows.

1. We define and calculate the loss unit (LU) and the total number of loss units in the portfolio. For a given reference entity i the loss given default (LGD_i) is equal to the notional $\times$ $(1 - \text{recovery})$ – for example, on the iTraxx portfolio with 1,000m total notional, then each individual notional is 8m and LGD_i is 4.8m assuming a 40% recovery. Calculate the LGD_i for each name in the pool then find the largest common denominator of all the LGD_i – 4.8m for iTraxx (since they are all the same). This defines the loss unit.
2. Calculate the number of loss units for each reference entity $\text{LU}_i = \text{LGD}_i/\text{LU}$ and the total number of loss units $\text{TLU} = \text{sum}(\text{LGD}_i)/\text{LU}$. Note that, in general, the LU will be a small number if recoveries and/or notionals differ and TLU will be a large number. For the iTraxx pool with 40% recoveries TLU = 125. If TLU is more than 1,000 then calculation times can become very slow, so it is usual to approximate the LU (and LGD_i) so that TLU is a workable number. This is the approximation underlying the method and no other significant approximation is required.
3. The basic idea is to think about adding another name to the pool, then (using a) the conditional probabilities of survival to any date are independent and

$$\begin{aligned}&\text{prob}(n \text{ units loss/larger pool}) = \text{prob}(n \text{ units loss/smaller pool})\\&\quad\times \text{prob}(0 \text{ loss on added name}) + \text{prob}(n - m \text{ units loss/smaller pool})\\&\quad\times \text{prob}(m \text{ units loss on added name}).\end{aligned}$$

   This (conditional) loss distribution is used to calculate the tranche loss.
4. Under a one-factor correlation model the conditional survival probability for a name i is given by Normal_probability$((C[i] - \sqrt{\rho} \times F) / \sqrt{(1 - \rho)})$ where F is between (say) -5 and $+5$ and is taken from a normal distribution and C is the distribution obtained from the CDS curve. This gives rise to a 'convolution integral' over the Normal distribution (M) which is generally performed with 40 to 80 steps.

Equation (20.9) relates the conditional Normal random variable (Z_i) to the common factor and unconditional variable and equation (18.3) can be restated as the probability that default time T_i is equal to the cumulative Normal probability of:

$$P(T_i) = \Phi(Z_i) \tag{20.13}$$

From the CDS calibration we can find the probability that name i defaults before some time T, say p_i, and this will correspond to $Z_i < F^{-1}(p_i)$. Equation (20.19) can then be rewritten as

$$c(X_i|f) = \Phi(X_i) = \Phi([\Phi^{-1}(p_i) - \sqrt{\rho}\Phi]/[\sqrt{(1-\rho)}]) \quad (20.14)$$

where c means the conditional probability of X.

In calculation we loop over the variable F (the 'convolution integral') and for each value of F we loop over the life of the trade, typically making calculations at each premium date and maturity. At each calculation date we use (20.14) and the process described in approach (3) above to calculate the conditional loss distribution and the expected tranche loss, and we assume (with a slight approximation) that any losses occur at the mid-time point.

For sensitivities it is best to think about the sensitivity formula obtained from the valuation formula and an SCF implementation of that rather than repeated calls to the valuation formula.

20.3 CORRELATION

20.3.1 Sources of Correlation

The Normal Copula – and other Copula models discussed in Chapter 23 – is based on parameters we cannot observe directly. Where does the correlation matrix come from and what are the implications of whatever approach is taken?

In the following sections, bear in mind that the result of the process is a *default time* correlation matrix. Some calculations are based on observable data – say, equity returns or spreads, and those correlations of course are equity return or spread correlations – but as soon as that matrix is used in the pricing model described in Chapter 22 they are being used as default time correlations. The model presented in Chapter 22 produces correlated default times, and the inputs are the drivers for that process whether they were taken from equity data, from spread data, or are arbitrary assumptions.

Sources of Default Time Correlation

Understanding the source of the default time correlation is likely to provide some help in deriving the correlations themselves.

Look again at Figure 2.1 – default rates over time. As default rates rise in a recession (and expected default times fall – equation (18.1)) then we see more defaults occurring together. RiskMetrics Group (Gupton *et al.*, 1997) use such data to derive the correlation of defaults within a year (a **binomial correlation**[6]), and find correlations in the 0 to 6% range. Such binomial correlations correspond to default time correlations on the 0 to 30% range.

The health of the economy is certainly a major factor[7] driving default time correlation. We shall also see (Figure 22.5) that industries (or geographical regions) may experience dramatic widening or narrowing of their spreads while other parts of the world economy are largely

[6] Binomial correlation is difficult to work with (Li, 2000). Also if we look at a longer period than a year the correlations starts to increase, and will become unity for a very long interval.

[7] 'Factor' can be interpreted here in the general sense or in the sense of an external data series driving the default rates (and times) – see section 20.3.6.

unaffected. We may think of using **equity data** as a rough indicator of moves in the economy and, possibly, moves in default times. Alternatively we may argue that we can identify key **factors** or **tags** (industry, region, etc.) and use a variety of sources to guess at the default time correlation matrix.

Equation (18.1) also tells us that default time correlation and default rate correlation are intertwined. We shall look at a correlated default rate model (with and without jumps – section 22.4) and see that developing such a model has few practical attractions. Nevertheless, CDS premiums or bond spreads (closely related to default rates and default times) may provide a valuable source of data for default time correlation. Secondly the hedging results in section 22.2 separately underline the importance of **spread correlation** in N2D/CDO product pricing.

In the following sections we investigate the above and related issues. Our conclusion is that, for the purposes of marking to market and risk-managing CDO positions 'implied correlation' is in principle the only valid correlation to use. Where a source of implied correlation is not available, we shall see that spread correlation is the more attractive alternative – although there is quite strong evidence that correlations based on data histories do not give market prices for CDO tranches (section 22.4).

20.3.2 Constraints: What Makes a Correlation Matrix?

Imagine you have three coins, A, B and C. If you toss A and it comes up heads, then B comes up tails – A and B are perfectly negatively correlated. Suppose B and C are also perfectly negatively correlated, what does this imply for A and C? If A comes up heads, then B comes up tails so C comes up heads – A and C are perfectly (positively) correlated. Any other 'correlation' (a number between -1 and $+1$) does not produce results consistent with the other correlation numbers.

We saw a slightly more realistic and less obvious example in Chapter 6. A more relevant example is the following. Consider a 10 name F2D portfolio. It is not possible that every name is *minus* 20% correlated with every other name. For a 100 name portfolio it is only possible for every name to have a tiny negative correlation (around 1%) with every other name.

What properties does a correlation matrix have? Every number on the diagonal must be unity (everything is 100% correlated with itself); every number off the diagonal must be between 1 and -1; the matrix must be symmetric – the correlation of A with B is the same as the correlation of B with A. But that is not enough. A further condition is that the matrix must be **invertible**.

Suppose we have a three name F2D. Two banks (one US and one EU) are highly correlated – say 0.9 – and the EU bank and a US software company have very low correlation – say 0. What are the restrictions on the third correlation, x, in the matrix below?

$$\begin{pmatrix} 1 & .9 & 0 \\ .9 & 1 & x \\ 0 & x & 1 \end{pmatrix}$$

It turns out that x must be between plus or minus 43%. Suppose now we have a 100 name correlation matrix, and we did not know 10 of those correlations. How do you find the limits on those remaining correlations?

We conclude the following.

1. Any method used to produce the default time correlation matrix must satisfy the invertibility constraint.
2. Any process using the correlation matrix – such as a 'correlation stress' or correlation sensitivity calculations – which involves changing values in the matrix must be sure to satisfy the invertibility constraint for the altered matrix.
3. It may be hard to produce a correlation matrix.

There are two simple ways to ensure the third requirement is met and we give names to these two matrices.

One-Factor Matrix

Note that a matrix which has unity on the diagonal, and the same number ρ where $0 <= \rho < 1$ everywhere else in the matrix, is invertible. Any stress of ρ to ρ_2, where $0 < \rho_2 < 1$ will also produce a correlation matrix.

Data Matrix

Also note that any correlation matrix derived from a consistent set of data (say daily returns for every Monday to Friday between 1 January 1997 and 31 December 2004 – with no gaps or omissions) is indeed a correlation matrix whether or not the underlying correlation data is of good quality. However, it may not be possible to stress this matrix by raising every correlation other than unity by the same number, delta (for any value of delta), in order to calculate correlation sensitivity.

20.3.3 Spread Correlation

Spread correlation has some attraction as a source of correlation data since spread correlation and volatility have an impact on trade value, and we shall see that the impact is zero (in expectation) if the spread correlation and the pricing correlation are the same numbers.

Calculation techniques are described in section 20.3.6. We confine our remarks here to data.

The appropriate correlation for a 5-year CDO deal, for example, will be a weighted average of the correlation over the future life of the deal. The weighting is unclear – the capital value of errors will be greater at longer outstanding life. Additionally, as the deal shortens the correlation changes from correlation of 5-year instruments to correlation of short-dated instruments. So it is not clear what historical correlation we should be using.

The CDO may be cash, referring to specific deliverable bonds, or may be synthetic, referring to CDS deals. In the former case we may be tempted to use bond spread data. Bond spreads suffer from a variety of problems:

(1) any data for the actual reference instruments is not directly relevant to the future – shorter dated – instruments;
(2) idiosyncratic nature of a particular historical bond (because of repo influences, liquidity) which will differ from the reference asset;
(3) decreasing maturity of the bond used as the source of historical data;
(4) history is usually limited for any one issue;
(5) data is generally 'noisy'.

Generally CDS data has some advantages for historical analysis.

1. It is relatively clean (assuming it is available)
2. It has a constant maturity
3. It refers to a greater liquidity instrument.

Once a data source is chosen there will be gaps in terms of the name coverage (there may be no data for some names) and also variable time periods covered (some names may have become liquid at a certain date and there may be no, or infrequent data, before that date).

20.3.4 Asset and Equity Correlation

Equity data is attractive because it is generally of good quality and often there are long data series. Historical data can be used to derive correlations (based on pairwise analysis or factor analysis – see section 20.3.6) and will produce a correlation matrix (in particular it is invertible) if a consistent time-period is used for all entities. Among investors, equity-based correlations are probably still the most popular means of deriving a default time correlation matrix for CDO analysis. Against the advantages of convenience, there are the following problems:

1. There is no theoretical equality between equity correlation and default time correlation. (On the other hand, there is a theoretical equality between 'asset correlation' and default time correlation – see below).
2. If historical data series are over different time periods, then pairwise derived correlations may not produce a (invertible) correlation matrix. Factor methods can help to avoid this problem.
3. Sovereigns and private companies, for example, do not have a readily available equity price.

The credit default models described in Chapter 9 and sections 20.1 and 20.2 above are referred to as '**reduced-form models**'. They are largely driven by market data (although recovery rates generally are modelled by looking at historical data). An alternative class of models – **structural models** – attempts to relate the value of the firm to the pricing of credit products. Such approaches are used by Moody'sKMV and CreditGrades™ (Finger, 2002) to derive a default probability using an asset model (and equity and balance sheet data). The basis of the model is the 'Merton firm model', and the relevant conclusion of the analysis is that, if the Merton firm model accurately describes the behaviour of the risk (subject to some assumptions about what constitutes default), then the asset correlation and the default time correlation are identical. The assets of a firm are impossible to measure on a real-time basis. Typically use is made of the fact that assets equal liabilities, and liabilities – consisting of equity and various forms of debt – are easier to measure. Usually this requires a combination of observed market data (the equity price), and balance sheet data to attempt to measure debt levels, so is still far from real time.

Attempts were made to use both equity and equity option data to imply asset values and obtain asset correlation, but these were eventually abandoned in favour of equity correlation or asset correlation derived from estimated asset data. The 'structural' approach has, at least initially, failed to produce an effective correlation model for credit derivatives and has left many market participants with the impression that equity correlation is an appropriate measure for default time correlation when, in fact, there is no such concrete link.

It is also worth noting that equity and asset based correlations do not produce market prices (using the Normal Copula) for liquidly traded CDO products but are a source of correlation numbers for 'fundamental valuation' – although the theoretical justification for using them is slight. Asset correlation is difficult if not impossible to measure but has a stronger theoretical basis for fundamental valuation.

20.3.5 Estimation from Historical Data

We discuss a few topics here and in the next section related to historical correlation and its estimation. Most of the following approaches are relevant to investors trying to perform a fundamental valuation, and also to the calculation of VaR for a portfolio along traditional lines.

These alternative approaches involve using market data histories (such as CDS premiums, bond spreads, or equity returns) and calculating the correlation from that data. Two approaches to historical volatility calculations are desired. As long as a consistent period of time is used for all the time series, and as long as every reference entity has a data value for every date in the time series, then either approach produces a valid correlation matrix.

Given two time series the correlation of these series is calculated. For economic time series we generally find that the number calculated by this process is very unstable over time if the time series is 'short', say, 3–6 months or less (Taylor, 1986). We would therefore typically take a 'long' time period (say 5–10 years). One advantage of this approach is that we can calculate some sort of error bar estimate around the correlation number as follows. If we divide the 5-year series into (say) 20 quarterly periods, we can calculate the short-term correlation for each quarter, and then calculate the (annualised) standard deviation of the correlation. We can then use this to get a confidence interval around our 5-year correlation estimate. We might alternatively use these standard deviation numbers to determine a stress to apply to the correlation matrix in order to get a stressed pricing or VaR calculation.

Imagine that we wish to perform VaR type analysis for a portfolio of 5 000 reference entities. There are approximately 12 500 000 time series to analyse for correlation. Many of these time series will be missing for at least part of the period. The chances of producing an (invertible) correlation matrix for all these names directly from pairwise historical analysis are minimal. Even if we manage to produce a correlation matrix with 25 000 000 entries, we undoubtedly will need to invert this matrix at some point in our application – and this will prove impractical.

20.3.6 Factor Analysis and Factor Correlation; Tag Correlation

An alternative data analysis approach is to use a 'factor analysis[8]' of the data. This is claimed to give more stable correlation estimates (it is clear that indeed it should do so since it estimates far fewer parameters). It is also the approach followed by RiskMetrics Group and Moody'sKMV for their estimation of correlation. Not only is this a method of analysis of data but 'factor correlation' – in particular 'one factor correlation' – is a key element behind (a) speeding up

[8] 'Factor analysis', 'principal component analysis (PCA)' and 'independent component analysis' cover a range of related methods, only some of which are relevant here. Our initial objective is to derive the correlation matrix – so methods starting with the correlation matrix, such as PCA, are not useful. (Hyvarinen *et al*., 2001; Jae-On Kim and Mueller, 1978)

simulation in the case where the correlation matrix has only one non-trivial number and (b) the semi-closed form implementation of the Copula model.

One approach is the following. Suppose we initially try to use a single factor to analyse the data. We set up a regression of each data series (5 000 series of, say 1 000 time points) on the (unknown) factor.

$$X_{i,t} = \alpha_i \cdot F_t + \varepsilon_{i,t} \tag{20.15}$$

where $\times$ are the $i = 1, \ldots, 5\,000$ data series, F is the factor series for $t = 1, \ldots, 1\,000$, α is the factor for series i, and ε are the error terms for each date and each series. The fitting process is to minimise the errors (in some way that we need not specify for the purposes of our discussion here).

The factor is defined by a single series of 1 000 (unknown) values. Each data series has a single regression parameter against the unknown factor. This gives only 6 000 parameters to estimate compared with the more than 12 million required for the approach of section 20.3.5. If our data represents equity returns, then the factor will probably turn out to be close to the market average return, in which case the 'factor' is the 'market' – but we did not know this in advance.

We can now add another factor to our regression formula above (keeping fixed the first factor and the regression parameters), and perform an estimation of this second factor in a similar way. Suppose our data series represent auto companies and telecoms companies. Our first factor may turn out to be the average return, and the second factor may turn out to be the out-performance of autos over telecoms. However the more factors we include the harder it is to identify what these factors represent.

Once we have identified sufficient factors, and we have decided that N provides a good model to the data but $N + 1$ does not add a sufficiently great improvement to make it is worth doing,[9] then we have N parameters for each (equity) data series and $5\,000 \times N$ parameters for all the series. In addition, there are N time series for the N factors, plus a residual error time series for each equity. We typically assume that the residual is insignificant or uncorrelated with the factors.

$$X_{i,t} = \sum_{j=1}^{N} \alpha_{i,j} \cdot F_j + \varepsilon_{i,t} \tag{20.16}$$

We can calculate the standard deviation and the correlation of the factors[10] (an $N \times N$ matrix). The pairwise correlation of two reference entities (if needed) can then be calculated from the factor correlations.

Once the factors have been derived (with, or without, an interpretation) then entities with short data series, unreliable or missing data, can be incorporated into the universe. If sufficient data is available, the regression factors may be estimated based on this, otherwise some assumption will be made (such as taking factors for a similar entity).

If we wish to simulate the portfolio on 5 000 credits, we simulating the N factors (and uncorrelated residuals) then calculate the simulated equity returns from the factor values and the regression parameters (a residual may also be simulated).

[9] There are many ways of doing this and different statisticians have different approaches. A 'scree' test (Jae-On Kim and Mueller, 1978) is one approach – where a chart of the residual error is plotted against the number of factors and if an abrupt change in the slope occurs then this point may be used to select the number of factors.

[10] The approach followed will lead to near orthogonal factors.

It has also been shown that a correlation matrix estimated from factor analysis is more stable over time than one estimated from pairwise histories.

It should be noted that the factor-based approach to calculation the correlations has nothing to do with the appropriateness of the correlation matrix to the task in hand, or to the invertibility of the resulting matrix. 'Factor correlations' are not 'better' if equity or spread correlation is the wrong data to be using. Both methods (20.3.5 and 20.3.6) estimate equity return correlations (or spread correlations depending on the underlying data) and invertibility of the matrix is guaranteed if a consistent time period is used. Neither matrix has anything to do in principle with CDO pricing from a risk-neutral perspective.

The approach described is rarely used in practice because of the non-linear estimation of the best-fit parameter values. Alternative approaches to the calculation of the factor areas follows.

1. Take $F(n,t)$ for $n = 0, 1, 2, \ldots$ to be orthonormal functions of time – for example the Legendre polynomials or Fourier series (http://encyclopedia.thefreedictionary.com/Legendre+polymonial). This approach is more suited to cases where some regular time dependency is suspected.
2. Assume that the factors are known (market return, sector return, geographical return, interest and FX rates, ...). This produces non-orthogonal factors but is the most common approach for equity return analysis.
3. If the dimensionality is small (a few hundred at most) we might calculate the pairwise equity correlations and then get the eigenvectors and values – these give us the orthonormal factors. The most important factors can be selected and more stable pairwise correlation estimates can be based on these factors.

The factors are now known and we continue as above to derive the regression and residuals for each series. Methods 1 and 2 have the advantage that the factors are meaningful and independently observable, although the process may be inefficient and in some ways less informative. Method (3) is only suitable for relatively low dimensionality problems.

Tag Correlations

It can be argued that the tag correlation approach (illustrated below) is equally valid (when compared to equity correlations) and has many practical advantages. It can produce the most stable correlation numbers (!) and is very quick to set up; it is also closely related to factor model approaches (Xiao, 2003). The approach is illustrated by the following.

We might identify industry (autos, finance, ...) and region (Americas, Europe, ...) as relevant factors. If two reference entities belong to the same region and to the same industry, then the correlation is 0.3 (say); if they belong to the same region but different industries, the correlation is 0.1; if they belong to the same industry but different regions then the correlation is 0.2; otherwise the correlation is zero.

A further potential problem is that, depending on the number of reference entities and the choice of numbers in the matrix, it may not be a correlation matrix (i.e. it may not be invertible). In addition it is unclear what the relevance of this matrix is to the calculation of CDO values on a risk-neutral or a fundamental basis.

Tag correlation can be implemented numerically in semi-closed form (one-factor correlation being a simple case and the one used in practice) but if we increase the number of tags, the dimensionality of the integral increases, and the time taken to calculate the numerical integral increases exponentially. Generally, if a one-dimensional integral requires 20 points for an accurate answer, then a five-dimensional integral may require around $20^5 = 3$ million points. Monte Carlo simulation is one means of performing numerical integrals and is often the fastest method for multidimensional cases (Fishman, 1995), often taken over at between 4 and 7 dimensions.

20.3.7 Impact on Hedging of Using Historical or Implied Correlations

Correlation estimates based on (say) a 6-month history will vary sharply as a trade matures and shortening time periods are used to estimate the current correlation. These correlation changes will affect the trade and hedges in place differently, causing rehedging of a position and potential profits or losses. Such rehedging is not tied to the market pricing of the CDO (which is driven by implied correlation) so any mismatch in the hedge ratios is spurious and will generate spurious losses (there being a bid–offer cost of rehedging).

Spurious rehedging can be minimised by minimising the variability of the historical estimates as described above.

20.3.8 Implied Correlation

There is only one valid approach to correlation (and other data) for mark-to-market and hedging purposes and that is – since correlation itself is not a traded variable – to use the implied correlation obtained from market prices using a ‘good’ model. The model currently used for this process is the Normal Copula model and implied correlation is obtained as a single number appropriate to the pricing of a particular tranche on its own.

Imagine that an equity options trader wishes to value a portfolio of options covering a range of strikes and maturities. Typically the trader uses the Black–Scholes model, and uses prices of exchange-traded options to derive implied volatilities, then interpolates these volatilities to the strikes and maturities needed, and then prices the portfolio using the Black–Scholes model. Incidentally, the trader may not use that model to help to determine his or her own trading strategy – a stochastic volatility model with jumps may be preferred but, since that is slow to evaluate, the policy is to mark to market using Black–Scholes for valuation purposes. Any other fundamentally different approach – such as using historical volatility estimates – is wrong since the trader needs to know the price at which a deal could be unwound in the market.

The Normal Copula model has come to take the role corresponding to the Black–Scholes model when it comes to pricing N2D and CDO tranches. Actively traded liquid instruments (iTraxx tranches for example) are used to calculate an implied correlation for the tranche. Implied correlations are then interpolated and adjusted to come up with pricing correlations for closely related products. We discuss the topic fully in the next chapter.

21
Correlation in Practice

This chapter looks at the variety of ways the correlated default time model can be applied in practice and the advantages and disadvantages associated with each approach. The final section also looks at other ways in which the default time model could be applied.

21.1 TRANCHE CORRELATION

21.1.1 Valuation and Key Features

We first illustrate some features of tranche correlation. The '**fair premium**' for a tranche is the premium that makes the net value of the tranche (calculated today) equal to zero: equation (20.12) is equal to zero if p is the fair premium. The fair premium changes as the life of the deal and the spreads, etc., on the underlying names change – it is unrelated to a fixed agreed premium on the tranche. Restating the above, the expected loss on the tranche (at any moment) is equal to the (then) valuation of the fair premium stream.

Table 21.1 and Figures 21.1–21.3 show the calculated fair premium on a 5-year 125 name reference pool with iTraxx tranching for the various tranches. The results are typical but change in detail according to the maturity of the deal, tranche attachment and detachment points, and spreads and recoveries on the underlying names.

The following features are typical:

1. The fair premium for the most junior tranche $[0, x]$ ($x = 3$ in this example) decreases monotonically as correlation increases.

   Think for a moment how the default time correlation works. At zero correlation the defaults of individual names are unrelated – intuitively thinking about a single simulation we would expect a relatively small number of defaults and the equity tranche to be hit by these defaults (and, occasionally the higher tranches). Contrast this with the case of 100% correlation – the defaults are ordered simply according to the hazard rates. If the first default (highest hazard rate) is beyond the maturity of the deal, then no tranches are hit. On the other hand, if the first default is very early in the life of the deal there will be many defaults before maturity, wiping out the junior and many other tranches and eating into the senior tranche. Thus, with high correlation the junior tranche may not be hit at all but may be completely wiped out.

   Low correlation means that the equity tranche will probably be hit by defaults, but high correlation means that in many cases the equity tranche will not be hit at all but occasionally will be wiped out. This leads to decreasing value of the equity tranche with increasing correlation.
2. The fair premium for the most senior tranche $[y, 100]$ ($y = 22$ in this example) increases monotonically as correlation increases.

   In the discussion above we see that the senior tranche is highly unlikely to be hit when correlation is zero, but with high correlation there is a high chance that no tranches will be hit. However, when they are there is a big impact on the tranches and the senior tranche is also hit.

Table 21.1 Fair premium vs. correlation

Corr	[0, 3]	[3, 6]	[6, 9]	[9, 12]	[12, 22]	[22, 100]
0	12,506.9	4,374.2	1,952.6	472.6	9.3	0.0
0.1	9,003.5	3,191.3	1,517.6	720.9	150.4	0.6
0.2	6,773.4	2,567.8	1,349.8	760.5	250.2	4.4
0.3	5,310.5	2,166.0	1,231.6	763.7	315.2	10.8
0.4	4,283.3	1,872.1	1,133.7	751.4	358.7	18.6
0.5	3,519.8	1,639.8	1,046.5	729.9	387.6	27.4
0.6	2,927.2	1,446.9	965.5	701.7	405.3	37.0
0.7	2,451.9	1,281.4	888.5	667.5	413.4	47.4
0.8	2,061.3	1,137.0	814.5	629.0	411.9	58.7
0.9	1,732.7	1,014.4	744.2	592.1	396.3	71.0
1	1,432.8	886.0	717.4	607.4	366.4	84.0

Remember that changing correlation has no impact on the default probability of a single name over the life of a deal – it merely changes the way these defaults clump together.

3. The fair premium for a $[a, a + b]$ tranche increases monotonically as a increases.

   This should be obvious since a lower attachment point means that tranche will be hit first, so the expected loss will increase as the attachment point gets closer to zero. In addition, the expected life of the tranche will decrease as the attachment point gets closer to zero – the tranche is going to be eaten into more than a more senior one. Hence the fair premium will also rise
4. For some tranches the fair premium curve is not monotonic (see Figure 21.1), unlike the equity and senior tranche. Given a fair premium there may be two correlation levels consistent with this. This is not just a feature of using fair premium, and Table 21.2 shows that the expected loss also shows a non-monotonic behaviour for some intermediate tranches. The same is true if we look at net values for a given fixed premium.

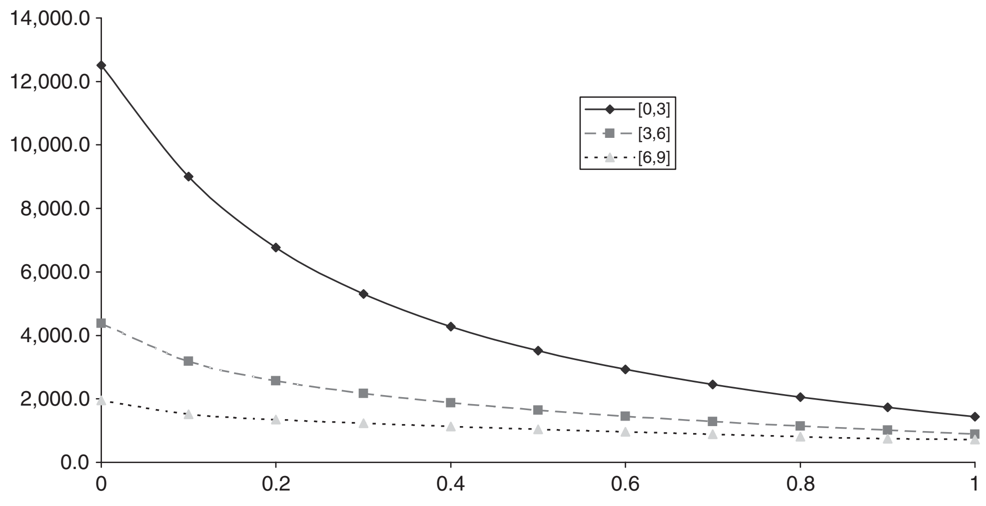

Figure 21.1 Fair premium vs. correlation – three junior tranches.

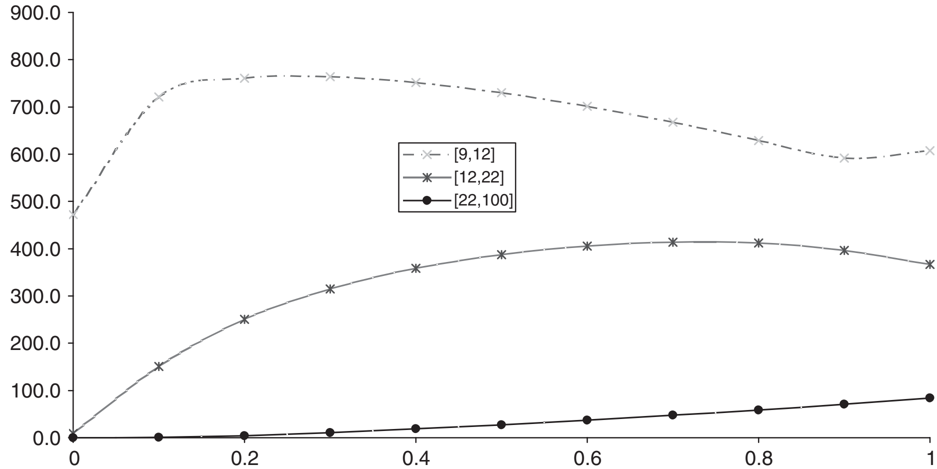

Figure 21.2 Fair premium vs. correlation – three senior tranches.

If we want to calculate the value of the tranche at the fair premium level, this is the same (zero) in both cases. But if we want to calculate sensitivities (and hedge ratios) for a specific reference entity in the underlying pool (described in Chapter 22) then we will get different answers according to the correlation we choose.

5. We should note that a tranche may trade on a premium such that no choice of implied correlation will give that amount as the fair value premium. Looking at the [9, 12] tranche in Table 21.1, if the quoted fair premium is more than about 765 or less than 472, then no (positive) correlation exists which is consistent with that premium. Such situations were common when there was no transparent market in CDO tranches, and have occurred again more recently (2008 and onwards). In such situations we conclude that either the tranches are above or below fair price (i.e. there is an arbitrage present), or the model is faulty.

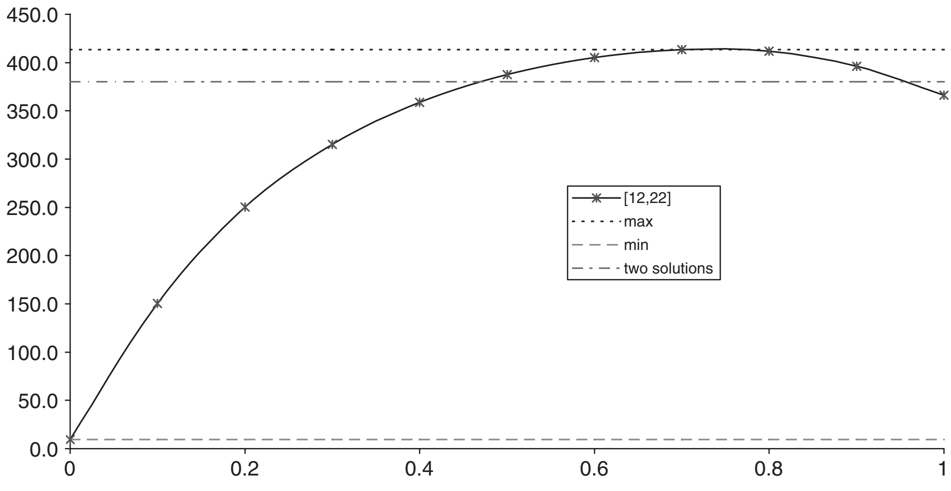

Figure 21.3 Fair premium vs. correlation – three senior tranches.

Table 21.2 Expected loss vs. tranche correlation

Corr	[0, 3]	[3, 6]	[6, 9]	[9, 12]	[12, 22]	[22, 100]
0	29.5	28.2	21.4	6.5	0.4	0.0
0.1	28.9	24.3	16.5	9.2	7.0	0.2
0.2	27.7	21.4	14.7	9.4	11.4	1.6
0.3	26.2	19.1	13.5	9.3	14.1	4.0
0.4	24.5	17.4	12.5	9.1	15.8	6.8
0.5	22.8	15.9	11.7	8.8	16.9	10.1
0.6	21.1	14.5	10.9	8.5	17.5	13.6
0.7	19.4	13.3	10.2	8.1	17.8	17.4
0.8	17.7	12.2	9.5	7.7	17.7	21.4
0.9	16.0	11.2	8.8	7.3	17.0	25.8
1	14.2	10.1	8.5	7.5	15.9	30.4

21.1.2 Implied Tranche Correlation

Tranches of the iTraxx and CDX portfolios are quoted consistently and on demand by market makers. For example, the 7–10% tranche of the CDX.IG may have a quoted current fair market premium of 150/170 bp.[1] In order to calculate an implied correlation from equation (20.12) we also need to know

(a) the relevant yield curve levels,
(b) the CDS premia on all the known underlying names and
(c) the assumed recovery rates or a recovery rate model.

The calculation process is iterative (like the calculation of a redemption yield from a bond's price) – guess a correlation and calculate the premium or net value, then make a better guess until the net value is zero with the fair premium, or equal to the up-front premium with the fixed ongoing premium.

If several implied correlations are consistent with a given premium, then the solution found will depend on the initial guess and also how the software seeks a solution. For example, for the [9, 12] tranche in Table 21.1 an initial guess of more than 0.5 will return an implied correlation of approximately 0.51, but an initial guess of under 0.1 will return approximately 0.11. An initial guess under somewhere in the region of 0.3 will also return 0.11, and over that will return 0.51. If more than two implied correlations exist, then the situation is even more complex.

A consequence of the 'implied tranche correlation' approach is that names A and B in the reference portfolio have a correlation of, say, 30% when we are looking at a senior tranche, but only 5% when we are looking at a junior mezzanine tranche. Each tranche must be priced using its own correlation, so we cannot generate a consistent set of default times across a universe of names which may appear in many of our trades. In particular, it is not clear how to price a CDO of CDOs, or a CDO with complex waterfall, consistently using implied correlations. Note that different implied tranche correlations lead to different loss distributions which are eventually used for pricing. Ideally we need a loss distribution that prices all market tranches. However,

[1] Of course the actual premium on the tranche of the standard structures is a fixed number and the tranches trade with a varying up-front premium. Both the up-front premium and the equivalent fair market premium are quoted.

Table 21.3 Bid–offer tranche premium and bid–offer correlation

Tranche	Market bid spread	Market offer spread	Bid correlation	Offer correlation
0–3	20%/500	24%/500	26.5	19.6
3–6	150	196	8.7	14.7
6–9	58	74	18.3	22.5
9–12	28	44	23.5	30.0
12–22	14	20	33.3	38.3

with a single correlation it is not possible to produce a loss distribution that calibrates to all quoted tranches.

Correlations implied from tranche price data often show a 'smile' – a non-monotonic curve initially falling to a minimum in the mezzanine area then rising on the senior tranches.

21.1.3 CDS Curve Adjustment

One practical problem with calculating implied correlations (*and this applies equally to tranche and to base correlation*) is that typically the user is required to estimate what the current CDS levels are behind the market maker's quote. This will lead to some (small) uncertainty in the calculation of the implied correlation. Often the market maker will quote the consistent index premium and, given the CDS premium data they hold, the users can then find a factor or a shift that has to be to those curves so that the calculated premium for the index agrees with the market maker's quote. Those adjusted curves are then used to calculate the implied correlation. This is known as 'single names' versus 'index basis' due to different supply and demand effects. Market makers will eventually adjust either the single names spreads or the index to avoid the possibility of arbitrage, or market participants will take any arbitrage opportunities until they are no longer available.

We shall encounter an issue later if we are trying to value a bespoke deal containing reference entities in iTraxx, CDX IG and HY portfolios, and also names not in any of these portfolios. If, to match the index premium, we have to adjust iTraxx spreads by 1.002, CDX.IG by 0.995 and CDX.HY by 1.03, then how should we adjust the remaining names in the reference pool of the bespoke deal before valuation?

21.1.4 Bid–Offer Impact

The impact of the premium 'bid to offer' on the implied correlation is indicated in Table 21.3, based on some wide bid–offer data for early TRAC-X trades (more recently bid–offer spreads have been narrow with consequently little impact on implied correlation). Typically, calculations are performed on the basis of mid tranche premium and mid correlation.

21.2 BASE CORRELATION

Two types of base correlation are currently found:

1. Maturity Matched Base Correlation – each maturity (e.g. 5 year and 3 year iTraxx) is treated independently of any earlier maturities. Thus, if the [0, 9] 3-year base correlation is 0.3 and the 5-year is 0.35, then in valuing the 3-year deal all correlations at all times are 0.3, and in the 5-year all correlations are 0.35.

2. Term Structure Base Correlation – the shortest maturity (3 year currently) defines the base correlations for the period up to the 3-year maturity, and the 5-year base correlation then applies a new number for the 2-year period from the maturity of the 3-year deal.

We shall describe maturity-matched base correlations in detail – no new principles are introduced by term structure correlations. The reader is reminded that the principles of 21.1.3 apply equally to data adjustment for base correlation calculations.

21.2.1 Valuation and Key Features

Base correlation was defined in section 20.2.2; here we explain how base correlations are applied. JP Morgan (12 March 2004) introduced the concept of base correlation, and defined an implementation using their 'standardised large pool model' (JP Morgan, 6 May 2004). One of the main aims in introducing the base correlation approach to tranche pricing was to avoid the non-uniqueness of tranche correlation for certain intermediate tranches. Another was that it was hoped that the curve of implied base correlations would be monotonic, making correlation interpolation easier than if a smile existed.

The market has developed alternative implementations of the base correlation concept. Given market CDS data for the reference pool, one approach (used by Bear Stearns, 17 May 2004, and others) is described below. In the following V_x and P_x are used to mean the expected loss and value of a unit of premium of the base tranche [0, x] using the correlation ρ_x

A. The most junior tranche [0, a] is the first base tranche, is subject to the market premium p_a and gives rise to the first base correlation ρ_a (= the implied tranche correlation and is the correlation at which the net value of the tranche is zero using the current fair market premium). Repeating equation (20.12) we have

$$0 = V_a - p_a \times P_a \text{ calculations being performed with correlation } \rho_{\mathrm{a}} \tag{21.1}$$

B. Take the premium p_b ($p_b < p_a$) on the next tranche [a, b], and calculate the net value of the first base tranche subject to this premium using ρ_a: $N = V_a - p_b \times P_a$. This value is positive. [Why?]. The value of the second tranche (zero since we are using the fair premium) is the value of the second base tranche [0, b] using correlation ρ_b less N – thus

$$0 = (V_b - p_b \times P_b) - (V_a - p_a \times P_a) \tag{21.2}$$

(calculations being performed with ρ_a and ρ_b respectively)

C. The process continues iteratively to the higher tranches.

Table 21.4 shows the calculated fair premium on a [9, 12] tranche assuming a 20% [0, 9] base correlation and various levels of [0, 12] correlation.

Table 21.4 Fair premium for a [9, 12] tranche under base correlation

	corr [0, 12]						
corr [0, 9]	0.102	0.2	0.21	0.25	0.3	0.35	0.3945
0.2	1349.1	760.5	710.3	524.4	320.2	141.6	0.0

Table 21.5 Fair premium for a [9, 12] tranche under base correlation

	corr [0, 12]			
corr [0, 9]	0.7543	0.8	0.9	1
0.8	814.3	629.0	290.8	59.9

We note the following features of base correlation.

1. If the second (i.e. [0, b]) base correlation is equal to the first (i.e. [0, a]) base correlation, then the valuation of the [a, b] tranche is the same as under tranche correlation valuation and the fair premium is the same – say X.
2. If the second base correlation is increased with the first base correlation fixed, then the value of the second base tranche decreases, leading to an decrease in the value of the second tranche and a lower fair premium P_1 on the [a, b] tranche ($P_1 < X$). At some point the fair premium could become zero (and then negative at even higher correlations) although often a correlation of 100% on the [0, b] tranche gives a positive lower bound to the [a, b] premium.
3. If the second base correlation is decreased with the first base correlation fixed, then the value of the second base tranche increases, leading to an increase in the value of the second tranche and a higher fair premium P_2 on the [a, b] tranche ($P_2 > X$). At some point the fair premium may become greater than the fair premium on a more junior tranche, or the expected loss on the [a, b] tranche may exceed the expected loss on a more junior tranche of the same thickness – which is absurd.
4. Generally base correlations are increasing with attachment point, although this is not always the case.
5. The first column in the above tables shows the correlation at which the fair premium for the [9, 12] tranche is equal to the fair premium on the [6, 9] tranche.
6. Since the base tranches have monotonically decreasing value with correlation, the net value of the [a, b] tranche corresponds to a unique value of ρ_b given ρ_a – unlike the situation under tranche correlation.
7. For some tranches and some ranges of base correlation there will be an upper and a lower limit to the corresponding fair premium – as illustrated in Table 21.5, with a minimum premium of 59.9 bp on the tranche. As for tranche correlations we note that the actual premiums may be outside the range that can be fitted by the model. Typically, the final tranche premium is very tightly constrained. For example, the [15, 30] CDX9 tranche base correlation has been failing to calibrate after the credit crunch. However, one way around this issue – for people interested in super senior tranches only – is to use a similar concept to base correlation but instead of calibrating to equity tranches, one can calibrate to [x, 100] tranches since the [30, 100] tranche is quoted.

21.2.2 Implied Base Correlation

Given market data (for example, fair premiums on the CDX or iTraxx tranches) then the process of section 21.1.2 gives us the implied base (= implied tranche) correlation on the [0, 3] tranche. Equation (21.2) can now be used for the second tranche to find the unique base correlation for that tranche corresponding to the fair premium. The process is then repeated for higher tranches.

In practice, these standard structures (iTraxx and CDX HY and IG) are used to derive implied base correlations. These implied correlations are then used to interpolate base correlations to price bespoke deals – deals whose reference entities may be different from the standard indices and whose tranching and maturities may be different from the standard tranching of the indices.

'Apex' Correlation

Working upwards from the nearest detachment point to zero, we may find a base tranche where there is no base correlation consistent with lower base correlations and the tranche premium (point 7 above). Instead of base tranches we could look at tranches detaching at 100% ('apex' tranches), and if there is a price for, say, the [30, 100] tranche we could use this to find the implied correlation for that tranche. The next apex tranche [15, 100] could then be used under an analogous process to that for implied base correlations to find an implied [15, 100] apex correlation. This method may allow access to implied correlation on senior tranches when base correlation gets stuck on a lower tranche. Of course, there may still be one or more tranches for which an implied base or apex correlation cannot be found.

Note that if a full set of implied base correlations exists, then a full set of implied apex correlations also exists.

21.2.3 Interpolating Base Correlation

In practice a bespoke CDO may differ from standard CDO structures in a variety of ways and we need a variety of interpolation methods to handle them. Interpolation is nothing more than educated guesswork; some methods and approaches are more secure than others and there are question marks over the validity of some of these methods – in particular, reference entity mapping and interpolation. Specifically, the following differences are typical:

1. Maturities intermediate between or outside the range of the standard deals – maturity interpolation
2. Tranche attachment and detachment points different from the standard – base correlation interpolation for detachment point.
3. Reference entities may differ from the standard pool – entity mapping and cross pool interpolation methods.

We shall describe some of the standard methods in turn.

Maturity Interpolation

For the moment think about an iTraxx structure differing only from the standard in its maturity – say 4 years. Implied correlations derived from 5-year CDO trades may not be appropriate for a 4-year CDO. Data for 3-year and 5-year CDOs are helpful when looking at an intermediate life CDO, and we could propose alternative simple interpolation methods – for example, linear. If term structure base correlation is used, then this has its own built-in maturity interpolation method.

For short-life CDOs it would be useful to get indications from the market for tranche valuations. One might also be tempted to look at current spread correlation over the recent period compared with the longer term average. There are some offsetting effects that lessen the importance of the correlation number at shorter maturities.

Exercise 1 Show that the correlation sensitivity of a tranche reduces as the outstanding maturity falls to zero (with no defaults on the underlying portfolio).

Detachment Point Interpolation

Suppose we wish to price or value a [5, 7] tranche of a CDO whose reference portfolio forms the basis of a regularly traded CDO (say, the iTraxx). Through the procedure described above we can calculate the implied base correlations for the [0, 3], [0, 6] and [0, 9] tranches. We can interpolate these to get base correlations for [0, 5] and [0, 7] tranches – one standard interpolation procedure uses expected losses on the bespoke and standard deal to interpolate base correlations as follows.

- For the standard deal, calculate the base tranche expected loss (EL) and express this as a percentage of total EL. Repeat this for the other tranches.
- Find the detachment points using the bespoke portfolio that correspond to the standard deal EL percentages (if the reference pool and maturity date are the same as the standard deal, then these detachment points are the standard ones).
- The standard deal base correlations are now associated with these non-standard detachment points.
- We use the actual detachment points to interpolate the appropriate base correlation – cubic spline interpolation being most favoured.

Entity Mapping and Interpolation

1. Where names in the reference pool are all EUR investment grade, or all USD investment grade, we would typically ignore the fact that the reference entities themselves are different and simply use the correlations derived from the iTraxx or CDX.IG deals.
2. If the names are a mix of EUR and USD investment grade names, we could calculate a weighting in each region (notional or better expected loss) and use this to interpolate between correlations.
3. This also applies if we have a mix of EUR investment grade names, USD investment grade and US high yield.
4. In other cases (for example, EUR high yield) we have to make a judgement as to whether the iTraxx or the CDX.HY is likely to be more relevant and map the name appropriately.

21.3 CORRELATED RECOVERIES

One feature of the implementations of the Normal Copula model described so far is the assumption of fixed recoveries on each reference entity. We have discussed earlier models for stochastic recovery and we can easily check (by running simulation pricing) that the impact of stochastic recoveries on tranche pricing is very small. However, the implementation assumes that recoveries are uncorrelated with the number of defaults occurring.

During 2008 the pricing of some (senior) CDX.IG tranches became inconsistent with the base correlation model as described so far. We noted in section 21.2.1 that tranche correlations could be in a region in which base correlations might not be able to match and, indeed, this occurred. A further complication arose when the market started quoting a [60, 100] tranche. Under the then standard recovery assumption of 40%, without stochastic recovery this would have zero value.

Two solutions were found to the problem. The first was a temporary fix, but related to the following solution. It transpired that if the assumed recoveries were 'marked down' to zero (or some value above zero, but sufficiently below the then standard 40% recovery), then it became possible to fit the fair premium on the tranche. The second set of solutions was to make recoveries stochastic, but now to correlate the number of simulated defaults occurring in the life of the deal with the recovery level. This was done in a variety of ways (Amraoui and Hitier, 2008; Ech-Chatbi, 2008; Krekel, 2008, and others). It is important to note in the following that recovery rates are not correlated with hazard rates, as this would invalidate the pseudo-hazard rate model we have developed for CDS. Instead, the number of credit events simulated in a specific time period is correlated with recoveries.

We shall describe the method of Krekel since this is closest to, and consistent with, the stochastic recovery model already described, which is based on observable (historic) data.

The first step is to create a stochastic recovery model consistent with the assumed mean recovery rate. In particular, the expected recovery rate conditional on default is the mean recovery rate that we are assuming for CDS calibrations. Such a model has been described in Part II and alternative recovery rate models are also consistent with the approach. The advantage of the model we have already described is that it is parsimonious – it reflects assumed mean recovery and standard deviation with only three possible recovery rates being simulated.

The second step is to correlate the recovery rates to the common factor in such a way that recovery is low when a higher than average number of defaults is triggered.

The impact of these changes is to make the base correlation model capable of generating higher premiums on the more senior tranches.

21.4 CORRELATION REGIME CHANGE AND OTHER MODELLING APPROACHES

We briefly mention two other modelling approaches that can be taken within the Normal Copula default time model framework.

21.4.1 Stochastic Correlation: Regime Change Models

We mentioned in Chapter 1 that correlation can become high in a credit crunch (and at other times). One way to handle this is to make correlation itself a stochastic variable. The term structure base correlation approach makes correlation a deterministic function of time – the next step is to make it stochastic. One simple implementation is known as a 'regime change' model. In any given time period the correlation is a constant, and in the next time period it is a constant, but in each time period the correlation number is randomly chosen from a low number with a high probability and a high number with a low probability. Such models are sometimes used in VaR analysis of a portfolio of risks, and application to CDO pricing is a natural extension of this.

21.4.2 Spread Factor

Comparing the correlated default time model to the Black–Scholes model, the former has two key deficiencies.

1. The key pricing parameter (default time correlation) is unobservable.
2. Unlike the Black–Scholes model where, given an option price above the intrinsic value, there is always a volatility that matches the price, the correlated default time model is not capable of matching any given tranche premium.

One simple way to resolve the latter problem (but not the former) is to change the model parameters to use a fixed recovery distribution and fixed default time correlations – perhaps derived from spread correlations – and find a scaling factor which, when applied to all CDS curves, matches the tranche premium. A different scaling factor would be needed for each tranche but this approach would match any positive tranche premium.

22 Valuation and Hedging

In this chapter we concentrate on risk management and hedging of portfolio structures, and model errors.

22.1 VALUATION EXAMPLES

The aim of this section is to present some graphical and numerical results to show the impact of different correlation on the pricing of portfolio CD products, and to show the impact of various other factors on CDO pricing. Numerical results quoted in this section are based on 10 000 simulations for a 100 name portfolio: otherwise it is generally immaterial whether a Monte Carlo model or an SCF pricing approach is followed.

22.1.1 F2D Baskets

We have already seen the 0% correlation and the 100% correlation results for the fair value premium on a F2D. Figure 22.1 shows the pattern of fair value spread for a two name basket as correlation varies from 0 to 1. We assume the CDS premiums for the names are 82 bp and 123 bp, and we assume both names have the same recovery. Some further pricing results specific to standard traded CDO structures appear in section 22.4.

Exercise 1

1. What happens at negative correlation on the two name basket? If correlation is minus 100% what does that mean for the simulated random numbers? If the simulated random number means that one name defaults early, show that the other name defaults late. Hence show that the curve in Figure 22.1 continues rising as correlation becomes increasingly negative.
2. At minus 100% correlation the above shows that you would pay more than 82 bp + 123 bp for the F2D protection. Intuitively, why is this the case? [*Hint:* (a) What is the hedge for the bought protection? (b) If one name defaults within the life of the deal, when does the other default? (c) For how much can can the sold protection be bought back?
3. In reality the recovery rates are unknown. What does the chart look like in general terms if we take into account recovery uncertainty?

22.1.2 CDO Pricing: Change of Correlation

First take a 5-year portfolio of 100 names with an average CDS premium 100 bp and a standard deviation of premiums across the names in the portfolio of 29 bp. We assume a recovery of 40% for all names. Table 22.1 and Figure 22.2 show the fair value premiums as correlation changes.

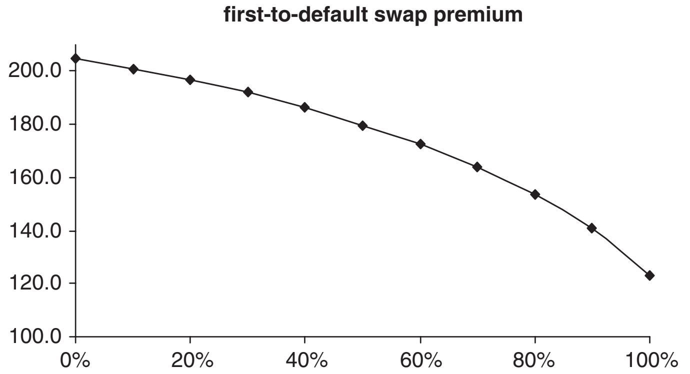

Figure 22.1 Fair value F2D premium on a 5-year two name basket.

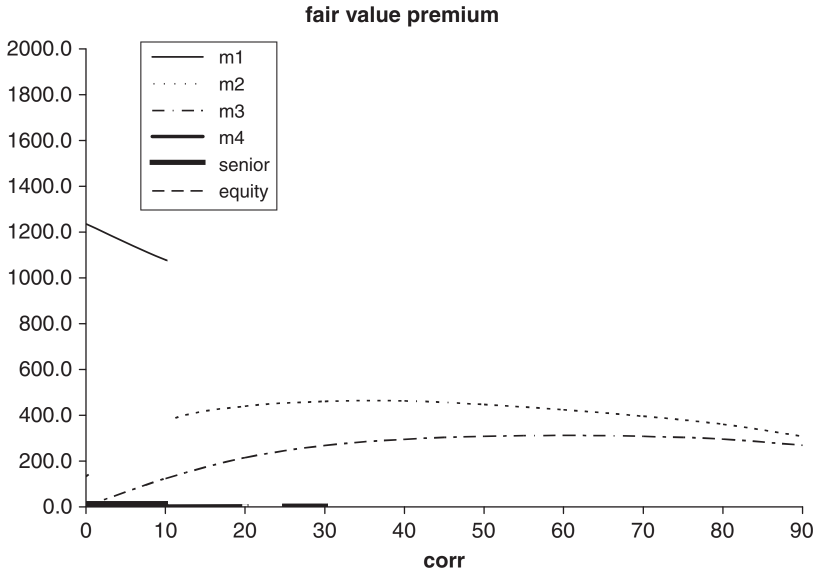

Figure 22.2 Fair value tranche premiums and correlation.

Table 22.1 Fair value tranche premiums and correlation

		Correlation premium									
Tranche	Size	0	10	20	30	40	50	60	70	80	90
equity	3	5,302.4	3,775.9	2,908.2	2,315.0	1,872.3	1,517.5	1,226.9	971.5	739.1	530.3
m1	3	1,236.6	1,079.5	971.1	879.5	790.5	704.4	624.0	543.3	462.8	375.7
m2	3	132.7	370.5	439.3	460.6	463.0	446.9	424.4	395.7	360.7	308.4
m3	3	3.0	123.2	214.7	268.5	294.8	308.2	312.0	308.2	295.9	268.7
m4	10	0.0	17.0	58.1	101.1	137.4	166.2	188.0	203.5	211.9	212.1
senior	78	0.0	0.0	0.7	2.9	6.7	12.0	18.8	26.7	36.3	48.7

Note the following:

1. The higher risk tranches have higher fair value premiums.
2. At low correlation the equity tranche premium is over 50% of notional per annum, and the premium for the senior tranche is only a fraction of a basis point.
3. At high correlation the senior premium is close to the lowest CDS premium on the portfolio, and the equity tranche premium is close to the highest CDS premium.[1]
4. As correlation increases, the fair value premium of the equity tranche drops monotonically.
5. As correlation increases, the fair value premium of the senior tranche increases monotonically.
6. One tranche ('m2') has a premium which initially increases, then decreases as correlation increases. In particular, if the actual tranche premium is 400 bp, there are two correlation figures ('implied correlation') consistent with this premium – one about 15% and another about 68%.

Note: Where the equity (or other) fair value premiums are 'large', the market generally deals using a combination of a 500 bp premium (per annum) plus a non-refundable up-front premium. This has significantly helpful hedging implications.

Exercise 2 As correlation increases, value shifts from the equity tranche towards the senior tranche. Why is this?

Hints:

1. At zero and 100% correlations what is the appearance of the simulated default times?
2. At zero correlation how many defaults would you expect to see? How much variation would you expect to see?
3. At 100% correlation how many defaults would you expect to see? (Why is this the same as in (2)?) How much variation would you expect to see? (Why is this much greater than in (2)?)

22.1.3 CDO Pricing: Change of Tranching

With the same underlying portfolio Table 22.2 shows the effect of reducing the tranche width and making the tranches more junior. Fair value spread on each tranche increases as the tranche becomes more junior.

A key feature in CDO product design, when producing a bespoke CDO, is the optimisation of tranche sizes and placement in order to obtain the best combination of tranche ratings, cost in the market to obtain protection on each tranche (not the same as fair value premium), and tranche sizes – to minimise the total cost of servicing all tranches of protection.[2]

[1] This is similar to the N2D result. The precise behaviour for CDO tranches will vary according to the reference portfolio and size of the CDO tranches.

[2] The cost of servicing the tranche is the premium actually paid times the value of a 1 bp premium stream – as calculated in equation (22.2). In cases where a junior tranche is retained the capital allocated (notional of the tranche) can be taken as the equivalent up-front premium for the tranche.

Table 22.2 Tranche size and premium

20% correlation				
Tranche	Size	Premium	Size	Premium
equity	2	3,737.6	3	2,908.2
m1	2	1,516.8	3	971.1
m2	2	838.7	3	439.3
m3	2	489.5	3	214.7
m4	4	245.9	10	58.1
senior	88	7.4	78	0.7

22.1.4 CDO Pricing: Change of Underlying

A change in the average spread of the underlying portfolio can be compensated for, to some extent, by changing the tranche attachment points. Table 22.3 shows a changed tranching for a high-spread portfolio compared with the portfolio above. In both cases we express the premium as a multiple of the average CDS. In practice the equity piece would be much larger with the aim of getting investment grade ratings on the senior tranches.

Introduction of a few high-spread names

Table 22.4 shows the impact of introducing 5 names with 1 000 bp CDS premium. With unchanged tranching the fair value premium on all tranches increases.

Exercise 3 Assuming zero default time correlation, what is the expected number of defaults from these high-spread names?

If we increase the equity tranche size by 2%, and reduce the senior tranche by the same amount, we find that the spreads on the new tranching for the new portfolio is now very similar to the old one.

Change in recovery assumption

Let us take the 100 bp average spread portfolio, but instead of assuming a single average 40% recovery for every name, let us assume that recoveries are 0%, 40% and 80% (evenly

Table 22.3 100 bp and 500 bp portfolios: tranche premium divided by average CDS

20% correlation				
Tranche	Size	Premium/100 bp	Size	Premium/500
equity	3	29.08	3	27.02
m1	3	9.71	7	8.86
m2	3	4.39	6.5	4.20
m3	3	2.15	10	1.89
m4	10	0.58	8.5	0.64
senior	78	0.01	65	0.03

Table 22.4 The impact of 5 wide spread names

	100 bp avg portfolio		Five 1,000 bp names		
20% correlation	Tranche	Premium	Premium	New tranching	Premium
equity	3	2,908.2	4,350.1	5	2,895.8
m1	3	971.1	1,437.7	3	842.1
m2	3	439.3	652.8	3	402.0
m3	3	214.7	315.9	3	200.4
m4	10	58.1	84.9	10	52.9
senior	78	0.7	1.0	76	0.6

Table 22.5 Flat 40% recovery assumption versus 0/40%/80% recovery assumption

20% correlation	Size	R = 0.4	R = 0.4/0.8
equity	3	2,908.2	3,007.2
m1	3	971.1	948.9
m2	3	439.3	407.6
m3	3	214.7	191.5
m4	10	58.1	48.1
senior	78	0.7	0.5

distributed across the names and across spread levels). We then find that there is a significant (at the 95% confidence level on each tranche) shift in value from senior tranches to the equity tranche (see Table 22.5).

22.1.5 CDO Pricing: Change of Maturity

With the 100 bp average portfolio, 40% recovery, and the same tranching, the fair value premiums on senior tranches drop as the CDO shortens *assuming that* no defaults and no changes in spread occur. Table 22.6 shows the fair value premiums at shorter maturities for the same underlying portfolio.

22.1.6 Managed CDO: Substitution and Change of Subordination

If a substitution is made to the reference pool of a managed CDO, then it is typically the case that the spreads on the new and old names are different. In addition, if several substitutions are made, there may be differences in individual notional. These changes will typically change the value of the tranche. A feature of many managed CDOs is that, on a substitution of one or more names by other reference entities, the subordination level (i.e. the attachment point) is changed so that the value of the tranche is the same as before the substitutions were made. In making this change certain allowable expenses – X – can be deducted from the tranche value. Let N be the pre-substitution tranche value and let $[a, b]$ be the tranche attachment and detachment points, then post-substitution the tranche becomes an $[a + x, b + x]$ tranche such that the new tranche value is $N - X$. Note that x can be positive or negative.

If valuation is performed using a fixed tranche correlation, then the calculation process is simple, requiring only iteration on the attachment point to match the value with the new reference pool to the agreed sum. However, if implied correlations are being used (tranche or

Table 22.6 Fair value premiums at several maturities assuming no defaults

20% correlation		No defaults 5-year premium	3-year premium	1-year premium
equity	3	2 908.2	3 019.5	3 210.4
m1	3	971.1	802.2	424.9
m2	3	439.3	298.4	97.3
m3	3	214.7	124.0	27.3
m4	10	58.1	26.9	2.5
senior	78	0.7	0.1	0.0

base) then the process is more involved. With the initial $[a, b]$ tranche, valuation will change not only because of the direct impact of the new names in the reference pool but also owing to the indirect impact on the calculation of implied correlation that these names have – altering the reference entity interpolation process and producing slightly different implied correlations. When a first iteration is made, changing the tranche to $[a + z, b + z]$, this will further change the implied correlation calculation because of the change in the attachment points. The process is involved but straightforward as long as the substitutions are relatively minor.

22.2 SENSITIVITY CALCULATION AND HEDGING

In this section we consider the calculation of a variety of risk (sensitivity) numbers and simplistic hedging methods. We return to this topic again when we look at some specific deals and consider correlation book management. The principles and methods described here are appropriate to both CDO tranches and N2D baskets. The key sensitivities we can calculate from the pricing model are as follows:

1. Time decay (often called 'theta') – a one-week shortening of life often being used to calculate the change in value of deals.
2. Interest rate sensitivities – shift and bumps to the interest rate curve.
3. Default event sensitivities – for example, the impact on the value of an $[a, b]$ tranche if reference entity 27 has a credit event (one such sensitivity being calculated for each reference entity).
4. Credit shift and bump sensitivities – again for each reference name.
5. Recovery sensitivities.

On a 125 name reference pool this can mean approaching 1 500 valuations. Repeated calls of the same valuation routine – even if a semi-closed form approach is used – can be slow and, in the case of Monte Carlo, impractical. Where an SCF implementation is used, it helps to incorporate the sensitivity calculations into the valuation calls – which speeds up the process considerably. Monte Carlo sensitivities, using repeated valuation calls, require much higher accuracy than a single valuation calculation because differences are then taken: so MC is only practical for time decay and aggregate shift sensitivities unless tricks are used in the valuation calculation to obtain credit sensitivities.

It should be borne in mind that SCF approaches are generally only approximations – it is easy to test the accuracy of the pricing by comparing it to Monte Carlo pricing, but it is less easy to test the accuracy of the hedging calculations.

Apart from the comments above, it is largely immaterial in the following whether an SCF or Monte Carlo implementation is used.

22.2.1 Dynamic Hedging: Spread Risk

Principles

Pick a name in the reference pool. If the CDS premium on the name changes from one day to the next, then the portfolio product (CDO tranche or N2D basket) will also change in value. If we own unit amount of the portfolio product and α of the CDS on the reference name, then the portfolio value is

$$P(x,t) = V(x,t) + \sum_{i=1..n} \alpha_i C_i(x_i,t) \tag{22.1}$$

and the change in the value of the portfolio is zero if

$$\alpha_i = -\frac{\partial V}{\partial x_i} \Big/ \frac{\partial C_i}{\partial x_i} \tag{22.2}$$

where V is the value of the CDO tranche, and C is the value of the CDS. A discrete approximation replaces the derivatives by finite differences with Δ denoting the change in value. We may take the differential form – the hedge ratio is the rate of change in the value of the tranche with respect to the CDS – or we may take a finite change such as 10 bp up on the CDS (or we may take the average of up and down moves).

The change in the value of the tranche for a given change in the premium on an underlying name is referred to as the '**name (spread) sensitivity**', 'the **name [or micro] delta [or PV01]**'. If we shift all names in the portfolio simultaneously, the change in the tranche value is sometimes referred to as the '**macro delta**'. The change in name sensitivity as spreads change is the '**name (spread) convexity**' or '**micro convexity**', while the change in macro delta as spreads change is sometimes called the '**macro (spread) convexity**'.

Implementation

The CDS or bond sensitivity may be derived from a formula (Part II section 9.3). Generally the tranche or N2D sensitivity is calculated numerically (even under 'semi-closed form' solutions).

Bumping Method

For each reference name we first calculate the sensitivity of the hedge (CDS or bond) instrument by shifting the premium or spread, recalibrating (to give two sets of hazard rates for each name – the 'base' hazard rate and the 'shifted' hazard rate), and revaluing the CDS or bond.

Valuation of the tranche depends on the method used for pricing – simulation or closed form.

1. *Simulation*. For each set of correlated random numbers generating N default times (where N is the number of names in the reference pool) we also generate other N default times (with the same random numbers) using the shifted hazard rate. Each set of default times gives rise to $N + 1$ values for the tranche – the base value where the base hazard rate curves are used for all names, and N values where each name's base curve is, in turn, replaced by

its shifted curve. The sensitivity of the tranche to a shift in a reference name's spread is then found by differencing the two relevant tranche valuations. Note that

(a) It is important to use the same random numbers to simulate the base and the shifted default time;
(b) Since we are taking differences in Monte Carlo simulated values, we need to use a greater number of simulations to get an accurate answer. If 10 000 simulations are required for a pricing (which might take 1 second), 1 million simulations may be needed for sensitivity calculation, which takes nearly 3 hours for a 100 name portfolio.

2. *Semi-closed form.* Semi-closed form methods use numerical integration but can only be applied when the correlation matrix has a certain form (in particular, a single non-trivial correlation number will do) and the CDO does not have a complicated waterfall. Then the tranche is priced using the base and shifted hazard rate curves and a difference in values calculated. Calculation time for the sensitivities is reduced to a few seconds. This has been one of the main practical reasons (but not the only reason – more complex correlation matrices do not necessarily give better prices) for quoting implied correlations based on a trivial correlation structure.

Approximate Method

The same procedure, described in detail for the single name CDS case in (Part II section 18.2 equations (18.6) onwards, and Figure 18.2), can be applied to get the sensitivities for each nth-to-default basket to the rth reference entity. The procedure is only a little more complicated. All existing N2D events triggered by the rth name contribute to an increase in value of the N2D if the hazard rate on name r is increased. We obtain an estimate of an increase in the value of the N2D arising from new events by looking at cases where the (N – 1)2D is triggered and the N2D is not triggered, and finding those simulations where the rth name lies within Δ of the life of the trade.

The procedure is implemented in the MathCad sheet 'chapter 22 N2D CDO CLN price and hedge.mcd' (for the case of assumed equal recoveries) for nth-to-default baskets.

Hedge Ratios for CDO Tranches

The hedge ratios for two CDOs which differ only in that, for the former CDO, the equity tranche receives a 1,200 bp premium, while the latter receives 500 bp. The reference portfolios for each are 125 investment grade names, and the tranching is the same. Hedge ratios are calculated using a flat 20% correlation across all tranches. Tables 22.7 and 22.8 show the minimum, average and maximum hedge ratios applicable to each name (expressed as a percentage of the notional of that name in the reference pool). Note that all three figures are higher for the former portfolio than for the equity tranche. The tables also show the hedge ratio for each name to each tranche (for a subset of the reference pool). The former portfolio has an average hedge ratio around 117%, while the latter is around 106%. In particular, the hedge ratios do not sum to 100%. Equivalently, a spread-hedged CDO is not event hedged.

Note also that the higher spread names have a higher hedge ratio to the equity tranche, and a lower hedge ratio to the senior tranches, while the lower spread names show the opposite phenomenon.

Why do the hedge ratios not sum to 100?

Table 22.7 CDO tranche hedges – high premium equity tranche

		Hedge ratio (% name notional)						
TRANCHE ->	Total	[22, 100]	[12, 22]	[9, 12]	[6, 9]	[3, 6]	[0, 3]	
min		0.1	2.6	4.2	10.6	27.2	53.2	
average		0.3	4.7	6.3	14.0	30.3	62.1	
max		0.5	7.0	8.3	16.7	31.4	72.2	
Reference entity								name CDS
Lafarge SA	117.9	0.2	4.2	5.9	13.5	30.1	64.0	50
Linde AG	117.8	0.3	4.9	6.6	14.4	30.7	61.0	35
Lloyds TSB Bank Plc	117.2	0.5	7.0	8.3	16.7	31.4	53.2	12
Marks & Spencer Group Plc	116.8	0.1	2.6	4.2	10.6	27.2	72.1	121
Metro AG	117.9	0.2	4.0	5.7	13.2	29.9	64.8	55
Muenchener Rueckversicherungs AG	117.7	0.3	5.3	6.9	14.9	31.0	59.2	28
National Grid Transco Plc	117.9	0.3	4.6	6.3	14.0	30.5	62.2	41
Pearson Plc	117.9	0.3	4.5	6.2	13.9	30.4	62.6	43
Peugeot SA	117.8	0.3	4.7	6.4	14.1	30.5	61.8	39
Pinault-Printemps-Redoute	117.2	0.1	2.6	4.2	10.6	27.5	72.2	119
Portugal Telecom International Finance BV	117.8	0.3	5.0	6.7	14.6	30.8	60.5	33
RWE AG	117.7	0.3	5.4	7.0	15.0	31.0	59.0	28

Table 22.8 CDO tranche hedges – 500 bp premium equity tranche

		Hedge ratio (% name notional)						
TRANCHE ->	Total	[22, 100]	[12, 22]	[9, 12]	[6, 9]	[3, 6]	[0, 3]	
min		0.1	2.6	4.2	10.6	27.2	43.2	
average		0.3	4.7	6.3	14.0	30.3	51.0	
max		0.5	7.0	8.3	16.7	31.4	59.9	
Reference entity								
Lafarge SA	106.6	0.2	4.2	5.9	13.5	30.1	52.7	50
Linde AG	106.8	0.3	4.9	6.6	14.4	30.7	50.0	35
Lloyds TSB Bank Plc	107.2	0.5	7.0	8.3	16.7	31.4	43.2	12
Marks & Spencer Group Plc	104.4	0.1	2.6	4.2	10.6	27.2	59.8	121
Metro AG	106.5	0.2	4.0	5.7	13.2	29.9	53.4	55
Muenchener Rueckversicherungs AG	107.0	0.3	5.3	6.9	14.9	31.0	48.5	28
National Grid Transco Plc	106.8	0.3	4.6	6.3	14.0	30.5	51.2	41
Pearson Plc	106.7	0.3	4.5	6.2	13.9	30.4	51.5	43
Peugeot SA	106.8	0.3	4.7	6.4	14.1	30.5	50.8	39
Pinault-Printemps-Redoute	104.9	0.1	2.6	4.2	10.6	27.5	59.9	119
Portugal Telecom International Finance BV	106.9	0.3	5.0	6.7	14.6	30.8	49.6	33
RWE AG	107.1	0.3	5.4	7.0	15.0	31.0	48.3	28

If the premium for protection on all CDO tranches was paid up front, and the premiums on CDS deals was also paid up front, then the sum of the two (all CDO tranches, and all CDSs) would be equal. We would also find that the hedge ratios sum to 100%. The capital flows from the CDO and the hedging CDS deals are identical no matter how the premiums are paid if the CDO tranches are in aggregate hedged by a 100% notional position in each reference CDS. However, a standard CDO tranche has a premium that is paid quarterly in arrears. Let us now consider the equity tranche – with a premium of 1,200 bp. As defaults occur this premium stream rapidly drops, whereas the sum of the CDS premiums falls much more slowly. There is a difference in the premium profile between a CDO and a portfolio of CDSs, and this causes hedging differences. Reducing the ongoing equity premium brings the sum of hedge ratios down (it may even be less than 100% if the equity premium is zero). Traders who prefer a statically hedged synthetic CDO will either

(a) retain the equity piece – ongoing premium of zero, or
(b) buy equity protection paying a substantial up-front premium and paying a much lower ongoing premium.

These hedge differences can be very substantial – on long-dated high-yield reference portfolios the delta hedging portfolio initially may be 50% larger than the notional size of the CDO. Reducing the ongoing equity premium brings the sum of hedge ratios down (it may even be less than 100% if the equity premium is zero).

Hedging in Practice

We illustrate some conclusions that can be drawn from the concept of hedging spread sensitivities and the above formulae applied to an F2D product. Similar results apply to CDO tranches.

Time Decay

Suppose we are long 10m F2D protection on two names, so that CDS hedges will be sold. The hedge ratios might be 70% in one name and 60% in another – i.e. we sell 7m CDS protection on the former name and 6m on the latter. The income from the hedge will generally not be equal to the premium on the F2D. As the trade matures we find that the hedge ratios increase (assuming constant spreads) and generate an increasing income from larger sold CDS positions (albeit at declining CDS levels arising from the slope of the typical CDS curve). Figure 22.3 shows the net premium income from the dynamic hedge (the initial position is the rhs of the figure at 5 years to maturity).

In the early years more is being paid for the F2D protection than is the expected cost of that protection – and the situation reverses in later years. This is an example of '**time sensitivity**', '**time decay**' or '**theta**' for a portfolio product. The time decay can be defined as the change in value of a portfolio CD as the time to maturity reduces (by a specified interval).

Spread Correlation and Volatility Rrisk

For simplicity, take a two name basket: at 70% correlation the premium is around 80% of the sum of CDS spreads. We can examine the hedges in two pairs of situations: if spreads move up or down simultaneously, or if one moves up and one down.

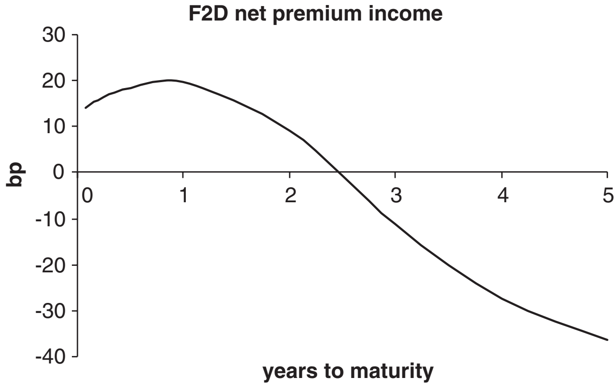

Figure 22.3 First-to-default protection and hedges: net premium income.

1. Simultaneous 10 bp moves cause a change in the hedge ratio in the opposite direction. If spreads rise we find that as we have smaller CDS hedge positions, we have to buy back hedges at higher spreads. This leads to a loss on rehedging whether CDS premiums rise or fall, as long as they move together. The greater the move, the greater the loss.
2. Opposite moves cause an increase in the hedge on the name whose premium has risen, and a reduction on the one that has fallen. We have to buy back some of the CDSs whose premium has fallen – leading to an immediate profit – and sell some more of the CDSs where the premium has risen[3] – leading to a profit again.

The supposed spread changes and impact on the hedge ratios is summarised in Table 22.9.

Dynamic spread hedging of a N2D or CDO tranche is subject to a risk to spread correlation and spread volatility – the higher the correlation and volatility the greater the loss (on long F2D positions), the lower the correlation the higher the profit.

Why does spread correlation and volatility not figure in the pricing model?

Table 22.9 F2D hedges and changes

		Spread changes	Hedge ratios (%)
Day 1	Name 1		70
	Name 2		70
Day 2	Name 1	+10	68
	Name 2	+10	68
	Name 1	−10	72
	Name 2	−10	72
	Name 1	+10	76
	Name 2	−10	66
	Name 1	−10	66
	Name 2	+10	76

[3] One might sell more notional at the same premium – below market levels – and take an up-front premium in addition to compensate. Thus a positive cashflow is generated. If premiums remain at expected levels, this cashflow will turn into profit.

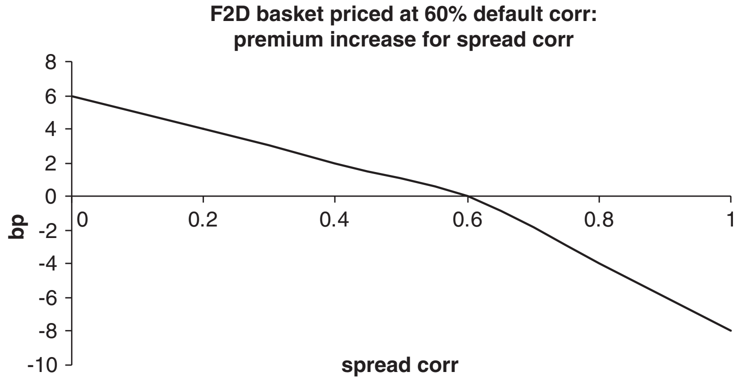

Figure 22.4 Long F2D dynamically spread hedged in different spread environments.

Figure 22.4 show the results of simulated dynamic (costless) hedging where the hedges are calculated under the default time Normal Copula model, and where hazard rates follow a lognormal random walk process. The pricing correlation for the two name F2D is 60%, and the spread correlation under the simulations is between 0 and 100%. The expected rehedging profit is expressed as a corresponding error in the premium (a positive meaning the premium charged on the F2D was too high and led to a profit). A 75% spread volatility was used.

We find that the expected rehedging cost is zero if the spread correlation and the pricing correlation are the same; if the spread correlation turns out to be lower than the pricing correlation, we make an expected profit on long F2D and equity CDO tranches through dynamic hedging; and if the spread correlation turns out to be higher, we make a loss. This is often used as an argument that the default time should be the spread correlation – if it is not, then there is a significant pricing element missing from our model.

Note that individual simulation paths can lead to substantially different P&L results from the expected value. *In other words, a trader spread hedging a book which is predominantly long of protection on F2D or equity CDO tranches, using CDS contracts as the hedge, is running a substantial risk to spread correlation* (even if the products are correctly priced). This risk is effectively captured by the book's (default time) correlation risk – see below.

Also note that rehedging is done daily: changes in short-term spread correlation can have a big impact on P&L.

Default and Spread Jump Risk

Consider, for example, a long protection F2D position on two names hedged by short positions (say 70%) in each CDS. According to the default time correlation model, on the early default of name 1, the spread on name 2 jumps substantially. 70 nominal of bonds delivered into the short CDS protection are passed (together with a further 30 bought in the market) into the long F2D protection – resulting in a capital gain of 30 × (1 – recovery). However, the short CDS on name 2 has to be unwound. Within the model world this should have jumped in spread. The unwind therefore leads to a capital loss. The expected value of this capital loss will be similar to the expected capital gain on the defaulted name – but there is not an exact match.

More importantly, does this pattern of behaviour occur in practice? To the extent that the model says the jump in spread occurs at the instant that the correlated name defaults, this is unrealistic. But consider a basket on Telecoms bought in 2000. As telecom names suffered a rapid increase in spreads, a long F2D basket would have suffered substantial rehedging losses from the delta rehedging in the highly correlated environment. The default of one of the names in the basket may then lead to no further widening in spreads, but would give rise to a capital profit on the defaulted name (as long as the hedge ratio was less than 100%). This profit would offset, to a greater or lesser extent, the loss realised in the run up to default. It can therefore be argued that the default time correlation model captured correctly the broad features of the correlated default process, although the timing is wrong.

22.2.2 Static Hedging: Default Event Risk

In the context of single name default swaps a 'default event hedge' is simple to understand. For example, a 5-year long USD 10m protection position on GMAC can be default event hedged by selling USD 10m of a 3-year CDS on the same name. On a credit event, debt delivered into the short position can be delivered into the long position, and the USD 10m received on the long position can be used to pay off the short.

Unfortunately 'default event hedging' does not have such a clear meaning in the context of credit portfolio products. We illustrate the problem with some examples and consider two alternative means of defining and calculating 'default event risk'.

'Back-to-Back' CDO

Typically a bank which sets up a CDO structure and obtains protection on all the tranches sells protection on all the underlying CDS contracts to the full notional amount in the reference portfolio. A credit event on a reference name will lead to a loss of (say) 6m on a 10m notional exposure, and protection on this loss is obtained from the tranches. This is often called a 'back-to-back' hedge, and is also a '**default event hedge**' in the sense that *capital flows are perfectly hedged*. It is also a static hedge – the CDS hedge size is the fixed notional in the reference pool, and is the constant amount on which the CDO provides credit event protection.

Event Risk on a Single Name

Another way of thinking of default event risk is as follows. Consider a specific name in the reference pool – name 1. If name 1 defaults, what is the change in the value of all the tranches? We can then choose a notional exposure for the hedge, so that the change in value of the hedge equals the change in value of the tranches. (We could apply this idea equally to a single tranche rather than an entire portfolio, or we could apply it to an *n*th-to-default position.)

This definition has some surprising consequences. Think of a CDO and all the tranches of protection and the reference pool of CDS, and suppose that all trades are done at fair value premiums. If we make a reference name default, then the CDS on that name in the reference pool has a jump in value – approximately the loss given default if the premium is low. The equity piece has a jump in value arising from the loss given default and now the premium on what remains on the tranche is below the fair market premium. So the hedge ratio for the equity tranche alone is greater than 100%. The remaining tranches have all now

Table 22.10 CDO 'default event' hedge ratios (impact of immediate default)

	Hedge ratio (% reference notional)	
Tranche	250 bp name	50 bp name
0–3	136	138
3–6	36	39
6–9	18	20
9–12	10	11
12–22	11	14
22–100	2	3

degraded – their lower attachment point has effectively dropped in relation to the remaining reference pool. Thus, the hedge ratio for all these tranches is also positive.

Table 22.10 illustrates the above concept for a CDO. The reference pool is 100 names with CDS premiums ranging from 55 bp to 250 bp and recoveries from 15% to 40% (20% correlation is assumed). Tranches are assumed to be paid the fair value premium. Table 22.10 shows the 'default event' hedge ratios for each tranche for a 55 bp name and a 250 bp name calculated by assuming immediate default of that name (and no change in the remaining CDS premiums) and recalculating tranche values. Note that the hedge ratios, and the total, are quite strongly dependent on the correlation used and the tranche premiums.

In addition, there is the problem with the approach in that we cannot force a specific name to default immediately, leaving other spreads unchanged, and be consistent with our pricing problem.

The above definition of default event risk may be useful for the calculation of a risk number but is not suitable for hedging purposes. The following approach is preferred.

Expected Event Risk

The 'back-to back' CDO had the property that any capital cashflow triggered by a default event in the reference pool is matched by a cashflow in the tranches. Thinking of the default time simulations, for each name we can see whether (and by how much) the default hit the equity tranche, the junior mezzanine tranche, etc. We can calculate the expected value of the impact of default of name 1 on each of the tranches. We know the sum of these expected values is equal to that for the CDS (since cashflows are identical). The value for each event is the discounted value of the LGD less the discounted value of the premium stream up to the default date (for narrow spread names we could ignore the latter term without much loss of accuracy since it is small compared with the LGD).

The following definition of default event risk is consistent with the model and has more natural properties. It can be thought of as the expected impact of the default event on each tranche compared with the impact on the underlying CDSs.

The 'default event risk' for the kth tranche (or, similarly, kth to default contract) to the ith name in the basket is given by

$$DEHR(k,i) = \frac{\sum\limits_{simulations} TDB(k,i)}{\sum\limits_{simulations} CDSB(i)}$$

Table 22.11 CDO 'default event' hedge ratios (impact of expected default)

Tranche	Hedge ratio (% reference notional)	
	250 bp name	50 bp name
0–3	29	38
3–6	22	25
6–9	16	15
9–12	12	9
12–22	16	11
22–100	5	2

where

- *CDSB*(i) denotes the value of the capital payoff less the premium stream arising from the CDS for those cases where name i defaults before maturity, and
- *TDB*(k,i) denotes the value of the capital payoff less the premium stream arising from the kth tranche for those cases where name i defaults before maturity.

For example, for a CDS, each simulation gives rise to the following valuation term

$$\begin{aligned} CDSB(i) &= (1 - R_i) \cdot \exp(-r \cdot T(i)) - p_i \cdot \frac{(1 - \exp(-r \cdot T(i)))}{r} \quad \text{if } T(i) < \text{contract life} \\ &= 0 \text{ otherwise} \end{aligned}$$

where R_i is the assumed recovery on name i and $T(i)$ is its time of default. The formula for *TDB*(k,i) is similar (we need to take into account the amount by which the tranche is hit). Such an approach is very fast to implement under a simulation environment.

Table 22.11 shows the expected event hedge ratios for the same portfolio as in Table 22.10.

The default event hedge ratios sum to 100% – as does the 'back-to-back' hedge – and there is a marked difference between high- and low-spread names and the event sensitivity of each tranche.

The default sensitivity of a tranche (to the default of a single name) is sometimes referred to as the '**omega**'.

22.2.3 Correlation Risk

Figure 22.2 and Table 22.1 show that the fair value premium depends on the pricing correlation – correspondingly, valuation of a portfolio deal will depend on the correlation. There is a sensitivity of value to the pricing correlation. This sensitivity varies according to the tranche – and is typically strongest (per unit of nominal) for equity tranches. Some (mezzanine) tranches may be quite insensitive to even substantial correlation changes. Table 22.12 shows the correlation sensitivities of the portfolio used for Table 22.7 (and assuming a 1bn reference portfolio). Correlation risk is calculated assuming a 1% increase, and is expressed in EUR.

At this level of correlation (a flat 20%) there is a significant level of risk on all the tranches shown. The equity tranche has an opposite sensitivity to all the others, as increasing correlation takes value from the equity tranche and adds it to the others.

Correlation risk describes the sensitivity of the tranche value to a change in the pricing correlation. There is also an unmeasured exposure to spread correlation and spread sensitivity.

Table 22.12 Correlation sensitivity of CDO tranches (EUR per 1bn)

Tranche	+1% correl sensitivity
[22, 100]	5 069
[12, 22]	51 737
[9, 12]	46 987
[6, 9]	64 962
[3, 6]	31 851
[0, 3]	−216 106

A tranche spread hedged along the lines of section 22.3.1 (using single name CDSs) may leave a substantial correlation risk, and the only way to hedge correlation products is with other correlation products.

A deal in a CDO tranche (perhaps a bespoke portfolio and structure for a client) is best hedged by hedging the correlation risk first – for example, using appropriate tranches of standard liquid CDO products (such as CDX and iTRAXX tranched portfolios). Once the correlation risk is eliminated, the spread risk can be addressed. A good correlation hedge will usually largely eliminate (bucketed) spread risk, although individual cross-bucket spread risks may remain to be hedged. For example, the bespoke portfolio may have no autos but an excess of finance exposures; when hedged with an iTraxx tranche there will be a net long protection in autos and short in finance. This could be hedged using the iTraxx auto and finance index products (not tranches). Residual large exposures in certain names could then be hedged with CDSs.

Following this approach, exposure to correlation changes (and to spread correlation and volatility) is largely eliminated.

Correlation sensitivity is sometimes referred to as '**rho**'.

22.2.4 Recovery Risk

Table 22.5 showed *sensitivity to a change in the standard deviation* of assumed recovery rates. In addition, if we change the expected recovery rate there is a greater sensitivity of tranche values – tranches of a CDO have digital risk. Table 22.13 shows the sensitivity to a 5% increase in assumed recovery rates across all names.

Recovery rate risk has to be hedged by other digital exposure. Hedging CDO tranches initially using other similar CDO tranches will tend to introduce equal and opposite recovery risk, leaving a typically small residual, but this will not be achieved by spread hedging alone.

Table 22.13 Tranche sensitivity to recovery rates

Tranche	5% recovery rate sensitivity
[22, 100]	−5 124
[12, 22]	−49 291
[9, 12]	−44 774
[6, 9]	−63 858
[3, 6]	−45 980
[0, 3]	186 455

Table 22.14 Spread convexity: change in hedge ratio for a 100 bp jump in spreads

		Hedge ratio (% name notional)					
TRANCHE ->	Total	[22, 100]	[12, 22]	[9, 12]	[6, 9]	[3, 6]	[0, 3]
min		3.7	17.4	9.8	8.0	−9.8	−33.4
average		4.9	18.9	8.7	5.3	−9.3	−36.7
max		8.4	23.6	8.0	2.7	−8.9	−43.2
Reference entity							
Lafarge SA	−8.7	4.4	18.2	8.9	5.9	−8.5	−37.6
Linde AG	−8.2	4.9	18.9	8.6	4.9	−9.7	−35.9
Lloyds TSB Bank Plc	−7.8	5.4	18.3	7.1	2.6	−11.2	−30.0
Marks & Spencer Group Plc	−8.0	3.7	17.5	9.9	8.6	−4.7	−43.1
Metro AG	−7.4	5.4	20.7	9.6	6.1	−9.4	−39.9
Muenchener Rueckversicherungs AG	−8.0	5.2	19.1	8.4	4.4	−10.3	−34.7
National Grid Transco Plc	−7.9	5.2	19.8	9.0	5.3	−9.8	−37.4
Pearson Plc	−8.2	4.9	19.2	8.9	5.4	−9.3	−37.3
Peugeot SA	−8.5	4.7	18.5	8.6	5.2	−9.3	−36.2
Pinault-Printemps-Redoute	−7.3	4.4	19.4	10.5	8.7	−5.6	−44.6
Portugal Telecom International Finance BV	−8.7	4.6	18.0	8.3	4.8	−9.4	−34.9
RWE AG	−8.4	4.9	18.4	8.1	4.3	−10.0	−34.1

22.2.5 Convexity Risks

Credit spreads are not well behaved –jumps in spreads occur on individual names (Xerox, Marconi, and many other examples exist of large overnight spread moves without an accompanying credit event), or on sectors or geographical regions (Telecoms, Autos, Asia), and this introduces a large exposure to **spread convexity** risk. Table 22.14 shows convexity risk for a junior mezzanine tranche calculated by shifting all CDS premiums up simultaneously by 100. An alternative measure would shift spreads individually and by sectors.

Again Figure 22.2 shows **correlation convexity** risk. Fortunately implied correlation does not appear to show marked short-term changes. On the other hand, investors using historically calculated correlation matrices, especially where unstable calculation methods are employed, will experience large profits or losses arising from correlation convexity risk.

Consider the 5–8% tranche of an investment grade CDO referencing 50 European and 50 Asian entities. This is hedged by appropriate nominals of the 5–8% tranche of two separate CDOs: one referencing the same 50 European names, and the other referencing the same 50 Asian names. Suppose 5 European and no Asian names default over the next year. The tranche of the European CDO will probably have made capital payments, whereas neither the 100 name nor the Asian CDO tranches have been hit (although the 100 name trance is now closer to being hit). In this case there is **name convexity** where the trade is subject to a risk to an imbalance in the way defaults occur.

Once more hedging initially using other similar correlation products greatly reduces the convexity risks.

22.3 PRICING MORE COMPLEX STRUCTURES

Managed CDOs are generally valued as if they were static. Managers argue that they add value through the management process but, although there is evidence that they are skilled in avoiding defaults, there is no clear evidence that they add value compared to a static pool. A manager may remove a name only after it has fallen to the recovery value, and may remove names that have fallen in value but subsequently do not default and recover in value. The simplest assumption is to assume that the manager is neutral in his or her actions – covering the additional costs of the managing process – and treat the pool at any given time as if it were static.

22.3.1 CDO Squared

Among the reference risks in CDOA may be tranches of CDOB. If we are performing a fundamental valuation with (say) pairwise correlations derived from historic data, then this can be handled by simulating together the universe of all names in the CDO structures and calculating the impact of a set of simulations on CDOB – calculating its maturity amount – and hence calculating the impact on the valuation of CDOA of that same set of simulated defaults. Note that this approach – whether using pairwise, tag, factor or base correlation – is consistent and capable of handling ragged maturities of the reference assets but may simply not be consistent with the mark-to-market pricing of the referenced CDOs.

The above approach cannot be used for risk-neutral pricing where correlations are implied from standard deals. If the correlation required to price the tranche of CDOB is ρB, but the correlation appropriate to price a similar CDO to CDOA (but one that does not have a reference to CDOB) is ρA, then it is not possible to get a consistent set of simulated default times across all names. How do we value the CDO squared? There is no consistent answer for mark-to-market valuation.

As a common approach to CDO squared valuation, first note that a CDO tranche does not default and, if funded, simply returns a below par amount at maturity if it is not completely wiped out, and may even return zero. We can value the tranche of a CDOB and now, in the reference pool of CDOA, replace the tranche by a single name reference entity (as if it were a corporate name) assuming zero recovery and with a fair CDS premium equal to the calculated current fair premium for the CDOB tranche. This is perhaps the closest we can get to being able to perform a consistent 'risk-neutral' valuation.

Another practical problem commonly met is that the reference pool for underlying tranches may not be known if it is a dynamic pool at the time a valuation is required, and the maturities of the CDOs referenced may not coincide with the maturity of the CDO squared.

The lack of popularity of these structures, after the initial interest, partly reflects the inability to mark them to market in a reliable way. The situation gets more complicated with CDO cubed (where the referenced assets are themselves CDO squared). Again, using a single correlation matrix for all the underlying reference names across all the CDOs is internally consistent, but not necessarily consistent with the market pricing of the individual CDOs.

22.3.2 Cashflow CDOs

We first review some of the differences between a general cashflow CDO and synthetic CDOs. Imagine during this discussion that we are performing some form of period by period simulation (we shall specify more clearly what is needed at the end).

Assets

The reference pool may contain callable bonds or loans. Not only do we have to simulate the default or survival over a specific period, but we also need to simulate whether or not the bond or loan matures early. Floating rate instruments will be driven partly by spread and expenses of refinancing (although loans will also be driven by the covenants), but fixed coupon instruments will also be driven by interest rate levels.

OC Tests (etc.) and Reserve Accounts

Simulation for cashflow CDOs should include the income generated by the reference pool which accumulates as it is received in the cashflow account (itself receiving interest). At appropriate dates payments are made from the cashflow account following the waterfall structure to the tranches and other liabilities.

Incorporation of OC, IC and other tests in the simulation is straightforward in principle. At each test date – based on the reference pool as it is simulated to exist at that date – the test is applied. If the test fails, a cashflow is transferred to the reserve account and payments to the junior tranches are reduced by the transfer to the reserve. At appropriate dates the reserve account is applied to restore tranche exposures, make early redemption payments, etc.

Rating-Dependent Premium

The premium for a tranche may be rating dependent, or waterfall tests may be rating dependent, or the management rules may require substitutions to be made if ratings on individual entities fall below a certain level.

Since rating a tranche is, in principle, a judgemental rather than a formulaic exercise, this feature can only be handled approximately; access to the rating agencies' models helps although these models may only apply at initiation of the trade and may be applied infrequently once the deal has been floated. A proxy formula, or a call to the rating agencies' routines, can be programmed to operate periodically to calculate the rating of the tranche, based on the simulated pool, and adjust the tranche premiums. Such a proxy formula will typically need the forward rating of the reference names. In addition, rating-dependent management rules or waterfall tests will need a simulation of rating.

Cashflow CDO Simulation

It is clear that a period by period simulation is required, potentially involving a simulation of interest rate levels, individual reference entity ratings and spreads, and recovery rates. One approach is to tie the rating to spread and simulate the rating using a transition matrix approach, as described in Chapter 6 in Part I. This has the advantage that it builds in a jump process in spreads but ignores spread volatility where rating remains constant – additional spread volatility within rating is required and also average spreads related to a specific rating change over time. Approaches along these lines are slow to execute – a pricing simulation for a synthetic CDO may take a few seconds, but the same reference pool simulated along the above lines can take 15 minutes for the same number of simulations.

A simpler, albeit crude but quicker, alternative is to value the tranche on the assumption that the rating is unchanged throughout its life and also that a rating change is triggered at various forward dates. Separate simulations of rating changes can then be used to find weightings to apply to the above simulated values.

22.4 MODEL ERRORS AND TESTS; ALTERNATIVE MODELS

We conclude this chapter with a review of the model, the captured and hidden risks, and some alternative possible modelling approaches.

22.4.1 Captured and Hidden Risks

The model has captured, and can explicitly calculate, the following risks:

1. Interest rate risk
2. Spread risk on individual reference names
3. Recovery rate risk
4. Event risk
5. Default time correlation risk
6. Convexity risk.

The model has also identified, but does not capture

7. Spread correlation risk
8. Spread volatility risk.

Additionally there is model risk related to

9. Choice of correlation
10. Choice of Copula
11. The general form of the model

Items 1 to 5 have been explicitly addressed above, while 6 to 8 have been implicitly addressed. We looked at correlation in detail in Chapter 21, and we shall look at alternative Copulas in Chapter 23. In the remainder of this section we briefly look at spread models – revisiting a point touched on in Part II – and discuss the setting of reserves.

22.4.2 Spread Models

Suppose we wish to price a first-to-default product on a basket of two names. We might initially assume that a stochastic hazard rate process, as described in Part II section 9.4 would be appropriate. We can use this model to price the individual CDSs and a F2D on the two names. If the two names are investment grade and the F2D is a 5-year deal, we surprisingly find that the fair value premium for the F2D is the sum of the two CDS premiums – no matter what the spread correlation (assuming a spread volatility of around 100%). Only if we look at a long-dated F2D on two initially wide spread names do we find any significant impact from the spread volatility and correlation. Figure 22.5 shows the results for such a model on a 25-year basket of 500 bp and 800 bp names. We conclude that a diffusion model for spreads is incapable of explaining the basket prices we observe in the market place.

Indeed spreads do not appear to follow a diffusion process – Figure 22.6 shows three telecom spreads and a relatively common 'explosive' price move (Andreasen, 2001).

Suppose, instead, spreads are driven by several external factors – world recession or an industry funding crisis (Telecoms, Asia) as well as individual name factors. Suppose also that these external factors can be modelled by a jump process. The factor can jump up a large amount at a low rate, or down by smaller amounts but at a higher frequency, so the expected

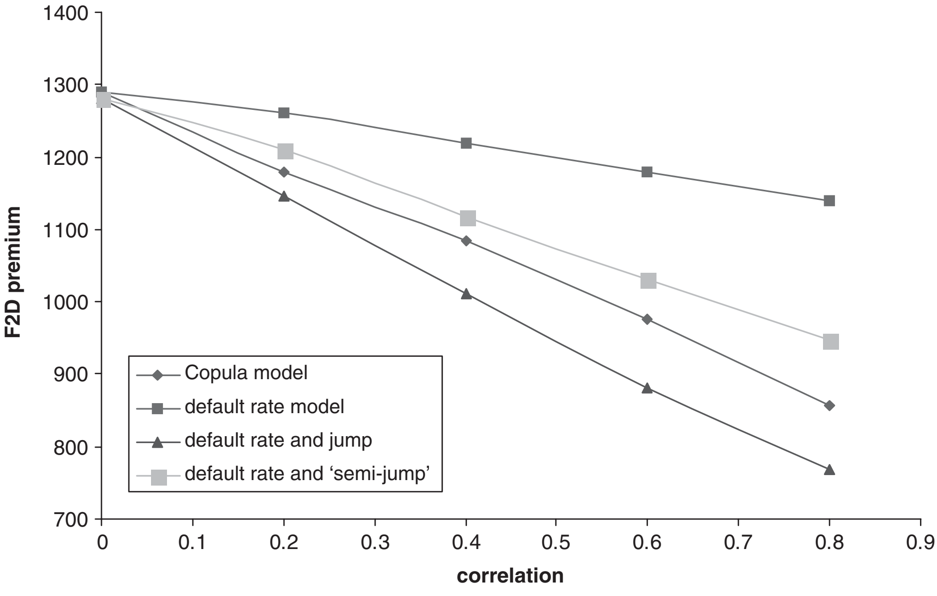

Figure 22.5 Model comparison 25-year F2D premium: Copula, spread diffusion and spread jump models.

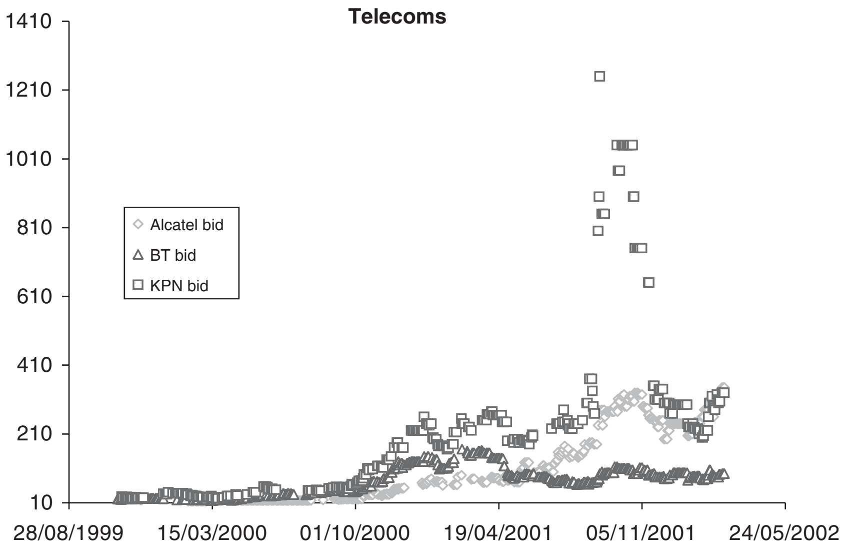

Figure 22.6 Telecom CDS bid spreads.

hazard rate change is zero. The factor is applied to all default rates for names for whom that factor is relevant. The implied jump process in default rates gives rise to a 'default time' correlation. Figure 22.6 compares the jump process model with Copula prices and prices from the spread diffusion model.

The Copula model prices lie between two sets of 'jump model' prices (one using an arbitrarily chosen large jump, the other with half that jump), and these three price curves are significantly steeper than that derived from the default rate (diffusion) model. Clearly a jump process model is capable of producing similar results to the Copula model. One could argue that the jumps are independently observable, therefore some real world estimates are available. However, calibration of the model is complex – we have to work back from CDS data using the stochastic (jump) process model to find the initial hazard rate. The end result is something that is indistinguishable from the Copula model – both curves have strong downward slopes, the absolute levels may differ but that would just mean that the market-implied correlations on the two models are different. There does not appear to be any significant advantage in this route, against which there are substantial calculation disadvantages.

22.4.3 Reserves

For a general CDO tranche marking to model requires the following.

1. Estimation of the marking correlation: this in turn will depend on
   (a) the underlying reference portfolio;
   (b) the tranche attachment points; and
   (c) the maturity of the deal.
2. Single name data and assumptions.

For portfolio products these reserves therefore include the following:

1. A bid–offer reserve if model values are based off mid.
2. A recovery (mean and standard deviation) reserve.
3. The underlying portfolio may differ from the portfolio used to estimate implied correlation – a name correlation reserve.
4. The portfolio tranching may differ from the portfolio tranching on the benchmark portfolio – a tranching correlation reserve.
5. The portfolio maturity may differ from benchmark maturity – a maturity correlation reserve.
6. A model reserve.

Even on liquid products – such as standard CDO tranches – this may mean marking to mid and having a **bid–offer reserve** to effectively mark to the worse of bid or offer. Some institutions take the view that marking to mid on a balance portfolio does not need such a reserve since a portfolio trade would be done closer to mid.

The **reference pool reserve** is chosen to reflect the impact on the tranche correlation of having a different reference pool from the CDO structure used to derive implied correlation. For large (60+ names) diversified (diversity score > 30) pools of investment grade names, there seems to be little variation in correlation for comparable tranches. Examination of the implied correlations on the CDX and iTraxx portfolios (and predecessor portfolios) shows differences of only a few percent (after adjusting for tranche attachment point differences) for portfolios that have totally different reference entities.

The **interpolation reserve** allows for errors that might be introduced in interpolating a correlation for an 8–11% tranche based on implied correlations on 6–9% and 9–12% tranches. Alternative interpolation routines typically give under 2% correlation errors for intermediate tranches. Equity tranche correlations are more sensitive for homogeneous portfolios, but the equity price may be driven by particular wide spread names rather than the correlation on other portfolios. The senior tranche has relatively little correlation sensitivity over the likely range of correlation, although it is typically a much larger size so the EUR sensitivity can be large. However, there is a more active market in senior tranches generally, so this tranche may be capable of being priced directly.

The deal on the books may differ from the maturity of the CDO being used to derive the implied correlation. This will give rise to a **maturity reserve**. Comparisons of the 5- and 10-year iTraxx or CDX quotes show relatively little difference. However, the situation may be quite different for shorter-dated portfolios where there are currently no market-traded instruments. The development of active shorter-dated indices and tranches would be useful for marking mature portfolios to market.

Finally, a **model reserve** may be applied. It can be argued that the Normal Copula is a standard pricing tool and that such a reserve is not necessary. But consider a CDO tranche which is spread hedged CDS only. One known model deficiency is the lack of spread correlation and volatility in the model, and the trading strategy is open to this risk.

Other reserves – such as a **convexity reserve** – may be raised. The choice and level of reserves to include depends very much on the trading and hedging strategy of the institution. A book of CDO tranches hedged with similar tranches faces far fewer risks, and needs less reserving than a directional (in correlation) book.

23
Alternative Copulas

We look at other Copulas for three main reasons:

1. Producing correlated default times based on Student's t-distribution is commonly talked about in the literature and often used in practice.
2. Having an alternative Copula is a worthwhile model test and risk-control measure.
3. Some other Copulas – in particular Clayton and the 'Archimedean' class of Copulas – give a different perspective on the Normal Copula and its application, and are also discussed frequently in the literature.

Student's t-distribution is a simple extension of the Normal case, and some useful results that require relatively little effort are contained in this chapter. Sections 23.2 to 23.4 are more specialised, but the general reader is recommended to read the 'implications' at the end of section 23.3 at least.

23.1 STUDENT'S t-DISTRIBUTION

Student's t-distribution is similar to the Normal when the 'degrees of freedom' are large. It is a simple extension of the Normal model in terms of implementation, and is widely discussed in the literature as an alternative – and an improvement – on the Normal Copula.

Random numbers from Student's t-distribution with d degrees of freedom

1. Generate N correlated random numbers, $\boldsymbol{x}$, from the Normal distribution as already described, and d independent Normal random numbers X_j.
2. Calculate an independent $s = \sum_{j=1}^{d} X_j^2$ (which is from the Chi-squared distribution with d degrees of freedom).
3. Calculate the cumulative probabilities $t_d(y_i)$ under the Student distribution[1] of $y_i = \sqrt{\frac{d}{s}} \cdot x_i$, where $i = 1, \ldots, N$.

The cumulative probabilities in item 3 give us the correlated default times

$$T_i = \frac{-\ln(t_d(y_i))}{\lambda}.$$

The code to implement the above is given in the addendum to this chapter.

Applications and Results

If the $N \times N$ correlation matrix in the Student's Copula is a one-factor correlation matrix, then we still have two parameters to fix – the correlation and the degrees of freedom. Student does

[1] Press et al. (2004)

Table 23.1 Traded US CDO pricing (June 2004) using equity factor-based correlations under Student's Copulas

Tranche (spread)	Mid premium	Premium using factor-based equity corr and df = 3	Premium using mean equity correlation and df = 7
0–3% (150 bp)	45.75% + 500 bp	45.75% ± 20 bp	45.75% + 105 bp
3–7% (40 bp)	430 bp	460 bp	490 bp
7–10% (20 bp)	160 bp	250 bp	245 bp
10–15% (10 bp)	69 bp	150 bp	140 bp
15–30% (5 bp)	17.5 bp	60 bp	40 bp

not help if we want to follow the 'implied correlation' per tranche description (we need to fix the degrees of freedom).

Does the use of Student's *t* help in the pricing of the TRAC-X portfolio if we use equity-driven correlations?

We have one further parameter to play with (the degrees of freedom), so in some sense we should be able to get a better fit than with Normal. Table 23.1 shows results with 3 and 7 degrees of freedom (and note that under the Normal Copula we would get the same result as under Student for a large number of degrees of freedom (around 20 and above)). We see that as the degrees of freedom drop the equity tranche and the senior tranche are increasingly mispriced – the equity premium dropping even more below market price and the senior premium increasing more and more above market price. Only the junior mezzanine tranche gets closer to the market price.

We conclude that Student's *t* Copula offers no practical benefits over the Normal Copula.

23.2 COPULAS IN GENERAL

Generally a Copula function is a means of generating a distribution for correlated variables (default probabilities) when all we know is the partial distribution (the single name survival probabilities). The general definition has not been necessary for an understanding and use of the Normal and Student's Copulas – indeed their application is quite different from the general approach. More detail on Copulas can be found in Nelsen (1998). Copula functions are a standard tool in actuarial science and other areas, but not common in derivatives until Li (2000) introduced the approach in the mid-1990s.

If $F(x_1, x_2, \ldots)$ is a known joint distribution with partials $F_1(x_1), F_2(c_2), \ldots$ then the Copula function (which is a cumulative probability distribution in its own right) is given by Sklar's theorem:

$$C(u_1,\ u_2, \ldots) = F[F_1 \text{ inv}(u_1),\ F_2 \text{ inv}(u_2), \ldots] \tag{23.1}$$

In the general case we do not know F but, instead, we can choose any increasing function C on the positive unit interval subject to certain constraints – including

$$\begin{aligned} C(u, \ldots,\ 0, \ldots) &= 0, \\ C(u,\ 1,\ 1, \ldots) &= u. \end{aligned} \tag{23.2}$$

Once we have found a Copula function we need a general method to generate correlated random numbers using the Copula. One procedure is as follows:

1. Generate independent uniform random numbers u, v.
2. Convert these to correlated variables u, t through the conditional distribution of v given u, i.e.

$$v = c_u^{-1}(t) \tag{23.3}$$

where $c_u(v) = \dfrac{\partial}{\partial u} C(u, v)$

For multivariate Copulas this requires the partial derivatives up to the order of the distribution.

We follow through the general process described above in the case of the Clayton Copula in section 23.3

In the case of the Normal Copula we know F – it is the multifactor correlated cumulative normal distribution. F_1 is the one-dimensional cumulative Normal distribution so, if u_1 is a randomly chosen number between 0 and 1 then F_1 inv(u_1) is between $-\infty$ and $+\infty$. If we were to write out C as a function of the uniform variables, we would have a very complex (and unhelpful) integral equation. Instead of following this route we skipped the step of generating the u's and generated the F_1 inv(u_1) directly from the uncorrelated random Normal number generator. We also do not follow the process described above to get the correlated Normal random variables, but instead use a trick based on the Cholesky decomposition of the correlation matrix. In summary, the general Copula description is unhelpful in the case of the Normal and Student's Copulas.

23.3 ARCHIMEDEAN COPULAS: CLAYTON, GUMBEL

Archimedean Copulas can be generated from a function of a single function $\varphi(x)$, which has certain properties (basically a continuously decreasing convex function mapping [0, 1] to [infinity, 0]). Given such a '**generating function**' we find that

$$C(u_1, u_2, \ldots) = \varphi^{-1}(\varphi(u_1) + \varphi(u_2) + \ldots) \tag{23.4}$$

is a Copula function. Note that a single generating function can be used to generate a Copula function for any required number of correlated variables.

Archimedean Copulas have the property that hazard rates change by a common factor when one name defaults (Schönbucher and Schubert, 2001). Unfortunately this is an unrealistic property for credit risks – a single 'shock' to the financial system does not cause a credit blow-out in all names in the same proportion. Nevertheless, the Clayton Copula in particular can be a worthwhile alternative for small portfolios, and is used later as a further model test.

Clayton Copula

The function

$$\varphi(t) = \frac{1}{\theta}(t^{-\theta} - 1)\ \theta \geq 0 \tag{23.5}$$

is a generating function, and definition (23.4) leads to the Clayton Copula

$$C(u_1, u_2, \ldots) = \max([u_1^{-\theta} + u_2^{-\theta} + .. - n + 1]^{-\frac{1}{\theta}}, 0) \tag{23.6}$$

In order to obtain the nth correlated random number u_{n-1} given the $n-1$ correlated numbers u_0 to u_{n-2}, and an independent uniform random number t, we need to calculate the inverse of the $(n+1)$th derivative. We find

$$u_{n-1} = \left(t^{-\frac{\theta}{(n-1)\theta+1}} \cdot \left(\sum_{i=0}^{n-2} u_i^{-\theta} - n + 2 \right) + n - 1 - \sum_{i=0}^{n-2} u_i^{-\theta} \right)^{-\frac{1}{\theta}} \tag{23.7}$$

Exercise 1

1. Check the values of the function φ at 0 and 1, and check the monotonic nature of the function.
2. Show that the inverse is $\varphi^{-1}(s) = (s \cdot \theta + 1)^{-\frac{1}{\theta}}$
3. Hence derive the Copula function. Also show that C is a probability distribution. [*Hint:* Take the one-dimensional case and show that the integral of C over [0, 1] is unity.]
4. Calculate the derivative of C with respect to the first variable and find the inverse of this function – show that it is consistent with (23.7). Repeat for the second, ...

The spreadsheet 'chapter 23.xls' implements the above for two names and calculates correlated default times.

Gumbel–Hougaard Copula

The generating function is

$$\varphi(t) = (-\ln t)^{\theta} \quad \theta \in [1, \infty) \tag{23.8}$$

leading to the Gumbel–Hougaard Copula

$$C(u_1, u_2, \ldots) = \exp(-[(-\ln u_1)^{\theta} + (-\ln u_1)^{\theta} + ..]^{\frac{1}{\theta}}) \tag{23.9}$$

A closed form for the inverse of the nth derivative does not exist and numerical methods have to be used.

Implications

The Clayton (and Gumbel) Copulas are both driven by a single parameter – whether there are two names in the reference portfolio or 200. Also, in the case of Clayton, the only constraint on the parameter is that it is positive. That parameter clearly is not a correlation. If the parameter to a Copula cannot be interpreted directly as a correlation (or any other quantity directly driven by the reference names), how do we derive its value? The obvious route would appear to be to imply the parameter from the prices of some traded structures.

Once we have accepted the idea that Copula parameters may have to be implied from market prices, we should ask 'What Copula would be most useful?'

If we are pricing a CDO on 100 reference names with seven tranches, there is one constraint: the sum of the values of the expected losses on all the tranches must be equal to the sum of the corresponding values for the reference name CDS. If we have a Copula with six parameters (and generally, the number-of-tranches-minus-one parameters) then we could fit the market values of all the tranches exactly with one function. The Clayton Copula has only one parameter (too few), and the Normal has 4,950 unless we impose some other restrictions (far too many). Such Copula functions, which are tractable, are not known, and the Normal Copula remains the simplest flexible approach.

23.4 CLAYTON AT $\theta = 0$ AND $\theta = \infty$

We know that (see Part III section 1.3) for zero and 100% correlated variables, the F2D premium is independent of the Copula. In fact no Copula is needed at all (in the former case independent random numbers give us independent default times, and in the latter case a single random number generates all the correlated default times). It is worth while confirming that the Clayton Copula reproduces these results in the limit as θ tends to zero and to infinity.

Consider the use of the Clayton Copula to derive the second correlated random variable, v, given two uncorrelated uniform random variables u and t. Formula (23.7) is (setting $n = 2$)

$$v = \left(t^{-\frac{\theta}{\theta+1}} \cdot u^{-\theta} + 1 - u^{-\theta}\right)^{-\frac{1}{\theta}} = \left(1 + \left(t^{-\frac{\theta}{\theta+1}} - 1\right) \cdot u^{-\theta}\right)^{-\frac{1}{\theta}} \tag{23.10}$$

Using $x^y = \exp(y \ln(x))$ and the fact that $0 < u$ and $t < 1$, the right-hand side can be shown to tend to $t - t \cdot \theta \cdot \ln(t) \cdot (\ln(u) + 1)$ as θ tends to zero (see, for example, Hardy (1999)). We can see that v tends to t – so v and u are independent in the limit as θ tends to zero.

As θ tends to infinity, note that the term $1 + (t^{-\frac{\theta}{\theta+1}} - 1) \cdot u^{-\theta}$ is dominated by $u^{-\theta}$ so v tends to $(u^{-\theta})^{-\frac{1}{\theta}} = u$ and the second random variable v is then identical to the first, as required.

23.5 MODEL RISK

The key model assumption is that the default time correlation drives the pricing of N2D baskets and CDO tranches. We have identified one major drawback in the model – the lack of spread correlation and spread volatility. Another problem is the arbitrary choice of correlation (unless we use 'implied'), and the arbitrary choice of Copula. One test[2] we can perform is to compare pricing under two Copulas. Alternatively, we could match market prices using the implied parameters and then compare hedge ratios under two Copulas. In this section we see how prices derived using the Normal Copula compare with those derived from the Clayton Copula.

The Normal correlation parameter varies from 0 to 1, whereas the Clayton parameter varies from 0 to ∞. In order to plot the results on the same chart, we map the Clayton Copula parameter θ to the range [0, 1] by 1 – $\exp(-\alpha \cdot \theta)$ where we choose α so that the F2D prices agree at $\theta = 0$ and $\rho = 0.3 = 1 - \exp(-\alpha \cdot \theta)$. Figure 23.1 shows that the pattern of prices is very similar, except at high 'correlation'. Note that, for a portfolio of two names, Clayton is capable of producing the same range of values as the Normal Copula, with the appropriate choice of parameter.

Of course, if we have more than two names in the portfolio, then the Normal model – with a non-degenerate correlation matrix – will be capable of producing a wider range of values.

We shall see in Part IV that the Clayton Copula produces some larger differences in values for counterparty CDSs compared with Normal.

[2] Another route would be to use the multinomial BET model outlined in section 2.2.5.

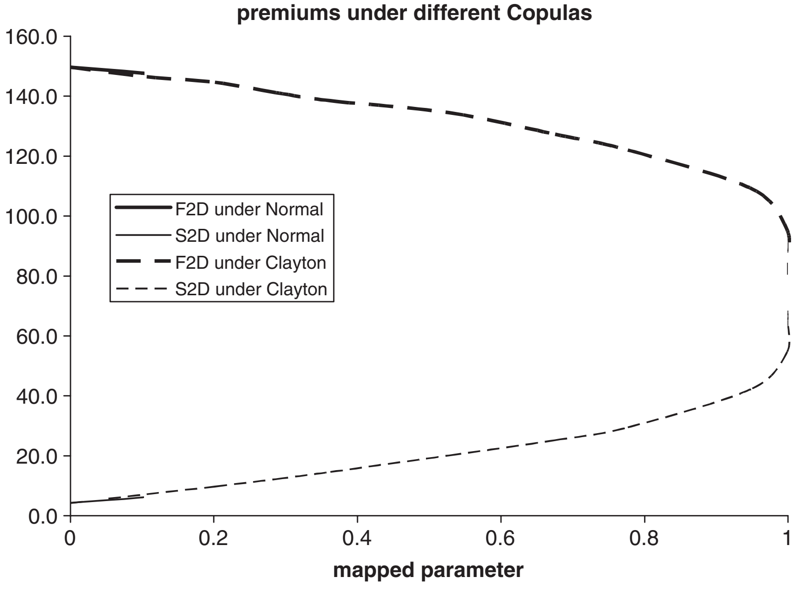

Figure 23.1 F2D and S2D prices under Normal and Clayton Copulas.

ADDENDUM

Code for the cumulative Student's distribution. (Copyright © 1987–1992 Numerical Recipes Software)

We need the cumulative function

$$t_d(y) = 1 - I_{\frac{d}{d+y^2}}\left(\frac{d}{2}, \frac{1}{2}\right)$$

where $I_x(a, b)$ is the incomplete beta function, given by

$$I_x(a, b) = \frac{1}{B(a, b)} \int_0^x t^{a-1}(1-t)^{b-1} dt,$$

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \text{ is the beta function,}$$

and $\Gamma(z) = \int_0^\infty t^{z-1} \exp(-t) dt$ is the Gamma function.

```
double CMaths::CumStudent(const double y, const double df)
{
    if (y>0) return 1.0-0.5*Betai(df/2.0,0.5,df/(df+y*y));
    else return 0.5*Betai(df/2.0,0.5,df/(df+y*y));
}
double CMaths::Betai(const double a, const double b, const
double x)
```

The incomplete beta function $I_x(a, b)$

```
{
    double bt;
    if (x<0.0 || x>1.0) return(0);
    if (x==0.0 || x==1.0) bt=0.0;
    else bt = exp(gammln(a+b)-gammln(a)-gammln
(b)+a*log(x)+b*log(1.0-x));
    if (x < (a+1.0)/(a+b+2.0)) return bt*betacf(a,b,x)/a;
    else return 1.0-bt*betacf(b,a,1.0-x)/b;
}
double betacf(const double a, const double b, const double x)
```

continued fraction evaluation – used to calculate the incomplete beta function rapidly

```
{
    double FPMIN = 1.0e-30;
    double aa, c=1.0, del, h, qab=a+b, qam=a-1.0,
qap=a+1.0, d=1.0-qab*x/qap;
    if (fabs(d) < FPMIN) d = FPMIN;
    d = 1.0/d;
    h=d;
    for (int m = 1; m<100; m++)
    {
        int m2 = 2*m;
        aa = m*(b-m)*x/((qam+m2)*(a+m2));
        d = 1.0+aa*d;
        if (fabs(d) < FPMIN) d = FPMIN;
        c=1.0+aa/c;
        if (fabs(c) < FPMIN) c = FPMIN;
        d=1.0/d;
        h *= d*c;
        aa = -(a+m)*(qab+m)*x/((a+m2)*(qap+m2));
        d = 1.0+aa*d;
        if (fabs(d) < FPMIN) d = FPMIN;
        c=1.0+aa/c;
        if (fabs(c) < FPMIN) c = FPMIN;
        d = 1.0/d;
        del = d*c;
        h *= del;
        if (fabs(del-1.0) < 3.0e-7) break;
    }
    return h;
}
```

24
Correlation Portfolio Management

The 'correlation book' contains risks arising from portfolio products including N2D baskets, CDO tranches, and portfolio volatility products.

24.1 STATIC AND DYNAMIC HEDGES

Statically (Default Event) Hedged Book

CDO deals may be default event hedged on a static basis, as described in Chapter 23. We saw that a statically hedged deal is not risk free – there is residual delta risk which will give rise to mark-to-market P&L changes. This residual may be minimised through product design, otherwise it can reach high values (equivalent to a naked spread position in 50% of the reference pool notional on a long-life high-yield portfolio).

If the 'static hedge' is incorporated with other deals in the correlation product book then the residual risk will be automatically and actively delta hedged. For this reason statically hedged deals need to be segregated from the rest of the correlation book to avoid the unwanted P&L arising from rehedging. The owner of the position remains exposed to the spread risk.

Dynamic Delta Hedged Book

We examine the delta hedging of several types of trades, and the consequences of this approach.

Example 1 Long F2D Protection, Long Equity Protection A position long of protection in a F2D basket is delta hedged by selling single name protection on the individual names in the basket – say around 50–60% notional for each name in the basket, depending on the correlation and depending on spread. This brings a strong (short) correlation risk. In addition – as we saw in section 23.5 – it brings a large exposure to spread correlation and spread volatility through future rehedging activity. Similar results apply to an equity tranche – hedge ratios are typically smaller but still substantial. A strategy of forcing the correlation book to have zero single name exposure requires day-to-day rehedging with the single name book resulting in substantial bid–offer payments unless interbook trades take place at mid-market prices. In the former case the single name book profits from correlation book activity; in the second case the single name book manages emerging single name risks at a potential loss – so some form of P&L sharing or fee payment becomes necessary.

Default and Spread Jump Risk

Consider again the Telecom basket example of 23.1. Long F2D protection positions will suffer substantial rehedging loss as spreads widen out in a highly correlated way. Default of a name in the basket may come as a welcome relief – giving risk to a profit on the mismatch in F2D notional and hedge notional – but cannot be relied on.

Example 2 Short Senior TDB Protection The same principles apply – the position is short correlation and short single name protection.

Senior and equity tranches are also exposed to **extreme event risk** – many defaults on a reference portfolio have a very substantial impact on the value of these tranches. Also, both tranches with their delta hedges are exposed to spread jump risk and delta **convexity risk**, and on a sudden large move in spreads the 'delta' hedge ration is incorrect.

Example 3 Mezzanine TDBs Mezzanine tranches may be long or short correlation depending on the placement and size of the tranche, the underlyings and the level of correlation. There are some intermediate (mezzanine) TDBs that are relatively insensitive to correlation. This is one way of controlling the correlation risk in the books – deal in those mezzanine tranches that are relatively insensitive to correlation. Clearly this is not appropriate to all mezzanine tranches; however, this approach does not eliminate the spread correlation risk arising from the delta hedges.

Example 4 Put option on a CDO Tranche The delta exposure to the reference pool names is subject to further risk – the poorly defined implied volatility for spreads while the product is relatively illiquid. Both correlation risks, and the volatility risk, are present but not resolved by the delta hedging approach.

24.2 CORRELATION BOOK MANAGEMENT

The above examples emphasise that delta hedging only leaves the following risks unhedged:

- time correlation
- spread correlation
- spread volatility
- delta convexity
- extreme event risk.

in addition to other less serious risks (see section 23.6). Hedging correlation in an integrated correlation book is still a much better strategy to protect the portfolio. This will leave only residual single name delta risks, will largely eliminate not only the spread correlation and volatility risk (except for option products) but also the delta convexity risk, and can also reduce extreme event risk. If the correlation book is allowed large limits on single names but tight limits on delta risk within certain buckets (for example, A autos) then the book can run name risk but spread hedge within buckets and avoid delta rehedging problems. How do we hedge correlation and minimise delta and other risks? Consider the following examples.

Example 1 Correlation Hedged Equity Tranche and F2D Positions A long position in an equity tranche is short correlation – its capital value falls as implied correlation rises. Conversely, a short position in a F2D basket is long correlation. We can hedge the correlation risk on the long equity tranche by selling F2D baskets – *if the correlations move together* – on some underlying names (or on some different names). This will leave long delta risks arising from the equity tranche, and short delta risks on the F2D positions. If these risks fall in the same (single name risk) buckets, then they will net, leaving only a residual delta risk.

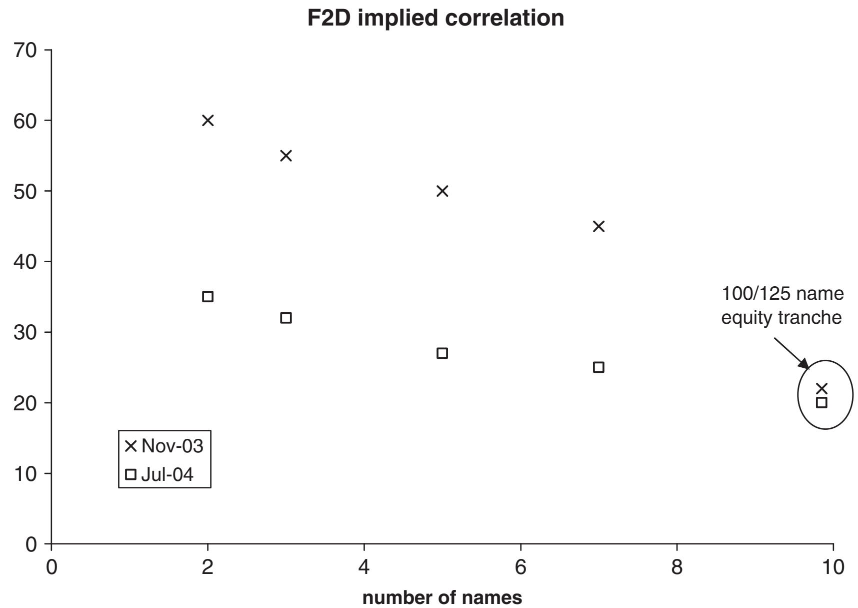

Figure 24.1 Implied correlation on F2D and equity CDO tranches.

Example 2 Correlation Hedged Long Equity and Long Senior Tranches A long position in a senior tranche is long correlation – its capital value increases as correlation increases. We could hedge the long equity tranche with long senior tranches and – *if the correlations move together* – the correlation risk will be greatly reduced. However, both positions would give risk to positive delta exposures to the underlying names and exacerbate the delta hedging problem.

There are two further risks in the types of hedges suggested above: basis difference between basket and CDO correlations, and the basis between different tranche correlations. Figure 24.1 illustrates how the implied correlation varied between baskets (of intuitively uncorrelated names) and CDO equity tranches as at December 2003 and July 2004.

We can see that the implied correlation depended quite strongly on the size of the underlying portfolio – F2D baskets on 2–10 names typically having implied correlation in the 40–60% range, whereas 100 name CDOs typically have FLP implied in the 20–30% range. In addition the relationship changed markedly over the time period with quite large declines in F2D correlations.

Figure 24.2 shows the correlation 'smile' as of June 2004. A hedge across materially different tranches can be compared to a curve trade – if correlations shift in parallel there is a hedge, but in reality there is the risk that correlations change differently senior and junior tranches.

Hedging of correlation risk by tranche buckets is much less risky than hedging with very different types of tranches or products. In practice this means hedging junior CDO tranches with (opposite positions) other junior tranches, senior tranches with other senior tranches, and F2D products with other F2D products. Within each correlation bucket, net correlation risk is

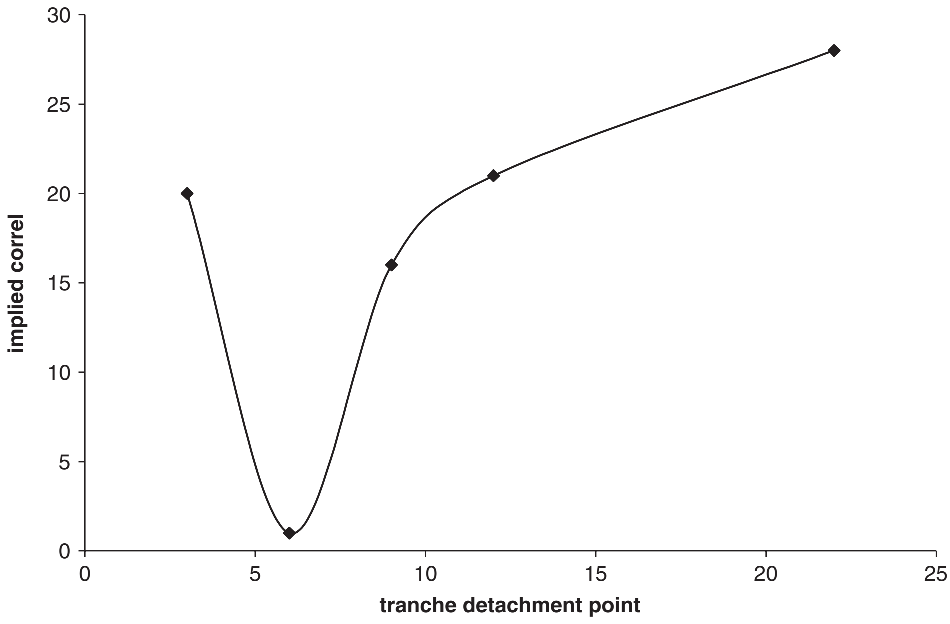

Figure 24.2 The correlation 'smile'.

calculated. The net delta risk – not by name but by delta bucket (rating and industry, etc.) – is calculated across all correlation buckets and needs to be hedged.

The growth of index-based CDOs and individually traded TDB tranches (iTraxx and CDX currently) is of major benefit to the correlation risk hedger. Structuring desks can arrange tailored deals for particular clients and hedge correlation risk using appropriate tranches of these index CDOs. The underlying portfolios may differ but large (50+ names) diversified (diversity score 30+) investment grade portfolios tend to be priced very similarly (see sections 21.4, 21.6 and 22.2). Tranche differences can also be handled (section 22.2).

There are additional problems with equity tranches related to the granularity of the particular reference portfolio. The equity piece is driven by high spread names (section 21.3) and may begin to resemble a F2D basket on these names. In these circumstances an F2D may be a much better hedge than an index CDO based on similar spread names. A special case is where a single name has a very high spread. This will be a key driver of the equity piece: the tranche with a delta hedge on that name then becomes a much better candidate for correlation hedging with index products.

In addition, liquidity in the F2D baskets is now at a level where the active hedging of correlation on small baskets is a realistic possibility.

24.3 CREDITVaR AND COUNTERPARTYVaR

Portfolio VaR

We conclude this section with some comments on portfolio VaR related to credit correlation and portfolio credit derivatives.

An objective may be to estimate the worst change in portfolio value with a certain level of confidence ('Total VaR'). Deal values should be estimated by simulation changes in the pricing parameter and repricing. The TM approach to simulate correlated spreads and defaults can be used to obtain the resulting portfolio levels at the VaR calculation date. In order to value CDOs we also need to simulate changes in the implied correlations (probably based on some historical analysis of variability, and changes in the shape of, the implied correlation curve[1]). Each CDO (tranche) can then be revalued using the pricing parameters at the VaR calculation date.

Alternative approaches to the calculation exist.

1. We may use a single correlation matrix (typically factor based equity correlations or a one-factor matrix), but this fails to correctly mark the deal to market initially and also therefore incorrectly estimates changes in value.
2. We may use a historical TM, but this fails to mark individual risks (bonds, CDS) correctly to market so fails to estimate change in value correctly. Furthermore, as the implied volatility of spreads in the historical TM fails to capture the way volatility is priced in the market place, it will incorrectly assess the risk associated with that volatility.

Suppose we have performed a factor analysis of spreads and identified (say) 20 factors. We could simulate the development of these factors to the VaR calculation date, which then gives us the simulated spreads for each name in a way that is consistent with the factor-based correlation analysis.

An alternative approach would be to replace the actual portfolio by (appropriately weighted) representative entities where there is a one-to-one mapping between reference entity and representative entity, and simulate these. All deals on actual reference entities are replaced by deals on the representative entity.

Either approach gives a manageable correlation matrix and portfolio calculation.

[1] Typically a single implied correlation curve would be simulated for (say) 5-year life investment grade CDOs. The correlation for the tranche would be obtained from the curve according to the tranche size and placement. A different maturity correlation curve would typically be deterministically driven by the 5-year curve.

Part IV

Default Swaps Including Counterparty Risk

25 ‘Single Name’ CDS

We saw in section 20.1 that, in the case of an F2D where the reference entities have uncorrelated spreads (hazard rates), the valuation formulas are relatively simple. We apply the similar arguments to a CDS with uncorrelated counterparty risk in section 25.1, and with 100% correlation in section 25.2.

We also saw in Part II section 9.8 that, even where name and counterparty hazard rates are correlated, the impact of high correlation and high spread volatility is minimal (except at high spreads and very long-life CDSs) if the hazard rate process is a diffusion without jumps. However, the default time model in Chapter 20 used to price baskets and CDOs, carries the implication that hazard rates follow a jump process. We also saw in Part III (section 22.4) that portfolio products exhibit significant default time correlation, therefore we cannot reasonably ignore the name and counterparty risk without more detailed investigation. Fortunately we can use the correlated default time model to value CDSs including counterparty risk (we shall illustrate some results in section 25.3).

We revisit the question of alternative Copulas in section 25.4. We look at the effects of collateralisation versus non-collateralised trades in section 25.5, and at CDSs on a counterparty to a CDS trade (CCDSs) in 25.6.

25.1 NON-CORRELATED COUNTERPARTY

If, under a default time correlation model, two names are uncorrelated, then the expected default time of one name conditional on the other having defaulted is the same as the unconditional expectation. In other words, there is no expected jump in the hazard rate on one name when an uncorrelated name defaults (see section 20.1). Let us assume that we buy protection on a reference name from an uncorrelated counterparty. In the event that the counterparty defaults there is no expected change in the spread on the reference name. The expected replacement cost of the CDS (now that the counterparty has defaulted) is the same as it was previously, hence deals on reference names with uncorrelated counterparties have no jump risk on counterparty default. If the deal is collateralised, then this also eliminates the risk contingent on changes in the reference entity spread.

If we assume that the (pseudo-)hazard rates on the name and counterparty are constant, and that the counterparty defaults, then we can replace the CDS on the reference name with another (uncorrelated but otherwise similar) counterparty, at the same spread as before. As long as an uncorrelated counterparty is always available, we know (on our assumption) that we can always get protection at the same price as before. The uncorrelated counterparty is therefore effectively the same as a risk-free counterparty, so the premium we pay to an uncorrelated counterparty must be the same as the premium paid to a risk-free counterparty.

This result is not exactly true (but it is generally a good approximation) in the case where the hazard rates have a term-structure. If the (pseudo-)hazard rate of the reference name is upward sloping, then default of the counterparty will be at time when the replacement cost is above the previous deal cost. This additional cost can be calculated and weighted by the

Table 25.1 CDS premiums for a reference entity with risk-free and uncorrelated counterparties (40% recovery)

Reference entity hazard rate	Counterparty hazard rate (flat)	CDS premium – risky counterparty	CDS premium – risk-free counterparty
Flat 2%	0.5%	120.0	120.0
2% now rising to 4% in 5 years	0.5%	178.3	178.5
2% now rising to 4% in 5 years	2%	177.5	178.5
Flat 10%	4%	600	600
20% now falling to 10% in 5 years	4%	947.6	937.8

probability of counterparty default at each date to get an estimate of the impact of dealing with an uncorrelated counterparty in this case. Implementation of formula (9.22) effectively achieves this. The formula for the fair value premium from an uncorrelated counterparty (name 2) on a reference entity (name 1) is (where we ignore the writer's default risk, and D represents the discount factor)

$$\frac{\int_0^T (1-R_1)\cdot\lambda_1(t)\cdot\exp\left(-\int_o^t \lambda_1(s)ds\right)\cdot\exp\left(-\int_o^t \lambda_2(s)ds\right)\cdot D(t)dt}{\int_0^T \exp\left(-\int_o^t \lambda_1(s)ds\right)\cdot\exp\left(-\int_o^t \lambda_2(s)ds\right)\cdot D(t)dt} \tag{25.1}$$

If the name hazard rate is constant we see that the integral in the numerator is just a constant times that in the denominator, so the integrals cancel and we get exactly the same premium as in the risk-free counterparty case (as expected), but for general hazard rate curves this is no longer true. Table 25.1 shows the correct CDS premium when dealing with a correlated counterparty and when the reference entity has a non-constant hazard rate curve. (Calculations are implemented in the MathCad sheet 'chapter 25.mcd'.)

25.2 100% CORRELATION

Recall that if the default time correlation for a correlated default time model is 100%, then, on every simulation, the default times are in the inverse order of the hazard rates for the names in the product. The highest hazard rate name defaults first, and the lowest hazard rate name defaults last.

Consider a CDS on a reference entity written by a 100% correlated counterparty (and assume that the buyer is risk free). We immediately see the following:

1. If the counterparty has a higher hazard rate than the reference entity, then the counterparty will always default before the reference entity and the CDS will be worthless.
2. If the counterparty has a lower hazard rate than the reference entity, then the reference entity will always default before the counterparty and the CDS payment will always be made. A CDS written by a 100% correlated but low-risk counterparty is as valuable as a CDS written by a risk-free counterparty! (See section 25.4 and Figures 25.1 and 25.2.)

We saw in section 20.1.5 that the F2D premium is not the highest CDS spread but the spread on the name with the highest hazard rate. Uncertainty over recovery means that we are not

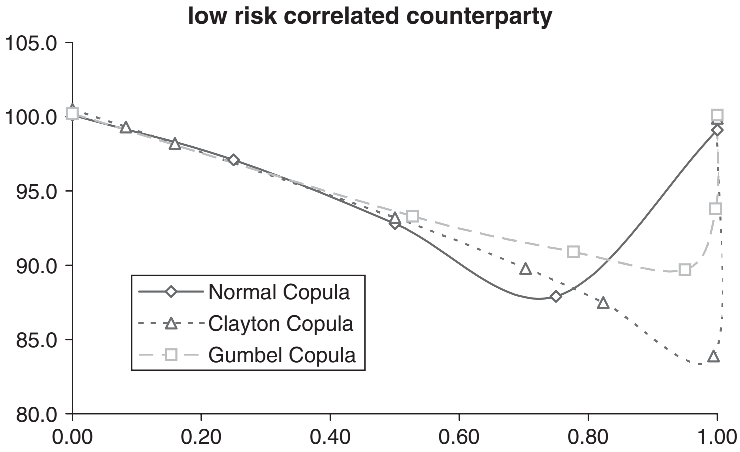

Figure 25.1 CDS protection on a high-risk name from a low-risk counterparty.

sure which name this is. Likewise, we cannot be sure whether the counterparty is higher risk or lower risk than the reference entity, so we cannot be sure whether the CDS is worthless or worth as much as from a risk-free counterparty!

As in section 20.1 we can model recovery rates on the two names allowing for the uncertainty in the recovery level. Assume that the expected recovery is 40% and the standard deviation (uncertainty) of recovery is 20%. We can model this (see the Excel spreadsheet 'chapter 9.xls') by assuming that recovery can take the value 0 with probability 0.11, 0.4 with probability 0.8, and 0.9 with probability 0.09, giving a 40% mean and 20% standard deviation. We can take all recovery pairs, calibrate each CDS (reference entity CDS = 100 bp, counterparty CDS = 50 bp, say, from a risk-free counterparty) to get the hazard rate, and then decide whether the CDS should be priced at zero or the risk-free level.

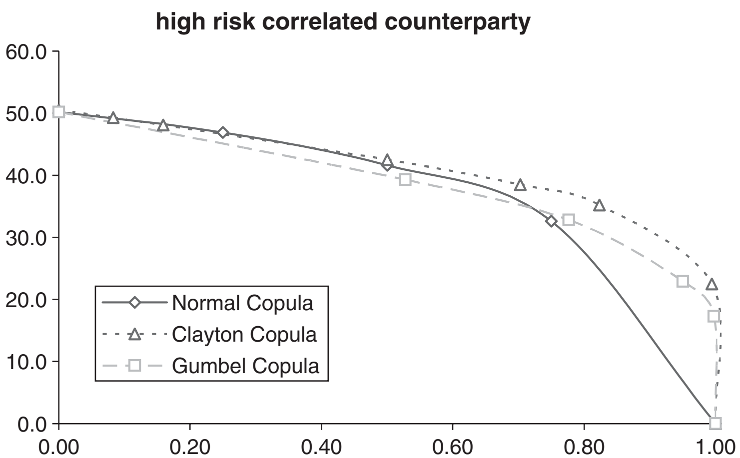

Figure 25.2 CDS protection on a low-risk name from a high-risk counterparty.

Exercise 1 Show that the expected level of the CDS is 91.9 bp.

Hint: How many possible recovery pairs are there? Enumerate these and calculate the probability associated with each pair. Calibrate the risk-free CDS and decide whether the risky CDS is worth 100 bp or 0 bp – then weight the results.

25.3 CORRELATED COUNTERPARTY: PRICING AND HEDGING

The pricing model for a CDS with correlated counterparty risk is very similar to the F2D model. We generate default times for all the parties to the deal in exactly the same way. Given the default times, a capital payment is made if the reference entity defaults before the maturity of the trade and before the writer and buyer of the CDS. Premiums are paid until the first default or the maturity of the trade (if earlier). We generally assume that trade counterparties have zero recovery.

An implementation of the simulation model for counterparty risk is contained in the file 'chapter 22 N2D CDO CLN price and hedge.mcd'.

Table 25.2 shows some results for a low-risk reference entity (100 bp CDS spread) and counterparty (50 bp CDS; we also assume the writer is risk free). Results are given at a range of correlation levels; we assume 40% recovery for the reference entity and 0% recovery for the counterparty. We also show the hedge ratios to each name: a long CDS position with the risky counterparty is hedged by selling very nearly 100% of the CDS on the reference name and also by buying a small amount of protection on the counterparty. Results are based on 1m simulations, showing a premium standard deviation of 0.3 bp.

Table 25.3 shows the results a high-risk reference entity (500 bp) and counterparty (200 bp). We assume 20% recovery for the reference entity, and 0% recovery for the counterparty. Results are based on 1m simulations, showing a premium standard deviation of 0.9 bp.

Table 25.2 CDS premiums with correlated risky counterparty (low-risk entities)

Correlation	CDS	Hedge ratio ref (%)	Hedge ratio counterparty (%)
0	100.1	99	0
0.2	98.7	98	2
0.4	96.9	97	5
0.6	94.6	98	10
0.8	92.9	100	17
1	100.1	100	0

Table 25.3 CDS premiums with correlated risky counterparty (high-risk entities)

Correlation	CDS	Hedge ratio ref (%)	Hedge ratio counterparty (%)
0	499.6	95	1
0.2	487.0	95	4
0.4	474.2	95	10
0.6	463.2	97	17
0.8	461.0	102	22
1	499.4	100	0

Note that the CDS premium cost plus the cost of protection on the counterparty is approximately offset by the CDS income on the protection sold. The hedge is dynamic – hedge ratio on the reference entity tending ultimately to 100% and that on the counterparty tending to 0.

25.4 CHOICE OF COPULA

The Clayton Copula is a less unnatural choice where there are only two reference names, such as is the case for counterparty risk where the writer's default risk is ignored. We can value a CDS on a risky reference entity, from a lower risk counterparty, and vice versa, and let the 'correlation' vary from 0 to 1 under the Normal Copula, the Clayton Copula and the Gumbel Copula. We find rather larger differences than in the case of F2D and S2D products. Figures 25.1 and 25.2 show the impact of changing correlation and Copula under the two cases, where:

(1) the reference entity hazard rate is greater than the counterparty hazard rate;
(2) the reference entity hazard rate is less than the counterparty hazard rate.

Note that all three Copulas agree at the 0 and 100% correlation limits. The Normal Copula approaches the 100% correlation limit more rapidly than the other two – from about the 75% area there is a significant divergence in trend and values emerging.

Since recovery is unknown, the actual values will be a blend of the above two results as described earlier.

25.5 COLLATERALISED DEALS AND CDS BOOK MANAGEMENT

Typically risky counterparties will be subject to a collateralisation and netting agreement. This will remove mark-to-market exposures as spread levels change, but does nothing to eliminate the theoretical jump risk.

Pricing

Table 25.4 compares the premium for an uncollateralised deal on a 600 bp reference entity from a 160 bp counterparty, with those for a collateralised trade. The results are displayed in Table 25.4 and Figure 25.3. Calculations have been done assuming that a collateralised position is equivalent to a rolling two-week uncollateralised deal.

It makes sense that the premium for a collateralised deal should be closer to the premium from a risk-free counterparty – but this is not the case at high correlations. Why not? Embedded in the Normal Copula model is a correlated spread model: spreads can widen out together, can widen a long way, or can narrow together. The latter must occur more frequently since the narrowing is limited and the expected forward spread is fixed. If spreads widen, collateral is received, and if the counterparty defaults there is a loss on the position because the reference entity spread jumps – but in this case the jump in market value of the CDS is from a wide spread and is limited by the recovery rate. On the other hand, if spreads have narrowed, capital is paid to the counterparty, and if the counterparty then defaults there is a large loss on the CDS position as the reference entity spread jumps from a narrow value.

It should be borne in mind that market prices for CDS contracts relate to deals between professional counterparties – i.e. low-risk names with a collateral agreement in place. Different

Table 25.4 CDS premiums from a risky counterparty: collateralised and uncollateralised deals

Correlation	Uncollateralised deal	Collateralised deal
0	600	600
0.1	594	599.5
0.2	584	598.5
0.3	574	596.5
0.4	563	593
0.5	553	587
0.6	543	579
0.7	537	569
0.8	539	556
0.85	544	546
0.9	556	540
0.95	578	546
0.99	600	600

industry F2D swaps tend to have implied correlations below the 40% level – so the market CDSs for, say, an auto reference entity written by a bank will be indistinguishable from the risk-free CDS premium. Where there is strong correlation – banks writing CDSs on banks, for example – this will not be the case even for low risk counterparties.

Book Management

There are two aspects to the risk assessment of trades on the books taking counterparty risk into account. The first is pricing and valuation of the deals. In applying the methodology

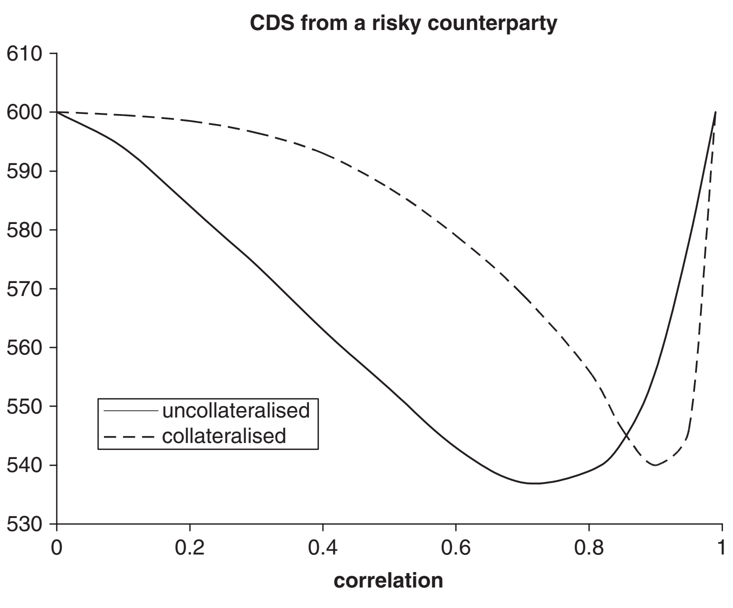

Figure 25.3 CDS premiums from a risky counterparty: collateralised and uncollateralised deals.

described above, counterparty jump risk is explicitly taken into account. Valuation is forward looking and reflects the expected value of the deal to the books, but does nothing to report counterparty risk.

The second aspect is monitoring (and potentially hedging) the counterparty exposure. The valuations above report the counterparty exposure synthetically via the hedge ratio. If the deal is off-market, then this value will be large – assuming we are long protection and the reference entity spread has widened out since purchase. For collateralised counterparties the deal hedge ratio will overstate the counterparty exposure since collateral will have been received. In these cases counterparty exposure needs to be calculated using a notional trade where the current market premium is being paid (equivalently reducing the hedge ratio by the collateral received). The counterparty exposure can then be obtained by adding the individual deal (counterparty) hedge ratios referencing that name. This exposure can either be charged for internally (a counterparty reserve) or protection can be obtained.

26
Counterparty CDSs

26.1 PRICING

In Part II we looked at variable exposure CDSs – related to interest rate swap or foreign exchange exposures. We now wish to consider default protection on the counterparty to a trade, from another counterparty. This might include protection on the exposure under an interest rate swap with a counterparty C1, or protection on the mark-to-market exposure arising on a CDS on a reference name N written by a counterparty C1, where we wish to obtain protection from counterparty C2. In the former case there is (assumed to be) correlation between the two counterparties only. In the latter case there are three correlated entities (see Figure 26.1).

The former case has been covered theoretically in section 25.3. The only material difference is the variable exposure under the default swap and the counterparties to this trade are likely to be highly correlated – typically banks or wider financial institutions such as insurance companies and hedge funds. The latter case is a three name portfolio. If the trade arises because C1 is 'risky' but C2 is a high-quality counterparty, then we may ignore the C2 risk and the deal becomes a variable exposure CDS with C1 counterparty risk.

In the general case, if we think of correlated default times generated under the Normal model then:

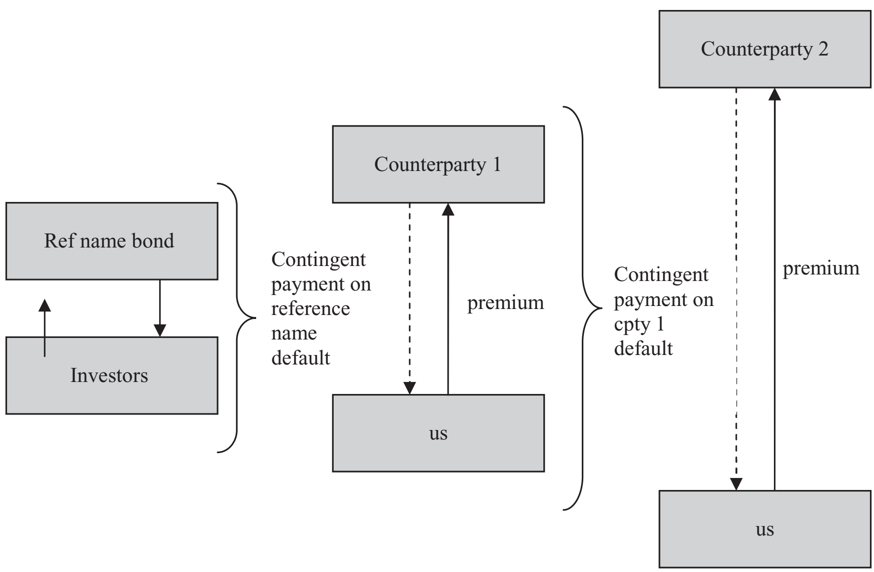

Figure 26.1 Counterparty Credit Default Swap.

1. We may wish to use a two-factor correlation matrix to explicitly take account of the high correlation between C1 and C2, and the much lower correlation with the reference name N.
2. If we generate default times T_1, T_2 and T_N, then a payoff under the contract arises if
   (a) $T_1 < T_2$ (the C1 counterpart defaults before the C2 counterparty),
   (b) $T_1 < T_N$ (the C1 counterpart defaults before the reference name), and
   (c) T_1 is less than the maturity of the underlying trade.

The payoff in this circumstance is the mark-to-market value of the CDS on the reference name. Valuation is then similar to other portfolio product valuation.

For item 2(a) to be the case as often as possible, we wish to choose a counterparty C2 whose hazard rate is considerably lower than that of counterparty C1. This is the case if its borrowing spread (or CDS premium) is much lower (intuitive), and if its marketable assets are less (non-intuitive). The latter means that the recovery rate is likely to be lower; hence the hazard rate will be lower.

26.2 COUNTERPARTY CDS (CCDS) BOOK MANAGEMENT

The bank's own counterparty exposures may be the first counterparty risks to be put in a separate book. Generally these are simple variable exposures with little or no correlation between the underlying variable and the counterparty.

Where the underlying contracts are not collateralised, there is substantially greater exposure to the counterparty per trade, being the same as the exposure on a collateralised deal plus the in-the money value of the trade. As we saw in Part I, Chapter 5 even collateralised deals still carry counterparty risk for two reasons.

1. If the counterparty fails, it takes time to realise that collateral is no longer being posted on the deal, to verify why, and to take steps to close down the deal and declare the counterparty in breach (and set up a replacement deal if necessary). This time period will depend on the institution but a period of around 10 working days seems to be a typical view of how long this process takes. During this period the market is moving. If (not very conservatively) we assume that the variable driving the value of the underlying contract moves on a random walk, then we anticipate $\sqrt{10/250} = 20\%$ of the annual volatility, which is not insignificant.
2. If the counterparty and the underlying variable are correlated, then default of the counterparty may cause a jump in the related variable and a sudden large movement in the exposure. For example, if the underlying contract is itself a CDS then, according to the correlated default time model, a default of the counterparty causes a jump in the hazard rate on the correlated reference entity.

Collateralised deals (and uncollateralised deals) are subject to possibly significant jump risk. Thus the CCDS book shares risks of the correlation book (correlated name jump risk) together with the vanilla CDS book (significant name exposure).

The second step is to seek protection on the net exposure to these counterparties. This may be crudely based on the maximum (estimated) exposure over the life of the trades, or may be a true variable exposure driven by the actual variable and unknown exposures on the underlying deals.

Risk management procedures appropriate to the correlation book and the CDS book should both be applied. In addition to measuring the single name exposures, the jump risk (on CDS underlyings) can be measured on each trade by looking at default simulations in which the counterparty defaults in the near term, and finding the value of the remaining exposure at the default date of the counterparty. Different underlyings (for example, interest rates, exchange rates or equity values) need a model of the jump risk between the reference entity default and the underlying. In the case of interest rates and exchange rates, this will typically be zero, and equity underlyings can be analysed with the same correlated default model as CD products.

These CCDS contain strong correlation between the counterparty who writes protection (C2) and the counterparty to the underlying deal (C1) – in addition to further possible correlation with the underlying asset or reference name. This strong correlation means that the counterparty (C1) exposure will be significant during the life of the trade whether the deal is collateralised or not. The objective within the book is to minimise counterparty exposure but, because of the strong correlation, it is never going to be possible to eliminate this. The conclusion is that CCDSs do not form a complete alternative to good counterparty risk management through diversification.

Part V

Systems Implementation and Testing

Chapters 27 and 28 discuss the implementation of a credit derivatives system. Much of what is covered here is relevant to other systems and is written in as general a form as possible. On the other hand, some aspects are less relevant for some applications – for example, the implementation of market-accepted models for single name CDS valuation required less emphasis on the testing of the mathematical model than a model developed for a bespoke securitisation.

27

Mathematical Model and Systems Validation

27.1 TESTING PROCEDURES

The following describes current practice within risk-control departments of banks and large hedge funds and is commonly used with regard to internal (self-written) and external (off-the-shelf or bespoke) systems and models used for valuation and risk reporting. Without approval at this level, trading desks, for example, are generally not allowed to trade using the trading desks systems and models. An exception applies at start-up of a new desk – generally a limited trading approval is given which may restrict the number of deals, their size, duration of trading period, and other aspects of the transactions. This limited trading approval gives the desk and other departments within the bank time and experience to set up management procedures and ensure that transactions are properly booked, reported and controlled throughout all aspects of the deal. It should also be borne in mind that the testing process is not an attempt to annoy traders and stand in the way of product development – although it may seem so from the point of view of the trading desk – it is a response to requirements imposed by regulators and a responsibility to shareholders and investors.

The testing process typically takes 3 to 6 months for a new 'product' of some complexity (such as a bank starting in CDS or CDO products). This process is usually (but not necessarily) undertaken by someone remote from the trading desk – such as a risk control or 'quant' group – and follows certain formal steps which help to separate the analysis from the pressure of the business. The process would include the following major steps.

1. Write a description of the product – in this case, the underlying business and contracts; how the products work, what the cashflows are and what triggers the cashflows. This should include example term-sheets and final contracts.
2. Write a verbal description of the mathematical model to be used to value and hedge mature deals (this may simply be references to standard works or papers or may be original).
3. Write a mathematical description of the model (formulas).
4. Write a description of input data required for the model and any data assumptions.
5. Write a summary of the key elements of the model and its shortcomings, and key data items and assumptions.
6. Develop (or purchase) a software implementation of the mathematical model.
7. Perform a series of tests on the software, which would include the following (the list is not exclusive and in practice expands as testing is undertaken and as further reasonable test ideas present themselves).
   (a) Perform tests on independent components of the system where possible – for example tests of the discounting (interest rate) component and tests on the survival curve.

(b) Perform simplified case tests – for example reducing the portfolio of contracts to a single contract and comparing the system results against manual calculation;
(c) Build an independent pricing system (typically in Excel or a '4th generation language' such as Matlab, MathCad or Maple). Typically this would be a simplified implementation and in this case might include key contract types. The independent model is then used to value a deal and is compared against the software supplied. Any significant differences would have to be resolved before the independent software could be regarded as acceptable.
(d) Perform stress tests. Extreme data values would be input (such as high/low interest rates, high/low default rates, etc.) to see if the software continued to function and that the results produced were in line with expectations.
(e) A variety of practical tests (does the software fall over if a contract has matured?) and tests on a real portfolio of data help to catch potential problems that might not have come to mind and been tested explicitly.

8. In addition if valuation uses historical rather than market implied data (typically the case for many structured asset backed deals) then this data should be subject to additional testing and analysis. In particular:
   (a) Spot checks on the consistency of data supplied – e.g. if summary values such as means and standard deviations are supplied these should be spot checked.
   (b) Bearing in mind the underlying contracts and business some analysis of trends (for example of default or termination rates) over time might be appropriate.
   (c) Specific questions on data supplied are likely to arise out of this analysis and, in particular, how applicable is the past data to the future?

   In addition to the mathematical and software review described above a 'business systems' review would need to be implemented. In particular, this should cover the following (in considerably more detail).
9. What are the interactions between the desk and its transactions and other departments of the institution (internal interaction) and other bodies (external interaction)?
10. What data flows/reports are needed to support these interactions?
11. Decide on the best way to implement the required reporting and control procedures.

Items 1 to 5 above may constitute a substantial and complex document – similar to this book in the case of complex new products – and would require considerable effort and dedication of the part of both the trading desk and the risk control department. The business review would additionally require the cooperation of many departments in the institution and would be likely to be managed by a business systems specialist rather than (but not remote from) the risk control function. Such a review is a precursor to the full-scale trading in the new product possibly as a global business area. Smaller operations and objectives would have more concise or abbreviated testing processes commensurate to the risk being undertaken by the business.

As part of the first part of this process a more open-ended section attempting to cover (a) the key elements of the model, (b) the shortcomings of the model and the business, and (c) key data items and assumptions. For example, what business risks does the introduction of this product bring – both for the institution and for the financial market place as a whole? Once a business had started, what risks are there to the continuation of the business (e.g. market data, liquidity, staff risks).

27.2 IMPLEMENTATION AND DOCUMENTATION

A live implementation is likely to be integrated into the institution's systems. Two key aspects need to considered initially and on an ongoing basis.

1. Documentation
2. Implementation testing

Documentation on the models and implementation needs to cover aspects discussed in the section above and, in addition:

(a) a user manual – how to operate the system
(b) systems/programming documentation.

The former is needed for new staff; the latter – often forgotten – is needed to enable the system software to be maintained.

Continued implementation testing is needed if – as is usually the case – there are fixed assumptions incorporated into the software that may have a bearing on the results. For example, it is often the case that certain calculations are performed by numerical integration. This may be done in such a way that it is self-checking – several calculations being performed until the result converges to the required accuracy – or a fixed number of integration steps may have been decided at the outset (which has the advantage of implementation speed). In the latter case, regular checks need to be performed on such system parameters to establish that they continue to be sufficiently accurate. In particular, if spreads widen significantly it may be that such fixed parameters cease to give sufficiently accurate results. Of course the model and system documentation should have identified the presence of any such parameters. More problematic assumptions are often hidden ones. For example, on large pools of data a zero correlation model may have been used. Such assumptions can and do fail with catastrophic results – for example, the prepayment of fixed rate mortgages or underlying asset values are highly correlated phenomena but are seldom explicitly incorporated in MBS models.

28
System Implementation

Robert Baker, Partner in Reoch Credit Partners

This chapter discusses the implementation of a credit derivatives system from an IT perspective. The focus is on data elements, data capture and flows, functionality and interactions, practical problems occurring in valuation runs, and reporting. The chapter concludes with an overview of the special features of a credit derivatives system and the relative attractions of in-house versus purchased systems.

28.1 ANATOMY OF A CDO

In order to understand the provision of an IT system to manage CDOs it is instructive to consider the composition of a CDO as outlined in Figure 28.1. We shall then look at how the composite elements change over time and how their data is sourced.

28.1.1 Reference Pool Data

The reference pool of assets may change over time. There are two principal causes of change:

(a) Default of a name in the reference pool will automatically cause it to be removed from the CDO.
(b) The CDO manager may substitute a name in the pool. This substitution will involve some or all of that name's notional being removed and a new name replacing it. In the following it is worth bearing in mind two examples – the iTraxx series 9 CDO (say) and a managed bespoke CDO. The first structure is static, it is a data source and may also exist as an actual transaction; the latter is a transaction.

Each reference entity in the pool contains static, semi-static and market data. Note that reference entities may disappear from the pool or new ones appear. In particular, merges may lead to one or more names disappearing and consequent changes to the tranching structure.

Static Data (rarely changes)

- Reference entity (its legal name) and other identifiers such as RED code
- Parent name
- Sector
- Country
- Region
- Notional holding of this name in the reference pool
- Currency
- Seniority
- Contract clauses – such as restructuring clause.

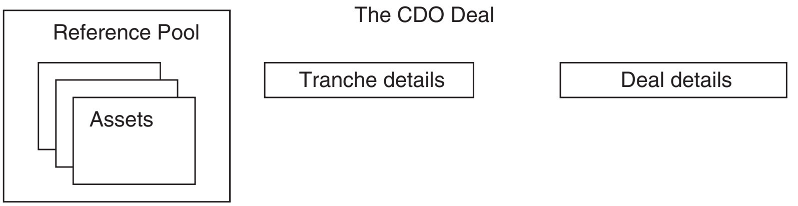

Figure 28.1 Anatomy of a CDO deal.

Semi-static Data (changes but not on a frequent basis)

- Rating assigned by one or more rating agencies
- User's own rating ('internal rating' – sometimes defined as a function of rating agency data)
- Recovery rate (prior to 2008 this would have been semi-static for investment grade names; post-2008 US names moved to dynamic recovery rates ('market data' – below))

Market Data (changing on a daily or frequent basis)

- CDS spreads at various tenor points
- Recovery rates (for some names – see above)

28.1.2 Tranche Data

Static Data

- Attachment point (except for managed CDOs where it is semi-static)
- Detachment point (as above)
- Maturity
- Currency
- Reference curve (fixed coupon, floating: LIBOR plus a fixed spread)
- Payment details – day count, day roll, frequency, holiday centre

Semi-static Data

- Rating (initial rating is fixed; bespoke deals may be re-rated)

Market Data

- Tranche price (spread)
- Correlation

28.1.3 Deal Details

- Start date
- End date
- Size
- Security ID
- Rating

28.2 MANAGEMENT

A CDO management system would be expected to provide the following information in addition to audit trails.

28.2.1 What is Happening?

CDOs are complex instruments comprising several underlying structures across different asset classes. The user will want to know the current state of the CDO both from an absolute viewpoint and relative to a previous point in time, usually yesterday, last week or last month.

The system must provide an assessment of the overall current position of the CDO or CDOs it is managing. Common techniques for monitoring the CDO are: composition measures, weighted average scores, individual winners and losers.

Composition Measures

This analyses the entire reference pool and provides measures such as mean, maximum and minimum for:

- Industry sectors.
- Rating buckets. (These could be the actual ratings or user-defined groupings such as investment grade, sub-investment grade.)
- Geographies (usually country or rating agency-defined region).
- Notionals.
- Diversity. (The rating agencies have devised methods for assessing the diversity of assets within the reference pool, usually by analysing the composition of industries. The Diversity Score is one such measure.).

Weighted Average Scores

This is a group of calculations, each of which finds the mean across the reference pool of the individual score of the asset multiplied by the asset's notional holding in the CDO. Typical scores are:

- WAR (Weighted Average Ratings). Generally each rating is given a score. 1 for AAA, 2 for AA+, etc.
- WARF (Weighted Average Rating Factor). The rating agencies ascribe a factor to each rating to reflect the relative safety of higher ratings and the more risky nature of lower ratings.
- WARR (Weighted Average Recovery Rate)
- WAS (Weighted Average Spread). Five-year credit default spreads are the most liquid credit instruments in the CDO.
- WAM (Weighted Average Maturity)

Individual Winners and Losers

A list of individual names in the reference pool for which one of the following data items has changed significantly, or a list of the best and worst *n* performing assets for that data item.

- Rating change, including a watch being imposed or removed.
- CDS spread – generally at the most liquid 5-year instrument and a move of more than $x\%$ or y bp overnight/week/month.
- Credit event
- Change in recovery rate of more than z overnight/week/month

28.2.2 What Has Happened?

Trustees, CDO managers and investors in CDOs need to be able to track the historical progress of a CDO. They will need to receive periodic reports on a variety of data connected to the CDO from its inception to the present. Usually these reports are generated monthly. They contain narrative together with statistical breakdowns in tabular and graphical formats. Typical reports comprise:

Composition Changes

How the composition measures outlined above have changed over time

Weighted Average Score Changes

How the various weighted average scores have changed over time

Substitutions

A historical record of all substitutions noting the substitution made, the reason, the immediate affect of the substitution on the portfolio and the cost of substitution in terms of fees paid or received.

28.2.3 What is Likely to Happen and What is the Worst that can Happen?

If a CDO system is functioning correctly and data is being input regularly and accurately, it should be possible to alert users to imminent events and reduce the element of surprise.

Easy Predictions

The system can keep a diary of future expected events such as expected cashflow payments. This would give the user a view on future coupon payments, tranche fees, interest payments, etc. If the user is required to take action or make a decision on a certain day, the diary is an easy method of giving early and final warnings to ensure good management of the CDO.

Harder Predictions

A future credit event is obviously difficult to predict but will have significant consequences for the CDO. A sophisticated CDO system could stress test the portfolio for a range of scenarios and report the effects of, say, a given number of downgrades or defaults. As these tests may take time to run, they could be batched up and processed at quiet times such as overnight or at the weekend. The user should also be given the ability to see the affect of a scenario of his choosing, such as a particular name suffering a credit event. Remember that, in scenario testing, the user will be keen to see the net effects of change across all portfolios being held so there will need to be aggregation of results for all CDOs.

It is important that a system should have the flexibility to accommodate user-specified stress tests and changing requirements over time.

28.2.4 What Opportunities do I Have?

A CDO manager sees the world of assets as being divided into those that he owns in one or more CDO and those that he doesn't. The user should be able to view information about assets in these two groups. In practice, this means keeping lists of

- The universe of all assets
- All those assets held in one or more CDO (held universe)
- All those assets not currently held (non-held universe)

Obviously, the first set is the union of the other two.

Like for Like Identification

If a currently held name becomes undesirable, or a non-held name becomes desirable, the user may require a means of identifying similar names across the held and non-held universes. The similarity might be defined by rating, CDS spread, industry, country or some other means.

Trading One Feature for Another

Alternatively, he may want to be offered the cost of changing the risk profile. For example, he may want to improve the credit quality by switching an asset rated BBB+ for one rated A– and be informed how much such a change will cost and which is the cheapest asset matching his new criteria.

System Suggested Substitutions

A sophisticated system might be able to suggest substitutions on a regular basis by comparing the current portfolio with alternatives in the currently non-held universe. This requires an algorithmic method of scoring a given group of names and then systematically choosing different t names and re-scoring.

What Would Have Been?

Knowing the full substitution history allows the system to inform the user of what his CDO would look like had he not performed his substitutions. It is informative to track the progress of the original portfolio against the evolving one over time, assuming the original was still in existence. This allows the user to see if his substitutions were beneficial and provides information on which factors have affected performance.

28.2.5 Reporting

As mentioned above, a key piece of functionality in any CDO management system is reporting. Likely reporting requirements will be

- Daily reports for managers
- Ad-hoc progress reports for business managers, potential clients, etc.
- Trading reports for middle office, operations, and finance departments.
- End of month formal reports for investors, trustees, etc.
- End of year financial reports
- Regulatory reports

Some of these will have fixed formats and be generated at known times. Others will be individual, ad-hoc queries. A flexible system will need to provide:

- Capture of all market, static and trading data at the atomic level
- A user interface allowing the user to specify the query on the data
- The ability to slice, dice and aggregate the atomic data to satisfy the query
- A sufficiently comprehensive formatting of output to satisfy all of the reporting requirements.

28.2.6 Limits

Many CDOs constrain their managers by imposing portfolio limits on the reference pool. These limits are often based on the composition measures and weighted average calculations described above. Examples are

- Weighted average spread to be no higher than 350 bp
- Not more than one region to have greater than 15% of notional holding
- The sum of the two most concentrated industrial sectors to be no more than 40% of portfolio.

To cope with limits and tests, the CDO management system will need to provide the following functionality:

Ability to specify a new limit or amend an existing limit

This is not trivial because there is a vast array of possible limit tests. Generally, however, there are two aspects to each test:

- Defining the unit of the test – e.g. WARF, average spread, region concentration
- Defining some combination of unit scores, e.g. top two, average, minimum, etc.

Calculation of Tests and Limits

The system must be capable of applying the full set of portfolio tests to a particular CDO and discovering how close each test score is to its limit. The calculation may need to be run

- on a periodic basis, such as every morning
- on demand, when the user is particularly concerned about the state of the portfolio, or
- after a given action on the portfolio such as a substitution.

The user may also want to know the affect of a substitution on the portfolio tests before it is performed.

Alerts

The user will want to know when a limit is approaching and when it has actually been breached. He will then need to take action to bring the portfolio back within the limit or, for internal limits, takes steps to get the limit changed temporarily or permanently. Alerts could be transmitted by various means such as on-screen messages, reports or machine-generated e-mails to interested parties.

28.3 VALUATION

The purpose of this section is to highlight some issues relating to CDO valuation and risk management of which the designer and builder of an IT solution for credit derivatives must be aware. Refer to the *Anatomy of a CDO* for the various data requirements before valuation can be performed.

Missing Data

It is common for some input market data such as CDS spreads to be missing on an occasional or regular basis. Here are some suggested solutions to the problem:

- Prompt the user to enter a value
- Use the previous day's value
- Engineer a value based on data from similar inputs. For example, in the case of an automobile spread missing, one could calculate the average movement of all automobile spreads from the previous day and apply that change to the previous day's spread of the missing name.

Another simple example is to use a 'proxy' name – i.e. RTW Engineering is assumed to have the same spread as BMW.

Generally the user will want a report when any piece of market data has changed considerably or when it has been stale for a period of time. A dramatic change could indicate that something important is happening that will affect the entire CDO or it could be an error. Stale data could be caused by a reference identity changing or an error on the part of the supplier. Missing data may arise because the name has defaulted or been taken over.

Pre-valuation

CDO valuations can take a long time to process. The optimum time for this is overnight. If, however, data is missing or wrong, the user will want to be told beforehand to ensure that he doesn't arrive the following morning to find that nothing has valued. One way to check that market data is complete and within an acceptable tolerance is to perform a quick valuation on the portfolio. A simple, price-only valuation or reference entity calibration is generally quick enough to be run at the end of day before the users leave the office and will reveal most errors that will prevent a full valuation.

Calibration

CDO valuation models have a dependency on all assets they are using to be calibrated to produce a survival curve (or hazard rate). The production of these survival curves is called the calibration process and uses the asset's CDS spreads at various maturities to derive its curve.

An individual reference entity only needs to be calibrated once, no matter in how many CDO reference pools it sits. Furthermore, unless CDS spreads change (and other data such as discount factors), the calibration process need not be re-run for different scenarios or simulations. For these reasons, it is more efficient to divide the valuation process into two – the first being for calibration and the second for CDO valuation, using the survival curves. This division requires the calibrated survival curves to be held in memory, and that each one has a unique identifier.

Registration

Due to the compartmentalised nature of CDOs it is often useful to be able to build individual components in memory and refer to them in later stages of the valuation process. This is particularly useful in a spreadsheet, where one might have:

- individual sets of discount factors for each currency
- a full universe of calibrated reference entity survival curves
- a full universe of stressed reference entity survival curves (for sensitivity analysis)
- individual CDOs holding reference pool and tranche data
- standard deals (such as iTraxx and CDX.IG) with their market data spreads or correlations.

A common approach is to register each component in memory and return an identifier. The valuation process can then use these references as parameters which means that

- less data needs to be passed as input in one go
- it is easier to track which part of the process is currently being run
- revaluation makes use of already registered units making it quicker
- if data is changed, fewer components need rebuilding.

Bespoke and Standard Reference Deals

The system must make a distinction between reference deals such as iTraxx and CDX.IG (which may not exist as actual transactions) and the user's own CDO transactions. Reference deals provide market data inputs to the valuation by either quoting price or correlation. The quote will be for a given series, maturity and tranche (e.g. iTraxx series 8, 7-year, 6–9%).

A bespoke deal may be a transaction on one or more reference deals or may have different reference entities and tranching structure. When the bespoke deal contains reference entities that differ from those in the iTraxx and CDX pools, the user will have to provide a mapping between the reference entity and the standard deal whose correlation structure is considered most appropriate. The default would be European names linked to iTraxx and US investment grade names to CDX.IG and US high-yield names to CDX.HY.

Common Valuation Requests

- Defined and implied data. We think of the valuation of a CDO as deriving the tranche price from its correlation but very often the user may wish to reverse this process; input a tranche price and view the derived correlations.
- Valuation plus sensitivity analysis.
- Change in subordination. Before a user makes a substitution he may want to see how the subordination will change in order to keep the value of the CDO the same.
- Aggregation. The user may want to aggregate a CDO with several individual CDS trades. In addition, he may have bought standard tranches of reference deals for hedging and may want to see the combined aggregation of risk between his bespoke CDO and the bought tranches.

28.4 IT CONSIDERATIONS

The following is a discussion of salient points related to the IT provision of a credit derivatives business

28.4.1 Why are Credit Derivatives Different?

Mix of Market Data Sources

Credit derivatives combine data from interest rates, foreign exchange, standard quoted credit instruments such as credit default swaps and the more specialist index-quoted CDO series such as iTraxx and CDX, and volatility data. In addition, most credit derivative systems require inputs of rating agency data such as ratings (current, outlook and warnings) plus historical recovery rates and transition matrices. The diversity of market data means dealing with many different suppliers; presenting both business acquisition issues such as purchasing contracts and technical concerns such as a variety of input media.

Dependency upon Accurate Static Data

As with most credit products, accurate static data is required. Many models use static data from individual assets to derive correlation assumptions – industry sector, country, legal name and relationship to any other child or parent entities. The IT solution will need to source the original static data and keep it up to date.

Complex Valuation

As described elsewhere in this book, the pricing and risk management of credit derivatives is far from easy and a variety of analytic solutions is available. Many organisations require CDOs to be valued using multiple models and many variations of input parameters. Valuation is time consuming. A sophisticated solution may require the use of parallel processing or grid technology to make optimal use of available hardware. This involves identifying atomic units of valuation that can be distributed, ensuring that each unit has all the input data required, and then amalgamating all the unit results to produce the correct overall valuation.

Scenarios

After generating a basic valuation result, it is often required to test the CDO valuation in various scenarios. Here are some examples:

- Choose particular historical market data combinations (Black Wednesday, 9/11, etc.).
- Make some virtual substitutions to the reference pool.
- Take user-defined correlation or recovery rate assumptions and override those from data suppliers.

High Sensitivity to Assumed Data Inputs

Perhaps what makes credit derivatives especially different from most financial products are their sensitivity to two inputs which are difficult to measure or derive from the market – asset recovery rates and correlation.

The problem of dealing with these two unreliable inputs to valuation means that the software solution must allow the user

- to experiment with different combinations and see the effect of input parameter changes upon the final output;
- to ascribe to every output (including stored and reported results) the input assumptions that were made.

Valuation results are meaningless if the underlying assumptions are unknown or unclear.)

Cashflow Waterfalls

Cash CDOs require a waterfall definition. These definitions are unique to each CDO. This presents a challenge to the IT solution

- to allow the user to specify the waterfall to the computer and
- for the machine to implement the algorithm when there is no generic set of inputs or processes.

28.4.2 Spreadsheet

Many financial institutions use a suite of spreadsheets to handle their credit derivative requirements. The advantages of using a spreadsheet-based solution are

- complete flexibility – particularly appropriate to 'one-off' deals such as securitisations;
- easy to convert disparate inputs;
- easy to plug into different valuation models;
- widely familiar to all users in the financial institution, so minimum of training and customisation is required.

28.4.3 Software Application

Alternatively some institutions use a full application developed, especially for the credit derivative business. The advantages being

- Full control of all user activities.
- Data can be checked to ensure inputs are realistic.
- Calculations can be scheduled and optimised.
- Outputs have integrity because they can be matched against known inputs.
- Easy to store inputs against time, so that calculations can be run for historical dates.
- Security – can control and monitor which users have access to which bits of functionality.
- A full audit of all system activities can be kept.

28.4.4 Buy versus Build

There are three general approaches to the problem of providing an IT solution for credit derivatives.

Buy a Vendor Solution

This places the onus of development and support on the third party vendor who will normally charge a yearly subscription. No in-house development will be required. In theory this approach should be cheaper. If 10 clients are using broadly the same system, the vendor can charge approximately one-tenth of the cost (plus profit) to each client and the vendor can maximise his core expertise of delivering software while leaving the client to concentrate on his own business.

The following points, however, need careful consideration before choosing this route:

- There will be an internal use of resources to evaluate a solution, test it and use it when it is purchased. There is also the configuration of the system to make it work in house. This will involve both accepting input data from market and static data suppliers, and feeding outputs to downstream reporting systems. Although the vendor will sometimes offer to configure his system at minimal cost, there are usually hidden costs associated with changing to any new system. New vendor releases may not come at a time when it is convenient to perform tests before release.
- A vendor solution is (in principle at least!) less responsive to changed requirements and is inflexible. If new features are required, the client will have to wait their inclusion in the vendor's next version of the product. The client becomes dependent on a release and priority cycle beyond his control. Also, there is a danger that if the request is particularly bespoke to one client, it will either be given low priority or there will be an additional charge for that client. On the other hand, if a client seeks a market advantage in a new piece of functionality, he may not want it publicised to his competitors who may share the same vendor product.

In-House Development

This requires technical expertise within the firm and knowledge of the complexities of credit derivatives. An in-house solution provides a much greater ability to respond to the business requirements and priorities. However, communication and understanding between the providers (programmers and quantitative analysts) and the users (traders, structurers and CDO managers) is of paramount importance. It can also be a costly solution as the business must pay the full cost of development, gaining none of the economies of scale outlined above with a third-party vendor solution. On the other hand, a '90/10' approach of achieving smooth processing of 90% of the products with 10% handled by exception can be quick and economic.

Mix of In-House and Vendor

Very often a client will want to develop his own analytical engine but to plug it into a generic framework. In practice the development effort in producing a valuation algorithm is less than that of integrating it with a user interface, database and other components of the business solution. Also the valuation is where the competitive advantage resides so it makes sense to buy a platform and plug in the in-house analytics.

Sometimes, the converse is true. The credit derivative owner or trader has his own software infrastructure, usually in the form of a system of spreadsheets. However, if he lacks the expertise to develop his own models or wants to use externally validated pricing algorithms he will hence choose to buy an analytical library and maintain the other components himself.

Part VI

The Credit Crisis

29

Cause and Effect: Credit Derivatives and the Crisis of 2007

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29.1 THE CREDIT MARKETS PRE-CRISIS

For most of history the availability of credit has been limited to two markets: the banking sector and the rest – a plethora of informal, unregulated direct lending mechanisms (person to person, credit cooperatives, micro lending, etc.). While a significant proportion of retail and small business lending may have gone through the informal markets, the majority of all borrowing and almost all large corporate borrowing has gone through the banking market.

As the global capital markets have evolved over the last 30 years, a new source of credit has grown exponentially and this is generally referred to as 'bank disintermediation'. The whole corporate bond market effectively disintermediated the banks by directly pairing off non-bank providers of liquidity with corporate and sovereign borrowers. The banks themselves benefited from the growth of non-bank bond investors by tapping this sector for its own senior and subordinated liquidity needs. And, having established a huge investor base of non-bank credit investors, the next step in the bank disintermediation process was to allow assets traditionally funded on bank balance sheets (corporate loans, mortgages, etc.) to be moved into separate companies and be financed by these same non-bank liquidity providers.

It is this last development that has seen huge growth over the last decade as bank loans, bonds, credit derivatives and a growing array of retail ABS are packaged into securitisation structures and sold to non-bank investors. To understand the drivers for this change, we need to look at the motivations of the banks and the fixed income investors as well as developments such as the growth of SIVs and such factors as liquidity.

29.1.1 Bank Motivation

If one looks at the basic economics of lending, the focus would be on the returns generated by raising money from depositors, the inter-bank or wholesale debt markets and lending it at a higher margin to corporate and retail. A certain amount of equity capital is needed to allow a bank to do this and, given the high operating costs of running a bank, the credit losses associated with lending and the competitive bank environment, the return to equity investors from pure lending is typically around 10%. The calculations are, of course, complex, but to appreciate this, one only has to look at Basel I capital needs. For every \$100 of loans a bank must have at least \$8 of capital. If the margin on the loan is 1%, this translates into a return on regulatory capital of 12.5%, and this is before any expenses associated with lending which are normally in excess of 50% of revenues.

Historically banks have pushed this return into the high teens by supplementing the low-return high-risk business of lending with a high-return low- or no-risk business of advisory and transactional services such as cash management, FX, M&A, etc. However, to enjoy these high-margin businesses the bank has to have client relations, and these normally come with a requirement to lend. So the high margins of non-lending are normally depleted by the low margins of the supporting loan book. Enter bank disintermediation and suddenly the need to hold the assets that are the expensive catalyst of high margin businesses disappears. For the first time in banking history, banks can enjoy all the benefits of their franchise without some of the crippling balance-sheet cost.

The banks that were originating loans to risky borrowers (so-called 'Leveraged Loans') during 2006 and 2007 were faced with an exceptionally challenging pricing environment. Owing to the imbalance between supply of and demand for fixed income products, assets prices were artificially high with risk premium more resembling liquidity premiums. The differential between assets prices within differing rating categories was very small. It reached a stage where the ability of the banks to be competitive with pricing was severely constrained. As an alternative to reducing the risk premium to completely unrealistic levels, the banks decided to relax the legal covenants contained with a loan agreement. This had the effect of the banks not having such a tight control over borrowers: restrictions on controls of the balance sheet were loosened and the mechanisms for spotting corporate problems were less robust.

The loosening of loan covenants became the new competitive edge and started a market for so-called 'cov-lite' loans. Market commentators were generally critical of this development and talked of a growing tidal wave of credit problems in years to come. Nevertheless, the securitisation machine was running well, and the demand for all debt products, including leveraged loans, was robust. Not only were large volumes of cov-lite loans issued but the banks were comfortable to hold them on balance sheet pending the next available securitisation coming off the production line.

29.1.2 Fixed Income Investors

The revolution in the banking sector just described was not one sided. The growth of the bond market, ABS and then CDO market created a huge non-bank fixed income investor base with expectations for growing supply and attractive economic returns. During the decade leading up to the crisis, a number of changes had contributed to the growth in demand for credit fixed income. There had been a small rebalancing of portfolios away from the equity markets; government bond returns had fallen and remained low; and corporate borrowing had not grown at a sufficient rate to meet increased demand. The gap had been filled by the ABSs, CDOs and SIVs – the so-called structured credit markets. Historically this risk was distributed via bank balance sheets and was available to fixed income investors in the form of bank senior and subordinated debt. However, in comparison to the tailored and innovated structures employed in the structured credit markets, bank debt is a very blunt instrument. There is no transparency as to the underlying portfolio, a bank is not a pure credit play and there are essentially only two levels of risk participation. It was very easy to persuade investors to use the more flexible and transparent structured credit solutions, and this was ideal for a banking sector keen to have more flexibility to separate risk from franchise.

29.1.3 Risk Traders versus Risk Absorbers

It is important to note that many of the investors who hold assets in SIVs or as an investment in CDOs are set up as 'traders'. Their risk management systems assume that if they cannot fund the assets for whatever reason, they will be able to sell them. They trade and treat the assets as ratings with yields. They 'outsource' the monitoring and understanding of the risks to the rating agencies, so they have no independent view of the risks. They are not risk absorbers.

A risk absorber needs to have the capacity to take a different view about the risks than the market place. To do this they would not depend on market liquidity and would have a credit monitoring capacity that was independent of market prices. Investors with long-term liabilities like pension funds and insurance companies are natural risk absorbers. Their ability to be risk absorbers, however, was curtailed by regulations that required or encouraged pension funds and insurance companies to mark-to-market their assets and respond to short-term changes in asset prices.

29.1.4 Structured Investment Vehicles

The use of Conduit Financing Vehicles (CFVs) or Structured Investment Vehicles (SIVs) was extensive and was established in the late 1980s. The structures disintermediate banks by enabling a range of long-dated debt instruments to be financed by short-term debt. If short-term financing becomes unavailable, the structures use a back-stop facility provided by a bank. This secondary source of financing is designed to plug any financing gaps and generally will only be used when liquidity is in short supply. As a result of the back-stop, the ratings agencies are prepared to give the short-term debt a high credit rating.

A SIV[1] is structured as an SPV that buys long-dated assets such as debt instruments, and finances the purchase by issuing short-term debt such as commercial paper. The debt instruments that the SIV buys can be corporate debt (bonds and loans), or retail debt (pools of mortgages, credit card receivables, car loans, etc.). The SIV normally enjoys a healthy profit due to the fact that short term borrowing rates (what the SIV pays) are normally less than the return on the longer dated assets. SIVs can be very profitable because they are highly leveraged and the returns on the small amount of equity may therefore be very high.

SIVs are also called conduits because they create a channel through which the long-term debt in which they invest can be funded by short-term debt. The business model (borrow short term, lend long term) is very similar to that of a bank, but by conducting its business through capital markets (rather than taking deposits), and by being an off-shore entity, it escapes the regulation to which banks are subject. Their risk can be summarised as follows:

- *Credit Risk:* The assets in the SIV are risky in that they are exposed to issuer default risk. Where the asset is synthetic (for example, a CDS) then, in addition to issuer risk, there is also counterparty risk. Although the assets in a SIV are generally of high quality, the SIV will need (a) processes for credit decisions and portfolio monitoring, and (b) limits to control exposure by company, sector, country and rating. Minimum standards for the latter are normally set by the rating agencies to give the SIV debt the best possible rating.
- *Liquidity Risk:* The SIV has a continual refinancing need as the maturity of the assets is often significantly longer than the tenor of the SIV's debt. A number of safeguards are employed:

[1] A special purpose vehicle (SPV) or special purpose company (SPC) is a company that is created solely for a particular financial transaction or series of transactions. It may sometimes be structured as a trust. SPVs are often used to make a transaction tax efficient by choosing the most favourable tax residence for the vehicle. They can also remove assets or liabilities from a bank's or corporate's balance sheet, transfer risk and (in securitisations) allow the effective sale of future cashflows. The management of an SPV is structured according to accounting standards that require a company controlled by another to be consolidated as a subsidiary; normally the objective is for the SPV not to consolidate, and is easy to achieve as SPVs require very little in the way of management.

the highest credit rating is maintained to ensure access to a wide range of investors; diversity of investor is achieved by using multiple debt programmes throughout the global capital markets; controls ensure that the funding requirement at any point in time does not exceed a set limit; a back-stop facility is in place to fill any funding gaps; and, normally, the SIV would expect to be able to quickly liquidate the highest quality assets to raise additional funds.

- *Market Risk:* To the extent that the SIV is exposed to interest rate or exchange rate risk, limits are put in place and active hedging ensures that these risks are minimised. The objective is to make the SIV insensitive to market risks.
- *Operational Risk:* A SIV would normally have procedures, control and necessary reporting to address operational risk: the possibility of losses resulting from sub-standard or failed internal processes. The combination of rating agency and auditor reviews and reporting is considered adequate to mitigate these risks. A key success factor is the SIV's platform that needs to both book a variety of assets and monitor the portfolio and the liabilities. As important however is its independent control functions that ensure the maintenance of its strong credit rating.

Pre-crisis the use of SIVs was extensive and they were a large component of bank disintermediation. Following their catastrophic performance during the early days of the crisis most structures were unwound and renewed usage seems unlikely in the near term.

29.1.5 Market Liquidity

As noted above, and contrary to common belief, the liquidity of a market rests not so much on its size (as measured by market capitalisation or turnover) but in the diversity of its participants. It is easier for observers to see this distinction in the midst of a crisis than during the quiet time before a crisis when liquidity appears high and capitalisation is galloping ahead.

In many markets there are many different types of market participants like hedge funds or pension funds, and within each type there are many different investment strategies. But diversity is often richer in appearance than in the reality of behaviour. A key measure of the critical degree of diversity required for liquid markets is how differently market participants respond at times of stress to short-term price declines. A market where, for whatever reason, falling prices trigger selling by most players and generate few buyers, is one that may be large and appear liquid in quiet times, but will be fragile and illiquid in stressful times. This is a stylised description of what has occurred in the global credit markets.

Many investors have long-term liabilities that do not require sensitivity to daily market moves. Examples would be pension funds, insurance companies, Sovereign Wealth Funds (SWFs) or any other investor where funding or liabilities are long term. These investors can earn a liquidity premium versus other investors who require short-term liquidity. From a systemic point of view, these investors act as a liquidity absorber during times of stress.

Investors who have short-term funding, or are forced to follow short-term solvency or stop-loss rules, or who intended to trade an asset and so are not incentivised to understand it sufficiently to hold on to losing positions – will be forced to sell assets when they fall sharply in price. Indeed, they are incentivised to try to be the first to sell assets before other investors do. Liquidity disappears in this rush for the exit. These liquidity-hungry investors act not as risk absorbers, but risk amplifiers.

At one time, many regulators argued that the transfer of risk from one bank's balance sheet to several investors was a desirable spreading of risk. But what matters is not the number or name of those to which risk is transferred, but their behaviour. The transfer of risk from banks was a transfer from a risk absorber to entities that acted as risk traders or amplifiers. This did not spread risk, it concentrated it. Supervisors ignored or misunderstood the distinction between risk traders and risk absorbers, and the need for heterogeneity.

29.2 THE EVENTS OF MID-2007

The events leading up to and during the credit crisis were a complicated combination of real losses and distressed valuations causing and being caused by liquidity problems for hedge funds, SIVs and banks. To understand the full sequence of events we need to start where the real losses started – the sub-prime mortgage market.

29.2.1 Sub-prime Mortgages

The process of originating and securitising mortgages is well established, especially in the US market where it has long been accepted that mortgages could be bundled together and sold to specialist mortgage companies. These companies raised the necessary funding from an array of investors who were given the choice of participating in the risk of the mortgage pool at a junior, mezzanine or senior level. There are a number of classifications, credit scores and other underwriting tests that provided the investors with some comfort for the risk of the pool to which they were exposed. It is beyond the scope of this book to describe the securitisation of mortgages in detail; however, there are two important features that help to explain the extensive confusion surrounding the size and extent of losses in the sub-prime sector.

Recovery of Losses by the Mortgage Servicer

Although mortgages may be moved into specialist financing companies (MBS) to allow the risk to be spread far and wide, the actual management of the mortgage relationship with the borrower is through mortgage servicing companies. They are required to make payments to the MBS which would normally be covered by back-to-back payments from the borrower. If the borrower misses a payment, then the servicer pursues payment culminating, in the worst case, in a foreclosure on the mortgage and taking possession of the house to sell it and repay the loan. Normally the servicer can only reclaim losses from the MBS when the last legal remedy of foreclosure has been reached. If, for some reason, foreclosure cannot happen, due perhaps to government intervention that prohibits foreclosure to prevent a social crisis, then the servicer will be making cash payments to the MBS that are not covered by the borrower. Even without the intervention by a body of the state, it is easy to imagine a protracted process before the servicer can declare legal foreclosure and reclaim losses from the MBS. Given the high geographic concentration of losses, it is likely that the court process will be another obstacle – it will not be equipped to deal with hundreds or even thousands of such claims. As a result there is a bottle-neck in the foreclosure process which adds considerable confusion to the process of estimating actual losses in the sub-prime mortgage space.

The Underwriting Process

The second problem is related to the actual underwriting process. Some of the tests that allow a mortgage to be placed in an MBS were enforced by the local lender. Checking the credit scores of the borrowers, checking that the borrowers actually live in the house that is to be mortgaged, and ensuring that the borrower makes the first few mortgage payments, are all open to abuse. Feedback from those close to the industry suggests that there had been fraudulent activity in this process: the credit score of one borrower might be represented as being the average credit score of a borrowing couple (when in fact the other borrower has a poor score); checks on whether a property is the borrower's primary residence were poor or non-existent; and, it was not unusual for the mortgage arranger to offer to make the first mortgage payment as a 'sweetener' for the deal. Whether this actually happened may be revealed as investigations proceed, but there is some certainty that many mortgages would never have been transferred into MBSs if the underwriting rules had been rigidly enforced.

By the time these pools of sub-prime mortgages had been financed by different risks tranches, and some of these tranches had been placed in CDOs to be further tranched, it should come as no surprise to discover that the CDO investors had little clue as to the risk and therefore value of their positions. Cashflows are vague due to uncertainties at the servicer level and the average risk of the portfolio is uncertain owing to possible flaws in the underwriting process.

The credit-modelling tools used by the mortgage markets may have been adequate for valuing individual mortgage-backed tranches in normal market conditions but proved ineffective (if used at all) to revalue CDO tranches that were exposed to mortgage-backed tranches in distressed market conditions. To add to the confusion, it is likely that many CDO investors (a) didn't know whether they were exposed to the MBS sector and (b) even if they knew of MBS positions in the portfolio they didn't know the quality of such positions.

It is easy to blame those who originated the mortgages for being sloppy and perhaps fraudulent in the underwriting process, or those who packaged and rated the pools of mortgages for not performing better due diligence, or those who distributed the securities for being irresponsible in their assessment of this risk. However, it is impossible to point the finger of blame at anyone other than all of those involved in the process. Everybody was motivated by the rewards: from the commission for originating a mortgage right through to the higher yields paid to end investors. Those involved early in the chain were never going to bear the risk, and those later in the chain were not aware of the risk. The former were perhaps negligent in their regard for the latter, and the latter were negligent in not questioning the seemingly endless supply of high-yielding low-risk products from the former.

29.2.2 Investor Impact

The impact of the uncertainty described in the previous section was a significant downward revaluation of all CDO tranches with little distinction between underlying portfolio asset mix. Even those deals with no sub-prime risk were marked down, which had the following impact on different types of investor:

- *Hedge Funds:* For the funds that were long CDO tranches, the revaluation resulted in margin calls. For some funds the margin calls exceeded available collateral and the hedge funds were forced to de-leverage. This could only be achieved by actually selling or terminating the CDO positions that the market was struggling to value. Not surprisingly, the termination values were poor, reflecting the market's reluctance to take on tranched credit risk.

- *Structured Investment Vehicles:* It was well known that SIVs invested in the AAA tranches of CDOs and, hence, when such tranches started being revalued at lower levels, there was an increased degree of nervousness in the asset-backed commercial paper market. Supplies of liquidity slowed, forcing some SIVs to liquidate parts of their holdings. This added to the downward pressure on asset prices and was compounded by the problems being faced by the hedge fund industry.
- *Mark-to-Market Investor:* Most financial institutions that held CDO investments were required to mark their positions to market. The significant revaluations that started to appear as the liquidations in the hedge fund and SIV markets gained momentum, caused material mark-to-market losses for many investors. In some cases the losses triggered an informal decision or a formal requirement to sell. This was particularly frustrating where the investment was taken on with a hold-to-maturity mentality, and the underlying position was probably no riskier than it was pre-crisis.
- *Nervous Investor:* For most of 2007 the financial press covered the structured credit market with a level of detail never seen in the market's 13-year history. The coverage during the crisis was very detailed but frequently misleading. For example, the highest rated CDO tranches – the AAA tranches – were often referred to as one group with little attempt to distinguish between the different underlying pools of risk. This may have contributed to all AAA tranches being revalued at much lower prices even where the underlying portfolio was 100% investment grade corporate risk.
- *Hold-to-Maturity Investor:* The group of investors who provided much needed stability during the period of CDO revaluation and liquidation were those who were not affected by liquidity needs; they either did not mark their positions to market or were able to absorb significant mark-to-market movements and were sophisticated enough to assess the true meaning of the market's revaluations.

29.2.3 Bank Impact

The impact on banks was more complicated, given their involvement in so many areas of the origination and distribution process. Asset due to be securitised unexpectedly remained on the balance sheet; unsold CDO risk positions fell in value; some collateralised credit lines failed; SIV back-stops were called; and some SIV and CDO deals had to be brought back onto the balance sheet. These are discussed in more detail below.

Warehousing of Assets for CDOs

By mid-2007 the CDO machine was in full swing with a broad range of asset classes being securitised. In order to speed up the process it was common for banks to 'warehouse' assets that were destined for a CDO. Since there was a reasonable certainty that any debt held in this way was only going to be on the balance sheet for a short period of time, it was possible that the attention paid to hold limits or underwriting standards was less than it would have been if the debt was destined to be held for a longer period of time.

The warehousing of leveraged loans proved to be particularly hazardous when the credit crunch started. Not only were the issuers of such loans considered risky due to their rating being the wrong side of a 'flight to quality' threshold; but because of the their cov-lite label the traditional takers of such loans were less keen to invest. Many banks, perhaps emboldened by the previous capacity of the CDO repackaging machine, held considerable overall positions and considerable exposures to individual borrowers. Once it became apparent that these loans were not moving as fast as expected, it was necessary to declare holdings and revaluations as part of the defensive steps taken by banks to reassure other banks of balance sheet strength.

A Fall in Value for Unsold Positions

The credit crisis left banks holding two sorts of CDO positions: those it had always planned to hold (mainly senior positions), and those it was unable to sell. Both had to be written down in value and it is likely that the former created the largest problem because of the sheer size of senior positions held by banks. A *CreditFlux* article[2] revealed that just in the CDO of ABS market, the size of super senior tranches of deals done in 2006 and 2007 was close to USD 240bn. It quoted an estimate that two-thirds of super senior risk was retained by arrangers with the rest transferred to monoline insurance companies. Data in the article showed nine arrangers responsible for more than USD 10bn of super senior tranches each, with one dealer arranging five times that amount.

Bringing Risk Back on Balance Sheet

Arrangers in the structured credit market had indirect exposure to CDO products due to their involvement with SIVs, hedge funds and CDOs themselves:

- *SIVs:* As explained in section 29.1.4 most SIVs had a back-stop credit facility provided by a bank. As traditional investors in SIV-issued commercial paper backed away, banks had to step in to provide financing. In a number of cases this resulted either in banks effectively taking control and having to consolidate the SIV, or in banks voluntarily taking the SIV on balance sheet to ease investor concerns.
- *Hedge Funds:* Where a hedge fund or other leveraged investment fund pursues a structured credit trading strategy, the first loss for the strategy is provided by the fund's equity investors, while the second loss is provided by the bank acting as swap counterparty. To the extent that MTM requirements exceed the fund's available equity, the bank was exposed to that excess and any further credit deterioration. If the fund ultimately failed, that MTM loss would be crystallised.
- *CDOs:* Aside from the direct exposure that an arranger had to a CDO, there may have been an *indirect* exposure if an arranger had to consolidate the assets of a CDO. One example is a reconsideration event that occurs when an institution acquires extra interest in an entity.

Accounting Changes

FAS 157 clarifies the definition of fair value, how to measure fair value and the broader disclosures about fair value measurements. It was intended to improve financial reporting and provide users of financial statements with better information. A challenge for auditors as they implemented FAS 157 was determining the values of 'Level 3' assets, which included those that do not trade in active markets or were unique. Their value could be determined by mathematical models. Pre-crisis CDOs had been treated as 'Level 2' assets, i.e. those that are not actively traded but have observable inputs.

[2] *CreditFlux*, December 2007, page 4: 'Fears grow over ABS CDO arrangers' exposures to super senior tranches'.

There was an increase in the use of the Level 3 classification with company filings from three US broker dealers, showing at least a 50% increase in total Level 3 assets from Q1 to Q3 2007. Such a change in classification required a higher level of disclosure; it also required companies to take into account what a position would realise if it were sold today. This could force some large write-downs even though the assets haven't incurred any actual losses and are unlikely to give the quality of underlying credit risk.

29.2.4 The Failure of Lehman Brothers and the Bailout of AIG

In 2004 the US Securities and Investment Commission relaxed a rule limiting the amount of leverage that Lehman and other investment banks could take. Lehman Brothers proceeded to take on substantial exposure to low-quality mortgages by way of asset-backed securities and credit derivatives. As a result of the fall in the market for sub-prime mortgages and securitised mortgages, Lehman faced massive losses and was forced to raise USD 6bn through a share issue. After further losses the US government decided not to bailout the bank and no buyer could be found, so on 15 September 2008 the company filed for insolvency in the US and soon after in the UK.

Lehman's 10-Q filing with the SEC on 13 May 2008 reported the fair value of its derivatives contracts outstanding at USD 729bn. The company chose fair value over notional value as it claimed that the latter can overstate expected payout. By contrast, Bear Stearns reported notional value of its derivatives at USD 13 trillion (November 2007), JP Morgan at USD 89 trillion (March 2009), and Bank of America and Citigroup both at USD 37 trillion (March 2009). Notional values give a more accurate picture of operational difficulties and although they inflate overall risk perceptions they are the correct metric for counterparty risk. Assuming that every transaction has been correctly margined and collateralised, the net number should be a small fraction of the gross.

Lehman's OTC CDS book had a gross notional of USD 72bn and there was initially considerable uncertainty and complexity in identifying close-out positions and replacing defaulted trades, particularly due to widespread rehypothecation[3] by Lehman of securities posted as collateral under CDS contracts. In spite of this, Lehman's net exposure to OTC CDS contracts was estimated to be about USD 5.2bn. In addition to CDS contracts, where Lehman was a counterparty, there were a large number of CDS contracts referencing Lehman – estimated to be around USD 400bn.

By contrast with its traditional, relatively low-risk business of life insurance and retirement services, from 1998 AIG diversified into more risky investments, taking massive exposures across mortgage-backed securities, CDS contracts and particularly CDOs on asset-backed securities (ABSs), many of which were based on bad asset classes such as sub-prime debt. In spite of AIG's huge exposures to these products, its financial models only estimated the risk of default of the underlying obligations. It was only in the second half of 2007 that AIG began to model the risk of the value of underlying obligations falling (as distinct from their default) and so started having to post collateral to counterparties and make write-downs on its balance sheet. However, by this time it was too late, with major counterparties demanding billions of dollars of collateral as a result of worsening credit markets. Needless to say, AIG did not have access to the liquidity to finance these payments.

[3] The pledging of securities in customer margin accounts as collateral for a brokerage's bank loan.

To exacerbate the problem, on 17 September 2008 AIG's long-term credit ratings were downgraded by S&P from AAA to A and by Moody's from AAA to A2. AIG was suddenly obliged to post more than USD 13bn in collateral as a result of that downgrade. On 17 September 2008, with Lehman having gone into insolvency two days before and credit markets freezing, the US Treasury decided it could not let AIG take the same route, and offered a bail-out in the form of an USD 85bn credit facility, later extended to 153bn, in return for a 79.9% stake in the group.

It is worth noting that the majority of AIG's actual losses occurred on CDOs of ABS. While its CDS book had a notional value of USD 270bn its actual losses were only USD 3bn, whereas its CDOs of ABS had a notional value of USD 300bn and an actual loss of USD 46bn.

All market observers have consistently exhibited great surprise at AIG's involvement in the structured credit markets. However, as the extracts from a Fitch 2006 survey show, it was well known in September 2006 that AIG's activity in this sector dwarfed its peers and many of the banks:

> Insurance companies and financial guarantors continued as large net sellers of protection at USD 645bn (up 16% on 2004), reflecting the dominant footprint of AIG Financial Products Corp (AIG) in this market. Excluding AIG, however, global net exposure was relatively stable at USD 277bn sold.
>
> The global insurance/reinsurance sector (excluding financial guarantors) continued as a growing seller of protection, recording an aggregate gross sold position of USD 514bn, up 30% from USD 397bn in 2004 and USD 258bn in 2003. By product, synthetic CDOs represented 71% of the total and cash CDOs made up an additional 20%. In terms of quality, of the total global gross sold position by insurers, 91% was in the super-AAA segment. However, as in the past, these results are distorted by AIG Financial Products' dominant 'footprint' in this market and their focus on synthetic CDOs.
>
> Excluding AIG Financial Products, the global insurance/reinsurance global net sold exposures actually fell 71% to USD 15bn (USD 29bn gross sold) – a continuation of a trend started in 2004 and discussed further below. By product, positions were more evenly distributed, with synthetic CDOs at 29%, cash CDOs at 24%, indices at 24% and single name CDSs at 17%. Similarly, the credit profile of the rest of the global industry was more varied, with A representing 33%, AA at 16%, BBB at 15%, equity/unrated at 14%, AAA at 10% and super-AAA at only 5%.
>
> AIG Financial Products is a major provider of credit protection on designated portfolios of loans and debt securities. The magnitude of its business dwarfs that of traditional insurance companies. Therefore, the overall North American insurance and reinsurance survey results are dominated by this company.

29.3 ISSUES TO BE ADDRESSED

29.3.1 A Different Rating Agency Process

Credit rating agencies were critical of the growth of securitisation, CDOs and SIVs. Banks would have found it hard to profitably sell-on individual loans of borrowers that generally were denied direct access to the capital markets. Packaging and structuring slices of the loans together into a CDO or SIV created a focus on credit risk and a diversification of credit risks that were attractive to a wider group of investors, but it also made the instruments more obtuse and so, for the market to work, investors required a credit rating.

Increasing attention has been placed on the inherent conflicts of interest in the relationship between banks, credit ratings and investors. The credit ratings were for the use of investors, but were paid for by the arrangers – the banks. The arrangers made more profit the more they could convince the rating agencies to give a higher credit rating to packages made up of risky instruments.. The rating agencies made more profit the more banks gave them packages to rate. The more discerning the rating agency was over what it rated, and the less generous its ratings, the less business the rating agencies would do. At the eventual cost of investors, were the rating agencies overly influenced by the banks who were paying them to give the packages a higher rating than they deserved? An analogy is with the bank analysts during the dotcom bubble, who, it is alleged, were encouraged to give generous ratings to companies in order to help them to win lucrative corporate finance mandates.

It is clear that investors have, yet again, paid a price. It is also clear that the ratings appeared generous and that these credit ratings have become so mistrusted that the assets did not have the liquidity expected of instruments with good ratings. Proving that credit rating agencies connived with banks to push up ratings is less clear.

It has been argued that one of the problems was that investors did not understand what the ratings meant. It is possible that they were confused between credit risk (the risk of default or some credit event), which is what the ratings were for, and liquidity risk (the risk that the assets could not be sold on to someone else). Some have suggested that the rating agencies should also have provided a liquidity rating. However, credit and liquidity analyses are very different. A credit view can be taken by an analyst working at home with publicly available data on an entity's liabilities. Once established, it is a view that is likely to change only slowly, especially for the highest rated entities. Liquidity conditions can change rapidly and require an appreciation of market liquidity, which is not easily gained outside the market places. Liquidity judgements are best left to market participants.

But the degree of confusion over what ratings mean is worrying. It is not clear that this confusion was deliberate, but there were commercial forces that made it likely to happen. Rating agencies were trying to extend their 'brand' from corporate bond ratings, where they have an established presence, to structured credit. To support the brand they wanted to use rating labels that had a resonance with their existing ratings.

At first sight it would appear that the principal solution to the concern over the veracity of ratings is to ban arrangers and issuers from paying for ratings as a result of the conflict of interest and lack of independence. The problem is that this neglects the fact that ratings are a 'public good' in that they only have value to issuer, investor and the market if all investors are aware of them, and if they are aware of them, it is difficult to get any individual investor to pay for them. Rating agencies have tried. If investors will not pay, the only alternative is issuers or government. But the public sector should steer clear from impinging on how ratings are derived. At best this would lead to a perception that the government was morally obliged to protect investors from losses that were the result of following government-funded or approved ratings. At worse, there could be a loss of confidence in ratings if the little scope for innovation and competition that was left in the industry's oligopoly structure was eroded by prescriptive and burdensome regulation.

29.3.2 Standardised Nomenclature for Credit Ratings

The three main credit rating agencies each has its own similar but different rating nomenclature, which they augmented in similar but different ways to extend ratings to credit derivatives and structured finance instruments. In quiet times these differences appeared insignificant and were easily ignored by investors. In stressful times these differences, and the confusion around them, became important.

The rating agencies should be invited to consult with each other – without fear of being accused of anti-competitive behaviour – to develop standard definitions and nomenclature. How rating judgements are decided, and the basis for applying them to specific entities, will remain the preserve of the individual rating agencies. Public coordination around the definition of ratings should not lead to any implicit public approval or guarantee of ratings.

29.3.3 Keeping a Percentage of Originated Risk on Balance Sheet

One of the compelling features of the Balance Sheet CDO is the market's requirement for the arranger to keep some or all of the first loss tranche. This is based on the belief that there needs to be a financial incentive to ensure that a bank continues to monitor lenders whose debt has been transferred into the CDO. To the extent that bank-originated debt is transferred into a CDO, this first loss ownership also ensures that underwriting standards remain robust.

The bulk of deals done in the last few years have been Arbitrage CDOs, and in the deals where recently originated debt (e.g. Leveraged Loans) is transferred into CDOs, there is no requirement for the arranger to keep a first loss position. The only mechanisms for ensuring the quality of deals sold to investors are: compliance and reputation risk. The former is meant to ensure that the investor is sufficiently sophisticated and has all of the risks explained; the latter puts the arranger on notice that bad deals will come back to haunt you.

It has been suggested that if originators kept a percentage of all of their originated debt instruments, this would ensure a layer of quality control. To a certain extent this happens already – in addition to the comments above about a Balance Sheet CDO, banks will rarely hedge (via CDSs, CDOs or otherwise) their entire holdings of a borrower's debt. Hedging is expensive and banks normally have capacity to hold a certain amount of a client's risk. If they have no capacity to hold a client's risk, one wonders whether the bank should maintain the client at all.

An extension of this discipline into the structured credit market might ensure a better quality product, but it would be difficult to achieve in practice. The implications of holding a prescribed or recommended amount of every deal would be an operational challenge. It would also transform a bank's structuring function into a risk-taking group with necessary allocations of capital, etc. And, unfortunately, even if some of the risk was retained by a visible holding of a bond, the actual risk could be subtly removed by an appropriate hedging strategy.

29.3.4 Undrawn Credit Facility Capital Charge

One of the great unknowns of the credit crisis was the size of the back-stop facilities granted by banks to facilitate the financing of SIVs. In order to secure a high rating for the commercial paper that is financing the bulk of a SIV's balance sheet, there is normally a bank back-stop in place to act as lender of last resort. Should short-term financing ever be unavailable, then the SIV may liquidate some of its assets (hence reducing its need for financing) or may call on its back-stop facility to fill any financing gap.

Owing to the structure of the back-stop facility, one cannot say for sure that the SIV will always be able to call on it when financing is needed. Hence, as the credit crisis unfolded it became usual to talk about the legal or moral duty of banks to pay out on back-stop credit facilities. The latter would appear to apply to the situations where the back-stop cannot be called for mainly technical reasons, and to the minority of cases where the SIV does not actively have a back-stop in place.

Whatever the duty of a bank to provide a back-stop credit facility to a SIV, the current crisis has revealed two factors which doubtless will question the allocation of capital against the facility. The first is that the numbers involved are huge, even in the context of the world's largest banks; the second is that the SIVs' needs for financing proved to have a high level of correlation due to the fact that an entire sector of the financial markets was affected.

In assessing the level of capital held against a back-stop facility, there should be a realisation that the borrower is likely to be somewhat distressed at the time of drawing on the facility. For most corporate borrowers there are many sources of finance available, and drawing on the back-stop is likely to be a last resort. The second consideration – probably not applicable to standard corporate back-stops but now shown to be highly applicable to the SIVs market – is the likelihood that the very economic conditions that bring about the drawing on one SIV back-stop facility are likely to trigger multiple drawings. In effect, if SIV exposure could be considered to be one 'collective borrower', then additional capital may need to be set aside if total exposure exceeds some 'large exposure' maximum.

29.3.5 The Future of CDOs

The development of credit derivatives and the CDO market has had a significantly positive impact on risk bearing and risk management in financial markets. It has also produced a systemic crisis of massive proportions. The problem is clearly to develop a market structure that enjoys the benefits of credit risk transfer and, at the same time, mitigates the downside risks.

A major difficulty is the lack of comprehensive information on the transfer of risk. If the location of risk could be clearly mapped by the regulatory authorities, then measures could be taken (for example, by means of capital charges) to disincentivise the accumulation of risk holdings that pose systemic risk.

29.3.6 Mitigating the Negative Impact of Mark-to-Market

The spread of mark-to-market requirements (whether internally developed within firms or required by regulators) has been a major stimulus to the development of risk trading at the expense of risk absorption. This is another area in which there is a need to balance the positive benefits with negative systemic effects. Without the requirement of mark to market the serious deficits in many pension funds would not have been revealed, and the funding of pensions today would be far weaker and less secure. Perhaps there should be a better balance between providing more detailed information about liabilities and risks, but doing so less frequently.

An analysis of the structures currently being set up to invest in distressed structured credit assets gives some hint of what the market is trying to do to combat the joint perils of mark-to-market losses and forced sales. Private equity funds, by their (original) nature, do not respond to short-term market movements, and lock up the funds committed to them for the duration of their investment cycle. In an area of financial service directly involved in trading, hedge funds are now typically launched with lock-ups. Where the investment universe includes illiquid securities, lock ups can be as long as 2 to 3 years.

But going beyond these market developments, there is a need for regulators to assess the systemic impact of their mark-to-market requirements and to manage them in such a way as to reduce the likelihood of disruptive forced sales. This will require regulators to take a wider view of macroeconomic conditions than they have previously done.

29.4 MARKET CLEARING MECHANISMS

In recent decades, the structure of financial markets has shifted from a bank-based to a market-based financial system. Financial intermediation has moved from institutions into markets, and financial crises are now manifest in markets rather than in institutions. Accordingly, the analytical interest has moved from bank runs to 'market gridlock' as a source of systemic risk. The market-oriented systemic crisis is a breakdown in the functioning of markets for traded assets. What has become apparent in recent years is that market-oriented systemic risk can arise because of underperforming and poorly regulated clearing and settlement systems. Settlement problems have been particularly visible in derivatives trading, where regulators have been very concerned about deficiencies in 'back-office' procedures and about settlement delays.

The turmoil of the last two years has brought new concerns to the fore. Although the existing infrastructure for clearing and settling derivatives did not fail during the recent crisis, regulators and others clearly believe that the management of risk could be improved.

One of the reasons for regulatory interest in centralised clearing is a hoped for improvement in pricing and price transparency. When confidence collapsed in CDOs and CDSs, this was in part because there was no knowledge of what their value might be. It was this lack of any basis of valuation that led to the questioning of the value of bank balance sheets. No one knew the value of assets held on the balance sheet, and hence no one knew whether the banks were solvent. The Turner Review (FSA, 2009) argued:

> The complexity and opacity of the structured credit and derivatives system, built upon a misplaced reliance on sophisticated mathematics, which, once irrational exuberance disappeared, contributed to a collapse in confidence in credit ratings, huge uncertainty about appropriate prices, and a lack of trust that published accounting figures captured the reality of emerging problems.

This view reflects a concern about complexity and a lack of transparency, hindering the ability of regulators and investors to properly assess risk. In addition to their custom-made character, a typical problem with CDOs and the underlying of CDSs was their complexity, not their transparency. Securitised instruments typically come with many pages of elaborate documentation, describing the character of the asset in detail – but even those who attempt to read it all seldom understand it. Complexity meant that few understood the structure of particular assets, and could not value them. Liquidity vanished because it was impossible to sell OTC assets, or to use them as collateral.

It is also argued that, due to the bilateral nature of OTC derivatives, the risk of a counterparty defaulting before the contract expires is relatively high – particularly for credit derivatives that generally have long maturities – making it more likely that a purchaser will be left unprotected. In addition, collateralisation provisions in CDS contracts are not standardised and normally do not take account of how credit enhancement on one transaction affects risk exposures on related transactions.

In the light of the leveraging practices of institutions like Lehman, there are major concerns about the CDS market having grown to many times the size of that of its underlying credit instruments. The Bank for International Settlements estimated that, at the end of June 2008, the notional value of CDSs outstanding in major financial markets was USD 57.3 trillion. It was only in late 2008 that data released on the DTCC website revealed that the net transfer of risk in the CDS market was a fraction – typically about 10% – of the notional value of outstanding CDSs. The data confirmed the need for the current focus on operational risk (trading volumes are very high) but removed some concerns about the sheer size of the credit derivatives market, if netting can be performed efficiently.

A further concern is that, due to the nature of CDS contracts involving the referencing of other credit instruments and the posting of more collateral as default probability increases, a downturn is likely to cause a downwards spiral of payouts and defaults, the type of which triggered the collapse of AIG. Pro-cyclicality is ‘hard-wired’ into the financial system. It is these difficulties that have led to the now generally accepted policy conclusion that as many assets as possible should be forced into clearing systems, facilitating price discovery and liquidity, and encouraging the development of simpler ‘plain vanilla’ assets.

29.4.1 Central Credit Counterparty

An exchange provides a platform for trading assets that are relatively simple, with well-defined structures, such as government bonds, company shares, commodities, many foreign exchange contracts, and so on. The minimum transaction size is small and there are few barriers to entry. Margin calculations are standardised and public and, crucially, all participants on the exchange transact with a central credit counterparty (CCP).

The CCP stands between the participants on each side of a transaction thereby replacing counterparty credit risk with the risk of the CCP. The CCP is normally protected by a guarantee structure provided by its members. The CCP is party to every transaction and so would stand between the buyer and seller in a credit default swap and the payer and receiver in an interest rate swap. If one of the external parties fails, then the other is protected because their counterparty is the CCP.

CCPs reduce complexity by significantly reducing both the probability of a counterparty failing and the correlation between a trading or investment portfolio and a derivative counterparty. To illustrate the point, a buyer of protection which hedged GM risk with Lehman was exposed both to Lehman default risk and to its correlation with GM. In theory a well-capitalised CCP would exhibit far less of both of these risks. In addition, a CCP offers a framework that embraces mark to market, collateralisation and monitoring of all positions on a daily basis, multilateral netting, mutualisation of risk, and a default fund.

However, a CCP is different from an exchange. As long as the CCP has the resources to analyse a structure and, hence, is content to stand in the middle of two end-user trades, then the complexity of a product does not really matter. Valuations get harder with complexity and, hence, so do margin calculations. Should one end-user fail, the CCP is faced with a challenging hedging or unwinding task. But these complexities are known to the CCP at the time the trade is recommended, and if they are uncomfortable with the structure they can refuse the transaction. The key point is that many such transactions are only known to the CCP and the two end-users. This is clearly not the case with an exchange where all products are available to all members, and the need to simplify products is far more compelling.

29.4.2 Centralised Clearing and Systemic Risk

The move to centralised clearing offers benefits such as transparency, liquidity and reduced counterparty risk. But by concentrating risk in a CCP, systemic risk may not be reduced.

Given the structure of most CCPs and exchanges, solvent members are penalised for non-performance by insolvent members: the guarantee provided by members is neither permanent nor infinite. Furthermore, complex OTC derivatives remain bilateral and these derivatives are believed to have contributed most to the financial crisis.

A CCP absorbs the credit and market risk fluctuations that are produced when OTC derivatives are traded. Rather than this risk being buried in multiple one-to-one relationships throughout the market, it is concentrated in fewer one-to-many relationships. This means that when one market participant fails, only one relationship is affected and the size of the problem is quickly realised.

A CCP can result in there being far fewer outstanding transactions since back-to-back transactions can often be netted down to a one net long or short position. This also means that settlement risk is reduced if all the trades done on one day are netted real-time or at day end. Should some termination event occur (such as a credit event in a CDS) then the termination payments and related settlement will also be reduced. While the close-out of Lehman's ISDA documented derivatives was considered to be a successful process, without the market disruption that many were expecting, there is no doubt that it would have been easier if many of the transactions could have been replaced by the CCP and transferred to other members of the CCP.

It is, however, questionable whether CCP for just CDS trades will necessarily make a substantial contribution to the diminution of systemic risk. Much will depend on the institutional structure of the CCP. A CCP dedicated to a single instrument may concentrate rather than disperse risk, imposing potentially ruinous costs on the central counterparty. Moreover, while the central counterparty may have a wider perception of risk than the individual firm or trader, the systemic risk created by the chains of relationships within a disintermediated market may nonetheless fail to prevent the accumulation of systemic risk and excessive leverage will remain a key issue. Moreover, because a CCP can only deal in relatively standard transactions, it would not have captured any of the AIG trades.

29.4.3 A Dedicated CCP for CDSs Alone

Through the nature of their activity, gross positions at banks and dealers are large and many times net positions, when netted by counterparty or by instrument. This picture is dramatically different for end investors: dealers' trade; investors position. In a systemic crisis, when there is great uncertainty about the survivability of counterparties, market participants worry about gross exposures, and dealers and banks become a great source of systemic risk.

The potential benefit for netting exposures of banks and dealers multilaterally across counterparties and instruments is large and many times the potential benefit for netting exposures bilaterally. There are, therefore, both macro and micro benefits to be had by developing a small number of central counterparties that net across a wide range of instruments. It is inevitable that margin requirements will be pro-cyclical (in an environment of market stress, higher margin requirements will be on those holding distressed assets, forcing them to sell these and other assets, thus spreading the distress). However, this is minimised by the greater the amount of netting and by the more dealers and banks that are subject to counter-cyclical capital requirements.

It is possible that, in the process of enforcing the move of trading to CCPs, systemic risk may be increased. CCPs separated by national boundaries are much less efficient than one pan-regional CCP, but they may well be more efficient that a pan-regional CCP in just one instrument, such as CDSs. CCPs for one instrument reduce netting possibilities across other instruments and is only sensible if that instrument dominates all others. The maximum netting opportunities would be through a single global CCP across the widest number of instruments.

Reducing the systemic risk associated with the 'grossing' of bank positions in a crisis is the key problem to solve, especially given the banks' systemically important role. However, commentators often lump this problem together with matters relating to issuers and investors.

The netting benefits to issuers and investors are small. The difference between gross and net positions is not so large. Having these transactions centrally cleared will bring less private benefit. However, there may be a tendency for these transactions to be structured so that the investor may sell-on his exposure to another investor and have that transaction centrally cleared and settled. The inter-dealer market may set the standard that others will follow in order to ensure the liquidity of instruments.

29.4.4 Conclusions

It is overly simplistic to point blame for market turmoil at an individual financial product. To the extent that credit derivatives, and particularly CDSs, were implicated in the current crisis, it was largely because they wrapped underlying exposures that were simply not sustainable.

Centralised clearing may improve transparency and may be appropriate for certain standardised CDS contracts that are traded in high volumes. However, it would not be appropriate for those transactions where parties are seeking to hedge risks in ways that are tailored to their specific needs.

Reform proposals have clearly been driven by market turmoil. To the extent that some types of credit default swaps were implicated in the financial crisis and the collapse of, for example Lehman and AIG, it is important to distinguish them from other credit derivatives and OTC derivatives in general. Interest, foreign exchange and equity derivatives were not involved, yet are often caught up in proposals for centralised clearing. Among credit derivatives, CDOs on ABSs, particularly those backed by sub-prime mortgages, caused huge liabilities when the market for their underlying obligations collapsed. However, exposures under single name CDS contracts fared well, as indicated by the successful close-out and settlement of the CDS transactions on Lehman's books.

Appendix
Markit Credit and Loan Indices

The following is a description of the main credit indices as of July 2009 and is quoted with permission from Markit Group Limited's material.

Credit and Loan Indices – Markit CDX, Markit iTraxx, Markit LCDX, Markit iTraxx LevX, Markit iTraxx SovX and Markit MCDX – provide the market standards for investing, trading and hedging in the credit markets (Figure A.1).

PRODUCT DESCRIPTIONS

Markit iTraxx

Markit iTraxx indices are the standard European and Asian tradeable credit default swap family of indices. The rules-based Markit iTraxx indices comprise the most liquid names in the European and Asian markets. The selection methodology ensures that the indices are replicable and represent the most liquid, traded part of the market. Markit iTraxx indices are easy and efficient to trade – investors can express their bullish or bearish sentiments on credit as an asset class, and portfolio managers can manage their credit exposures actively.

The benchmark Markit iTraxx Europe index comprises 125 equally-weighted European names. A Markit iTraxx HiVol index consisting of the 30 widest spread non-financial names and three sector indices are also published. The Markit iTraxx Crossover index comprises the 50 most liquid sub-investment grade entities.

The Markit iTraxx Europe indices trade 3-, 5-, 7- and 10-year maturities and a new series is determined by dealer liquidity poll every 6 months. In its role as calculation agent, Markit calculates the official mid-day (11 am GMT) and end-of-day (4 pm GMT) levels for the Markit iTraxx Europe suite of indices on a daily basis.

The Markit iTraxx Asian indices comprise two Markit iTraxx Asia ex-Japan indices (a 50 equally-weighted investment grade and a 20 equally-weighted high-yield CDS index of Asian entities), a Markit iTraxx Australia index (25 equally-weighted Australian entities) and the Markit iTraxx Japan index (50 equally-weighted CDS of Japanese entities).

The Markit iTraxx Asia/Pacific indices typically trade on a 5-year maturity and a new series is determined by dealer liquidity poll every 6 months. In its role as calculation agent, Markit calculates the official end-of-day levels for the Markit iTraxx Asia suite of indices on a daily basis.

Markit iTraxx LevX

The Markit iTraxx LevX indices are the first European indices on Leveraged Loans CDS. They are constructed from the universe of European corporates with leveraged loan exposures. The Markit iTraxx LevX Senior Series 4 index comprises the 40 most liquid first-lien credit agreements traded in the European Leveraged Loan CDS market.

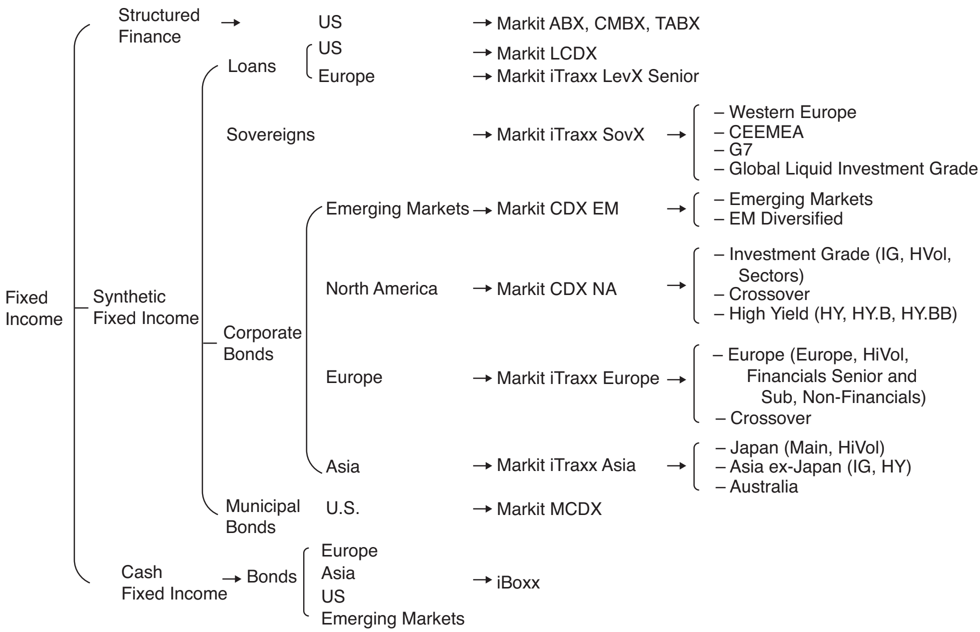

Figure A.1 Markit Credit and Loan Indices overview

Markit iTraxx SovX

The Markit iTraxx SovX indices are a family of sovereign CDS indices covering countries across the globe. The indices have 5- and 10-year maturities and the underlying currency is USD.

Markit CDX

The Markit CDX indices are a family of credit default swap indices covering multiple sectors. The main indices are: Markit CDX North American Investment Grade (125 names), Markit CDX North American Investment Grade High Volatility (30 names from CDX IG), Markit CDX North American High Yield (100 names), Markit CDX North American High Yield High Beta (30 names), Markit CDX North American Emerging Markets (14 names), and Markit CDX North American Emerging Markets Diversified (40 names). The Markit CDX indices roll semi-annually in March and September.

Markit LCDX

Markit LCDX is a tradeable index with 100 equally-weighted underlying single name loan-only credit default swaps (LCDS). The index attempts to cover the more liquid names in the LCDS space. The default swaps each reference an entity with first-lien loans that trade in the secondary leveraged loan market. Each constituent of an Markit LCDX index needs to have current loans listed on the Markit Syndicated Secured List. From series 10 onwards, the index trades with a maturity of both 3 years and 5 years. The Markit LCDX indices roll semi-annually in April and October.

Markit Syndicated Secured List

The Markit Syndicated Secured List (SSL) covers the more liquid and high-profile names in the LCDS space. The SSL provides a list of deliverable loan obligations for the reference entity on which the LCDS contract is written. To qualify, the listed loans must be syndicated, secured and trade with a particular seniority. First-lien, second-lien and third-lien loans can all be included.

The SSL is a trading standard and not a legal one. The reference entities and loans are added and removed from the list on the basis of a polling process. Markit sources the information and administers the process. Poll participants consist of all the LCDX dealers.

Markit MCDX

The Markit MCDX index comprises 50 CDS contracts referencing municipal issuers as the Reference Entity.

COMPARATIVE ANALYSIS

Differences between bond and loan-only CDS indices

	Bonds CDS indices	Loan-only CDS indices
Deliverable	Bonds or Loans	Loans only
Cancellability	CDS trade is not cancellable in cases where all the underlying debt is called or matures	CDS trade is cancellable if an issuer repays all its secured debt without issuing new relevant debt
Valuation	Duration is not adjusted for cancellability	Duration of the index needs to be reduced to account for the cancellability of the loans
Credit Event	The three most commonly used credit events are failure to pay, bankruptcy and restructuring. Also defined but rarely seen are Obligation Acceleration, Repudiation / Moratorium	Bankruptcy, Failure to Pay, and – for Markit iTraxx LevX only – Restructuring

Differences between Markit LevX and Markit LCDX

	LevX	LCDX
Roll Date	March and September 20th	April and October 3rd
Region	Europe	North America
Reference	Reference Obligation	Reference Entity
Currency	EUR	USD or EUR

(*continued*)

(*continued*)

	LevX	LCDX
Credit Event	Bankruptcy, Failure to Pay, Restructuring	Bankruptcy, Failure to Pay
Deliverables	Markit iTraxx LevX Senior includes a Senior (First Lien) index	LCDX only includes First Lien loans
Loan Criteria Eligibility	Senior – min EUR500 million deal	Loans must be on the Markit Syndicated Secured List *
Entities	Senior – 40 names	100 names
Business Days	London and TARGET Settlement Day	USD – New York and London EUR – London and TARGET Settlement Day

* Markit's Syndicated Secured List (SSL) is a database of syndicated secured loans traded in the primary and secondary markets including information about the priority of such loans gathered from market participants and other information. Markit RED maintains the SSL.

Differences between Markit iTraxx and Markit CDX

	iTraxx	CDX
Region	Europe and Asia/Pacific	North America and Emerging Markets
Credit Event	Bankruptcy, Failure to Pay, Modified Restructuring	Bankruptcy, Failure to Pay
Currency	Europe – EUR Japan – JPY Asia ex-Japan – USD Australia – USD	USD, EUR
Reference Entities	Liquidity – A liquidity poll decides inclusions and exclusions	Dealer Poll – Dealers select reference entities to be added and removed
Business Days	London and TARGET Settlement Day	USD – New York and London EUR – London and TARGET Settlement Day

KEY BENEFITS

Credit indices have expanded dramatically in recent years, with volumes rising, trading costs decreasing, and a growing visibility across financial markets. Benefits of using CDS indices include:

- *Tradeability:* Credit indices can be traded and priced more easily than a basket of cash bond indices or single name CDSs

- *Liquidity:* Significant liquidity is available in indices and has also driven more liquidity in the single name market
- *Operational efficiency:* Standardised terms, legal documentation, electronic straight-through processing (STP)
- *Transaction costs:* Cost-efficient means to trade portions of the market
- *Industry support:* Credit indices are supported by all major dealer banks, buy-side investment firms, and third parties (for example, Markit offers transaction processing and valuations services)
- *Transparency:* Rules, constituents, fixed coupon, daily prices are all available publicly

PARTICIPANTS

There are five main parties involved in credit indices:

- **Banks** – Banks trade indices on their own behalf and provide liquidity for their clients.
- **Institutional Investors** – Investors can hedge their positions, or express views on a specific market segments via credit indices
- **Markit** – Markit owns and operates the indices: including licensing, marketing, administration, and calculation. Markit publishes prices daily on its website
- **ISDA** – Markit and banks have worked with ISDA to create globally approved legal documentation for the Markit CDX, Markit LCDX, Markit iTraxx, Markit iTraxx SovX and Markit iTraxx LevX indices
- **Third parties** – Third parties have made trading credit indices easier by integrating them into their platform. For example, Markit Trade Processing allows buyside and sellside firms to communicate and confirm trade details with counterparties, including industry matching utilities like MarkitSERV, a joint venture between Markit and DTCC.

RULES OF CONSTRUCTION

General Rules

Indices roll every 6 months – a new series of the index is created with updated constituents. The previous series continues trading although liquidity is concentrated on the on-the-run series. The roll consists of a series of steps which are administered by Markit:

Liquidity Poll (iTraxx Indices)

- Each market maker submits to Markit a list of entities based on the following criteria:
  - Those entities with the highest single name CDS trading volumes (the 'most liquid' entities), as measured over the previous 6 months. The list is ranked according to trading volumes, i.e. the entity with the highest trading volume is ranked no. 1.
  - Volumes for financial entities are based on Subordinated (Lower Tier 2) transactions only.
  - Internal transactions (e.g. those with an internal proprietary trading desk) are excluded from the volume statistics.
  - Volumes for entities that fall under the same ticker, but trade separately in the CDS market, are summed to arrive at an overall volume for each ticker. The most liquid entity under the ticker qualifies for index membership.

	Index	# Entities (I)	Roll Dates	Maturity in years (2)	Underlying
iTraxx Europe	Europe	125	3/20–9/20	3, 5, 7, 10	Top 125 single name CDS contract by volume
	Non Financials	100	3/20–9/20	5, 10	
	Senior Financials	25	3/20–9/20	5, 10	
	Sub Financials	25	3/20–9/20	5, 10	
	Crossover	50	3/20–9/20	3, 5, 7, 10	Sub-investment grade reference entities
	High Volatility	30	3/20–9/20	3, 5, 7, 10	Top 30 highest spread names from iTraxx Europe
iTraxx Asia	Japan	50	3/20–9/20	5	
	Asia ex-Japan IG	50	3/20–9/20	5	
	Australia	25	3/20–9/20	5	
	Asia ex-Japan HY	20	3/20–9/20	5	
iTraxx SovX	Western Europe	15	3/20–9/20	5, 10	Top 15 Sovereign entities that trade on Western European documentation
	CEEMEA	15	3/20–9/20	5, 10	Top 15 govereign entities that trade on Emerging European documentation
	Global Liquid IG	11–27	3/20–9/20	5, 10	Most liquid high grade global sovereign entities
	G7	Upto 7	3/20–9/20	5, 10	Most liquid industrialised countries
iTraxx LevX	LevX Senior	40	3/20–9/20	5	European First Lien Syndicated Loans

(1) All indices are equally weighted
(2) Exact maturity is 3 months past the years after anniversary (coinciding with coupon payment dates and IMM roll dates)

- **Dealer Polls (CDX Indices)** Dealers select the most liquid names, adjusting for sector diversification based on the exclusion list. Markit then verifies all names that have been selected to ensure they meet the ratings criteria and have bonds outstanding in the underlying company hierarchy; thereby they can be selected as a reference obligation to be published in the annex (more liquid bond, longer maturity, larger outstanding)

	Index	# Entities (1)	Roll Dates	Maturity in years (2)	Underlying	Sub-Indices
LCDX	LCDX	100	4/3–10/3	3, 5	North American First Lien Senior Secured Loans	HVol – 30 names in IG with High Volatility Sectors
CDX	IG	125	3/20–9/20	1, 2, 3, 5, 7, 10	Investment Grade	HY.B, HY.BB, HB
	HY	100	3/27–9/27	3, 5, 7, 10	High Yield	
	XO	35	3/20–9/20	3, 5, 7, 10	Cross-Over (7B or 6B)$^{(3)}$	
	EM	14 (can vary)	3/20–9/20	5, 10	Emerging Markets (Sovereign)	
	EM Div	40	3/20–9/20	3, 5, 7, 10	Emerging Markets Diversified (Sovereign and Corporate)	
MCDX	MCDX	50 credits	4/3–10/3	3, 5, 10	US Municipal Bonds	

(1) All indices are equally weighted – Except for EM, which is decided by poll
(2) Exact maturity is 3 months past the years after anniversary (coinciding with coupon payment dates and IMM roll dates)
(3) 7B – a rating the BBB/Baa rating category by one of S&P, Moody's or Fitch and in the BB/Ba rating category by the other two 6B – a rating the BB/Ba rating category by S&P, Moody's and Fitch

- *Reference obligation assignment:* Markit RED reviews reference obligations for entities in the previous series and updates them if necessary, and proposes new reference obligations for new entities. Selection of the reference obligations are subject to a dealer review and vote.
- *Fixed rate determination:* Licensed Dealers determine the spread for each index and maturity. For both CDX and iTraxx the fixed coupon are set based on the then prevailing fixed coupons used for single name CDS.
- *Assumed recovery rate:* In addition to the fixed rate, Licensed Dealers also vote on an assumed recovery rate used for up-front calculations (not for the determination of cashflows in the case of a credit event). This applies particularly for indices trading in spreads.
- *Annex:* The final annex, stating the composition of each index, and fixed rate is published by Markit. The annex is attached to each index trade confirmation.
- New Series starts trading.

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